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On the Hardness of 4-to-1 Games with Perfect Completeness

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Abstract

We prove that for all $\varepsilon > 0$, there exists a positive integer k such that given a 4-to-1 game Ψ with alphabet size at most k, it is **NP**-hard to distinguish between the case that $\mathrm{val}(\Psi) = 1$ and the case that $\mathrm{val}(\Psi) \leqslant \varepsilon$. This confirms the 4-to-1 Games Conjecture from [Khot, *CCC 2002*]. Previously, the best known result, due to [Dinur, Khot, Kindler, Minzer, Safra], established the almost-perfect completeness version (but applied to the stricter problem of 2-to-1 games). Using results from the literature, we get the following implications:

(1) For all $k \in \mathbb{N}$, given a 3-colorable graph G, it is **NP**-hard to find a proper k-coloring.

(2) For all $\delta > 0$, given a 2-colorable 3-uniform hypergraph G, it is **NP**-hard to find in it an independent set containing at least δ fraction of the vertices.

Our proof is a three-step construction that builds on the two-step framework of [Dinur, Khot, Kindler, Minzer, Safra]. In the outer-PCP step, we use quadratic equations to gain perfect completeness. We then construct a new middle PCP that performs low-rank tests while preserving a key covering property. Finally, we construct a new inner PCP based on a tensor of the standard Grassmann encoding with its low-rank variant due to [Golowich, FOCS 2023].

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1 Introduction

The PCP Theorem [FGL+96, AS98, ALM+98] is a fundamental result in theoretical computer science with many consequences. Most relevant to the current paper are the applications in the area of hardness of approximation, in which the PCP theorem often serves as the starting point for reductions. The combination of Fourier analysis with the long-code (or, in some cases, other error correcting codes) has proven itself to be a very successful recipe, leading to many strong hardness of approximation results [Fei98, Hås01, DS05, Zuc07]. However, this paradigm appears to be incapable of establishing optimal results for many other problems, and below are a few notable examples:

(1) The Max-Cut problem: given a graph $G = (V, E)$, the goal is to partition the vertex set into two sides A, B so as to maximize the number of edges crossing between A and B. A polynomial time algorithm due to [GW95] achieves the approximation ratio $\alpha_{GW} \approx 0.878$, and the best known **NP**-hardness result [TSSW00] has ratio $\frac{16}{17} + \varepsilon$, for any $\varepsilon > 0$.

(2) The Vertex-Cover problem: given a graph $G = (V, E)$, the goal is to find the smallest set of vertices $A \subseteq V$ that touches all edges of E. Known polynomial time algorithms achieve approximation ratio 2, and the best known **NP**-hardness result [KMS25, DKK+25, DKK+21, KMS23] has ratio $\sqrt{2} - \varepsilon$, for any $\varepsilon > 0$.

(3) The 3-coloring problem: given a graph $G = (V, E)$ promised to be 3-colorable, the goal is to find a proper k-coloring of the vertices of G with k as small as possible. Polynomial time algorithms for this problem have been studied extensively over the years [Wig83, KMS98, BK97, ACC06, Chl07, KT17, KTY24, BHL26], and the state-of-the-art algorithm achieves $k = |V|^{0.2-\varepsilon}$ for a small explicit $\varepsilon > 0$. The **NP**-hardness of this problem has also been studied [KLS00, GK04, BBKO21], and the state-of-the-art **NP**-hardness result is for $k = 5$.

To make progress on these problems, Khot [Kho02b] introduced the Unique-Games Conjecture and the d-to-1 Games conjectures, which we present next.

1.1 Label-Cover, Unique Games and d-to-1 Games

It is convenient to discuss PCPs in the language of the label-cover problem, defined as follows.

Definition 1.1. An instance Ψ of the label-cover problem consists of a bipartite graph $G = (L \sqcup R, E)$,[1] alphabets Σ_L, Σ_R, and a collection of constraints $\{\Phi_e \subseteq \Sigma_L \times \Sigma_R\}_{e \in E}$. For any given positive integers d and d', a constraint $\Phi_e \subseteq \Sigma_L \times \Sigma_R$ is called a d-to-d' constraint if there is a partition $\Sigma_L = \Sigma_{L,1} \sqcup \cdots \sqcup \Sigma_{L,t}$ into sets of size d and a partition $\Sigma_R = \Sigma_{R,1} \sqcup \cdots \sqcup \Sigma_{R,t}$ into sets of size d' such that

$$\Phi_e = \bigcup_{i=1}^{t} \Sigma_{L,i} \times \Sigma_{R,i}.$$

Given a label-cover instance Ψ and assignments $A_L \colon L \to \Sigma_L$, $A_R \colon R \to \Sigma_R$, we define

$$\mathrm{val}_\Psi(A_L, A_R) = \frac{|\{e = (u, v) \in E \mid (A_L(u), A_R(v)) \in \Phi_e\}|}{|E|},$$

that is, the fraction of constraints that are satisfied by A_L, A_R. We define the value of Ψ as

$$\mathrm{val}(\Psi) = \max_{A_L, A_R} \mathrm{val}_\Psi(A_L, A_R).$$

[1] We always allow parallel edges, so E is a multi-set.

For parameters $0 < s < c \leqslant 1$, an input to the promise problem $\mathsf{GapLC}(c, s)$ is a label cover instance Ψ that either has $\mathrm{val}(\Psi) \geqslant c$ or $\mathrm{val}(\Psi) \leqslant s$, and the goal is to distinguish between the two cases. The basic PCP theorem [FGL+96, AS98, ALM+98] combined with the parallel repetition theorem [Raz98] gives:

Theorem 1.2. *For all $\varepsilon > 0$, there exists a positive integer r such that $\mathsf{GapLC}_r(1, \varepsilon)$ is* **NP***-hard. Here GapLC_r is the GapLC problem in which both alphabets have size at most r.*

Khot conjectured that versions of Theorem 1.2 continue to hold even when the constraints are d-to-1 for a small d. Towards presenting these conjectures, we make the following definition.

Definition 1.3. Let d and d' be positive integers. An instance Ψ of label-cover is called a d-to-d' game if all of its constraints are d-to-d'. In the case that $d = d' = 1$, Ψ is called a unique-game. We write $\mathsf{GapUG}_r(c, s)$ for the restriction of $\mathsf{GapLC}_r(c, s)$ to unique-game instances, and Gap-d-to-$d'_r(c, s)$ for its restriction to d-to-d' game instances.

It is easy to see that one cannot hope for Theorem 1.2 to hold for unique-games, and Khot [Kho02b] conjectured that allowing for imperfect completeness remedies that:

Conjecture 1.4. *For any $\varepsilon > 0$, there exists $r \in \mathbb{N}^+$, such that $\mathsf{GapUG}_r(1 - \varepsilon, \varepsilon)$ is* **NP***-hard.*

For $d \geqslant 2$ and $d' = 1$, he conjectured that the assertion of Theorem 1.2 should hold as is. More precisely:

Conjecture 1.5. *For any $\varepsilon > 0$ and any integer $d \geqslant 2$, there exists a positive integer r such that Gap-d-to-$1_r(1, \varepsilon)$ is* **NP***-hard.*

These conjectures, and especially Conjecture 1.4, have been shown to have remarkable implications, see for example [KR08, KKMO07, Rag08, KV15] and [Kho10b, Kho10a, Tre12, Kho14] for surveys, and significant effort has gone into studying them. On the algorithmic front, connections with semi-definite programming, sum-of-squares algorithms and graph expansion have led to the development of very powerful techniques and a number of algorithms for unique-games with non-trivial guarantees [CMM06, AKK+08, ABS15]. On the hardness front, a line of work [KMS25, DKK+25, DKK+21, KMS23] (with contributions from [BKS19, KMMS25]) established the imperfect completeness version of Conjecture 1.5, showing that for all $\varepsilon > 0$, there is $r \in \mathbb{N}^+$ such that Gap-2-to-$1_r(1 - \varepsilon, \varepsilon)$ is **NP**-hard. This implies that for all $\varepsilon > 0$ there exists $r \in \mathbb{N}^+$ such that $\mathsf{GapUG}_r(1/2, \varepsilon)$ is **NP**-hard, which to date is the best evidence towards Conjecture 1.4.

The difference between Conjecture 1.5 and its imperfect completeness version is significant in a number of ways:

(1) **Implications:** there are a number of hardness of approximation results that crucially rely on perfect completeness, most prominently graph coloring problems.

(2) **Techniques:** the approach taken by [KMS25, DKK+25, DKK+21, KMS23] inherently incurs imperfect completeness. Currently, constructions of low-soundness PCPs with perfect completeness follow composition of PCPs, for example [MR10, DH13, DHK15], which at the very least require constructions that already have perfect completeness and small soundness (with perhaps a large alphabet which is then reduced), or from parallel repetition [Raz98, Hol09, Rao11], which inherently increases the d-parameter rapidly.

1.2 Main Result

Our main result is that Conjecture 1.5 holds for $d = 4$ and $d' = 1$.

Theorem 1.6. *For all $\varepsilon > 0$, there exists $r \in \mathbb{N}^+$, such that Gap-4-to-$1_r(1, \varepsilon)$ is* **NP***-hard.*

1.2.1 Implications

We now discuss some implications of Theorem 1.6. Using the known connection between d-to-1 games and graph coloring/independent sets [DMR09, GS20], we get the following results.

Corollary 1.7. *For any integer $k \geqslant 3$, given a graph G, it is* **NP***-hard to distinguish between the following two cases:*

(1) ***Yes case:*** *G is 3-colorable.*

(2) ***No case:*** *G is not k-colorable.*

Previously, Corollary 1.7 was known [BBKO21] for $k = 5$, or in the case the condition in the yes-case only required a proper 3-coloring of $1 - \varepsilon$ fraction of the vertices [HMS23].

Remark 1.8. Under slightly super-polynomial time reductions, our techniques give versions of Corollary 1.7 for k growing with the instance size, but at a very slow rate. Namely, assuming **NP** has no algorithms running in time $n^{t(n)}$, we get that there are no polynomial time algorithms that color a given 3-colorable graph with $O(\log \cdots \log t(n))$ colors, where the number of logs is finite (probably at most 10, say).

In the next corollary, the condition in the completeness case is weaker, but the condition in the soundness case is stronger:

Corollary 1.9. *For all $\delta > 0$, given a graph G, it is* **NP***-hard to distinguish between the following two cases:*

(1) ***Yes case:*** *G is 8-colorable.*

(2) ***No case:*** *G has no independent set of relative size at least δ.*

Using [KS14], we get the following result for hypergraphs:

Corollary 1.10. *For all $\delta > 0$, given a 3-uniform hypergraph G, it is* **NP***-hard to distinguish between the following two cases:*

(1) ***Yes case:*** *G is 2-colorable.*

(2) ***No case:*** *G has no independent set of relative size at least δ.*

Using the results of [HMRW14], Corollaries 1.7 and 1.9 show hardness results for low-rank matrix completion. An instance in the (k, r, ε, c)-completion problem is a matrix $A \in (\mathbb{R} \cup \{\perp\})^{n \times n}$, such that there exists $M \in \mathbb{R}^{n \times n}$ satisfying:

(1) $|M(i,j)| \leqslant c$ and M has rank at most r.

(2) Letting $\Omega \subseteq [n] \times [n]$ be the set of (i,j) such that $A(i,j) \neq \perp$, we have that $|\Omega| \geqslant 0.9n^2$ and for each $(i,j) \in \Omega$ we have $M(i,j) = A(i,j)$.

The goal is to find a matrix $B \in \mathbb{R}^{n \times n}$ of rank at most k, such that $|B(i,j)| \leqslant c$ for all (i,j) and

$$\sum_{(i,j) \in \Omega} |A(i,j) - B(i,j)|^2 \leqslant \varepsilon n.$$

Applying [HMRW14, Theorem 1.8] on Corollary 1.7 gives:

Corollary 1.11. *For all $k > 3$ and $c > 1$, the $(k, 3, 0, c)$-completion problem is* **NP***-hard.*

Applying [HMRW14, Theorem 1.9] on Corollary 1.9 gives:

Corollary 1.12. *For all $k > 8$ and $c > 1$, the $(k, 8, \varepsilon, c)$-completion problem is* **NP***-hard for all $0 < \varepsilon < \frac{1}{(2ck)^2}$.*

1.3 Our Techniques

In this section we give a brief account of the components that go into the proof of Theorem 1.6. We refer the reader to Section 2 for a more thorough description.

For 2-to-1 games with almost-perfect completeness, the proof of [KMS25, DKK+25, DKK+21, KMS23] combines the following two components:

(1) An outer PCP with completeness close to 1, vanishing soundness, and a feature called the "covering property" (originally appearing in [KS13]). This game is very far from being a 2-to-1 game, and to turn it into a 2-to-1 game it is composed with an appropriate inner PCP. The goal of the covering property is to facilitate the composition step.

(2) An inner PCP with perfect completeness and a non-standard notion of soundness. The inner PCP is based on the Grassmann graph, the Grassmann encoding and the Grassmann agreement test. The Grassmann agreement test can be viewed as a local tester for the Grassmann code, but it turns out not to be sound in the traditional error correcting codes sense. Instead, it satisfies a weaker notion of soundness, which we refer to as non-standard soundness.

The outer PCP and inner PCP are then composed, and the covering property of the outer PCP is used in the soundness analysis of the composed PCP (even though the inner PCP only satisfies a non-standard notion of soundness).

The outer PCP construction of [KMS25, DKK+25, DKK+21, KMS23] is based on the hardness of approximation result of [Hås01] for solving linear equations, which inherently has imperfect completeness, and this propagates throughout their construction. To remedy that, a key difficulty is to come up with an outer PCP that has perfect completeness and simultaneously admits (some version of) the covering property.

We start our outer PCP construction based on hardness of solving a system of quadratic equations (which has perfect completeness). However, the Grassmann agreement test (which is still basically the only tool we have to achieve d-to-1-ness in the inner PCP) is inherently a "linear" test and cannot incorporate quadratic equations. In order for quadratic equations to be accommodated in a d-to-1 way, we add a layer of construction called the "middle PCP" between the outer and inner PCPs. In the middle PCP, we carry out a low-rank test that allows us to convert the quadratic equations into linear ones.

However, this linearization of quadratic equations destroys the covering property that is essential for composing with a Grassmann-based inner PCP. To remedy that, we have to come up with another two key ideas:

(1) We do a "lazy parallel repetition" in the middle PCP. This creates room for covering properties while preserving the soundness.

(2) The covering property we get from lazy parallel repetition is still not sufficient for composing with the Grassmann agreement test of [KMS25, DKK+25, DKK+21, KMS23]. As a workaround, we base our inner PCP on a combination of standard Grassmann graphs over matrix spaces $\mathbb{F}_2^{n\times n}$ with a sparser version of them, known as the low-rank Grassmann graph over $\mathbb{F}_2^{n\times n}$, recently defined by [Gol23].

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2 Proof Overview

In this section we give an overview of the proof of Theorem 1.6. It will be useful to first discuss the construction of [KMS25, DKK+25, DKK+21, KMS23] which achieves imperfect completeness. We then discuss why it fails to have perfect completeness, and then in Section 2.2 we explain preliminary attempts at fixing that, building towards our construction.

2.1 The Imperfect Completeness Construction

In this section we explain the components that go into the construction of [KMS25, DKK+25, DKK+21, KMS23].

2.1.1 The Outer PCP

It will often be easier for us to describe label-cover instances in the equivalent, active language of 2-prover-1-round games, which we introduce next. In a 2-prover-1-round game, a verifier samples a challenge $e = (u, v)$, sends u to a prover named Alice, sends v to a prover named Bob. Upon receiving their challenges, they respond with labels (they are not allowed to communicate once they get their challenges), and based on their responses the verifier decides to accept or reject. The value of the game is the maximum, over all strategies of Alice and Bob, of the probability that the verifier accepts. Identifying the individual challenges the players receive as vertices of a bipartite graph, pairs of challenges the verifier may sample with its edges, and the verifier's acceptance criteria with the constraints, one gets the equivalence. It is easy to see that the value of a game is exactly the value of the resulting label cover instance.

With this in mind, the reduction of [KMS25, DKK+25, DKK+21, KMS23] starts with the result of [Hås01] regarding the approximability of $\mathsf{3Lin}_{\mathbb{F}_2}$.

Definition 2.1. An instance of the $\mathsf{3Lin}_{\mathbb{F}_2}$ problem consists of a set of variables $\mathscr{V}$ and a set of equations $\mathscr{E}$, where each $e \in \mathscr{E}$ is a linear equation over $\mathbb{F}_2$ containing 3 distinct variables.

Håstad proved that for all $\varepsilon > 0$, $\mathsf{Gap3Lin}_{\mathbb{F}_2}(1-\varepsilon, 1/2+\varepsilon)$ is **NP**-hard, and the reduction of [KMS25, DKK+25, DKK+21, KMS23] starts off with (some regularized version of) this problem. More specifically, they use the parallel repetition theorem to prove **NP**-hardness for a certain type of label-cover instance $\Psi_{q,\beta,k} = \Psi_{q,\beta,k}(\mathscr{V}, \mathscr{E})$, which we present below in the active 2-prover-1-round game view, and refer to as the β-smooth k-fold repeated game. For technical reasons, this game contains an additional feature called "advice", which we encourage the reader to ignore in a first reading.

1. **Alice's challenge:** the verifier samples equations $e_1, \ldots, e_k$ from $\mathscr{E}$ uniformly and independently, and sets E to be the set of all variables that appear in them.
2. **Bob's challenge:** for each $i \in \{1, \ldots, k\}$ independently, with probability $1-\beta$, the verifier takes E'_i to be the set of variables in e_i, and with probability β the verifier chooses E'_i to consist of a single uniformly chosen variable from e_i. The verifier sets $E' = E'_1 \cup \ldots \cup E'_k$.
3. **Advice:** the verifier chooses $b^{(1)}, \ldots, b^{(q)} \in \mathbb{F}_2^{E'}$ uniformly, and then takes $a^{(1)}, \ldots, a^{(q)} \in \mathbb{F}_2^{E}$ where $a^{(i)}$ is obtained from $b^{(i)}$ by filling the missing coordinates with 0's. We write $B = \left(b^{(1)}, \ldots, b^{(q)}\right)$ and $A = \left(a^{(1)}, \ldots, a^{(q)}\right)$.
4. **Acceptance criteria:** the verifier sends (E, A) to Alice, and (E', B) to Bob. They respond with vectors $x \in \mathbb{F}_2^E$ and $y \in \mathbb{F}_2^{E'}$ respectively, and the verifier accepts if and only if x satisfies all of the equations $e_1, \ldots, e_k$ and $x|_{E'} = y$.

The works [KMS25, DKK+25, DKK+21, KMS23] prove that for all $\varepsilon > 0$, it is **NP**-hard to distinguish between the case that $\mathrm{val}(\Psi_{q,\beta,k}) \geqslant 1-\varepsilon$, and the case that $\mathrm{val}(\Psi_{q,\beta,k}) \leqslant 2^{-\Omega_q(\beta k)}$. Here and throughout, one should think of q as an absolute constant, and of β as slightly larger than $1/k$, say $\beta = k^{-3/4}$. The game $\Psi_{q,\beta,k}$ has several useful features in the context of 2-to-1 Games:

(1) **Simplicity of the constraints:** the constraints of $\Psi_{q,\beta,k}$ can all be seen to be linear equations in the responses of the provers. This is because the starting point, the $3\mathsf{Lin}_{\mathbb{F}_2}$, consisted of linear constraints, and on top of that we have the consistency constraints between the responses of Alice and Bob, which are also linear.

(2) **The covering property:** for a fixed E, the spaces $\mathbb{F}_2^{E'}$ living on Bob's side cover Alice's space $\mathbb{F}_2^E$ almost perfectly. For example, fixing E, taking E' randomly as described above conditioned on E, then $y \in \mathbb{F}_2^{E'}$ and lifting it to $x \in \mathbb{F}_2^E$ by filling in the missing coordinates with 0's, we have that the distribution of x is statistically close to the uniform distribution over $\mathbb{F}_2^E$ (this property first appeared in [KS13] in a somewhat different context).

The relevance of these properties will become clear in the next part, the inner PCP.

2.1.2 The Inner PCP

The 2-prover 1-round game $\Psi_{q,\beta,k}$ constructed in Section 2.1.1 is far from 2-to-1, because given a Bob-answer $y \in \mathbb{F}_2^{E'}$, there are many possible Alice-answers $x \in \mathbb{F}_2^E$ consistent with it. In order to make it a 2-to-1 game, in a single query we can ask the provers only for a smaller piece of information.

For each possible question (E, A) to Alice, we ask her to prepare her answer in "Grassmann encoding," and instead of requesting a full answer $x \in \mathbb{F}_2^E$, we ask her to return only the information in one location of the Grassmann encoding of x.

The Grassmann encoding scheme is based on the Grassmann graph $\mathsf{Gr}(\mathbb{F}_2^E, \ell)$, whose vertices are the ℓ-dimensional linear subspaces of $\mathbb{F}_2^E$. Two vertices L and L' are adjacent if and only if $\dim(L \cap L') = \ell - 1$. By a slight abuse of notation, we also use $\mathsf{Gr}(\mathbb{F}_2^E, \ell)$ to denote the vertex set of this graph. The Grassmann encoding of a vector $x \in \mathbb{F}_2^E$ stores, at each location $L \in \mathsf{Gr}(\mathbb{F}_2^E, \ell)$, the local linear function

$$F[L] : L \to \mathbb{F}_2 \text{ defined by } z \mapsto \langle z, x \rangle .$$

Similarly, we ask Bob to prepare his answer $y \in \mathbb{F}_2^{E'}$ in Grassmann encoding, where each location $L' \in \mathsf{Gr}(\mathbb{F}_2^{E'}, \ell - 1)$ stores the local function

$$G[L'] : L' \to \mathbb{F}_2 \text{ defined by } z \mapsto \langle z, y \rangle .$$

The final 2-to-1 game is the composition of the outer PCP with the Grassmann graph gadget. In the composed PCP, upon choosing (E, A) and (E', B) as in $\Psi_{q,\beta,k}$, the verifier chooses a space L' from $\mathsf{Gr}(\mathbb{F}_2^{E'}, \ell - 1)$ and a space $L \in \mathsf{Gr}(\mathbb{F}_2^{E'}, \ell)$ containing L'. The subspace $L \subseteq \mathbb{F}_2^{E'}$ will be regarded as a subspace of Alice's space $\mathbb{F}_2^E$ by appending 0's in the missing coordinates. The verifier then sends (E, A, L) to Alice and (E', B, L') to Bob, expecting to get linear functions $F[E, A, L] \colon L \to \mathbb{F}_2$ and $G[E', B, L'] \colon L' \to \mathbb{F}_2$. The verifier accepts only if $G[E', B, L']$ agrees with the restriction $F[E, A, L]|_{L'}$. This is essentially a 2-to-1 game because for any possible choice of $G[E', B, L']$, there are at most two choices of $F[E, A, L]$ consistent with it.

Covering property and consistency: consider a question (E, A) for Alice in the outer PCP (think of ℓ as much smaller than k), and the Grassmann graph $\mathsf{Gr}(\mathbb{F}_2^E, \ell)$. The covering property above asserts that if we sample E' as in the outer PCP and a vertex in the graph $\mathsf{Gr}(\mathbb{F}_2^{E'}, \ell)$, lifting it to $\mathsf{Gr}(\mathbb{F}_2^E, \ell)$ yields a nearly uniform vertex in the latter graph. This implies that if Alice's assignment $F[E, A, \cdot]$ has agreement ε' with an actual Grassmann codeword F, then with decent probability over E', Bob's assignment $G[E', B, \cdot]$ agrees with the projection of F to $\mathbb{F}_2^{E'}$. In short, the covering property ensures the consistency between decodings of assignments of the provers holds.

The linear equations side conditions: recall that in the outer PCP we want to ensure that Alice's response $x \in \mathbb{F}_2^E$ satisfies a collection of k linear equations. Note that these equations correspond to requiring that the function $z \mapsto \langle z, x \rangle$ attains specific values on k inputs. More precisely, letting the equations be $e_1, \ldots, e_k$, letting $s^{(i)} \in \mathbb{F}_2^E$ be the indicator vector of the variables appearing in e_i, and letting $t_i \in \mathbb{F}_2$ be the constant term of the equation e_i, we want to ensure that $\langle s^{(i)}, x \rangle = t_i$. To do so, we define the space $N_E = \mathrm{span}(s^{(1)}, \ldots, s^{(k)})$, and request Alice to give assignment $F[E, A, L + N_E] \colon (L + N_E) \to \mathbb{F}_2$ (instead of just $L \to \mathbb{F}_2$). The verifier then checks consistency between Alice and Bob similarly to the above, and additionally that $F[E, A, L + N_E]|_{N_E} = g_E$, where the linear function $g_E \colon N_E \to \mathbb{F}_2$ is defined by $g_E(s^{(i)}) = t_i$.

2.1.3 On the Non-Standard Notion of Soundness

One would wish that the Grassmann gadget in the inner PCP has the following type of soundness: suppose $F[L]$ is a linear map $L \to \mathbb{F}_2$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^E, \ell)$, such that

$$\mathop{\mathbb{P}}_{L_1, L_2 \in \mathsf{Gr}(\mathbb{F}_2^E, \ell)} \left[F[L_1]|_{L_1 \cap L_2} = F[L_2]|_{L_1 \cap L_2} \,\middle|\, \dim(L_1 \cap L_2) = \ell - 1 \right] \geqslant \delta \tag{2.1}$$

for some constant $\delta \in (0, 1)$, then there exists a linear function $f : \mathbb{F}_2^E \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathbb{F}_2^E, \ell)} \left[F[L] = f|_L \right] \geqslant \delta'$$

for some $\delta' \in (0, 1)$. However, this conclusion cannot be made, as we next explain. Take $\varepsilon > 0$ to be a small constant, and take a random set $X \subseteq \mathbb{F}_2^E \setminus \{0\}$ of size $2^{-\ell} \varepsilon \cdot 2^{|E|}$. For each $x \in X$ choose a random linear function $f_x \colon \mathbb{F}_2^E \to \mathbb{F}_2$, and set a table F on ℓ-dimensional subspaces L as follows:

- If L contains a unique point $x \in X$, define $F[L] = f_x|_L$.
- Otherwise, choose $F[L]$ as a random linear function on L.

Using the inclusion-exclusion formula, one can prove that for a random edge $\{L_1, L_2\}$ in the Grassmann graph, with probability at least $\Omega(\varepsilon)$ there is $x \in X$ such that $L_1 \cap X = L_2 \cap X = \{x\}$, and F satisfies such constraints. Thus, F passes the Grassmann test Equation (2.1) with $\delta = \Omega(\varepsilon)$. However, it only agrees on $o(1)$ locations with any given codeword from the Grassmann code.

This example tells us that while there might be no linear function that agrees with F globally, it may be possible to extract a linear function by focusing on certain substructures inside the Grassmann graph. Indeed, it turns out that what can be concluded from Equation (2.1) is something of this form: for at least $\exp(-O(\ell^2))$ fraction of $Q \in \mathsf{Gr}(\mathbb{F}_2^E, q)$, where q depends only on δ, there exists $W \in \mathsf{Gr}(\mathbb{F}_2^E, |E| - q)$ and a linear function $f : W \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathbb{F}_2^E, \ell)} \left[F[L] \equiv f|_L \,\middle|\, Q \subseteq L \subseteq W \right] \geqslant \delta'. \tag{2.2}$$

We refer to the spaces Q and W in Equation (2.2) as zoom-ins and zoom-outs, respectively, and encourage the reader to ignore W (it will be irrelevant for the majority of the discussion). In words, Equation (2.2) asserts that if F passes the Grassmann test with noticeable probability, then inside non-trivial fraction of zoom-ins of constant dimension q, it does have a global linear structure.

The proof of [KMS25, DKK+25, DKK+21, KMS23] has to deal with this non-standard notion of soundness, and this is the purpose of the advice feature in the game $\Psi_{q,\beta,k}$. More precisely, given Alice and Bob's tables of assignments F and G for the composed PCP, one needs to be able to derive a strategy for the original game $\Psi_{q,\beta,k}$. However, the strategy that one can derive using Equation (2.2) heavily depends on the choice of what zoom-in Q is used. The advice feature allows the two provers in $\Psi_{q,\beta,k}$ to use (essentially) the same Q for decoding — namely the span of their advice vectors — which guarantees consistency between their answers.

2.1.4 Difficulties in Achieving Perfect Completeness

The construction for 2-to-1 games with imperfect completeness can be illustrated using Figure 1.



Figure 1: The imperfect-completeness construction of [KMS25, DKK+25, DKK+21, KMS23]

The above construction loses perfect completeness right from the start: it uses the $3\mathsf{Lin}_{\mathbb{F}_2}$ problem, which inherently has imperfect completeness. Moreover, in some sense this seems necessary: the Grassmann encoding presented above only allows for linear checks in the encoded string $x \in \mathbb{F}_2^n$.

Thus, to achieve perfect completeness it is necessary to come up with an encoding scheme which is more powerful than the Grassmann encoding, in the sense that it allows for non-linear checks on the encoded string $x \in \mathbb{F}_2^E$. By itself this is not a hard task; for example, the long-code would work. However, the key difficulty is that one additionally needs the encoding scheme to satisfy a covering property, and then build an outer PCP with a suitable advice feature.

2.2 Our Techniques

In this section we discuss the new ideas that go into our construction.

2.2.1 A First Attempt: the Quadratic Grassmann Encoding

We begin by outlining a natural first attempt at a reduction achieving perfect completeness. It starts with the problem of quadratic equations, $3\mathsf{QS}_{\mathbb{F}_2}$, defined as follows.

Definition 2.2. An instance of the $3\mathsf{QS}_{\mathbb{F}_2}$ problem consists of a set of variables $\mathscr{V}$ and a set of equations $\mathscr{E}$, where each $e \in \mathscr{E}$ is a quadratic equation over $\mathbb{F}_2$ containing 3 distinct variables.

Using standard results, it can be proved that there is $s < 1$ such that $\mathsf{Gap3QS}_{\mathbb{F}_2}[1, s]$ is **NP**-hard. Starting with an instance $(\mathscr{V}, \mathscr{E})$ of $3\mathsf{QS}_{\mathbb{F}_2}$, one can construct the β-smooth k-fold repeated game corresponding to it, $\Psi_{\beta,k} = \Psi_{\beta,k}(\mathscr{V}, \mathscr{E})$ in the same way as in Section 2.1.1 (we ignore the advice for now). The main question is how to compose it with a Grassmann-style encoding so as to get to a d-to-1 Game.

In the game $\Psi_{\beta,k}$, the verifier performs two types of checks: consistency between the responses of Alice and Bob, and quadratic equations in the response of Alice. The Grassmann encoding is compatible with the former type of checks, but not with the latter type of checks, in the sense that we do not know how to incorporate checking quadratic equations on the encoded message into the Grassmann encoding itself. Instead, it is natural to consider the *quadratic Grassmann encoding*.

Let $x \in \mathbb{F}_2^E$ represent Alice's response in the game $\Psi_{\beta,k}$. In the quadratic Grassmann encoding, one considers the functions over $\mathbb{F}_2^{E\times E}$ defined by $z \mapsto \langle z, x \otimes x\rangle$, and takes its Grassmann encoding QF of x, which assigns to each ℓ-dimensional subspace $L \subseteq \mathbb{F}_2^{E\times E}$ the linear function

$$QF[L] : L \to \mathbb{F}_2 \text{ defined by } z \mapsto \langle z, x \otimes x\rangle .$$

The main benefit of the quadratic Grassmann encoding is that it indeed allows us to test quadratic equations as side conditions, as we explain next.

The quadratic equations side conditions: Given legitimate quadratic Grassmann codeword QF, it is possible to incorporate with it checks of quadratic equations on the encoded message. Indeed, suppose that $e_1, \ldots, e_k$ are quadratic equations over E, let $s^{(1)}, \ldots, s^{(k)} \in \mathbb{F}_2^{E\times E}$ be the indicator vectors of the pairs that appear in $e_1, \ldots, e_k$ respectively, and let $t_1, \ldots, t_k$ be the constant terms. Then the equation e_i can be written as $\langle s^i, x \otimes x\rangle = t_i$ for each $i \in \{1, \ldots, k\}$. As before, this can be folded into the encoding by defining $N_E = \text{span}(s^{(1)}, \ldots, s^{(k)})$, asking for the table to assign values to the sum $L + N_E$ (instead of to just L), and require that their projection onto N_E completely agrees with the linear function $g\colon N_E \to \mathbb{F}_2$ defined by $g(s^{(i)}) = t_i$. This way of incorporating quadratic side conditions will work for all of the constructions we present below, so for simplicity of presentation they will be omitted henceforth.

On the other hand, the quadratic Grassmann encoding has 2 very significant drawbacks:

(1) **Codeword testing:** Even ignoring the issue of non-standard-type soundness, the Grassmann agreement test from Section 2.1.2 is not actually a tester for the quadratic Grassmann code. Namely, if a table G that assigns each ℓ-dimensional subspace $L \subseteq \mathbb{F}_2^{E\times E}$ a linear function $G[L]\colon L \to \mathbb{F}_2$ passes it with non-negligible probability, then at best we will be able to argue that it has significant agreement with the Grassmann encoding of an arbitrary $x' \in \mathbb{F}_2^{E\times E}$, i.e., not necessarily of rank 1. We need an additional feature that will guarantee that x' has rank 1 (or at the very least small rank) and is symmetric.

(2) **The covering property:** The covering property does not hold for the Grassmann encoding. Namely, fixing a question for Alice E and looking at typical possible questions E' for Bob (conditioned on E), the (lifting of the) spaces $\mathbb{F}_2^{E'\times E'}$ do not cover $\mathbb{F}_2^{E\times E}$. Indeed, the number of such typical questions for Bob is roughly $\binom{3k}{2\beta k}$, and the size of Bob's space is roughly $2^{(3-2\beta)^2k^2}$, so in total they can cover at most

$$\binom{3k}{2\beta k} 2^{(3-2\beta)^2k^2} \leqslant 2^{O(\beta k \log(1/\beta))} 2^{9k^2-8\beta k^2} = o(2^{9k^2})$$

points in $\mathbb{F}_2^{E\times E}$, which is a tiny fraction.

The first issue, namely ensuring that the encoded message has a low rank, is significantly easier to handle, and to do so we incorporate tensor tests. More precisely, instead of asking for the value of the Grassmann encoding at a subspace $L \subseteq \mathbb{F}_2^{E\times E}$, we will also take a random subspace $U \subseteq \mathbb{F}_2^E$ of dimension ℓ, and then ask for the value at the subspace $L + U \otimes U$, requiring that the projection onto $U \otimes U$ is of rank 1. This yields the following 4-to-4 Grassmann agreement + tensor test. Upon receiving a table F assigning linear functions to spaces $L + U \otimes U$ as above, do the following:

1. Sample $L_1, L_2 \subseteq \mathbb{F}_2^{E \times E}$ of dimension ℓ intersecting at dimension $\ell - 1$, and $U_1, U_2 \subseteq \mathbb{F}_2^E$ of dimension ℓ intersecting at dimension $\ell - 1$.

2. Query $F[L_1 + U_1 \otimes U_1]$ and $F[L_2 + U_2 \otimes U_2]$. Accept if they agree on

$$L_1 \cap L_2 + (U_1 \cap U_2) \otimes (U_1 \cap U_2),$$

and their respective projections onto $U_1 \otimes U_1$ and $U_2 \otimes U_2$ are symmetric and of rank at most 1.

Given the (non-standard) soundness of the basic Grassmann agreement test, one can analyze the Grassmann agreement + tensor test and show that it satisfies a similar (non-standard) notion of soundness. Also, this test is 4-to-4: given a linear function on L_1, there are at most 2 linear functions on L_2 that agree with it, and given a symmetric rank-1 functional on $U_1 \otimes U_1$, there are at most 2 symmetric rank-1 linear functions on $U_2 \otimes U_2$ agreeing with it.

Ensuring symmetry is more straightforward. To do that, define for distinct points $i, j \in E$ the matrix $M_{i,j} \in \mathbb{F}_2^{E \times E}$ which is 1 only at locations (i, j) and (j, i), and set $N_{\mathsf{sym}} = \mathrm{span}(\{M_{i,j} \mid i \neq j\})$. Then, $x' \in \mathbb{F}_2^{E \times E}$ is symmetric if and only if the function $z \mapsto \langle z, x' \rangle$ vanishes on N_{sym}. Thus, instead of requesting an assignment on $L + U \otimes U$, the verifier requests an assignment on $L + U \otimes U + N_{\mathsf{sym}}$, and checks that its projection on N_{sym} vanishes.

The second issue, namely the fact that the covering property fails, is substantially more difficult to handle, and in the next section we begin the discussion about it.

2.2.2 Variations on the Quadratic Grassmann Encoding

From now on, it will be convenient for us to think of points from $\mathbb{F}_2^{E \times E}$ and $\mathbb{F}_2^{E' \times E'}$ as matrices. The covering property fails for the quadratic Grassmann encoding because if, letting Bob's question be denoted by E', we have $i \in E \setminus E'$, then i-th row and column will always be 0 in liftings from $\mathbb{F}_2^{E' \times E'}$ to $\mathbb{F}_2^{E \times E}$. On the other hand, typical matrices from $\mathbb{F}_2^{E \times E}$ do not contain fully-0 rows. This raises the question of whether we can replace the structure of $E \times E$ with a sparser structure that circumvents this atypical behaviour of liftings. Moreover, requiring a response from the provers for each pair of variables in their respective variable sets is not necessary, as the quadratic equations we wish to check only require access to $\Theta(k)$ pairs (out of the $\binom{k}{2}$ many ones that exist).

This view motivates the following general recipe: can we find a graph H with vertex set E and edge set $\mathcal{E}$, such that we can restrict the quadratic Grassmann encoding to $\mathbb{F}_2^{\mathcal{E}}$ in a meaningful way? The quadratic Grassmann encoding discussed above corresponds to choosing H to be a complete graph, but there are many other options for H. Below are a few simple choices we tried:

(1) Expander graphs: One can try to take H to be an arbitrary sparse expander graph. It can be shown that if the degree of H is small then now the covering property will hold. Unfortunately, an arbitrary expander graph does not support any form of tensor tests.

(2) Cliques in a sparse high-dimensional expander: one can take a sparse high-dimensional expander $X = (X(0), \ldots, X(k))$, where $X(0)$ is the vertex set, and for each i, $X(i) \subseteq \binom{X(0)}{i+1}$, and take H to be the graph underlying it. Because the degree of X is constant, the covering property holds. Also, we can perform local type of tensor tests: if $K \in X(k)$ is a $(k+1)$-face, then K is a clique in X, so we can choose a subspace $U \subseteq \mathbb{F}_2^{X(0)}$ fully supported in K, and have access to all of the entries in $U \otimes U$.

This observation initially seems very promising, but it has one serious issue. We can only afford to perform such a local tensor test for a single clique (or perhaps $O(1)$-many cliques if we

are fine with increasing the value of d for which we are proving Conjecture 1.5). Unfortunately, these localized tensor tests do not actually guarantee that the encoded function has a small rank, and Alice and Bob can use a cheating strategy in the following way. When Alice receives a space of the type $L+U\otimes U$, she extracts from it a rank-1 vector, and takes any element v in its support. Then it is highly likely that v belongs to K, and because X has constant degree, the neighbourhood $N(v)$ of v has constant size. Thus, Alice can choose a linear function on $\mathbb{F}_2^{X(1)}$ which is rank-1 on $\mathbb{F}_2^{X(1)\cap(N(v)\times N(v))}$ but otherwise arbitrary, and responds according to it. Bob proceeds in the same way, where they agree beforehand on what is the rank-1 function they will use, and what the arbitrary function they will use on the rest of the space.

We remark that another attempt at implementing tensor tests via high-dimensional expanders is based on coboundary-type expansion, and we have not explored this thoroughly.

In summary, the attempts above suggest that there are meaningful variants of the quadratic Grassmann encoding that enjoy the covering property, but they come at the cost of, at best, being able to ensure small rank locally. This type of guarantee seems insufficient, at least with an outer PCP along the lines of Section 2.1.1.

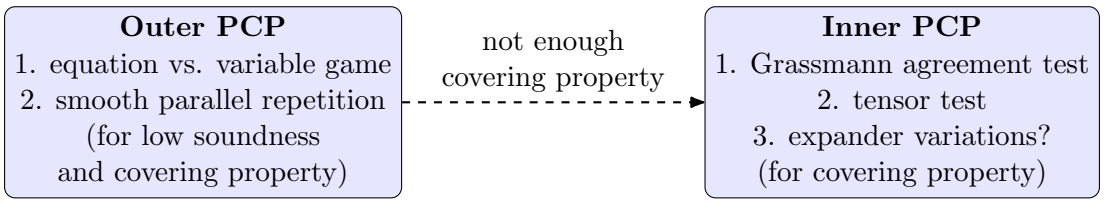


Figure 2: Our failed attempt for perfect completeness

2.2.3 A Second Attempt: Three-Step Construction

So far, the described attempts to create covering property for a quadratic-Grassmann-based inner PCP are not successful. In particular, it seems that the idea of "sparsifying" the quadratic-Grassmann encoding conflicts with tensor test. To address this conflict, we separate the tensor test from the inner PCP and name it the "middle PCP" (see Figure 3). Notice that the tensor test (proposed in Section 2.2.1) is a Grassmann structure over $\mathbb{F}_2^E$, instead of over $\mathbb{F}_2^{E\times E}$. Therefore, the tensor test by itself is compatible with the covering property provided by the smooth parallel repetition in the outer PCP. We only have to create more covering property for the inner PCP (which now consists only of a Grassmann agreement test) *after* incorporating tensor tests into the middle PCP. This will be provided by a "lazy parallel repetition" on the level of the middle PCP (see Figure 3).

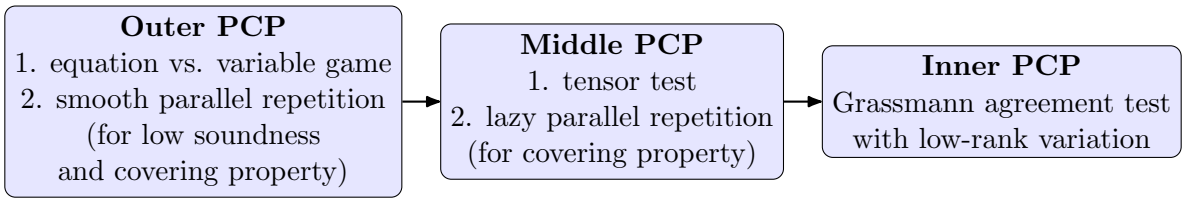


Figure 3: Our construction for perfect completeness

2.2.4 Our Outer PCP

Besides using quadratic equations to replace linear ones, the main difference between our outer PCP and the one in Section 2.1.1 is that we need to establish a stronger soundness condition. This is because the tensor tests we ultimately perform will only guarantee that the decoded assignment has low rank (as opposed to rank-1).

Given a collection of quadratic equations $(\mathscr{V}, \mathscr{E})$ over $\mathbb{F}_2$ and a solution $x \in \mathbb{F}_2^{\mathscr{V}}$, one can consider the rank-1 (or 0) matrix $x \otimes x \in \mathbb{F}_2^{\mathscr{V} \times \mathscr{V}}$ and view it as an assignment to $(\mathscr{V}, \mathscr{E})$. This proposes a broader set of solutions given by low-rank matrices, motivating the following definition:

Definition 2.3. Let e be a quadratic equation over $\mathscr{V}$ described by $\langle x, Ax \rangle = b$ for a matrix $A \in \mathbb{F}_2^{\mathscr{V} \times \mathscr{V}}$, and let $x^{(1)}, \dots, x^{(r)} \in \mathbb{F}_2^{\mathscr{V}}$ be assignments. We say that $x^{(1)}, \dots, x^{(r)}$ jointly satisfy e if

$$\sum_{i=1}^{r} \sum_{u,v \in \mathscr{V}} A_{uv} x_u^{(i)} x_v^{(i)} = b.$$

We remark that [KS17] use the same notion, which they refer to as "superposition complexity". With this definition in mind, the starting point of our reduction is the following result.[2]

Theorem 2.4. *For all $r \in \mathbb{N}^+$, there exists $\varepsilon > 0$ such that the following holds. Given an instance $\Psi = \Psi(\mathscr{V}, \mathscr{E})$ of $3\mathsf{QS}_{\mathbb{F}_2}$, it is* **NP***-hard to distinguish between the following two cases:*

(1) ***Yes case:*** $\mathrm{val}(\Psi) = 1$.

(2) ***No case:*** *for all $x^{(1)}, \dots, x^{(r)} \in \mathbb{F}_2^{\mathscr{V}}$, we have that $x^{(1)}, \dots, x^{(r)}$ jointly satisfy at most $1 - \varepsilon$ fraction of the equations in $\mathscr{E}$.*

As in Section 2.1.1, we can view the instances in Theorem 2.4 as an equation vs. variable game, and then apply a β-smooth k-fold parallel repetition to get a game $\Psi_{q,\beta,k}$.

The soundness condition in Theorem 2.4 translates to a variant of $\Psi_{q,\beta,k}$ where we ask the provers to return r vectors. In particular, upon receiving a question (E, A), Alice needs to return $x^{(1)}, \dots, x^{(r)} \in \mathbb{F}_2^E$ that jointly satisfy all the equations in E. It turns out that one can modify this game into a different game where we ask the provers return low-rank symmetric matrices. In particular, Alice now needs to return a symmetric matrix $X \in \mathbb{F}_2^{E \times E}$ that satisfies the equations in E, in the sense that for each e in E described by $\langle x, Ax \rangle = b$, we have

$$\sum_{u,v \in E} A_{uv} X_{uv} = b.$$

In Section 4.4, we will show that this conversion from joint satisfaction to low-rank-matrix answers does not affect the soundness by too much.

For the purpose of composing with the middle and inner PCPs, we view an Alice-answer $X \in \mathbb{F}_2^{E \times E}$ as a symmetric bilinear map $f_X : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ defined by

$$f_X(x \otimes y) = \sum_{u,v \in E} X_{uv} x_u y_v \qquad \text{for all } x, y \in \mathbb{F}_2^E.$$

Similarly, Bob should return a bilinear map $g : \mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'} \to \mathbb{F}_2$.

[2] A more formal version of Theorem 2.4 is stated in Theorem 3.4.

2.2.5 Our Middle PCP

In the outer PCP described in Section 2.2.4, the verifier (roughly speaking) carries out the following (cf. Section 2.1.1):

1. Sample (E, A) and send it to Alice. Sample (E', B) and send it to Bob.
2. Receive from Alice a symmetric bilinear map $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ of rank at most r, and from Bob a symmetric bilinear map $g : \mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'} \to \mathbb{F}_2$.
3. Accept if the restriction of f to $\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}$ is identical to g.

Local tensor check: the first goal of the middle PCP is to replace the global rank check "$\mathrm{rank}(f) \leqslant r$" with a local tensor check. We move to the game Ψ_{mid} where the verifier carries out the following:

1. Sample (E, A) and (E', B) as in the outer PCP game $\Psi_{q,\beta,k}$.
2. Sample a subspace $U \in \mathsf{Gr}(\mathbb{F}_2^{E'}, \ell_2)$ and a subspace $U' \in \mathsf{Gr}(\mathbb{F}_2^{E'}, \ell_2 + 1)$ containing U. The subspace $U \subseteq \mathbb{F}_2^{E'}$ will be regarded as a subspace of Alice's space $\mathbb{F}_2^E$ by appending 0's in the missing coordinates.
3. Send (E, A, U) to Alice and (E', B, U') to Bob.
4. Receive from Alice a symmetric bilinear map $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ such that $\mathrm{rank}(f|_{U \otimes U}) \leqslant 1$, and from Bob a symmetric bilinear map $g : \mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'} \to \mathbb{F}_2$.
5. Accept if the restriction of f to $\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}$ is identical to g.

In words, we have replaced the global rank check on Alice's response with a local rank check, that verifies that the rank of f on $U \otimes U$ is at most 1, where U is a (nearly uniformly chosen) ℓ_2-dimensional subspace given to Alice.

The completeness of this transformation is clear, but the soundness is much less so. Indeed, because we provide U to Alice, she knows where the rank test will be performed, and therefore she can potentially use that information to cheat. Nevertheless, we show that for all $\delta > 0$, if we fix r, q, ℓ_2 appropriately and let k be sufficiently large, then in the "no case" of Theorem 2.4 we have $\mathrm{val}(\Psi_{\mathsf{mid}}) \leqslant \delta$. The intuition here is as follows:

(1) If Alice did not use any cheating strategy, but her $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ was of rank more than r, then the probability that $f|_{U \times U}$ has rank at most 1 approaches 0 as ℓ_2 goes to infinity.

(2) The type of cheating strategies Alice may use is limited. To see that, fix strategies F, G for Alice and Bob that achieve value at least s, fix the question of Bob, and sample two questions for Alice that only differ on the tensor test, say (E, A, U_1) and (E, A, U_2). Using the Cauchy-Schwarz inequality, we get that with probability at least s^2, Alice and Bob win both of these challenges. In that case, Alice's functions $F[E, A, U_1]$, $F[E, A, U_2]$ have to agree with Bob's function, which is the same in both scenarios (his challenge has not changed). In particular, in that scenario we must have that

$$F[E, A, U_1]|_{\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}} = F[E, A, U_2]|_{\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}}.$$

In words, fixing (E, A), we get that Alice's responses (as a function of the tensor test U she receives) must pass a sort of Grassmann agreement test. If that test had the standard

notion of soundness, it would mean that Alice's responses have to basically follow an honest strategy, and a version of the argument from the first item will further guarantee that this honest strategy uses a low-rank function.

(3) Using the non-standard notion of soundness (which we know to hold), the discussion in Section 2.1.3 implies that after an appropriate zoom-in (and zoom-out, which we ignore for now), Alice's responses do behave like an honest strategy, and one can argue as in the first item. The implementation of the zoom-ins in the soundness analysis of Ψ_{mid} uses the advice A.

Lazy parallel repetition: next, we wish to turn the check that $f|_{\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}} = g$ from Ψ_{mid} into a 2-to-1 test, and towards this end we intend to use the Grassmann encoding. A naive approach would have been to apply the Grassmann encoding directly on Alice's space $\mathbb{F}_2^E \otimes \mathbb{F}_2^E = \mathbb{F}_2^{E \times E}$ and on Bob's space $\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'} = \mathbb{F}_2^{E' \times E'}$, but as discussed in Section 2.2.1 this fails due to the lack of the covering property. Our solution for this issue is to first hide the actual challenge within a large collection of trivial challenges, in an operation that we refer to as lazy repetition (we remark that it has appeared in PCP literature previously, see [Kho02a]). In the terminology of Section 2.2.2, this amounts to taking a graph H which is a union of a large number of cliques (with no edges between them at all).

To describe the operation, fix a parameter t thought of as much larger than any of the previously mentioned parameters. Consider the game Ψ_{lazy} where the verifier carries out the following:

1. Sample $c \in [t]$, and sample challenges (E_c, A_c, U_c), (E'_c, B_c, U'_c) as in Ψ_{mid}. We refer to c as the critical index.

2. For each $j \in [t] \setminus \{c\}$, sample an Alice-challenge (E_j, A_j, U_j) from Ψ_{mid} independently. We discard the obtained A_j and instead let A_j be a sequence of uniformly sampled vectors

$$a^{(1)}, \ldots, a^{(q')} \in \mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j}.$$

   Let $E'_j = E_j$, $B_j = A_j$ and $U'_j = U_j$.

3. We use calligraphic fonts to denote a sequence of objects. For instance, write $\mathcal{E} = (E_1, \ldots, E_t)$ and $\mathcal{U}' = (U'_1, \ldots, U'_t)$. Similarly define sequences $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{U}$. Let $\mathcal{V}(\mathcal{E}) = \bigoplus_{j=1}^t \left(\mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j} \right)$ and $\mathcal{V}(\mathcal{E}') = \bigoplus_{j=1}^t \left(\mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j} \right)$.

4. Send $(\mathcal{E}, \mathcal{A}, \mathcal{U})$ to Alice and $(\mathcal{E}', \mathcal{B}, \mathcal{U}')$ to Bob.

5. Alice must return a linear map $f : \mathcal{V}(\mathcal{E}) \to \mathbb{F}_2$ such that for each $j \in [t]$, the restriction of f to $\mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j}$ is symmetric and satisfies all equations in E_j, and $\operatorname{rank}(f|_{U_j \otimes U_j}) \leqslant 1$. Bob must return a linear map $g : \mathcal{V}(\mathcal{E}') \to \mathbb{F}_2$.

6. Accept if $f|_{\mathcal{V}(\mathcal{E}')} = g$.

Lazy repetition is easily seen to preserve perfect completeness. It also does not worsen the soundness, because Alice and Bob still have to answer the questions (E_c, A_c, U_c) and (E'_c, B_c, U'_c) planted in the critical index, independent of what they do in the non-critical indices.

The main benefit of the lazy repetition is that it improves the covering property. More precisely, consider a random $x \in \mathcal{V}(\mathcal{E})$ obtained by the following process:

1. Sample c and $\mathcal{E}'$ as in Ψ_{mid}.

2. Sample $y \in \mathcal{V}(\mathcal{E}')$, and lift to $x \in \mathcal{V}(\mathcal{E})$ by appending 0's in missing coordinates (which only happens in the critical index).

Then, provided that t is sufficiently large compared to k, the distribution of x is close to uniform in $\mathcal{V}(\mathcal{E})$. The intuition here comes from the following basic fact:

Claim 2.5. *Suppose that $(\Omega, \mathbb{P})$ is a discrete probability space, and $\Gamma \subseteq \Omega$ is an event such that $p = \mathbb{P}(\Gamma) > 0$. Then for each t, the following two distributions are $O_p(1/\sqrt{t})$-close in statistical distance:*

(1) Sample $x_1, \ldots, x_t \sim \mathbb{P}$ independently.

(2) Sample $c \in [t]$, sample $x_c \sim \mathbb{P}$ conditioned on $x_c \in \Gamma$, and for $j \neq c$ sample $x_j \sim \mathbb{P}$ independently.

In summary, the game Ψ_{lazy} has perfect completeness, small soundness, local tensor tests and the covering property. It can actually be composed with an appropriate version of the Grassmann graph but this is still not straightforward. In the next section we discuss why a naïve composition with the standard Grassmann encoding still fails, and explain how we circumvent this issue by taking an appropriate variant of the Grassmann encoding.

2.2.6 Our Inner PCP

A natural attempt at composing Ψ_{lazy} with the Grassmann encoding is to encode the functions $f\colon \mathcal{V}(\mathcal{E}) \to \mathbb{F}_2$ and $g\colon \mathcal{V}(\mathcal{E}') \to \mathbb{F}_2$ via the Grassmann encoding, and then check consistency via the Grassmann test. While this test does satisfy the same notion of non-standard soundness as discussed in Section 2.1.3, we do not know how to implement suitable advice vectors to facilitate the zoom-in coordination. Indeed, the Grassmann encoding via $\mathsf{Gr}(\mathcal{V}(\mathcal{E}), \ell_1)$ would require uniformly random advice vectors in $\mathcal{V}(\mathcal{E})$, and specifying a vector

$$x \in \mathcal{V}(\mathcal{E}) = \bigoplus_{j=1}^{t} \left(\mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j} \right)$$

would require $|E_j|^2$ bits of information for each index j. However, although the advice $\mathcal{A} = (A_1, \ldots, A_t)$ available in Ψ_{lazy} does provide quadratic-in-k amount of information for each non-critical index j via

$$A_j = \left(a^{(j,1)}, \ldots, a^{(j,q')} \right) \in \left(\mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j} \right)^{q'},$$

the advice available for the critical index $c \in [t]$ is

$$A_c = \left(a^{(1)}, \ldots, a^{(q)} \right) \in \left(\mathbb{F}_2^{E_c} \right)^{q},$$

which contains only $O_q(k)$ bits of information. Therefore, it is of interest to have a Grassmann-type structure on $\mathbb{F}_2^{E_c} \otimes \mathbb{F}_2^{E_c}$ with only $\exp(O_q(k))$ different zoom-in possibilities.

The low-rank Grassmann encoding: the idea is to consider not all, but only a special type of subspaces in $\mathsf{Gr}(\mathbb{F}_2^{E_c} \otimes \mathbb{F}_2^{E_c}, \ell_1)$. We only allow subspaces that consist of low-rank vectors in $\mathbb{F}_2^{E_c} \otimes \mathbb{F}_2^{E_c}$. By the rank-1 decomposition, any vector in $\mathbb{F}_2^{E_c} \otimes \mathbb{F}_2^{E_c}$ of rank at most s can be specified by $O(s|E_c|)$ bits of information.

In order to systematically construct linear subspaces consisting of low-rank vectors, we use the Hadamard code (the same idea has appeared in [Gol23]). Given a positive integer ℓ, let $\mathrm{Had}_\ell \in \mathbb{F}_2^{2^\ell \times (2^\ell - 1)}$ be the $\{0,1\}$-Hadamard matrix with the first column (the all-0 column) deleted. For any $\mathbb{F}_2$-vector space V and vectors $x^{(1)}, \dots, x^{(2^\ell-1)} \in V$, let

$$\mathrm{Lspan}\big(x^{(1)}, \dots, x^{(2^\ell-1)}\big) = \left\{ \sum_{j=1}^{2^\ell-1} \mathrm{Had}_\ell(i,j) x^{(j)} \;\middle|\; i \in \{1, 2, \dots, 2^\ell\} \right\}.$$

In words, for each row in Had_ℓ, the set $\mathrm{Lspan}\big(x^{(1)}, \dots, x^{(2^\ell-1)}\big)$ contains a vector which is the linear combination of $x^{(1)}, \dots, x^{(2^\ell-1)}$ with the coefficients given by that row. Because any linear combination of rows in Had_ℓ is a row in Had_ℓ, it follows that $\mathrm{Lspan}\big(x^{(1)}, \dots, x^{(2^\ell-1)}\big)$ is always a linear subspace, and if $x^{(1)}, \dots, x^{(2^\ell-1)}$ are linearly independent then that subspace has dimension exactly ℓ.

The upshot is that if $x^{(1)}, \dots, x^{(2^\ell-1)} \in \mathbb{F}_2^{E_c} \otimes \mathbb{F}_2^{E_c}$ are rank-1 vectors, then all vectors in the space $\mathrm{Lspan}\big(x^{(1)}, \dots, x^{(2^\ell-1)}\big)$ have ranks at most $2^{\ell-1}$. For the sake of reference, we present the following definition of [Gol23]:[3]

Definition 2.6 ([Gol23, Definition 17])**.** Suppose V is an $\mathbb{F}_2$-vector space of dimension at least 2^ℓ. We define $\mathsf{LGr}(V, \ell)$, called *the low-rank Grassmann graph*, to be the subgraph of the Grassmann graph $\mathsf{Gr}(V \otimes V, \ell)$ induced by the vertex set

$$\left\{ \mathrm{Lspan}\big(u^{(1)} \otimes v^{(1)}, \dots, u^{(2^\ell-1)} \otimes v^{(2^\ell-1)}\big) \;\middle|\; \begin{array}{l} u^{(1)}, \dots, u^{(2^\ell-1)} \in V \;\text{ linearly independent} \\ v^{(1)}, \dots, v^{(2^\ell-1)} \in V \;\text{ linearly independent} \end{array} \right\}.$$

Composing with middle PCP: we can now make Ψ_{lazy} into a 4-to-1 game by using the low-rank Grassmann encoding for the critical block $\mathbb{F}_2^{E_c} \otimes \mathbb{F}_2^{E_c}$, and using the ordinary Grassmann encoding for the non-critical blocks $\mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j}$ where $j \in [t] \setminus \{c\}$. We define Ψ_{final} to be the game where the verifier carries out the following:

1. Sample the critical index $c \in [t]$, and sample $\mathcal{E}, \mathcal{E}', \mathcal{U}, \mathcal{U}'$ as in Ψ_{lazy}.
2. For each $i \in \{1, 2, \dots, 2^{\ell_1+1} - 1\}$, sample

   $$x^{(i)} = x^{(i,1)} + \cdots + x^{(i,t)} \in \mathcal{V}(\mathcal{E}'),$$

   where $x^{(i,c)}$ is a uniformly random rank-1 vector in $\mathbb{F}_2^{E'_c} \otimes \mathbb{F}_2^{E'_c}$, while $x^{(i,j)}$ is a uniformly random vector in $\mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j}$ for each $j \in [t] \setminus \{c\}$.
3. Define the linear subspace $L' = \mathrm{Lspan}\big(x^{(1)}, \dots, x^{(2^{\ell_1+1}-1)}\big) \subseteq \mathcal{V}(\mathcal{E}')$. If $\dim L' < \ell_1 + 1$, give up on asking questions and accept immediately. Otherwise, let L be a uniformly random subspace of L' with dimension ℓ_1. We regard $L \subseteq \mathcal{V}(\mathcal{E}')$ as a subspace of $\mathcal{V}(\mathcal{E})$ by appending 0's in the missing coordinates (which only happens in the critical index).
4. Send the challenge $(\mathcal{E}, \mathcal{U}, L)$ to Alice and $(\mathcal{E}', \mathcal{U}', L')$ to Bob.

[3] Definition 2.6 will not be used explicitly in this paper.

5. Alice must return a linear map $f : \left(L + \bigoplus_{j=1}^{t} U_j \otimes U_j\right) \to \mathbb{F}_2$ such that $\operatorname{rank}(f|_{U_j \otimes U_j}) \leqslant 1$ for each $j \in [t]$. Bob must return a linear map $g : \left(L' + \bigoplus_{j=1}^{t} U'_j \otimes U'_j\right) \to \mathbb{F}_2$ such that $\operatorname{rank}(g|_{U'_j \otimes U'_j}) \leqslant 1$ for each $j \in [t]$.

6. Accept if the restriction of g to $\left(L + \bigoplus_{j=1}^{t} U_j \otimes U_j\right)$ is identical to f.

The game Ψ_{final} is our final composed PCP (modulo incorporating the side conditions checking satisfiability of the chosen quadratic equations, and the side conditions checking symmetry; this is done by adding the spaces N_{E_j} and N_{sym} as discussed in Section 2.2.1).

It is easy to see that in Ψ_{final}, for any response of Alice, there are at most 4 possible answers of Bob that can make the verifier accept. This is due to the following facts:

- Since $\dim L' = 1 + \dim L$, any linear function on L has two linear extensions to L'.
- For each $j \in [t] \setminus \{c\}$, we have $U_j = U'_j$. For the critical index c, we have $\dim U'_c = 1 + \dim U_c$, and hence any symmetric bilinear function on $U_c \otimes U_c$ of rank at most 1 has two symmetric bilinear extensions to $U'_c \otimes U'_c$ of rank at most 1.

The completeness of Ψ_{final} is clear by using the Grassmann encoding on the strategies of the provers from Ψ_{lazy}. For soundness, we show that a strategy for Ψ_{final} achieving value at least ε can be used to construct a strategy for Ψ_{lazy} achieving value at least $\Omega_\varepsilon(1)$. This argument is more elaborate, and we only give a high-level overview.

First, we describe a high-level overview where we pretend there was no need for zoom-outs in Equation (2.2). We show that for noticeable fraction of the partial challenges $(\mathcal{E},\mathcal{U})$ for Alice, the table $L \mapsto F[\mathcal{E},\mathcal{U},L]$ passes the standard Grassmann test (Equation (2.1)) over $\mathcal{V}(\mathcal{E})$ with noticeable probability. Applying Equation (2.2), we know that Alice can use the advice $\mathcal{A}$ in Ψ_{lazy} as the zoom-in and thus decodes a global linear function g on $\mathcal{V}(\mathcal{E})$ which has rank 1 on $U_j \otimes U_j$ for each j. We argue that for noticeable fraction of the challenges $(\mathcal{E}',\mathcal{U}')$ of Bob (conditioned on $\mathcal{E},\mathcal{U},\mathcal{A}$), the restriction $g|_{\mathcal{V}(\mathcal{E}')}$ is a global linear function for Bob that has noticeable agreement with his table $L' \mapsto G[\mathcal{E}',\mathcal{U}',L']$ under the same zoom in (constructed from his advice). Together, this shows that for a fixed successful advice, each one of Alice and Bob only has constantly many global linear functions that have noticeable agreement with their table, and there is at least one pair of choices between them which is consistent. Thus, if each one of them chooses a random such global linear function, they will produce a satisfactory answer to Ψ_{lazy} with constant probability.

The need for zoom-outs in Equation (2.2) complicates matters considerably, as the number of "successful" zoom-outs could be rather large. This issue was already handled in [DKK+21], and we will make use of some ideas from there. However, the fact that we use the low-rank Grassmann encoding introduces further difficulties. In [DKK+21] it is shown that if Bob's table has many zoom-outs of codimension w that admit a global linear function with noticeable agreement, then there is also a zoom-out of codimension $w-1$ that admits a global linear function with noticeable agreement. While this may be possible in our case, it requires extra features from the low-rank Grassmann graph which we do not establish. Instead, we show how to stitch together global linear functions from many different Bob-questions $(\mathcal{E}',\mathcal{U}')$ to a function of Alice, on which we can apply the result of [DKK+21] (since Alice's space has the standard Grassmann graph structure).

3 Preliminaries

In this section we set up basic notations and tools that will be used throughout the paper.

3.1 General Notations

In this subsection we summarize the general notational conventions used throughout the paper. Additional notation will be introduced as needed, typically within dedicated "Notation" environments.

Linear algebra. Given a finite dimensional $\mathbb{F}_2$-vector space V and a linear subspace $W \subseteq V$, the codimension of W with respect to V is denoted by $\mathrm{codim}_V(W) = \dim V - \dim W$. Given a finite dimensional $\mathbb{F}_2$-vector space V, we let V^* denote the dual space of V, which consists of all linear functionals $f : V \to \mathbb{F}_2$. For any finite-dimensional $\mathbb{F}_2$-vector spaces V_1, V_2 and any bilinear form $f : V_1 \otimes V_2 \to \mathbb{F}_2$, we define the linear map $T_f : V_1 \to V_2^*$ by letting $T_f(x)$ be the functional $y \mapsto f(x \otimes y)$.

As indicated in Sections 2.2.4 and 2.2.5 of the proof overview, bilinear forms will play an important role in our construction and analysis of 4-to-1 games. More linear algebraic preliminaries about bilinear forms can be found in Appendix A.

Grassmann graph. Given a finite dimensional $\mathbb{F}_2$-vector space V and a positive integer $\ell \leqslant \dim V$, the Grassmann graph $\mathsf{Gr}(V, \ell)$ has as vertices all ℓ-dimensional subspaces $L \subseteq V$, and two vertices are adjacent if they intersect in dimension $\ell - 1$. By a slight abuse of notation, we also use $\mathsf{Gr}(V, \ell)$ to denote the vertex set of this graph. We denote by $\mathcal{G}(V, \ell)$ the uniform distribution over the edges of $\mathsf{Gr}(V, \ell)$.

Probability. For a finite set Λ, we write $\mathbb{E}_{x \in \Lambda}[\cdot]$ and $\mathbb{P}_{x \in \Lambda}[\cdot]$ to denote expectation and probability, respectively, when x is drawn uniformly at random from Λ. If x is sampled according to a specific distribution $\mathcal{D}$ over Λ, we write $x \sim \mathcal{D}$ in place of $x \in \Lambda$ in the subscripts.

Assignment tables. In a 2-prover 1-round game, we often organize a prover's strategy as an *assignment table* $F[\cdot]$, where for each possible question Q from the verifier, the entry $F[Q]$ takes value from an alphabet of answers. It is often convenient to allow partial assignment tables, where each entry $F[Q]$ is allowed to take a special nullity value outside of the prescribed alphabet. We usually denote such a nullity value by the symbol `nil`. Whenever a `nil` entry is tested for agreement with other entries or objects, we always assume the test rejects (i.e. reports a disagreement). For instance, if L_1 and L_2 are linear subspaces of a vector space V, the entry $F[L_1]$ takes value in $(L_1)^* \cup \{\texttt{nil}\}$ and $F[L_2]$ takes value in $(L_2)^* \cup \{\texttt{nil}\}$, then the statement

$$\Big[\, F[L_1]|_{L_1 \cap L_2} = F[L_2]|_{L_1 \cap L_2} \,\Big]$$

evaluates to "false" whenever either of $F[L_1]$ or $F[L_2]$ is `nil`.

Empty sequences. A sequence of objects $a^{(1)}, \ldots, a^{(q)}$ is denoted by $\big(a^{(1)}, \ldots, a^{(q)}\big)$. If $q = 0$, the sequence is empty and is denoted by the symbol `empty`.

3.2 Sub-4-to-1 Games

In order to prove the main theorem (Theorem 1.6), it is convenient to work with a slight relaxation of the 4-to-1 requirement:

Definition 3.1. In the setting and notation of Definition 1.1, a constraint $\Phi_e \subseteq \Sigma_L \times \Sigma_R$ is called a sub-d-to-d' constraint if there is a partition $\Sigma_L = \Sigma_{L,1} \sqcup \cdots \sqcup \Sigma_{L,t}$ into sets of size d and a partition $\Sigma_R = \Sigma_{R,1} \sqcup \cdots \sqcup \Sigma_{R,t}$ into sets of size d' such that

$$\Phi_e = \bigcup_{i=1}^{t} \Gamma_{L,i} \times \Gamma_{R,i}, \qquad \text{for some subsets } \Gamma_{L,i} \subseteq \Sigma_{L,i} \text{ and } \Gamma_{R,i} \subseteq \Sigma_{R,i}.$$

The notion of sub-d-to-d' constraints motivates the following definition parallel to Definition 1.3.

Definition 3.2. Let d, d' and r be positive integers. An instance Ψ of label-cover is called a sub-d-to-d' game if all of its constraints are sub-d-to-d'. For parameters $0 < s < c \leqslant 1$, we write $\mathsf{GapS}\text{-}d\text{-}\mathsf{to}\text{-}d'_r(c, s)$ for the restriction of $\mathsf{GapLC}_r(c, s)$ to sub-d-to-d' game instances.

Accordingly, we have the following slight weakening of Theorem 1.6.

Theorem 3.3. *For any $\varepsilon \in (0, 1)$, there exists a positive integer r such that $\mathsf{GapS}\text{-}4\text{-}\mathsf{to}\text{-}1_r(1, \varepsilon)$ is* **NP***-hard.*

Although Theorem 3.3 is a relaxation of Theorem 1.6, we will show in Appendix B that Theorem 3.3 actually implies Theorem 1.6. The main body of this paper (Sections 4 to 6) will be devoted to proving Theorem 3.3.

3.3 Technical Ingredients from Prior Work

In this subsection, we record the main technical ingredients of this paper that are imported from prior works.

3.3.1 Approximate Coloring of Hypergraphs

As discussed in Section 2.2.4, our construction starts from the hardness of approximating satisfiable $3\mathsf{QS}_{\mathbb{F}_2}$ instances. However, limitations of the tensor tests used in our middle PCP require us to start from an **NP**-hard gap problem in which the "yes" and "no" cases impose conditions on *different* constraint satisfaction instances. Specifically, the soundness condition in Theorem 2.4 must assert that, even after the alphabet of the $3\mathsf{QS}_{\mathbb{F}_2}$ instance is suitably enlarged (from $\mathbb{F}_2$ to $(\mathbb{F}_2)^r$, roughly speaking), the resulting $3\mathsf{QS}_{(\mathbb{F}_2)^r}$ instance still has small value.[4]

Fortunately, the desired starting point in Theorem 2.4 is essentially implicit in the result of [DRS05] on the hardness of coloring 2-colorable 3-uniform hypergraphs.[5] Although their result is stated with the no-case condition that the hypergraph H is not c-colorable, a closer inspection of their argument yields the statement below.

Theorem 3.4 ([DRS05])**.** *For every fixed positive integer $c \geqslant 2$, there exists a constant $\varepsilon \geqslant \exp(-\exp(O(c^3)))$ such that given a 3-uniform hypergraph H, it is* **NP***-hard to distinguish between the following cases:*

(1) Yes case: H is 2-colorable.

(2) No case: in every c-coloring of H, at least ε fraction of the edges are monochromatic.

Since Theorem 3.4 differs slightly from the result proved in [DRS05], we provide a proof in Appendix C. Our proof is a minor adaptation of their argument, which in turn uses the PCP theorem [FGL+96, AS98, ALM+98] and parallel repetition [Raz98].

[4] This formulation is characteristic of promise constraint satisfaction problems (PCSPs); see, for example, [KO22].

[5] In Section 4.2, we will explain why 2-coloring a 3-uniform hypergraph can be viewed as solving a $3\mathsf{QS}_{\mathbb{F}_2}$ instance.

3.3.2 Grassmann Agreement Test

The main analytical/combinatorial tool used in this paper is the (non-standard) soundness of the Grassmann agreement test, informally stated in Equation (2.2). Under the notation specified in Section 3.1, given a table $F[\cdot]$ that assigns to each ℓ-dimensional subspace $L \subseteq \mathbb{F}_2^n$ either a linear function $F[L]\colon L \to \mathbb{F}_2$ or the nullity element $\texttt{nil}$, the passing probability of the Grassmann test is

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)}\left[F[L_1]|_{L_1\cap L_2} = F[L_2]|_{L_1\cap L_2}\right],$$

where the event does not hold if either $F[L_1]$ or $F[L_2]$ are $\texttt{nil}$. The following result is due to [KMS23, EKL23], but we use a slightly different formulation which we derive directly from their results.

Theorem 3.5 ([KMS23, EKL23])**.** *There is an absolute constant $C > 0$ such that the following holds. Let n, ℓ, q be nonnegative integers such that $n - q - 3 \geqslant \ell \geqslant q + 3$. Suppose $F[\cdot]$ is a table where $F[L] \in L^* \cup \{\texttt{nil}\}$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, such that*

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)}\left[F[L_1]|_{L_1\cap L_2} = F[L_2]|_{L_1\cap L_2}\right] \geqslant 2^{-q+5}. \tag{3.1}$$

Then for at least $2^{-q-\ell^2}$ fraction of subspaces $Q \in \mathsf{Gr}(\mathbb{F}_2^n, q)$, there exist a subspace $W \in \mathsf{Gr}(\mathbb{F}_2^n, n-q)$ containing Q and a linear map $f : W \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[F[L] = f|_L \;\middle|\; Q \subseteq L \subseteq W\right] \geqslant 2^{-Cq^2}.$$

A derivation of Theorem 3.5 from the formulation of [EKL23] will be presented in Appendix D.

We also need the following simpler result, which concerns not a table of local linear functions, but a table where every entry takes value from the same abstract alphabet. The following result asserts that if the "Grassmann equality test" passes with noticeable probability, then after zoom-ins and zoom-outs the entries take a certain constant value noticeably often.

Theorem 3.6. *There is an absolute constant $C > 0$ such that the following holds. Let n, ℓ, q be nonnegative integers such that $n - q - 3 \geqslant \ell \geqslant q + 3$. Suppose Σ is a finite set and $F[\cdot]$ is a table where $F[L] \in \Sigma \cup \{\texttt{nil}\}$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, such that*

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)}\left[F[L_1] = F[L_2] \neq \texttt{nil}\right] \geqslant 2^{-q+4}. \tag{3.2}$$

Then for at least $2^{-q-\ell^2}$ fraction of subspaces $Q \in \mathsf{Gr}(\mathbb{F}_2^n, q)$, there exist an element $\sigma \in \Sigma$ and a subspace $W \in \mathsf{Gr}(\mathbb{F}_2^n, n-q)$ containing Q such that

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[F[L] = \sigma \;\middle|\; Q \subseteq L \subseteq W\right] \geqslant 2^{-Cq^2}.$$

Proof. Deferred to Appendix D. □

3.3.3 List Decoding Bounds

When we use Theorem 3.5 we will often need to argue that given a zoom-in Q, the number of possible decodings (f, W) is not too large. In this section we precisely define this notion and state the bounds we will use, established by [DKK+25].

Definition 3.7 ([DKK+25])**.** Let n, ℓ, q, w be nonnegative integers such that $n - w \geqslant \ell \geqslant q + 1$. Fix parameters $\tau_0, \tau_1, \ldots, \tau_w \in (0, 1)$ and write $\vec{\tau} = (\tau_0, \ldots, \tau_w)$. Fix an $\mathbb{F}_2$-vector space V of dimension n and a subspace $Q \in \mathsf{Gr}(V, q)$. Suppose $F[\cdot]$ is a table where $F[L] \in L^* \cup \{\mathtt{nil}\}$ for each $L \in \mathsf{Gr}(V, \ell)$. For a linear subspace $W \subseteq V$ with $\operatorname{codim} W \leqslant w$ and a linear functional $f : W \to \mathbb{F}_2$, we say that the pair (f, W) is *$\vec{\tau}$-occurring in $F[\cdot]$ modulo Q* if $Q \subseteq W$ and

$$\mathop{\mathbb{P}}_{L \in \mathsf{Gr}(W,\ell)} \left[F[L] = f|_L \,\middle|\, Q \subseteq L \right] \geqslant \tau_i, \qquad \text{where } i = n - \dim W.$$

It is not possible to give a bound on the number of $\vec{\tau}$-occurring pairs that is independent of n, simply because the number of zoom-outs W can be very large. We have to restrict our attention to pairs (f, W) that "cannot be extended," as defined below.

Definition 3.8. In the setting and notation of Definition 3.7, we say that (f, W) is *$\vec{\tau}$-maximal in $F[\cdot]$ modulo Q* if

(1) (f, W) is $\vec{\tau}$-occurring in $F[\cdot]$ modulo Q, but

(2) for any linear subspace $W' \subseteq V$ that properly contains W and any extended linear functional $f' : W' \to \mathbb{F}_2, f'|_W = f$, the pair (f', W') is not $\vec{\tau}$-occurring in $F[\cdot]$ modulo Q.

The set of all $\vec{\tau}$-maximal pairs will be denoted by $\mathsf{LIST}^{\vec{\tau}}_Q(F)$.

We use the following bound on the size of maximal pairs proved by [DKK+25].

Lemma 3.9 ([DKK+25])**.** *There is an absolute constant $C \geqslant 40$ such that the following holds. Let n, ℓ, q, w be nonnegative integers such that $n - w \geqslant 2\ell \geqslant C(q + 1)$. Suppose $\vec{\tau} = (\tau_0, \ldots, \tau_w)$ is a vector such that*

$$\tau_0 \geqslant 2^{-\ell/C} \qquad \text{and} \qquad \tau_{i-1} \leqslant C^{-1} \tau_i^C \text{ for each } i \in [w].$$

Then for any table $F[\cdot]$ where $F[L] \in L^* \cup \{\mathtt{nil}\}$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, we have

$$\left| \mathsf{LIST}^{\vec{\tau}}_Q(F) \right| \leqslant \exp(C(w + 1)^2 \ell).$$

In a part of the analysis of our inner PCP, we are not able to use Lemma 3.9 as a black box, and we will instead draw from the proof of Lemma 3.9. A useful component of the proof by [DKK+25] is the notion of *sunflower* of zoom-outs, defined as follows.

Definition 3.10. Let V be a vector space over $\mathbb{F}_2$, and let $W_1, \ldots, W_m \subseteq V$ be subspaces of co-dimension w. We say that $(V, W_1, \ldots, W_m)$ forms a sunflower if $\operatorname{codim}_V(W_i \cap W_j) = 2w$ for each pair of distinct indices $i, j \in [m]$.

In words, a sunflower is a collection of subspaces $W_1, \ldots, W_m$ living inside a space V that have as small as possible pairwise intersections. The following result from [DKK+25] asserts that any sufficiently large collection of zoom-outs of constant codimension contains a large sunflower structure.

Lemma 3.11 ([DKK+25, Corollary 3.14])**.** *Let V be a finite dimensional $\mathbb{F}_2$-vector space, and let $W_1, \ldots, W_N \subseteq V$ be a list of subspaces of codimension w. Let m be a positive integer such that $m \cdot (m \cdot 2^w)^w \leqslant N$. Then, there exists a subspace $W \subseteq V$ containing m of the W_i's, say, $W_1, \ldots, W_m$, such that $(W, W_1, \ldots, W_m)$ forms a sunflower.*

3.4 Proof Organization

The proof of Theorem 3.3 spans Sections 4 to 6, during which we construct a sequence of five 2-prover 1-round games; see Figure 4.

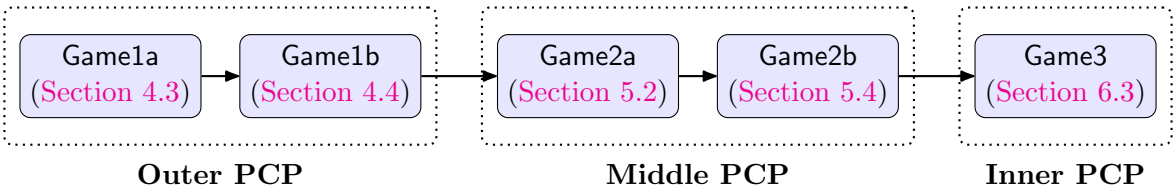


Figure 4: Proof organization

For each version label $\mathsf{xx} \in \{\mathsf{1a}, \mathsf{1b}, \mathsf{2a}, \mathsf{2b}, \mathsf{3}\}$, we will prove completeness and soundness guarantees of the following form:

(1) Completeness: if H is 2-colorable, then $\mathrm{val}(\mathsf{Gamexx}[H]) = 1$.

(2) Soundness: if every c-coloring of H makes at least ε fraction of the edges monochromatic, then $\mathrm{val}(\mathsf{Gamexx}[H])$ is small.

Completeness is straightforward and can be proved independently for each game, without invoking the completeness of preceding games. Most of the work lies in establishing soundness: for each game after the first, the soundness proof relies on that of its predecessor in the sequence. The final construction $\mathsf{Game3}[H]$ is a sub-4-to-1 game, whose completeness and soundness yield Theorem 3.3.

3.5 Parameter Relations

During the course of our construction of 4-to-1 games, there will be many parameters on different scales (they are all constant with respect to the size of the starting hardness instance in Theorem 3.4). We summarize the relations between these constants in the following condition.

Condition 3.12. Let $k, q_0, q_1, q_2, t, r, \ell_1, \ell_2, s$ be positive integers, let $\delta, \beta, \varepsilon \in (0,1)$, and let $\vec{\tau} = (\tau_0, \tau_1, \ldots, \tau_{2q_0})$ be a vector such that for a sufficiently large absolute constant $C > 0$, the conditions in Table 1 hold.

Parameter	Requirement	First appears in	Why require this
$\delta \in (0, 1/2)$		inner PCP	
$q_0 \in \mathbb{N}^+$	$q_0 \geqslant C \log_2(1/\delta)$	inner PCP	inner PCP
$\tau_{2q_0} \in (0, 1)$	$\tau_{2q_0} \leqslant \exp(-q_0^C)$	inner PCP	inner PCP
$\tau_{2q_0-1}, \ldots, \tau_0 \in (0, 1)$	$\tau_{i-1} \leqslant C^{-1}\tau_i^C,\ \forall i \in [2q_0]$	inner PCP	inner PCP
$\ell_1 \in \mathbb{N}^+$	$\ell_1 \geqslant C \log_2(1/\tau_0)$	inner PCP	inner PCP
$s \in \mathbb{N}^+$	$s = 3\ell_1$	inner PCP	inner PCP
$q_1 \in \mathbb{N}^+$	$q_1 = 2s(2^{2\ell_1} - 2^{2\ell_1 - q_0})$	outer PCP	inner PCP
$q_2 \in \mathbb{N}^+$	$q_2 \geqslant C\ell_1^2$	outer PCP	middle PCP
$\ell_2 \in \mathbb{N}^+$	$\ell_2 \geqslant Cq_2^2$	middle PCP	middle PCP
$r \in \mathbb{N}^+$	$r = 2q_2 + 1$	outer PCP	middle PCP
$\varepsilon \in (0, 1)$	$\varepsilon \leqslant \exp(-\exp(\exp(Cr)))$	outer PCP	outer PCP
$k \in \mathbb{N}^+$	$k \geqslant \varepsilon^{-C} \exp(\exp(\ell_2))$	outer PCP	outer PCP
$\beta \in (0, 1)$	$\beta = k^{-3/4}$	outer PCP	outer PCP
$t \in \mathbb{N}^+$	$t \geqslant \exp(Ck^5)$	middle PCP	inner PCP

Table 1: Parameter relations

Condition 3.12 is meant to be a consistent setting of parameters that we can invoke in Sections 4 to 6. Whenever the condition is invoked, the absolute constant $C > 0$ can be different from previous invocations. More precisely, a statement of the form

$$\Big[\text{Assume Condition 3.12. Then [XX] holds.}\Big]$$

should be interpreted as

$$\Big[\text{There exists an absolute constant } C > 0 \text{ such that the conditions in Table 1 imply [XX].}\Big]$$

4 The Outer PCP

In this section, we formally construct and analyze our outer PCP, following the outline in Section 2.2.4. We first introduce the necessary framework in Sections 4.1 and 4.2 before turning to the actual construction in Sections 4.3 and 4.4.

4.1 Preparation: Tripled Sets

As discussed in Sections 2.1.1 and 2.2.4, Alice's challenge in the outer PCP consists of a set E of variables together with an advice sequence. The variables in E are naturally grouped into triples, each consisting of the three distinct variables appearing in an equation of the $3\mathsf{QS}_{\mathbb{F}_2}$ instance. We call such a set E a *tripled set*, formally defined as follows.

Definition 4.1. A *tripled set* is a finite set E equipped with a partition into subsets of size 3. We refer to the parts of this implicit partition as the *triples* of E, and we write $\mathrm{Triples}(E)$ for the collection of its triples.

As discussed in Sections 2.1.1 and 2.2.4, our outer PCP construction applies (smooth) parallel repetition to a $3\mathsf{QS}_{\mathbb{F}_2}$ instance. Since the $3\mathsf{QS}_{\mathbb{F}_2}$ instance provided by Theorem 3.4 is represented by a 3-uniform hypergraph H, k-fold parallel repetition yields a 2-prover 1-round game played on the vertex-disjoint union of k copies of H.

Definition 4.2. Let H be a 3-uniform hypergraph with vertex set $\mathscr{V}(H)$ and edge set $\mathscr{E}(H)$. Given a positive integer k, we let $H^{\sqcup k}$ denote the k-fold vertex-disjoint union of H. More explicitly, the vertex set of $H^{\sqcup k}$ is $[k] \times \mathscr{V}(H)$, and we denote $\mathscr{V}(H,k) = [k] \times \mathscr{V}(H)$; the edge set of $H^{\sqcup k}$ is $\bigcup_{i=1}^{k} \mathscr{E}_i(H,k)$, where

$$\mathscr{E}_i(H,k) = \big\{\{(i,\mathsf{v}_1),(i,\mathsf{v}_2),(i,\mathsf{v}_3)\} \,\big|\, \{\mathsf{v}_1,\mathsf{v}_2,\mathsf{v}_3\} \in \mathscr{E}(H)\big\}.$$

We let $\mathscr{E}(H,k)$ be the collection of all vertex subsets $E \subseteq [k] \times \mathscr{V}(H)$ that result from picking one edge from each $\mathscr{E}_i(H,k)$. That is, it consists of E of the form

$$E = \{v_{i,j} \mid i \in [k],\, j \in [3]\} \qquad \text{such that} \qquad \{v_{i,1}, v_{i,2}, v_{i,3}\} \in \mathscr{E}_i(H,k) \text{ for all } i \in [k], \tag{4.1}$$

and we view it naturally as a tripled set, with $\mathrm{Triples}(E) = \big\{\{v_{i,1}, v_{i,2}, v_{i,3}\} \,\big|\, i \in [k]\big\}$.

In words, viewing the set of hyperedges in H as the challenges that a verifier samples, the edges in $H^{\sqcup k}$ are naturally viewed as the challenges that the verifier chooses in the k-fold repeated game.

Remark 4.3. For a tripled set E of the form (4.1), the k triples of E are naturally ordered. For each $i \in [k]$, we will call $\{v_{i,1}, v_{i,2}, v_{i,3}\}$ the i-th triple of E.

4.1.1 Thinning, and Thinning with Advice

We next introduce notation to formalize the smoothness in our application of parallel repetition. The following definition captures the structure of the challenges in each coordinate: with probability $1-\beta$, the two players receive identical challenges; otherwise, Alice receives the full challenge, while Bob receives only a single variable from it.

Definition 4.4. Let E be a tripled set and let $\beta \in [0,1]$. We write $E' \sim \mathrm{Thin}_\beta(E)$ for the random subset $E' \subseteq E$ obtained as follows. Independently for each triple $T \in \mathrm{Triples}(E)$, we include all elements of T in E' with probability $1-\beta$, and include a uniformly random element of T in E' with probability β. We stress that the output E' itself is only a set, not a tripled set.

In the outer PCP construction, each prover will receive *two* advice sequences A_{in} and A_{mid}, the latter supplying the Grassmann-type zoom-ins needed in the middle PCP, while the former is reserved for zoom-ins in the inner PCP. The next definition formalizes the process by which the verifier samples the two advice sequences.

Definition 4.5. Let E be a tripled set. Fix nonnegative integers q_1, q_2 and a parameter $\beta \in [0,1]$. We let $\mathcal{J}_{\mathsf{out}}(E, q_1, q_2, \beta)$ denote the output distribution of the following process:

1. Sample a β-thinning $E' \sim \mathrm{Thin}_\beta(E)$.
2. Sample $q = q_1 + q_2$ vectors $a^{(1)}, \ldots, a^{(q)} \in \mathbb{F}_2^{E'}$ independently and uniformly at random. We denote $A_{\mathsf{in}} = \big(a^{(1)}, \ldots, a^{(q_1)}\big)$ and $A_{\mathsf{mid}} = \big(a^{(q_1+1)}, \ldots, a^{(q_1+q_2)}\big)$.

3. Output the tuple $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$.

Remark 4.6. For a subset E' of a finite set E, we always view $\mathbb{F}_2^{E'}$ as a linear subspace of the ambient space $\mathbb{F}_2^E$. In particular, the vectors $a^{(1)}, \dots, a^{(q)}$ in Definition 4.5 are viewed also as elements of the vector space $\mathbb{F}_2^E$.

In the outer PCP, Alice will see a tripled set E and the two advice sequences A_{in} and A_{mid}, but not the β-thinning E' which is only sent to Bob. The next proposition states that, even conditioned on A_{in} and A_{mid}, the distribution of E' is still nicely structured. In particular, it is independent over the triples of E.

Proposition 4.7. Let E be a tripled set. Fix nonnegative integers q_1, q_2 and a parameter $\beta \in [0, 1)$. Fix arbitrary tuples

$$A_1 \in (\mathbb{F}_2^E)^{q_1} \qquad \text{and} \qquad A_2 \in (\mathbb{F}_2^E)^{q_2}.$$

Sample

$$(E', A_{\mathsf{in}}, A_{\mathsf{mid}}) \sim \mathcal{J}_{\mathsf{out}}(E, q_1, q_2, \beta)$$

conditioned on the event $\{A_{\mathsf{in}} = A_1\} \cap \{A_{\mathsf{mid}} = A_2\}$. Then the resulting marginal distribution of E' is independent over triples of E. Furthermore, each triple of E is entirely included in E' with conditional probability at least $1 - 4^{q_1+q_2}\beta$.

Proof. Deferred to Appendix F.1. □

As emphasized throughout the proof overview, "covering properties" are crucial in our construction of 4-to-1 games, and they will appear many times (in different forms) throughout Sections 4 to 6. They are important primarily for Grassmann encodings, and a job of our outer PCP construction is to prepare a strong enough "covering property" for the Grassmann-based tensor tests in the middle PCP to work. On the most basic level, we need that the distribution of vectors sampled uniformly from $\mathbb{F}_2^{E'}$ (where E' is a random β-thinning), once lifted to $\mathbb{F}_2^E$ as discussed in Remark 4.6, is close to uniform over $\mathbb{F}_2^E$. The next proposition (essentially from [DKK+25]) proves that this is the case if $\beta = o(|E|^{-1/2})$.

Proposition 4.8. Let E be a tripled set. Fix nonnegative integers q_1, q_2 and a parameter $\beta \in [0, 4^{-q_1-q_2})$. Fix an arbitrary tuple $A_1 \in (\mathbb{F}_2^E)^{q_1}$. Sample

$$(E', A_{\mathsf{in}}, A_{\mathsf{mid}}) \sim \mathcal{J}_{\mathsf{out}}(E, q_1, q_2, \beta)$$

conditioned on the event $\{A_{\mathsf{in}} = A_1\}$. Then the resulting marginal distribution of A_{mid} has total variation distance at most $4^{q_1+q_2}\beta\sqrt{|E|/3}$ to the uniform distribution over $(\mathbb{F}_2^E)^{q_2}$.

Proof. Deferred to Appendix F.1. □

We remark that although we include Proposition 4.8 here in Section 4, it will not be used until Section 5.

4.2 Preparation: The NAE Predicate

The reason that 2-colorability of a 3-uniform hypergraph can be viewed as a system of quadratic equations is that the ternary not-all-equal predicate on a Boolean alphabet is a "quadratic" constraint:

Definition 4.9. We define a function $\mathrm{NAE}_3 : \mathbb{F}_2^3 \to \mathbb{F}_2$ defined by

$$\mathrm{NAE}_3(x_1, x_2, x_3) = x_1 + x_2 + x_3 + x_1x_2 + x_2x_3 + x_3x_1.$$

We note that $\mathrm{NAE}_3(x_1, x_2, x_3) = 1$ if and only if x_1, x_2, x_3 are not all equal.

As discussed in Section 2.2.4, we will use the notion of "joint NAE-satisfaction", which we formalize below. Here we assume that an assignment to the variables in E is presented as a linear functional $f : \mathbb{F}_2^E \to \mathbb{F}_2$.

Definition 4.10. Let E be a tripled set and let r be a positive integer. Given r linear functionals $f^{(1)}, \dots, f^{(r)} : \mathbb{F}_2^E \to \mathbb{F}_2$, let $a^{(1)}, \dots, a^{(r)} \in \mathbb{F}_2^E$ be the unique vectors such that

$$f^{(i)}(x) = \sum_{v \in E} a_v^{(i)} x_v \qquad \text{for all } i \in [r] \text{ and } x \in \mathbb{F}_2^E.$$

If we have

$$\sum_{i=1}^{r} \mathrm{NAE}_3\big(a_{v_1}^{(i)}, a_{v_2}^{(i)}, a_{v_3}^{(i)}\big) = 1 \qquad \text{for all } \{v_1, v_2, v_3\} \in \mathrm{Triples}(E),$$

then the linear functionals $f^{(1)}, \dots, f^{(r)}$ are said to be *jointly* NAE-*satisfying*. In the case $r = 1$, then the functional $f^{(1)}$ is simply said to be NAE-*satisfying*.

We can now translate the "2-colorability" of a 3-uniform hypergraph H into the language of NAE-satisfaction. For the sake of future reference, the following proposition is stated for the k-fold disjoint union $H^{\sqcup k}$.

Proposition 4.11. Suppose H is a 2-colorable 3-uniform hypergraph and k is a positive integer. Then there exists a linear functional $f : \mathbb{F}_2^{\mathscr{V}(H,k)} \to \mathbb{F}_2$ such that for any $E \in \mathscr{E}(H,k)$, the restriction $f|_{\mathbb{F}_2^E}$ is NAE-satisfying.

Proof. Since H is 2-colorable, so is the k-fold vertex-disjoint union $H^{\sqcup k}$. Let $\varphi : \mathscr{V}(H,k) \to \mathbb{F}_2$ be a proper 2-coloring of $H^{\sqcup k}$. Let $f : \mathbb{F}_2^{\mathscr{V}(H,k)} \to \mathbb{F}_2$ be the linear functional defined by

$$f(x) = \sum_{v \in \mathscr{V}(H,k)} \varphi(v) x_v \qquad \text{for all } x \in \mathbb{F}_2^{\mathscr{V}(H,k)}.$$

For any $E \in \mathscr{E}(H,k)$ and any $\{v_1, v_2, v_3\} \in \mathrm{Triples}(E)$, since $\{v_1, v_2, v_3\}$ is an edge of $H^{\sqcup k}$, by definition we know that $\varphi(v_1), \varphi(v_2), \varphi(v_3)$ are not all equal. Therefore, it follows from Definition 4.10 that $f|_{\mathbb{F}_2^E}$ is NAE-satisfying. □

4.2.1 Linear Functionals vs. Bilinear Forms

In later stages of the outer PCP construction, an assignment to variables in E will be presented as a symmetric bilinear map $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$. The NAE-satisfaction notion generalizes to bilinear maps as follows.

Definition 4.12. Let E be a tripled set. Given a symmetric bilinear map $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$, let $X \in \mathbb{F}_2^{E \times E}$ be the unique matrix such that

$$f(x \otimes y) = \sum_{u,v \in E} X_{uv} x_u y_v \qquad \text{for all } x, y \in \mathbb{F}_2^E.$$

If we have

$$X_{v_1v_1} + X_{v_2v_2} + X_{v_3v_3} + X_{v_1v_2} + X_{v_2v_3} + X_{v_3v_1} = 1 \qquad \text{for all } \{v_1, v_2, v_3\} \in \text{Triples}(E),$$

then the bilinear map f is said to be NAE-*satisfying*.

In Section 4.4, we will need to decode from a low-rank symmetric bilinear map a collection of linear functionals. This process is essentially the standard "rank-1 decomposition" in linear algebra, which we formalize below.

Definition 4.13. Let V be a finite-dimensional $\mathbb{F}_2$-vector space and let r be a positive integer. Given a symmetric bilinear map $f : V \otimes V \to \mathbb{F}_2$ and r linear functionals $f^{(1)}, \dots, f^{(r)} : V \to \mathbb{F}_2$, we say that $\big(f^{(1)}, \dots, f^{(r)}\big)$ is a *tensor decomposition* of f if the following conditions hold:

(1) For each $i \in [r]$, we have $f^{(i)} \in \text{image}(T_f)$. In other words, there exists a vector $x^{(i)} \in V$ such that $f^{(i)}(y) = f(x^{(i)} \otimes y)$ for each $y \in V$.

(2) For each $x, y \in V$, we have

$$f(x \otimes y) = \sum_{i=1}^{r} f^{(i)}(x) f^{(i)}(y). \tag{4.2}$$

The following two propositions show that the NAE-satisfaction of symmetric bilinear maps is interchangeable with the joint-NAE-satisfaction of its tensor decompositions.

Proposition 4.14. Let E be a tripled set and let r be a positive integer. Suppose $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ is an NAE-satisfying symmetric bilinear map. Then any tensor decomposition $\big(f^{(1)}, \dots, f^{(r)}\big)$ of f is jointly NAE-satisfying.

Proof. Let $a^{(1)}, \dots, a^{(r)} \in \mathbb{F}_2^E$ be the unique vectors such that

$$f^{(i)}(x) = \sum_{v \in E} a_v^{(i)} x_v \qquad \text{for all } i \in [r] \text{ and } x \in \mathbb{F}_2^E.$$

Let $X \in \mathbb{F}_2^{E \times E}$ be the matrix defined by

$$X_{uv} = \sum_{i=1}^{r} a_u^{(i)} a_v^{(i)} \qquad \text{for all } u, v \in E.$$

Plugging the expansion of $f^{(i)}$ into (4.2) and changing the order of summation, we get that

$$f(x \otimes y) = \sum_{u,v \in E} X_{uv} x_u y_v \qquad \text{for all } x, y \in \mathbb{F}_2^E.$$

Since f is NAE-satisfying, for each triple $\{v_1, v_2, v_3\} \in \text{Triples}(E)$ we must have

$$X_{v_1v_1} + X_{v_2v_2} + X_{v_3v_3} + X_{v_1v_2} + X_{v_2v_3} + X_{v_3v_1} = 1,$$

and hence

$$\sum_{i=1}^{r} \Big(a_{v_1}^{(i)} a_{v_1}^{(i)} + a_{v_2}^{(i)} a_{v_2}^{(i)} + a_{v_3}^{(i)} a_{v_3}^{(i)} + a_{v_1}^{(i)} a_{v_2}^{(i)} + a_{v_2}^{(i)} a_{v_3}^{(i)} + a_{v_3}^{(i)} a_{v_1}^{(i)} \Big) = 1.$$

By Definition 4.9 we thus have

$$\sum_{i=1}^{r} \text{NAE}_3\big(a_{v_1}^{(i)}, a_{v_2}^{(i)}, a_{v_3}^{(i)}\big) = 1 \qquad \text{for all } \{v_1, v_2, v_3\} \in \text{Triples}(E),$$

and thus $f^{(1)}, \dots, f^{(r)}$ are jointly NAE-satisfying. □

Proposition 4.15. Let E be a tripled set and let r be a positive integer. Suppose $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ is a symmetric bilinear map and $f^{(1)}, \dots, f^{(r)} : \mathbb{F}_2^E \to \mathbb{F}_2$ are jointly NAE-satisfying linear functionals. If (4.2) holds for all $x, y \in \mathbb{F}_2^E$, then f is NAE-satisfying.

Proof. The proof follows the same lines as in Proposition 4.14, and we omit the details. □

The following straightforward corollary of Proposition 4.15 will be used in Lemma 5.19.

Corollary 4.16. *Let E be a nonempty tripled set. There exists an* NAE*-satisfying symmetric bilinear map $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ of rank 1.*

Proof. There clearly exists a nonzero linear functional $f^{(1)} : \mathbb{F}_2^E \to \mathbb{F}_2$ that is NAE-satisfying (by itself). We can then let $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ be the rank-1 bilinear map defined by $f(x \otimes y) = f^{(1)}(x) f^{(1)}(y)$ for all $x, y \in \mathbb{F}_2^E$, and apply Proposition 4.15 to show that f is NAE-satisfying. □

Combining Propositions 4.11 and 4.15 yields the following "bilinear" version of Proposition 4.11.

Corollary 4.17. *Suppose H is a 2-colorable 3-uniform hypergraph and k is a positive integer. Then there exists a rank-1 symmetric bilinear form $f : \mathbb{F}_2^{\mathscr{V}(H,k)} \otimes \mathbb{F}_2^{\mathscr{V}(H,k)} \to \mathbb{F}_2$ such that for any $E \in \mathscr{E}(H,k)$, the restriction $f|_{\mathbb{F}_2^E \otimes \mathbb{F}_2^E}$ is* NAE*-satisfying.*

Proof. By Proposition 4.11, there is a (nonzero) linear functional $f^{(1)} : \mathbb{F}_2^{\mathscr{V}(H,k)} \to \mathbb{F}_2$ such that $f^{(1)}\big|_{\mathbb{F}_2^E}$ is NAE-satisfying for every $E \in \mathscr{E}(H,k)$. We can then let

$$f : \mathbb{F}_2^{\mathscr{V}(H,k)} \otimes \mathbb{F}_2^{\mathscr{V}(H,k)} \to \mathbb{F}_2$$

be the rank-1 symmetric bilinear map defined by $f(x \otimes y) = f^{(1)}(x) f^{(1)}(y)$ for all $x, y \in \mathbb{F}_2^{\mathscr{V}(H,k)}$, and apply Proposition 4.15 to show that $f|_{\mathbb{F}_2^E \otimes \mathbb{F}_2^E}$ is NAE-satisfying for each $E \in \mathscr{E}(H,k)$. □

4.3 The Equation vs. Variable Game

We are now ready to start the construction of our outer PCP. The first version of our outer PCP is defined as follows (cf. Section 2.1.1).

Definition 4.18. Given positive integers k, r, nonnegative integers q_1, q_2, a parameter $\beta \in [0,1]$ and a 3-uniform hypergraph H, let $\mathsf{Game1a}[H, k, q_1, q_2, r, \beta]$ be the 2-prover 1-round game where the verifier executes the following:

1. Sample a uniformly random tripled set $E \in \mathscr{E}(H,k)$.
2. Sample $(E', A_{\mathsf{in}}, A_{\mathsf{mid}}) \sim \mathcal{J}_{\mathsf{out}}(E, q_1, q_2, \beta)$. Send $(E, A_{\mathsf{in}}, A_{\mathsf{mid}})$ to Alice[6] and $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$ to Bob.
3. Alice must return r linear functionals $f^{(1)}, \dots, f^{(r)} : \mathbb{F}_2^E \to \mathbb{F}_2$ that are jointly NAE-satisfying; Bob must return r linear functionals $g^{(1)}, \dots, g^{(r)} : \mathbb{F}_2^{E'} \to \mathbb{F}_2$. Accept if

$$f^{(i)}\Big|_{\mathbb{F}_2^{E'}} = g^{(i)} \qquad \text{for all } i \in [r].$$

[6] In Alice's view, A_{in} and A_{mid} are sequences of vectors in $\mathbb{F}_2^E$, instead of in $\mathbb{F}_2^{E'}$; it is important that Alice does not get to see E'.

Remark 4.19. We remark that the game $\mathsf{Game1a}[H, k, q_1, q_2, r, \beta]$ can be viewed as the k-fold parallel repetition of $\mathsf{Game1a}[H, 1, q_1, q_2, r, \beta]$, and the game $\mathsf{Game1a}[H, 1, 0, 0, r, 1]$ is the equation-vs.-variable game corresponding to the result of Theorem 3.4.

We now analyze the completeness and the soundness of $\mathsf{Game1a}$.

Lemma 4.20 (Game1a completeness)**.** *For any positive integer k, nonnegative integers q_1, q_2, parameter $\beta \in [0, 1]$ and 2-colorable 3-uniform hypergraph H, we have*

$$\mathrm{val}\Big(\mathsf{Game1a}[H, k, q_1, q_2, 1, \beta]\Big) = 1.$$

Proof. We obtain from Proposition 4.11 a linear functional $f : \mathbb{F}_2^{\mathscr{V}(H,k)} \to \mathbb{F}_2$, whose restriction on $\mathbb{F}_2^E$ is NAE-satisfying for any $E \in \mathscr{E}(H, k)$. We can then construct a strategy for Alice in the game $\mathsf{Game1a}[H, k, q_1, q_2, 1, \beta]$, as follows:

1. Receive the question $(E, A_{\mathsf{in}}, A_{\mathsf{mid}})$.
2. Return the functional $f|_{\mathbb{F}_2^E}$, which is guaranteed by Proposition 4.11 to be NAE-satisfying.

Similarly, the strategy for Bob is:

1. Receive the question $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$.
2. Return the functional $f|_{\mathbb{F}_2^{E'}}$.

It is clear that Bob's answer is always consistent with Alice's answer. Therefore, if Alice and Bob operate under such strategies, the verifier in the game $\mathsf{Game1a}[H, k, q_1, q_2, 1, \beta]$ accepts with probability 1. □

To show soundness of $\mathsf{Game1a}[H, k, q_1, q_2, r, \beta]$, we start by analyzing the soundness of the basic game $\mathsf{Game1a}[H, 1, 0, 0, r, 1]$, which is effectively the standard equation-versus-variable game with "superpositions."

Lemma 4.21. *Fix a positive integer r and a parameter $\varepsilon \in (0, 1)$. Suppose H is a 3-uniform hypergraph such that in every 2^r-coloring of H, at least ε fraction of the edges are monochromatic. Then we have*

$$\mathrm{val}\Big(\mathsf{Game1a}[H, 1, 0, 0, r, 1]\Big) \leqslant 1 - \frac{\varepsilon}{3}.$$

Proof. Let V be the vertex set of $H^{\sqcup 1}$ (which is isomorphic to H).

We fix an arbitrary strategy for Alice and Bob in the game $\mathsf{Game1a}[H, 1, 0, 0, r, 1]$, and we will prove that the verifier rejects with probability at least $\varepsilon/3$. Since there are no advice vectors in this game, the question for Alice is simply a random edge of $H^{\sqcup 1}$; and as the smoothness parameter is set to 1, the question for Bob will be a single vertex.

Our goal is to define a 2^r-coloring map $\varphi : V \to \mathbb{F}_2^r$. For each vertex $v \in V$, the strategy for Bob must assign an answer to the question $\{v\}$, and the answer must be r linear functionals

$$g_v^{(1)}, \ldots, g_v^{(r)} : \mathbb{F}_2^{\{v\}} \to \mathbb{F}_2.$$

Aggregating Bob's answers for all $v \in V$ yields r linear functionals

$$g^{(1)}, \ldots, g^{(r)} : \mathbb{F}_2^V \to \mathbb{F}_2.$$

We can then let $\varphi : V \to \mathbb{F}_2^r$ be the unique assignment map such that

$$\left(g^{(1)}(x), g^{(2)}(x), \ldots, g^{(r)}(x)\right) = \sum_{v \in V} \varphi(v) x_v \qquad \text{for all } x \in \mathbb{F}_2^V. \tag{4.3}$$

By the assumption of the lemma, for at least ε fraction of the edges $\{v_1, v_2, v_3\} \in \mathscr{E}(H, 1)$, we have $\varphi(v_1) = \varphi(v_2) = \varphi(v_3)$. Fix such an edge $E = \{v_1, v_2, v_3\}$, and let $f^{(1)}, \ldots, f^{(r)} : \mathbb{F}_2^E \to \mathbb{F}_2$ be Alice's answer to the question E. By Definition 4.18, $f^{(1)}, \ldots, f^{(r)}$ must be jointly NAE-satisfying. However, $\varphi(v_1) = \varphi(v_2) = \varphi(v_3)$ implies (via Definition 4.10 and Equation (4.3)) that

$$g^{(1)}\Big|_{\mathbb{F}_2^E}, g^{(2)}\Big|_{\mathbb{F}_2^E}, \ldots, g^{(r)}\Big|_{\mathbb{F}_2^E}$$

are not jointly NAE-satisfying. Therefore, there exists a vertex $v \in \{v_1, v_2, v_3\}$ with

$$\left(f^{(1)}\Big|_{\mathbb{F}_2^{\{v\}}}, f^{(2)}\Big|_{\mathbb{F}_2^{\{v\}}}, \ldots, f^{(r)}\Big|_{\mathbb{F}_2^{\{v\}}}\right) \neq \left(g_v^{(1)}, g_v^{(2)}, \ldots, g_v^{(r)}\right). \tag{4.4}$$

Conditioned on the event that Alice receives the question E, with probability $1/3$ the question for Bob is $\{v\}$, in which case their answers are not consistent due to (4.4). Since this is true for at least ε-fraction of the edges $E \in \mathscr{E}(H, 1)$, the verifier must reject with probability at least $\varepsilon/3$. □

Next, we incorporate the advice as well as constantly many repetitions, and show that the value does not increase by too much:

Lemma 4.22. *Fix positive integers k, r, nonnegative integers q_1, q_2 and parameters $\beta, \varepsilon \in (0, 1)$ such that $k \geqslant 2^{q_1+q_2}\beta^{-1}$. Suppose H is a 3-uniform hypergraph such that in every 2^r-coloring of H, at least ε fraction of the edges are monochromatic. Then we have*

$$\mathrm{val}\Big(\mathsf{Game1a}[H, k, q_1, q_2, r, \beta]\Big) \leqslant 1 - \frac{\varepsilon}{6}.$$

Proof. We write $q = q_1 + q_2$. As in Remark 4.19, we view $\mathsf{Game1a}[H, k, q_1, q_2, r, \beta]$ as the k-fold parallel repetition of the game $\mathsf{Game1a}[H, 1, q_1, q_2, r, \beta]$. Let $(E, A_{\mathsf{in}}, A_{\mathsf{mid}})$ and $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$ be the questions for Alice and Bob in the k-fold repetition game, respectively. We write

$$A_{\mathsf{in}} = \left(a^{(1)}, \ldots, a^{(q_1)}\right) \qquad \text{and} \qquad A_{\mathsf{mid}} = \left(a^{(q_1+1)}, \ldots, a^{(q_1+q_2)}\right).$$

We say a triple $\{v_1, v_2, v_3\} \in \mathrm{Triples}(E)$ is "critical" if we have

$$a_{v_1}^{(i)} = a_{v_2}^{(i)} = a_{v_3}^{(i)} = 0 \text{ for all } i \in [q] \qquad \text{and} \qquad \left|E' \cap \{v_1, v_2, v_3\}\right| = 1.$$

Let $I \subseteq [k]$ be the collection of indices $i \in [k]$ such that the i-th triple of E is critical (see Remark 4.3). Conditioned on the event that $I \neq \emptyset$, we pick a uniformly random $i \in I$ and note that the i-th "coordinate" of the parallel repetition is a copy of the equation vs. variable game $\mathsf{Game1a}[H, 1, 0, 0, r, 1]$ and is independent of the remaining coordinates. Therefore, Lemma 4.21 implies that conditioned on $\{I \neq \emptyset\}$, the verifier rejects with probability at least $\varepsilon/3$. Since

$$\mathbb{P}\left[I \neq \emptyset\right] = 1 - \left(1 - 2^{-q}\beta\right)^k \geqslant 1 - 2^{-k \cdot 2^{-q}\beta} \geqslant 1 - 2^{-1} = \frac{1}{2},$$

we conclude that in the k-fold repetition game $\mathsf{Game1a}[H, k, q_1, q_2, r, \beta]$, the verifier rejects with probability at least $\frac{1}{2} \cdot \varepsilon/3 = \varepsilon/6$. □

Finally, we apply the parallel repetition theorem for projection games by [Rao11] to get the soundness of Game1a.

Lemma 4.23 (Game1a soundness)**.** *Assume Condition 3.12. Suppose H is a 3-uniform hypergraph such that in every 4^r-coloring of H, at least ε fraction of the edges are monochromatic. Then we have*[7]

$$\mathrm{val}\Big(\mathsf{Game1a}[H, k, q_1, q_2, 2r, \beta]\Big) \leqslant \exp(-k^{1/5}).$$

Proof. Let $k_0 = \lceil 2^{q_1+q_2}\beta^{-1}\rceil$. By Lemma 4.22 we know that

$$\mathrm{val}\Big(\mathsf{Game1a}[H, k_0, q_1, q_2, 2r, \beta]\Big) \leqslant 1 - \frac{\varepsilon}{6}.$$

Since $\mathsf{Game1a}[H, k_0, q_1, q_2, 2r, \beta]$ is a projection game, we may apply [Rao11, Theorem 1.4] to conclude that its $\lfloor k/k_0 \rfloor$-fold parallel repetition has value at most $\exp(-c\varepsilon^2\lfloor k/k_0\rfloor)$, for some absolute constant $c \in (0,1)$. It is then easy to see (via Remark 4.19) that

$$\mathrm{val}\Big(\mathsf{Game1a}[H, k, q_1, q_2, 2r, \beta]\Big) \leqslant \exp(-c\varepsilon^2\lfloor k/k_0\rfloor) \leqslant \exp(-k^{1/5}),$$

where in the last transition we use the relation $k/k_0 \gg \varepsilon^{-2}k^{1/5}$ guaranteed by Condition 3.12. □

4.4 From Joint Satisfaction to Bilinear Forms

As hinted in Section 2.2.4 and Section 4.2.1, the last goal in the outer PCP stage is to change the format of the provers' responses to symmetric bilinear forms. More precisely, we will consider the following 2-prover 1-round game defined below. It is identical to $\mathsf{Game1a}[H, k, q_1, q_2, r, \beta]$, except that the provers are expected to provide a single bilinear form of low-rank (instead of r linear forms).

Definition 4.24. Given positive integers k, r, nonnegative integers q_1, q_2, a parameter $\beta \in [0,1)$ and a 3-uniform hypergraph H, let $\mathsf{Game1b}[H, k, q_1, q_2, r, \beta]$ be the 2-prover-1-round game where the verifier executes the following:

1. Sample a uniformly random tripled set $E \in \mathscr{E}(H, k)$.
2. Sample $(E', A_{\mathsf{in}}, A_{\mathsf{mid}}) \sim \mathcal{J}_{\mathsf{out}}(E, q_1, q_2, \beta)$. Send $(E, A_{\mathsf{in}}, A_{\mathsf{mid}})$ to Alice and $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$ to Bob.
3. Alice must return a symmetric bilinear form $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ that is NAE-satisfying and has rank at most r; Bob must return a bilinear form $g : \mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'} \to \mathbb{F}_2$. Accept if

$$f|_{\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}} = g.$$

We now analyze the completeness and the soundness of Game1b.

Lemma 4.25 (Game1b completeness)**.** *For any positive integer k, nonnegative integers q_1, q_2, parameter $\beta \in [0,1)$ and 2-colorable 3-uniform hypergraph H, we have*

$$\mathrm{val}\Big(\mathsf{Game1b}[H, k, q_1, q_2, 1, \beta]\Big) = 1.$$

[7] We have $2r$ instead of r as the rank parameter here, in order to prepare for applying Lemma 4.26 later.

Proof. We obtain from Corollary 4.17 a rank-1 symmetric bilinear map $f : \mathbb{F}_2^{\mathscr{V}(H,k)} \otimes \mathbb{F}_2^{\mathscr{V}(H,k)} \to \mathbb{F}_2$, whose restriction on $\mathbb{F}_2^E \otimes \mathbb{F}_2^E$ is NAE-satisfying for every $E \in \mathscr{E}(H,k)$. We can then construct a strategy for Alice in the game $\mathsf{Game1b}[H,k,q_1,q_2,1,\beta]$, as follows:

1. Receive the question $(E, A_{\mathsf{in}}, A_{\mathsf{mid}})$.
2. Return the symmetric bilinear map $f|_{\mathbb{F}_2^E \otimes \mathbb{F}_2^E}$, which is guaranteed by Corollary 4.17 to be NAE-satisfying. Furthermore, it must have rank at most 1 because $\mathrm{rank}(f) = 1$.

Similarly, the strategy for Bob is:

1. Receive the question $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$.
2. Return the symmetric bilinear map $f|_{\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}}$.

It is clear that under this set of strategies, the verifier accepts with probability 1. □

We now move on to the soundness. The following lemma establishes a connection between the soundness of $\mathsf{Game1a}$ and $\mathsf{Game1b}$, and more precisely it shows how to derive a strategy for the former from a strategy for the latter. Intuitively, this comes from the fact that a rank r bilinear form has at most $O_r(1)$ many decompositions into rank-1 bilinear forms, but some care is needed (specifically, the event that the rank of Alice's bilinear form drops in Bob's space is problematic, but thankfully it has low probability).

Lemma 4.26. *For any positive integers k, r, nonnegative integers q_1, q_2, parameter $\beta \in [0, 4^{-q_1-q_2})$ and 3-uniform hypergraph H, we have*

$$\mathrm{val}\left(\mathsf{Game1a}[H,k,q_1,q_2,2r,\beta]\right) \geqslant 4^{-r^2}\left(\mathrm{val}\left(\mathsf{Game1b}[H,k,q_1,q_2,r,\beta]\right) - 4^{q_1+q_2} r\beta\right).$$

Proof. We organize the proof into the following three steps.

Step 1: controlling rank drop. We denote

$$s = \mathrm{val}\left(\mathsf{Game1b}[H,k,q_1,q_2,r,\beta]\right).$$

Let $F[\cdot]$ and $G[\cdot]$ be the strategies of Alice and Bob, respectively, under which the verifier accepts with probability s in $\mathsf{Game1b}[H,k,q_1,q_2,r,\beta]$. For each $E \in \mathscr{E}(H,k)$ and $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$ in the support of $\mathcal{J}_{\mathsf{out}}(E,q_1,q_2,\beta)$, we have bilinear maps

$$F[E, A_{\mathsf{in}}, A_{\mathsf{mid}}] : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2 \qquad \text{and} \qquad G[E', A_{\mathsf{in}}, A_{\mathsf{mid}}] : \mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'} \to \mathbb{F}_2.$$

A pair of questions $(E, A_{\mathsf{in}}, A_{\mathsf{mid}})$ and $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$ for Alice and Bob is said to be "good" if

$$F[E, A_{\mathsf{in}}, A_{\mathsf{mid}}]|_{\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}} = G[E', A_{\mathsf{in}}, A_{\mathsf{mid}}];$$

it is said to be "excellent" if in addition to being good, it also satisfies

$$\mathrm{rank}\big(F[E, A_{\mathsf{in}}, A_{\mathsf{mid}}]\big) = \mathrm{rank}\big(G[E', A_{\mathsf{in}}, A_{\mathsf{mid}}]\big).$$

By definition, a random pair of questions is "good" with probability s. Furthermore, conditioned on each fixed Alice question $(E, A_{\mathsf{in}}, A_{\mathsf{mid}})$, since $\mathrm{rank}\big(F[E, A_{\mathsf{in}}, A_{\mathsf{mid}}]\big) \leqslant r$, we know from Corollary A.3 and Proposition 4.7 that

$$\mathop{\mathbb{P}}_{E'}\left[\mathrm{rank}\Big(F[E, A_{\mathsf{in}}, A_{\mathsf{mid}}]|_{\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}}\Big) = \mathrm{rank}\Big(F[E, A_{\mathsf{in}}, A_{\mathsf{mid}}]\Big) \,\Big|\, E, A_{\mathsf{in}}, A_{\mathsf{mid}}\right] \geqslant (1 - 4^{q_1+q_2}\beta)^r$$

$$\geqslant 1 - 4^{q_1+q_2} r\beta.$$

Therefore, the probability that a random pair of questions is "good" but not "excellent" is at most $4^{q_1+q_2} r\beta$. Consequently, a random pair of questions is "excellent" with probability at least $s - 4^{q_1+q_2} r\beta$.

Step 2: deriving a strategy for Game1a. Note that in $\mathsf{Game1a}[H, k, q_1, q_2, 2r, \beta]$ the joint distribution of questions to Alice and Bob is the same as in $\mathsf{Game1b}[H, k, q_1, q_2, r, \beta]$. We next derive strategies for the provers in the former game from the tables $F[\cdot]$ and $G[\cdot]$. The strategy for Alice is as follows:

1. Receive the question $(E, A_{\mathsf{in}}, A_{\mathsf{mid}})$, and let $f = F[E, A_{\mathsf{in}}, A_{\mathsf{mid}}]$. Note that $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ is symmetric and NAE-satisfying.
2. Use Lemma A.4 to obtain a tensor decomposition $\left(f^{(1)}, \ldots, f^{(2r)}\right)$ of f.
3. Return the linear functionals $f^{(1)}, \ldots, f^{(2r)} : \mathbb{F}_2^E \to \mathbb{F}_2$. They must be jointly NAE-satisfying, due to Proposition 4.14.

Accordingly, the strategy for Bob is:

1. Receive the question $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$, and let $g = G[E', A_{\mathsf{in}}, A_{\mathsf{mid}}]$.
2. For each $i \in [2r]$, select a linear functional $g^{(i)}$ from the subspace $\mathrm{image}(T_g) \subseteq (\mathbb{F}_2^{E'})^*$ independently and uniformly at random.
3. Return the linear functionals $g^{(1)}, \ldots, g^{(2r)} : \mathbb{F}_2^{E'} \to \mathbb{F}_2$.

Step 3: analysis of the strategy. We show that the above strategy answers every "excellent" pair of questions consistently with probability at least 4^{-r^2}, and thereby concluding the desired inequality

$$\mathrm{val}\left(\mathsf{Game1a}[H, k, q_1, q_2, 2r, \beta]\right) \geqslant 4^{-r^2}(s - 4^{q_1+q_2} r\beta).$$

Suppose $(E, A_{\mathsf{in}}, A_{\mathsf{mid}})$ and $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$ are an "excellent" pair of questions for Alice and Bob, and let

$$f = F[E, A_{\mathsf{in}}, A_{\mathsf{mid}}] \qquad \text{and} \qquad g = G[E', A_{\mathsf{in}}, A_{\mathsf{mid}}].$$

Let $\left(f^{(1)}, \ldots, f^{(2r)}\right)$ be an arbitrary tensor decomposition of f. By Definition 4.13, we know that $f^{(i)} \in \mathrm{image}(T_f)$ for each $i \in [2r]$. Furthermore, since $\mathrm{rank}(f) = \mathrm{rank}(g)$ (due to the excellence condition), it follows from Corollary A.2 that

$$f^{(i)}\Big|_{\mathbb{F}_2^{E'}} \in \mathrm{image}(T_g) \qquad \text{for all } i \in [2r].$$

Since the cardinality of $\mathrm{image}(T_g)$ is at most $2^{\mathrm{rank}(g)} \leqslant 2^r$, if $g^{(1)}, \ldots, g^{(2r)}$ are independently and uniformly sampled from $\mathrm{image}(T_g)$, we have

$$\mathbb{P}\left[f^{(i)}\Big|_{\mathbb{F}_2^{E'}} = g^{(i)} \text{ for all } i \in [2r]\right] \geqslant (2^{-r})^{2r} = 4^{-r^2},$$

as desired. □

Using Lemma 4.26, we can now prove the soundness of Game1b.

Lemma 4.27 (Game1b soundness)**.** *Assume Condition 3.12. Suppose H is a 3-uniform hypergraph such that in every 4^r-coloring of H, at least ε fraction of the edges are monochromatic. Then we have*

$$\mathrm{val}\Big(\mathsf{Game1b}[H,k,q_1,q_2,r,\beta]\Big) \leqslant k^{-1/2}.$$

Proof. Combining Lemmas 4.23 and 4.26, we have

$$\mathrm{val}\Big(\mathsf{Game1b}[H,k,q_1,q_2,r,\beta]\Big) \leqslant 4^{q_1+q_2} r\beta + 4^{r^2}\exp(-k^{1/5}) \leqslant k^{-1/2},$$

where the last transition is due to Condition 3.12. □

5 The Middle PCP

In this section we construct and analyze our middle PCP as discussed in Section 2.2.5. Sections 5.1 to 5.3 are devoted to the design and analysis of Game2a, where we incorporate local tensor checks. We then perform a lazy parallel repetition to get Game2b in Section 5.4.

5.1 Preparation: Thinning with Subspaces

In the middle PCP, we will use a Grassmann-type structure to perform "local tensor checks." Along with a question $(E, A_{\mathsf{in}}, A_{\mathsf{mid}})$ sent to Alice (as in the outer PCP), the verifier also sends a subspace $U \subseteq \mathbb{F}_2^E$ and checks whether the restriction of Alice's answer $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ to $U \otimes U$ has rank at most 1. The following augmentation of Definition 4.5 captures the process by which the verifier samples the tensor-check subspace U.

Definition 5.1. Let E be a tripled set. Fix a parameter $\beta \in [0,1]$ and nonnegative integers q_1, q_2, ℓ such that $\ell \geqslant q_2$. We let $\mathcal{J}_{\mathsf{mid}}(E, q_1, q_2, \beta, \ell)$ denote the output distribution of the following process:

1. Sample $(E', A_{\mathsf{in}}, A_{\mathsf{mid}}) \sim \mathcal{J}_{\mathsf{out}}(E, q_1, q_2, \beta)$.
2. Sample $\ell - q_2$ vectors $a^{(1)}, \dots, a^{(\ell-q_2)} \in \mathbb{F}_2^{E'}$ independently and uniformly at random. Let $A' = \big(a^{(1)}, \dots, a^{(\ell-q_2)}\big)$, and obtain a subspace $U = \mathrm{span}(A_{\mathsf{mid}}) + \mathrm{span}(A') \subseteq \mathbb{F}_2^{E'}$.
3. Output the tuple $(E', A_{\mathsf{in}}, A_{\mathsf{mid}}, U)$.

Remark 5.2. Similarly to Remark 4.6, the subspace $U \subseteq \mathbb{F}_2^{E'}$ in Definition 5.1 is viewed also as a subspace of $\mathbb{F}_2^E$.

Remark 5.3. Consider a sample $(E', A_{\mathsf{in}}, A_{\mathsf{mid}}, U) \sim \mathcal{J}_{\mathsf{mid}}(E, q_1, q_2, \beta, \ell)$ as defined in Definition 5.1. We note that the distribution of (E', A_{in}, U) does not depend on q_2. This is because after fixing E' and A_{in}, the advice sequence A_{mid} is uniformly random over $(\mathbb{F}_2^{E'})^{q_2}$.

The next proposition (which can be derived from Proposition 4.8) is the crucial "covering property" we need in the middle PCP: a random subspace of $\mathbb{F}_2^{E'}$ (where E' is a random β-thinning of E) lifted to $\mathbb{F}_2^E$ is close to a random subspace of $\mathbb{F}_2^E$ (even with conditioning on an advice).

Proposition 5.4. Let E be a tripled set. Fix a parameter $\beta \in [0,1)$ and nonnegative integers q_1, q_2, ℓ such that $|E| \geqslant \ell \geqslant q_2$. Fix arbitrary tuples

$$A_1 \in (\mathbb{F}_2^E)^{q_1} \qquad \text{and} \qquad A_2 \in (\mathbb{F}_2^E)^{q_2} \text{ such that } \dim \mathrm{span}(A_2) = q_2.$$

Sample

$$(E', A_{\mathsf{in}}, A_{\mathsf{mid}}, U) \sim \mathcal{J}_{\mathsf{mid}}(E, q_1, q_2, \beta, \ell)$$

conditioned on the event $\{A_{\mathsf{in}} = A_1\} \cap \{A_{\mathsf{mid}} = A_2\}$. Then the resulting marginal distribution of U has total variation distance at most

$$4^{q_1+\ell}\beta\sqrt{|E|/3} + 2^{\ell-|E|}$$

to the uniform distribution over the set $\left\{U \in \mathsf{Gr}(\mathbb{F}_2^E, \ell) \,\middle|\, \mathrm{span}(A_2) \subseteq U\right\}$.

Proof. Deferred to Appendix F.2. □

We also need the following simpler smoothness property of $\mathcal{J}_{\mathsf{mid}}$ (derived from Proposition 4.7).

Proposition 5.5. Let E be a tripled set. Fix a parameter $\beta \in [0, 1)$ and nonnegative integers q_1, ℓ. Fix an arbitrary tuple $A \in (\mathbb{F}_2^E)^{q_1}$ and an arbitrary subspace $U_0 \in \mathsf{Gr}(\mathbb{F}_2^E, \ell)$. Sample

$$(E', A_{\mathsf{in}}, \mathtt{empty}, U) \sim \mathcal{J}_{\mathsf{mid}}(E, q_1, 0, \beta, \ell)$$

conditioned on the event $\{A_{\mathsf{in}} = A\} \cap \{U = U_0\}$. Then for any subset $S \subseteq E$ with $|S| \leqslant 2$, the probability that $S \subseteq E'$ is at least $1 - 4^{q_1+\ell+1}\beta$.

Proof. According to Definition 5.1, when $(E', A_{\mathsf{in}}, A_{\mathsf{mid}}, U)$ is sampled from $\mathcal{J}_{\mathsf{mid}}(E, q_1, 0, \beta, \ell)$, a vector sequence $A' \in (\mathbb{F}_2^{E'})^\ell$ is generated, and then U is taken to be $\mathrm{span}(A')$. Let

$$\mathcal{B} = \left\{ \left(a^{(1)}, \ldots, a^{(\ell)}\right) \in (\mathbb{F}_2^E)^\ell \,\middle|\, \mathrm{span}\left(a^{(1)}, \ldots, a^{(\ell)}\right) = U_0 \right\}$$

be the collection of bases of U_0. The event $\{U = U_0\}$ is thus equal to the disjoint union of events $\bigcup_{B\in\mathcal{B}}\{A' = B\}$. It suffices to prove that for each $B \in \mathcal{B}$, conditioned on the event $\{A_{\mathsf{in}} = A\}\cap\{A' = B\}$, we have $\mathbb{P}[S \subseteq E'] \geqslant 1 - 4^{q_1+\ell+1}\beta$ for any two-element subset $S \subseteq E$. It is easy to see that the distribution of E' conditioned on $\{A_{\mathsf{in}} = A\} \cap \{A' = B\}$ is the same as in the following process (by comparing Definitions 4.5 and 5.1):

- sample $(E', A_{\mathsf{in}}, A_{\mathsf{mid}}) \sim \mathcal{J}_{\mathsf{out}}(E, q_1, \ell, \beta)$ conditioned on $\{A_{\mathsf{in}} = A\} \cap \{A_{\mathsf{mid}} = B\}$.

The desired conclusion thus easily follows from Proposition 4.7. □

The following notation will be used later in Section 6.5.2. Here the subscript "rev" stands for "reverse," meaning that we reversely generate a thinning E' after seeing a subspace U contained in $\mathbb{F}_2^{E'}$.

Definition 5.6. Let E be a tripled set. Fix a parameter $\beta \in [0, 1)$ and a nonnegative integer ℓ. Given any subspace $U_0 \subseteq \mathbb{F}_2^E$ with dimension at most ℓ, we let $\mathcal{J}_{\mathsf{rev}}(E, U_0, \beta, \ell)$ be the output distribution of the following process:

1. Sample $(E', \mathtt{empty}, \mathtt{empty}, U) \sim \mathcal{J}_{\mathsf{mid}}(E, 0, 0, \beta, \ell)$ conditioned on the event $\{U = U_0\}$.
2. Output E'.

5.2 The Local Tensor Check Game

We now construct the first version of our middle PCP. The 2-prover 1-round game constructed in the next definition is similar to Game1b but with one important distinction. Letting f be Alice's response, instead of requiring $\mathrm{rank}(f)$ is small, we require that $\mathrm{rank}(f|_{U\otimes U})$ is small, where U is (roughly) a random ℓ-dimensional subspace in Alice's space $\mathbb{F}_2^E$ which is given to Alice as part of her challenge. Additionally, as part of his challenge, we give Bob a random subspace U' of dimension (roughly) $\ell+1$ that contains U.

Definition 5.7. Given positive integers k, ℓ_2, nonnegative integers q_1, q_2 such that $\ell_2 \geqslant q_2$, a parameter $\beta \in [0,1)$ and a 3-uniform hypergraph H, let $\mathsf{Game2a}[H, k, \ell_2, q_1, q_2, \beta]$ be the 2-Prover-1-Round game where the verifier executes the following:

1. Sample a uniformly random tripled set $E \in \mathscr{E}(H,k)$.
2. Sample $(E', A_{\mathsf{in}}, A_{\mathsf{mid}}, U) \sim \mathcal{J}_{\mathsf{mid}}(E, q_1, q_2, \beta, \ell_2)$. Sample a random vector $x \in \mathbb{F}_2^{E'}$ and let $U' = U + \mathrm{span}(x)$.
3. Send (E, A_{in}, U) to Alice and $(E', A_{\mathsf{in}}, U')$ to Bob.
4. Alice must return an NAE-satisfying symmetric bilinear form $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ such that $\mathrm{rank}(f|_{U\otimes U}) \leqslant 1$; Bob must return a bilinear form $g : \mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'} \to \mathbb{F}_2$. Accept if

$$f|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}} = g.$$

Remark 5.8. In Definition 5.7, the verifier does not use A_{mid} after generating it. Therefore, the distribution of $(E, A_{\mathsf{in}}, U, U')$ does not depend on q_2. Consequently, the value of the game $\mathsf{Game2a}[H, k, \ell_2, q_1, q_2, \beta]$ does not depend on q_2 either. The purpose of A_{mid} in Definition 5.7 is solely for the convenience of its soundness analysis in Section 5.3.

Remark 5.9. In Definition 5.7, it is easy to see that the distribution of $(E', A_{\mathsf{in}}, A_{\mathsf{mid}}, U')$ is identical to $\mathcal{J}_{\mathsf{mid}}(E, q_1, q_2, \beta, \ell_2 + 1)$.

The completeness of Game2a is straightforward and is given in the following lemma.

Lemma 5.10 (Game2a completeness)**.** *For any positive integers k, ℓ_2, nonnegative integers q_1, q_2 such that $k \geqslant \ell_2 \geqslant q_2$, parameter $\beta \in [0,1)$ and 2-colorable 3-uniform hypergraph H, we have*

$$\mathrm{val}\Big(\mathsf{Game2a}[H, k, \ell_2, q_1, q_2, \beta]\Big) = 1.$$

Proof. It is clear that the provers can ignore the subspaces U or U' given to them and pretend that they are in the game $\mathsf{Game1b}[H, k, q_1, 0, 1, \beta]$. Since there is a strategy for $\mathsf{Game1b}[H, k, q_1, 0, 1, \beta]$ with 0 rejection probability due to Lemma 4.25, it follows that the same strategy also passes $\mathsf{Game2a}[H, k, \ell_2, q_1, q_2, \beta]$ with probability 1. □

The soundness analysis of Game2a, presented in the next subsection, is much more involved. Intuitively, if we did not supply Bob with the space U', the analysis would be relatively straightforward. Indeed, in that case, while in principle Alice's response could depend on the space U she receives, in actuality such strategies do not achieve a high value, as they have to be consistent with Bob (whose response does not depend on U at all). Thus, in that case Alice's response can be thought of as not being dependent on U, and hence the tensor test on U is random and faithfully checks whether Alice's response has low-rank globally.

To make our final game 4-to-1 we must supply Bob with the space U' (otherwise we would not be able to compose with the inner PCP), and the soundness analysis in that case becomes more complicated. Indeed, there are genuine cheating strategies that Alice and Bob can use, and our argument below analyzes them (using the soundness of the Grassmann test) and shows that they can still be converted into good strategies for $\mathsf{Game1b}$.

5.3 Soundness Analysis

In this subsection, we fix a 3-uniform hypergraph H, positive integers k, ℓ_2, nonnegative integers q_1, q_2 and a parameter $\beta \in (0,1)$. We consider the game $\mathsf{Game2a}[H, k, \ell_2, q_1, q_2, \beta]$ as defined in Definition 5.7. The following notational convention will be used throughout Section 5.3 (but *not* in later parts of the paper).

Notation 5.11. We use $\mathcal{D}_{\mathsf{mid}}$ to denote the joint distribution of $(E, A_{\mathsf{in}}, A_{\mathsf{mid}}, U, E', U')$ in the game $\mathsf{Game2a}[H, k, \ell_2, q_1, q_2, \beta]$, where (E, A_{in}, U) is the question sent to Alice and $(E', A_{\mathsf{in}}, U')$ is the question sent to Bob. We use the symbol $*$ to denote marginalization over the corresponding component. For instance, sampling from the marginal distribution of (A_{in}, U, E') under $\mathcal{D}_{\mathsf{mid}}$ is denoted by $(*, A_{\mathsf{in}}, *, U, E', *) \sim \mathcal{D}_{\mathsf{mid}}$.

Notation 5.12. We will often condition the distribution $\mathcal{D}_{\mathsf{mid}}$ on some subset of the six components $(E, A_{\mathsf{in}}, A_{\mathsf{mid}}, U, E', U')$ and consider the induced joint distribution of the remaining components. For instance, if E and U' are fixed, then the conditional joint distribution of (U, E') is denoted by

$$\mathop{\mathbb{P}}_{(E,*,*,U,E',U') \sim \mathcal{D}_{\mathsf{mid}}} \left[\cdot \,\middle|\, E, U' \right].$$

Notation 5.13. A table $F[\cdot]$ is called an Alice-table if for each possible Alice-question (E, A_{in}, U) under $\mathcal{D}_{\mathsf{mid}}$, the entry $F[E, A_{\mathsf{in}}, U]$ is either $\mathtt{nil}$ or an NAE-satisfying symmetric bilinear form

$$F[E, A_{\mathsf{in}}, U] : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2 \qquad \text{such that} \qquad \operatorname{rank}\left(F[E, A_{\mathsf{in}}, U]|_{U \otimes U} \right) \leqslant 1.$$

Similarly, a table $G[\cdot]$ is called a Bob-table if for each possible Bob-question $(E', A_{\mathsf{in}}, U')$ under $\mathcal{D}_{\mathsf{mid}}$, the entry $G[E', A_{\mathsf{in}}, U']$ is either $\mathtt{nil}$ or a bilinear form

$$G[E', A_{\mathsf{in}}, U'] : \mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'} \to \mathbb{F}_2.$$

In the following notation, the function $\mathrm{Vrf}_{F,G}(\cdot,\cdot,\cdot,\cdot,\cdot)$ is the indicator of whether the verifier accepts under the strategy given by $F[\cdot]$ and $G[\cdot]$. Since A_{mid} is irrelevant to the verifier's acceptance, the $\mathrm{Vrf}_{F,G}(\cdot,\cdot,\cdot,\cdot,\cdot)$ only has five arguments instead of six.

Notation 5.14. Fix an Alice-table $F[\cdot]$ and a Bob-table $G[\cdot]$. For any tuple $(E, A_{\mathsf{in}}, *, U, E', U')$ with nonzero probability under $\mathcal{D}_{\mathsf{mid}}$, we define

$$\mathrm{Vrf}_{F,G}(E, A_{\mathsf{in}}, U, E', U') = \begin{cases} 1, & \text{if } F[E, A_{\mathsf{in}}, U]|_{\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}} = G[E', A_{\mathsf{in}}, U'], \\ 0, & \text{otherwise.} \end{cases}$$

We will often sample $(E, A_{\mathsf{in}}, *, U, E', U')$ from $\mathcal{D}_{\mathsf{mid}}$ conditioned on some subset of its components. For instance, if E and U' are fixed, then we denote the corresponding conditional expectation of $\mathrm{Vrf}_{F,G}(E, A_{\mathsf{in}}, U, E', U')$ by

$$\mathrm{Vrf}_{F,G}(E, *, *, *, U') = \mathop{\mathbb{P}}_{(E,A_{\mathsf{in}},*,U,E',U') \sim \mathcal{D}_{\mathsf{mid}}} \left[F[E, A_{\mathsf{in}}, U]|_{\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}} = G[E', A_{\mathsf{in}}, U'] \,\middle|\, E, U' \right].$$

When none of the components is fixed, we write $\mathrm{Vrf}_{F,G} = \mathrm{Vrf}_{F,G}(*,*,*,*,*)$ for short.

5.3.1 Preparing for Grassmann Decoding

The core of the soundness analysis of Game2a is to use the Grassmann decoding theorem (Theorem 3.6). In order to make use of that theorem, we first need to convert the agreement between Alice's and Bob's tables into the condition that Alice's table *alone* passes the "Grassmann equality test" (see Equation (3.2)) with noticeable probability. This is achieved by the following lemma.

Lemma 5.15. *Assume Condition 3.12. Let $F[\cdot]$ be an Alice-table and $G[\cdot]$ be a Bob-table. For any $E \in \mathscr{E}(H,k)$ and any $A_{\mathsf{in}} \in (\mathbb{F}_2^E)^{q_1}$, we have*

$$\mathop{\mathbb{P}}_{(U_1,U_2)\sim\mathcal{G}(\mathbb{F}_2^E,\ell_2)}\left[F[E,A_{\mathsf{in}},U_1] = F[E,A_{\mathsf{in}},U_2] \neq \mathtt{nil}\right] \geqslant \mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},*,*,*)^2 - 2^{-\ell_2}.$$

Proof. We organize the proof into the following three steps:

Step 1: fixing E' and U'. Fix a subset $E' \subseteq E$ with size $|E'| \geqslant \ell_2 + 1$ and a subspace $U' \in \mathsf{Gr}(\mathbb{F}_2^{E'}, \ell_2+1)$ such that the tuple $(E, A_{\mathsf{in}}, *, *, E', U')$ has nonzero probability under $\mathcal{D}_{\mathsf{mid}}$. By Definitions 5.1 and 5.7, it is easy to see that sampling $(E, A_{\mathsf{in}}, *, U, E', U') \sim \mathcal{D}_{\mathsf{mid}}$ conditioned on E, A_{in}, E' and U' is equivalent to sampling a uniformly random subspace $U \in \mathsf{Gr}(U', \ell_2)$. Therefore, we have

$$\begin{aligned}\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},*,E',U')^2 &= \mathop{\mathbb{E}}_{(E,A_{\mathsf{in}},*,U,E',U')\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},U,E',U') \,\middle|\, E,A_{\mathsf{in}},E',U'\right]^2 \\ &= \mathop{\mathbb{E}}_{U_1,U_2\in\mathsf{Gr}(U',\ell_2)}\left[\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},U_1,E',U')\,\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},U_2,E',U')\right].\end{aligned} \tag{5.1}$$

Expanding the right-hand side of Equation (5.1) by definition, we get

$$\begin{aligned}&\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},*,E',U')^2 \\ =\ &\mathop{\mathbb{P}}_{U_1,U_2\in\mathsf{Gr}(U',\ell_2)}\left[F[E,A_{\mathsf{in}},U_1]|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}} = G[E',A_{\mathsf{in}},U'] = F[E,A_{\mathsf{in}},U_2]|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}}\right] \\ \leqslant\ &\mathop{\mathbb{P}}_{U_1,U_2\in\mathsf{Gr}(U',\ell_2)}\left[F[E,A_{\mathsf{in}},U_1]|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}} = F[E,A_{\mathsf{in}},U_2]|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}}\right].\end{aligned} \tag{5.2}$$

Step 2: fixing only U'. For any fixed $U' \in \mathsf{Gr}(\mathbb{F}_2^E, \ell_2+1)$ and $U_1, U_2 \in \mathsf{Gr}(U', \ell_2)$ such that $F[E,A_{\mathsf{in}},U_1] \neq F[E,A_{\mathsf{in}},U_2]$, one of the following two possibilities holds:

(1) At least one of $F[E,A_{\mathsf{in}},U_1]$ and $F[E,A_{\mathsf{in}},U_2]$ is $\mathtt{nil}$.

(2) Both of $F[E,A_{\mathsf{in}},U_1]$ and $F[E,A_{\mathsf{in}},U_2]$ are bilinear forms $\mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$, and their difference $f = F[E,A_{\mathsf{in}},U_1] - F[E,A_{\mathsf{in}},U_2]$ is nonzero. In this case, there exist two unit vectors $x, y \in \mathbb{F}_2^E$ (i.e. $|\mathrm{supp}(x)| = |\mathrm{supp}(y)| = 1$) such that $f(x \otimes y) = 1$. Therefore, for any subset $E' \subseteq E$ that contains $\mathrm{supp}(x) \cup \mathrm{supp}(y)$, we have $F[E,A_{\mathsf{in}},U_1]|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}} \neq F[E,A_{\mathsf{in}},U_2]|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}}$.

In either of the above two cases, it follows from Proposition 5.5 and Remark 5.9 that (note that U', U_1 and U_2 remain fixed in the following inequality)

$$\mathop{\mathbb{P}}_{(E,A_{\mathsf{in}},*,*,E',U')\sim\mathcal{D}_{\mathsf{mid}}}\left[F[E,A_{\mathsf{in}},U_1]|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}} = F[E,A_{\mathsf{in}},U_2]|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}} \,\middle|\, E,A_{\mathsf{in}},U'\right] \leqslant 4^{q_1+\ell_2+2}\beta.$$

Therefore, taking expectation of Equation (5.2) over the conditional distribution of E' yields (note that U_1 and U_2 are no longer fixed in the following inequality)

$$\begin{aligned}&\mathop{\mathbb{E}}_{(E,A_{\mathsf{in}},*,*,E',U')\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},*,E',U')^2\,\middle|\,E,A_{\mathsf{in}},U'\right]\\&\leqslant\mathop{\mathbb{P}}_{U_1,U_2\in\mathsf{Gr}(U',\ell_2)}\left[F[E,A_{\mathsf{in}},U_1]=F[E,A_{\mathsf{in}},U_2]\neq\mathtt{nil}\right]+4^{q_1+\ell_2+2}\beta.\end{aligned}\tag{5.3}$$

Step 3: both E' and U' unfixed. By Proposition 5.4 and Remark 5.9, we know that conditioned on the fixed E and A_{in}, the distribution of U' in the sample $(E,A_{\mathsf{in}},*,*,*,U')\sim\mathcal{D}_{\mathsf{mid}}$ is $\left(4^{q_1+\ell_2+1}\beta\sqrt{k}+2^{\ell_2+1-3k}\right)$-close to uniform over $\mathsf{Gr}(\mathbb{F}_2^E,\ell_2+1)$. Therefore, we have

$$\begin{aligned}&\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},*,*,*)^2\\&=\mathop{\mathbb{E}}_{(E,A_{\mathsf{in}},*,*,E',U')\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},*,E',U')\,\middle|\,E,A_{\mathsf{in}}\right]^2&&\text{(by definition)}\\&\leqslant\mathop{\mathbb{E}}_{(E,A_{\mathsf{in}},*,*,E',U')\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},*,E',U')^2\,\middle|\,E,A_{\mathsf{in}}\right]&&\text{(by Cauchy-Schwarz)}\\&\leqslant\mathop{\mathbb{E}}_{U'\in\mathsf{Gr}(\mathbb{F}_2^E,\ell_2+1)}\left[\mathop{\mathbb{E}}_{(E,A_{\mathsf{in}},*,*,E',U')\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},*,E',U')^2\,\middle|\,E,A_{\mathsf{in}},U'\right]\right]+\\&\quad 4^{q_1+\ell_2+1}\beta\sqrt{k}+2^{\ell_2+1-3k}.\end{aligned}$$

We then apply Equation (5.3) to the above and get

$$\begin{aligned}\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},*,*,*)^2\leqslant&\mathop{\mathbb{E}}_{U'\in\mathsf{Gr}(\mathbb{F}_2^E,\ell_2+1)}\left[\mathop{\mathbb{P}}_{U_1,U_2\in\mathsf{Gr}(U',\ell_2)}\left[F[E,A_{\mathsf{in}},U_1]=F[E,A_{\mathsf{in}},U_2]\neq\mathtt{nil}\right]\right]+\\&4^{q_1+\ell_2+1}\beta\left(4+\sqrt{k}\right)+2^{\ell_2+1-3k}.\end{aligned}\tag{5.4}$$

For any fixed $U'\in\mathsf{Gr}(\mathbb{F}_2^E,\ell_2+1)$, two independent samples $U_1,U_2\in\mathsf{Gr}(U',\ell_2)$ are equal to each other with probability $(2^{\ell_2+1}-1)^{-1}$. Furthermore, if $U'\in\mathsf{Gr}(\mathbb{F}_2^E,\ell_2+1)$ is sampled uniformly at random and $U_1,U_2\in\mathsf{Gr}(U',\ell_2)$ are sampled uniformly conditioned on $U_1\neq U_2$, the distribution of (U_1,U_2) is uniform over the edges of the Grassmann graph $\mathsf{Gr}(\mathbb{F}_2^E,\ell_2)$. Therefore, Equation (5.4) implies

$$\begin{aligned}\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},*,*,*)^2&\leqslant\mathop{\mathbb{P}}_{(U_1,U_2)\sim\mathcal{G}(\mathbb{F}_2^E,\ell_2)}\left[F[E,A_{\mathsf{in}},U_1]=F[E,A_{\mathsf{in}},U_2]\neq\mathtt{nil}\right]+\frac{1}{2^{\ell_2+1}-1}\\&\quad+4^{q_1+\ell_2+1}\beta\left(4+\sqrt{k}\right)+2^{\ell_2+1-3k}\\&\leqslant\mathop{\mathbb{P}}_{(U_1,U_2)\sim\mathcal{G}(\mathbb{F}_2^E,\ell_2)}\left[F[E,A_{\mathsf{in}},U_1]=F[E,A_{\mathsf{in}},U_2]\neq\mathtt{nil}\right]+2^{-\ell_2},\end{aligned}$$

where we used Condition 3.12 in the last transition. □

5.3.2 Invoking the Grassmann Decoding Theorem

We are now ready to apply the Grassmann decoding theorem to Alice's table. The advice A_{mid}, which played no real role in the definition of Game2a, helps us state the following lemma.

Lemma 5.16. *Assume Condition 3.12. Let $F[\cdot]$ be an Alice-table and $G[\cdot]$ be a Bob-table. For any $E\in\mathscr{E}(H,k)$ and any $A_{\mathsf{in}}\in(\mathbb{F}_2^E)^{q_1}$ such that $\mathrm{Vrf}_{F,G}(E,A_{\mathsf{in}},*,*,*)\geqslant 2^{-q_2/3}$, the following holds:*

with probability at least $4^{-\ell_2^2}$ *over* $(E, A_{\mathsf{in}}, A, *, *, *) \sim \mathcal{D}_{\mathsf{mid}}$ *conditioned on* E *and* A_{in}*, there exists an* NAE*-satisfying symmetric bilinear map* $f_A : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ *of rank at most* $2q_2 + 1$ *satisfying*

$$\mathop{\mathbb{P}}_{(E,A_{\mathsf{in}},A,U,*,*)\sim\mathcal{D}_{\mathsf{mid}}}\left[F[E, A_{\mathsf{in}}, U] = f_A \,\middle|\, E, A_{\mathsf{in}}, A\right] \geqslant 2^{-\ell_2^2}.$$

Proof. We organize the proof into the following three steps.

Step 1: Grassmann decoding. Since $\mathrm{Vrf}_{F,G}(E, A_{\mathsf{in}}, *, *, *) \geqslant 2^{-q_2/3}$, Lemma 5.15 implies

$$\mathop{\mathbb{P}}_{(U_1,U_2)\sim\mathcal{G}(\mathbb{F}_2^E,\ell_2)}\left[F[E, A_{\mathsf{in}}, U_1] = F[E, A_{\mathsf{in}}, U_2] \neq \mathtt{nil}\right] \geqslant 2^{-2q_2/3} - 2^{-\ell_2} \geqslant 2^{-q_2+4},$$

where we used Condition 3.12 in the last transition. Applying Theorem 3.6 to the table $F[E, A_{\mathsf{in}}, \cdot]$ yields[8] a collection of subspaces $\mathcal{Q} \subseteq \mathsf{Gr}(\mathbb{F}_2^E, q_2)$ such that:

(1) $|\mathcal{Q}| \geqslant 2^{-q_2-\ell_2^2} \left|\mathsf{Gr}(\mathbb{F}_2^E, q_2)\right|$;

(2) for each $Q \in \mathcal{Q}$, there exists a subspace $W_Q \in \mathsf{Gr}(\mathbb{F}_2^E, 3k - q_2)$ and an NAE-satisfying symmetric bilinear map $h_Q : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ satisfying

$$\mathop{\mathbb{P}}_{U\in\mathsf{Gr}(\mathbb{F}_2^E,\ell_2)}\left[F[E, A_{\mathsf{in}}, U] = h_Q \,\middle|\, Q \subseteq U \subseteq W_Q\right] \geqslant 2^{-Cq_2^2}, \tag{5.5}$$

where C is an absolute constant. For any $U \in \mathsf{Gr}(\mathbb{F}_2^E, \ell_2)$ such that $F[E, A_{\mathsf{in}}, U] = h_Q$, by definition (see Notation 5.13) we must have $\mathrm{rank}(h_Q|_{U\otimes U}) \leqslant 1$. It thus follows from Equation (5.5) that

$$\mathop{\mathbb{P}}_{U\in\mathsf{Gr}(\mathbb{F}_2^E,\ell_2)}\left[\mathrm{rank}(h_Q|_{U\otimes U}) \leqslant 1 \,\middle|\, Q \subseteq U \subseteq W_Q\right] \geqslant 2^{-Cq_2^2} > 2^{4-(\ell_2-q_2)/2},$$

where we used Condition 3.12 in the last transition. By Lemma A.5, we must have $\mathrm{rank}(h_Q) \leqslant 2q_2 + 1$.

For each $Q \in \mathcal{Q}$, since W_Q has codimension q_2 with respect to $\mathbb{F}_2^E$, we have

$$\mathop{\mathbb{P}}_{U\in\mathsf{Gr}(\mathbb{F}_2^E,\ell_2)}\left[U \subseteq W_Q \,\middle|\, Q \subseteq U\right] = \begin{bmatrix} 3k - 2q_2 \\ \ell_2 - q_2 \end{bmatrix}_2 \begin{bmatrix} 3k - q_2 \\ \ell_2 - q_2 \end{bmatrix}_2^{-1} \geqslant 2^{-\ell_2(q_2+1)}. \tag{5.6}$$

Combining Equation (5.5) and Equation (5.6) yields

$$\mathop{\mathbb{P}}_{U\in\mathsf{Gr}(\mathbb{F}_2^E,\ell_2)}\left[F[E, A_{\mathsf{in}}, U] = h_Q \,\middle|\, Q \subseteq U\right] \geqslant 2^{-Cq_2^2-\ell_2(q_2+1)} \geqslant 2^{1-\ell_2^2}, \tag{5.7}$$

where we used Condition 3.12 in the last transition.

[8] We apply Theorem 3.6 with Σ equal to the collection of NAE-satisfying symmetric bilinear maps from $\mathbb{F}_2^E \otimes \mathbb{F}_2^E$ to $\mathbb{F}_2$.

Step 2: good Q translates to good A. Now for any $A \in (\mathbb{F}_2^E)^{q_2}$, we let

$$f_A = \begin{cases} h_{\mathrm{span}(A)}, & \text{if } \dim \mathrm{span}(A) = q_2 \text{ and } \mathrm{span}(A) \in \mathcal{Q}, \\ 0, & \text{otherwise.} \end{cases}$$

If $\dim \mathrm{span}(A) = q_2$ and $\mathrm{span}(A) \in \mathcal{Q}$, then by Equation (5.7) we have

$$\mathop{\mathbb{P}}_{U \in \mathsf{Gr}(\mathbb{F}_2^E, \ell_2)} \Big[F[E, A_{\mathsf{in}}, U] = f_A \,\Big|\, \mathrm{span}(A) \subseteq U \Big] \geqslant 2^{1-\ell_2^2}. \tag{5.8}$$

By Proposition 5.4, we know that conditioned on the fixed E, A_{in} and $A_{\mathsf{mid}} = A$, the distribution of U in the sample $(E, A_{\mathsf{in}}, A, U, *, *) \sim \mathcal{D}_{\mathsf{mid}}$ is $\big(4^{q_1+q_2+\ell_2}\beta\sqrt{k} + 2^{\ell_2-3k}\big)$-close to uniform over the set $\big\{U \in \mathsf{Gr}(\mathbb{F}_2^E, \ell_2) \,\big|\, \mathrm{span}(A) \subseteq U\big\}$. Therefore, Equation (5.8) implies

$$\mathop{\mathbb{P}}_{(E, A_{\mathsf{in}}, A, U, *, *) \sim \mathcal{D}_{\mathsf{mid}}} \Big[F[E, A_{\mathsf{in}}, U] = f_A \,\Big|\, E, A_{\mathsf{in}}, A \Big] \geqslant 2^{1-\ell_2^2} - 4^{q_1+q_2+\ell_2}\beta\sqrt{k} - 2^{\ell_2-3k} \geqslant 2^{-\ell_2^2},$$

where we used Condition 3.12 in the last transition.

Step 3: abundance of good A's. It remains to show that

$$\mathop{\mathbb{P}}_{(E, A_{\mathsf{in}}, A, *, *, *) \sim \mathcal{D}_{\mathsf{mid}}} \Big[\dim \mathrm{span}(A) = q_2 \text{ and } \mathrm{span}(A) \in \mathcal{Q} \,\Big|\, E, A_{\mathsf{in}} \Big] \geqslant 4^{-\ell_2^2}.$$

By Proposition 4.8, it suffices to show

$$\mathop{\mathbb{P}}_{A \in (\mathbb{F}_2^E)^{q_2}} \Big[\dim \mathrm{span}(A) = q_2 \text{ and } \mathrm{span}(A) \in \mathcal{Q} \Big] \geqslant 4^{-\ell_2^2} + 4^{q_1+q_2}\beta\sqrt{k}.$$

But the left-hand side of the above display is clearly equal to

$$\mathop{\mathbb{P}}_{A \in (\mathbb{F}_2^E)^{q_2}} \Big[\dim \mathrm{span}(A) = q_2 \Big] \cdot \frac{|\mathcal{Q}|}{\big|\mathsf{Gr}(\mathbb{F}_2^E, q_2)\big|} \geqslant \frac{1}{4} \cdot 2^{-q_2-\ell_2^2} \geqslant 4^{-\ell_2^2} + 4^{q_1+q_2}\beta\sqrt{k},$$

where we used Condition 3.12 in the last transition. □

5.3.3 Deriving Strategy for the Outer PCP

We now use the structural results regarding Alice's strategy to complete the soundness analysis of Game2a. Before we do so, fixing an Alice-table and a Bob-table, we prune the Alice-table and only keep the responses on questions which have noticeable success probability. This will ensure that whatever functions we decode from Alice's table have noticeable consistency with Bob's table.

Definition 5.17. Let $\eta \in (0, 1)$ be a threshold parameter. Given an Alice-table $F[\cdot]$ and a Bob-table $G[\cdot]$ for $\mathsf{Game2a}[H, k, \ell_2, q_1, q_2, \beta]$, consider the Alice-table $F'[\cdot]$ defined by

$$F'[E, A_{\mathsf{in}}, U] = \begin{cases} F[E, A_{\mathsf{in}}, U], & \text{if } \mathrm{Vrf}_{F,G}(E, A_{\mathsf{in}}, U, *, *) \geqslant \eta, \\ \mathtt{nil}, & \text{otherwise.} \end{cases}$$

The pair of tables (F', G) is called the *η-pruning* of (F, G).

The following straightforward proposition shows that η-pruning drops the success probability of Alice and Bob by at most η.

Proposition 5.18. Let $\eta \in (0,1)$ be a threshold parameter. Given an Alice-table $F[\cdot]$ and a Bob-table $G[\cdot]$ for $\mathsf{Game2a}[H,k,\ell_2,q_1,q_2,\beta]$, let (F',G) be the η-pruning of (F,G). Then

$$\mathrm{Vrf}_{F',G} \geqslant \mathrm{Vrf}_{F,G} - \eta.$$

Proof. For any $E \in \mathscr{E}(H,k)$, any $A_{\mathsf{in}} \in (\mathbb{F}_2^E)^{q_1}$ and any $U \in \mathsf{Gr}(\mathbb{F}_2^E, \ell_2)$, from Definition 5.17 it is easy to see that

$$\mathrm{Vrf}_{F',G}(E, A_{\mathsf{in}}, U, *, *) \geqslant \mathrm{Vrf}_{F,G}(E, A_{\mathsf{in}}, U, *, *) - \eta.$$

Taking expectation over $(E, A_{\mathsf{in}}, *, U, *, *) \sim \mathcal{D}_{\mathsf{mid}}$ then yields the desired conclusion. □

We are now ready to state and prove the soundness of $\mathsf{Game2a}$.

Lemma 5.19 (Game2a soundness)**.** *Assume Condition 3.12. Suppose that in every 4^r-coloring of H, at least ε fraction of the edges are monochromatic. Then we have*

$$\mathrm{val}\Big(\mathsf{Game2a}[H,k,\ell_2,q_1,q_2,\beta]\Big) \leqslant 2^{-q_2/4}.$$

Proof. Assume on the contrary that the value of $\mathsf{Game2a}[H,k,\ell_2,q_1,q_2,\beta]$ is larger than $2^{-q_2/4}$. Then by definition, there are an Alice-table $F[\cdot]$ and a Bob-table $G[\cdot]$ for $\mathsf{Game2a}[H,k,\ell_2,q_1,q_2,\beta]$ such that $\mathrm{Vrf}_{F,G} > 2^{-q_2/4}$. Let (F',G) be the $(2^{1-q_2/3})$-pruning of (F,G). By Proposition 5.18 we have

$$\mathrm{Vrf}_{F',G} \geqslant 2^{-q_2/4} - 2^{1-q_2/3} \geqslant 2^{1-q_2/3},$$

where we used $q_2 \geqslant 24$ (as guaranteed by Condition 3.12).

Step 1: strategy for Game1b. For each $E \in \mathscr{E}(H,k)$ and each possible sample $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$ from $\mathcal{J}_{\mathsf{out}}(E,q_1,q_2,\beta)$, let $\mathcal{F}[E,A_{\mathsf{in}},A_{\mathsf{mid}}]$ be the collection of all NAE-satisfying symmetric bilinear maps $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ such that $\mathrm{rank}(f) \leqslant 2q_2+1$ and

$$\mathop{\mathbb{P}}_{(E,A_{\mathsf{in}},A_{\mathsf{mid}},U,*,*)\sim\mathcal{D}_{\mathsf{mid}}}\Big[F'[E,A_{\mathsf{in}},U] = f \,\Big|\, E, A_{\mathsf{in}}, A_{\mathsf{mid}}\Big] \geqslant 2^{-\ell_2^2}.$$

We now derive strategies for the provers in $\mathsf{Game1b}[H,k,q_1,q_2,r,\beta]$ from the tables $F'[\cdot]$ and $G[\cdot]$. The strategy for Alice is as follows:

1. Receive the question $(E, A_{\mathsf{in}}, A_{\mathsf{mid}})$.
2. If $\mathcal{F}[E,A_{\mathsf{in}},A_{\mathsf{mid}}]$ is empty, return an arbitrary NAE-satisfying symmetric bilinear map $f : \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ of rank 1 (see Corollary 4.16).
3. If $\mathcal{F}[E,A_{\mathsf{in}},A_{\mathsf{mid}}]$ is nonempty, return an arbitrary NAE-satisfying symmetric bilinear map $f \in \mathcal{F}[E,A_{\mathsf{in}},A_{\mathsf{mid}}]$. By the definition of $\mathcal{F}[E,A_{\mathsf{in}},A_{\mathsf{mid}}]$, we have $\mathrm{rank}(f) \leqslant 2q_2+1 = r$, where the last transition is due to Condition 3.12.

The strategy for Bob is:

1. Receive the question $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$.
2. Conditioned on E', A_{in} and A_{mid}, sample $(*, A_{\mathsf{in}}, A_{\mathsf{mid}}, *, E', U') \sim \mathcal{D}_{\mathsf{mid}}$.
3. Return the bilinear map $G[E', A_{\mathsf{in}}, U']$ (note that by definition, the table $G[\cdot]$ does not contain `nil` entries).

Step 2: analysis of the strategy. Since

$$\mathop{\mathbb{E}}_{(E,A_{\mathsf{in}},*,*,*,*)\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathrm{Vrf}_{F',G}(E,A_{\mathsf{in}},*,*,*)\right]=\mathrm{Vrf}_{F',G}\geqslant 2^{1-q_2/3},$$

we have[9]

$$\mathop{\mathbb{P}}_{(E,A_{\mathsf{in}},*,*,*,*)\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathrm{Vrf}_{F',G}(E,A_{\mathsf{in}},*,*,*)\geqslant 2^{-q_2/3}\right]\geqslant 2^{-q_2/3}. \tag{5.9}$$

For any $E\in\mathscr{E}(H,k)$ and $A_{\mathsf{in}}\in(\mathbb{F}_2^E)^{q_1}$ such that $\mathrm{Vrf}_{F',G}(E,A_{\mathsf{in}},*,*,*)\geqslant 2^{-q_2/3}$, we know from Lemma 5.16 that

$$\mathop{\mathbb{P}}_{(E,A_{\mathsf{in}},A_{\mathsf{mid}},*,*,*)\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathcal{F}[E,A_{\mathsf{in}},A_{\mathsf{mid}}]\neq\emptyset\;\middle|\;E,A_{\mathsf{in}}\right]\geqslant 4^{-\ell_2^2}.$$

Combining with Equation (5.9) yields

$$\mathop{\mathbb{P}}_{(E,A_{\mathsf{in}},A_{\mathsf{mid}},*,*,*)\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathcal{F}[E,A_{\mathsf{in}},A_{\mathsf{mid}}]\neq\emptyset\right]\geqslant 2^{-2\ell_2^2-q_2/3}. \tag{5.10}$$

Given a tuple $(E,A_{\mathsf{in}},A_{\mathsf{mid}})$ and an element $f\in\mathcal{F}[E,A_{\mathsf{in}},A_{\mathsf{mid}}]$, we have

$$\begin{aligned}
&\mathop{\mathbb{P}}_{(E,A_{\mathsf{in}},A_{\mathsf{mid}},*,E',U')\sim\mathcal{D}_{\mathsf{mid}}}\left[G[E',A_{\mathsf{in}},U']=f|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}}\;\middle|\;E,A_{\mathsf{in}},A_{\mathsf{mid}}\right]\\
\geqslant&\mathop{\mathbb{P}}_{(E,A_{\mathsf{in}},A_{\mathsf{mid}},U,E',U')\sim\mathcal{D}_{\mathsf{mid}}}\left[F'[E,A_{\mathsf{in}},U]=f\text{ and }\mathrm{Vrf}_{F',G}(E,A_{\mathsf{in}},U,E',U')=1\;\middle|\;E,A_{\mathsf{in}},A_{\mathsf{mid}}\right]\\
=&\mathop{\mathbb{E}}_{(E,A_{\mathsf{in}},A_{\mathsf{mid}},U,*,*)\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathbb{1}\left[F'[E,A_{\mathsf{in}},U]=f\right]\cdot\mathrm{Vrf}_{F',G}(E,A_{\mathsf{in}},U,*,*)\;\middle|\;E,A_{\mathsf{in}},A_{\mathsf{mid}}\right]\\
\geqslant&\,2^{1-q_2/3}\cdot\mathop{\mathbb{P}}_{(E,A_{\mathsf{in}},A_{\mathsf{mid}},U,*,*)\sim\mathcal{D}_{\mathsf{mid}}}\left[F'[E,A_{\mathsf{in}},U]=f\;\middle|\;E,A_{\mathsf{in}},A_{\mathsf{mid}}\right]\\
&\qquad\qquad\text{(because }(F',G)\text{ is the }(2^{1-q_2/3})\text{-pruning of }(F,G))\\
\geqslant&\,2^{1-\ell_2^2-q_2/3}.\qquad\qquad\text{(since }f\in\mathcal{F}[E,A_{\mathsf{in}},A_{\mathsf{mid}}])
\end{aligned}$$

Importantly, conditioned on E, A_{in}, A_{mid} and E', the distribution of U' in a conditional sample $(E,A_{\mathsf{in}},A_{\mathsf{mid}},*,E',U')\sim\mathcal{D}_{\mathsf{mid}}$ does not depend on E. Therefore, the above display implies that under the strategies presented in Step 1 of the proof, for any tuple $(E,A_{\mathsf{in}},A_{\mathsf{mid}})$ with any element $f\in\mathcal{F}[E,A_{\mathsf{in}},A_{\mathsf{mid}}]$, the following expectation is at least $2^{1-\ell_2^2-q_2/3}$:

$$\mathop{\mathbb{E}}_{(E,A_{\mathsf{in}},A_{\mathsf{mid}},*,E',*)\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathop{\mathbb{P}}_{\mathrm{Bob}}\left[\text{Bob returns }\;f|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}}\text{ on question }(E',A_{\mathsf{in}},A_{\mathsf{mid}})\right]\;\middle|\;E,A_{\mathsf{in}},A_{\mathsf{mid}}\right].$$

Combining with Equation (5.10), this implies that[10]

$$\mathop{\mathbb{E}}_{(E,A_{\mathsf{in}},A_{\mathsf{mid}},*,E',*)\sim\mathcal{D}_{\mathsf{mid}}}\left[\mathop{\mathbb{P}}_{\mathrm{Bob}}\left[\mathrm{Alice}(E,A_{\mathsf{in}},A_{\mathsf{mid}})|_{\mathbb{F}_2^{E'}\otimes\mathbb{F}_2^{E'}}=\mathrm{Bob}(E',A_{\mathsf{in}},A_{\mathsf{mid}})\right]\right]$$

is at least

$$2^{-2\ell_2^2-q_2/3}\cdot 2^{1-\ell_2^2-q_2/3}>k^{-1/2},$$

where we used Condition 3.12 in the last transition. Since the distribution of $(E,A_{\mathsf{in}},A_{\mathsf{mid}},E')$ in $\mathsf{Game1b}[H,k,q_1,q_2,r,\beta]$ is identical to the distribution of the sample $(E,A_{\mathsf{in}},A_{\mathsf{mid}},*,E',*)\sim\mathcal{D}_{\mathsf{mid}}$, it follows that our prescribed strategies pass $\mathsf{Game1b}[H,k,q_1,q_2,r,\beta]$ with probability

$$\mathrm{val}\Big(\mathsf{Game1b}[H,k,q_1,q_2,r,\beta]\Big)>k^{-1/2},$$

contradicting Lemma 4.27. □

[9] Here we use the fact that $\mathrm{Vrf}_{F',G}(E,A_{\mathsf{in}},*,*,*)$ always lies in $[0,1]$.

[10] Here we use $\mathrm{Alice}(E,A_{\mathsf{in}},A_{\mathsf{mid}})$ to denote Alice's answer to the question $(E,A_{\mathsf{in}},A_{\mathsf{mid}})$, and $\mathrm{Bob}(E',A_{\mathsf{in}},A_{\mathsf{mid}})$ denotes Bob's answer to $(E',A_{\mathsf{in}},A_{\mathsf{mid}})$.

5.4 Lazy Parallel Repetition

In this subsection, we apply a "lazy parallel repetition" to Game2a to get Game2b. More precisely, we consider the following 2-prover 1-round game, in which Alice and Bob get t-blocks of challenges, one of which is a true challenge from Game2a, whereas on each one of the rest of the blocks they receive the same randomly sampled challenge of Alice from Game2a. Additionally, on each block the players receive a list of advice vectors.

Definition 5.20. Given positive integers k, ℓ_2, t, nonnegative integers q_0, q_1, a parameter $\beta \in [0,1)$ and a 3-uniform hypergraph H, let $\mathsf{Game2b}[H, k, \ell_2, t, q_0, q_1, \beta]$ be a 2-Prover 1-Round game where the verifier executes the following:

1. For each $j \in [t]$, sample a uniformly random tripled set $E_j \in \mathscr{E}(H, k)$.
2. Sample a uniformly random index $c \in [t]$. Sample $(E'_c, A_c, \mathtt{empty}, U_c) \sim \mathcal{J}_{\mathsf{mid}}(E_c, q_1, 0, \beta, \ell_2)$. Sample a random vector $x \in \mathbb{F}_2^{E'_c}$ and let $U'_c = U_c + \mathrm{span}(x)$.
3. For each $j \in [t] \setminus \{c\}$, do the following:
   - Sample a uniformly random advice sequence $A_j = \left(a^{(j,1)}, \ldots, a^{(j,q_0)}\right) \in \left(\mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j}\right)^{q_0}$, i.e. $a^{(j,i)}$ is a uniformly random vector in $\mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j}$ for each $i \in [q_0]$.
   - Sample $(E_j^{\mathrm{ghost}}, \mathtt{empty}, \mathtt{empty}, U_j) \sim \mathcal{J}_{\mathsf{mid}}(E_j, 0, 0, \beta, \ell_2)$.
   - Let $E'_j = E_j$ and $U'_j = U_j$.
4. Denote sequences
$$\mathcal{E} = (E_1, \ldots, E_t), \qquad \mathcal{U} = (U_1, \ldots, U_t), \qquad \mathcal{A} = (A_1, \ldots, A_t), \qquad \text{and}$$
$$\mathcal{E}' = (E'_1, \ldots, E'_t), \qquad \mathcal{U}' = (U'_1, \ldots, U'_t).$$
5. Send $(\mathcal{E}, \mathcal{A}, \mathcal{U})$ to Alice and $(c, \mathcal{E}', \mathcal{A}, \mathcal{U}')$ to Bob.
6. Alice must return NAE-satisfying symmetric bilinear forms
$$f_j : \mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j} \to \mathbb{F}_2 \text{ such that } \mathrm{rank}(f_j|_{U_j \otimes U_j}) \leqslant 1 \qquad \text{for each } j \in [t];$$
   Bob must return bilinear forms
$$g_j : \mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j} \to \mathbb{F}_2 \qquad \text{for each } j \in [t].$$
   Accept if g_j is the restriction of f_j to $\mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j}$ for each $j \in [t]$.

If we ignore the lists A_j, the game we have above is exactly the lazy parallel repetition version of Game2a. Lazy parallel repetition is well-known to preserve completeness and soundness (but not improve them), as well as to improve the "smoothness" of a game. We include the lists A_j to enable the advice feature in the inner PCP, and we show that they do not hurt the soundness of the construction (they have no effect on other features of the construction).

The completeness of Game2b is again straightforward:

Lemma 5.21 (Game2b completeness)**.** *For any positive integers k, ℓ_2, t such that $k \geqslant \ell_2$, nonnegative integers q_0, q_1, parameter $\beta \in [0,1)$ and 2-colorable 3-uniform hypergraph H, we have*
$$\mathrm{val}\Big(\mathsf{Game2b}[H, k, \ell_2, t, q_0, q_1, \beta]\Big) = 1.$$

Proof. We fix strategies from Lemma 5.10 that achieve value 1. Alice ignores the lists A_j and answers each block $j \in [t]$ separately according to her strategy from Game2a. Bob answers according to Alice's strategy for Game2a on the noncritical blocks $j \in [t] \setminus \{c\}$, and according to Bob's strategy for Game2a on the critical block $j = c$. □

The following lemma establishes the soundness of Game2b.

Lemma 5.22 (Game2b soundness)**.** *Assume Condition 3.12. Suppose that in every 4^r-coloring of H, at least ε fraction of the edges are monochromatic. Then we have*

$$\mathrm{val}\Big(\mathsf{Game2b}[H, k, \ell_2, t, q_0, q_1, \beta]\Big) \leqslant 2^{-q_2/4}.$$

Proof. It suffices to show that

$$\mathrm{val}\Big(\mathsf{Game2b}[H, k, \ell_2, t, q_0, q_1, \beta]\Big) \leqslant \mathrm{val}\Big(\mathsf{Game2a}[H, k, \ell_2, q_1, 0, \beta]\Big),$$

after which the conclusion follows from Lemma 5.19 (combined with Remark 5.3).

Assume Alice and Bob have a strategy for $\mathsf{Game2b}[H, k, \ell_2, t, q_0, q_1, \beta]$, and we next show how to derive a strategy for $\mathsf{Game2a}[H, k, \ell_2, q_1, 0, \beta]$. In the latter game, Alice and Bob can first agree upon the following "shared data" using public randomness:

- a uniformly random critical index $c \in [t]$, and
- a uniformly random tuple (E_j, A_j, U_j) for each non-critical index $j \in [t] \setminus \{c\}$. More precisely, sample E_j from $\mathscr{E}(H, k)$ uniformly, sample A_j from $\big(\mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j}\big)^{q_0}$ uniformly, and sample U_j by drawing $(E_j^{\mathrm{ghost}}, \mathtt{empty}, \mathtt{empty}, U_j) \sim \mathcal{J}_{\mathsf{mid}}(E_j, 0, 0, \beta, \ell_2)$.

Given a question (E, A, U) in $\mathsf{Game2a}[H, k, \ell_2, q_1, 0, \beta]$, Alice can construct $(\mathcal{E}, \mathcal{A}, \mathcal{U})$ by planting (E, A, U) in the critical block c and using the shared data for the non-critical blocks. Similarly, given a question (E', A', U'), Bob can construct $(\mathcal{E}', \mathcal{A}', \mathcal{U}')$ by planting (E', A', U') in the critical block c and using the shared data for the non-critical blocks. The provers can then pretend that they are in $\mathsf{Game2b}[H, k, \ell_2, t, q_0, q_1, \beta]$, and use their strategy for $\mathsf{Game2b}[H, k, \ell_2, t, q_0, q_1, \beta]$ to answer the verifier in $\mathsf{Game2a}[H, k, \ell_2, q_1, 0, \beta]$. Whenever the strategy for $\mathsf{Game2b}[H, k, \ell_2, t, q_0, q_1, \beta]$ provides consistent answers to $(\mathcal{E}, \mathcal{A}, \mathcal{U})$ and $(c, \mathcal{E}', \mathcal{A}', \mathcal{U}')$, the consistency in the critical block implies that the derived answers to the questions (E, A, U) and (E', A', U') in $\mathsf{Game2a}[H, k, \ell_2, q_1, 0, \beta]$ are also consistent. Therefore, the verifier in $\mathsf{Game2a}[H, k, \ell_2, q_1, 0, \beta]$ accepts with probability at least the acceptance probability in $\mathsf{Game2b}[H, k, \ell_2, t, q_0, q_1, \beta]$. □

6 The Inner PCP

In this section we construct and analyze our inner PCP as discussed in Section 2.2.6. We first prepare some necessary framework in Sections 6.1 and 6.2. The final sub-4-to-1 game will then be defined in Section 6.3.

6.1 Preparation: Tripled Packets

In the inner PCP, we will add another layer of Grassmann-type structure onto Game2b, which is already somewhat complicated. To make future notations more concise, in this subsection we set up some notational framework for the objects featured in Game2b (Definition 5.20).

Recall that in Definition 5.20, for each index $j \in [t]$, Alice needs to return a bilinear map $f : \mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j} \to \mathbb{F}_2$ that is both symmetric and NAE-satisfying. The following definition combines the two conditions, exploiting the fact that both are "linear" conditions in f.

Definition 6.1. Let E be a tripled set. For any $v \in E$, we let $e_v \in \mathbb{F}_2^E$ denote the standard basis vector supported on v; that is, for $u \in E$ let $(e_v)_u = 1$ if and only if $u = v$.

(1) We define $\mathcal{N}_1(E) \subseteq \mathbb{F}_2^E \otimes \mathbb{F}_2^E$ to be the linear subspace spanned by all vectors of the form

$$e_{v_1} \otimes e_{v_1} + e_{v_2} \otimes e_{v_2} + e_{v_3} \otimes e_{v_3} + e_{v_1} \otimes e_{v_2} + e_{v_2} \otimes e_{v_3} + e_{v_3} \otimes e_{v_1},$$

where $\{v_1, v_2, v_3\}$ ranges in Triples(E).[11]

(2) We let $\mathcal{N}_2(E) \subseteq \mathbb{F}_2^E \otimes \mathbb{F}_2^E$ be the linear subspace spanned by all anti-symmetric vectors, i.e.

$$\mathcal{N}_2(E) = \operatorname{span}\left\{x \otimes y + y \otimes x \,\middle|\, x, y \in \mathbb{F}_2^E\right\}.$$

(3) Let $\mathcal{N}(E) = \mathcal{N}_1(E) + \mathcal{N}_2(E)$ be the sum of the two subspaces.

(4) A linear map $f : \mathcal{N}(E) \to \mathbb{F}_2$ is said to be SymNAE-*satisfying* if we have:

- $f(e_{v_1} \otimes e_{v_1} + e_{v_2} \otimes e_{v_2} + e_{v_3} \otimes e_{v_3} + e_{v_1} \otimes e_{v_2} + e_{v_2} \otimes e_{v_3} + e_{v_3} \otimes e_{v_1}) = 1$ for all triples $\{v_1, v_2, v_3\} \in$ Triples(E); and
- $f(x \otimes y + y \otimes x) = 0$ for all $x, y \in \mathbb{F}_2^E$.

Intuitively, the space $\mathcal{N}_1(E)$ from the first item above is the subspace of $\mathbb{F}_2^E \otimes \mathbb{F}_2^E$ on which the value of a bilinear form $f \colon \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ is fully determined if it is NAE-satisfying. The space $\mathcal{N}_2(E)$ from the second item above is the subspace of $\mathbb{F}_2^E \otimes \mathbb{F}_2^E$ on which the value of a bilinear form $f \colon \mathbb{F}_2^E \otimes \mathbb{F}_2^E \to \mathbb{F}_2$ is fully determined if it is symmetric. Thus, overall, whether f is both NAE-satisfying and symmetric is fully captured by its value on $\mathcal{N}(E)$, and this is the definition of SymNAE-satisfying.

The following proposition asserts that there is always a linear map on $\mathcal{N}(E)$ which is SymNAE-satisfying.

Proposition 6.2. For any tripled set E, there is a unique linear map $f : \mathcal{N}(E) \to \mathbb{F}_2$ that is SymNAE-satisfying.

Proof. This is clear because $\mathcal{N}_1(E) \cap \mathcal{N}_2(E) = \{0\}$. □

In order to aggregate the spaces $\mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j}$ in all blocks, we introduce the following notation. Here we use $\mathcal{E}'$ instead of $\mathcal{E}$ to emphasize that the sets $E'_1, \ldots, E'_t$ are not necessarily tripled sets.

Definition 6.3. For a sequence of (not necessarily tripled) finite sets $\mathcal{E}' = (E'_1, \ldots, E'_t)$, we let $\mathcal{V}(\mathcal{E}')$ denote the $\mathbb{F}_2$-vector space

$$\mathcal{V}(\mathcal{E}') = \bigoplus_{j=1}^{t} \left(\mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j} \right).$$

[11] We assume that for each triple $\{v_1, v_2, v_3\} \in$ Triples(E), there is a fixed ordering of the elements v_1, v_2, v_3.

Remark 6.4. Let H be a 3-uniform hypergraph. For $E_1, \dots, E_t \in \mathscr{E}(H,k)$ and $\mathcal{E} = (E_1, \dots, E_t)$, the space $\mathcal{V}(\mathcal{E})$ can be regarded as a subspace of the huge ambient space

$$\left(\mathbb{F}_2^{\mathscr{V}(H,k)} \otimes \mathbb{F}_2^{\mathscr{V}(H,k)}\right)^t.$$

Next, we define subspaces of $\mathcal{V}(\mathcal{E}')$ that correspond to the tensor checks we impose on Alice or Bob in Game2b.

Definition 6.5. Fix a sequence of (not necessarily tripled) finite sets $\mathcal{E}' = (E'_1, \dots, E'_t)$ and a sequence of linear subspaces $\mathcal{U} = (U_1, \dots, U_t)$ with $U_j \subseteq \mathbb{F}_2^{E'_j}$ for each $j \in [t]$. We let $\bigoplus \mathcal{U} \subseteq \mathcal{V}(\mathcal{E}')$ be the linear subspace defined by

$$\bigoplus \mathcal{U} = \bigoplus_{j=1}^{t} \left(U_j \otimes U_j\right) \subseteq \bigoplus_{j=1}^{t} \left(\mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j}\right) = \mathcal{V}(\mathcal{E}').$$

Combining the side conditions, symmetry and tensor checks gives rise to the space $\mathcal{N}(\mathcal{E},\mathcal{U})$ defined as follows.

Definition 6.6. A *tripled packet* is a pair $(\mathcal{E},\mathcal{U})$, where $\mathcal{E} = (E_1, \dots, E_t)$ is a sequence of tripled sets, and $\mathcal{U} = (U_1, \dots, U_t)$ is a sequence of linear subspaces with $U_j \subseteq \mathbb{F}_2^{E_j}$ for each $j \in [t]$. We let $\mathcal{N}(\mathcal{E},\mathcal{U}) \subseteq \mathcal{V}(\mathcal{E})$ be the linear subspace defined by

$$\mathcal{N}(\mathcal{E},\mathcal{U}) = \bigoplus_{j=1}^{t} \left(\mathcal{N}(E_j) + U_j \otimes U_j\right) \subseteq \bigoplus_{j=1}^{t} \left(\mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j}\right) = \mathcal{V}(\mathcal{E}).$$

Thus, whether $f : \mathcal{V}(\mathcal{E}) \to \mathbb{F}_2$ satisfies all side conditions, symmetry and tensor tests amounts to checking the following condition on $f|_{\mathcal{N}(\mathcal{E},\mathcal{U})}$.

Definition 6.7. Given a tripled packet $(\mathcal{E},\mathcal{U})$ where $\mathcal{E} = (E_1, \dots, E_t)$ and $\mathcal{U} = (U_1, \dots, U_t)$, a linear map $f : \mathcal{N}(\mathcal{E},\mathcal{U}) \to \mathbb{F}_2$ is said to be SNR-*satisfying* if we have:

- $f|_{\mathcal{N}(E_j)}$ is SymNAE-satisfying for each $j \in [t]$.
- $\operatorname{rank}(f|_{U_j \otimes U_j}) \leqslant 1$ for each $j \in [t]$.

6.1.1 Thinning of Tripled Packets

Recall that in Game2b, we plant the true challenge in one randomly chosen coordinate, and it plays a crucial role in our argument. We often refer to it as the *critical index/coordinate/block*, and make use of the following notations.

Definition 6.8. Let $(E_1, \dots, E_t)$ be a sequence of tripled sets. Fix an index $c \in [t]$ and a parameter $\beta \in [0,1]$. We write $\mathcal{E}' \sim \mathrm{Thin}_{c,\beta}(\mathcal{E})$ for the random sequence $\mathcal{E}' = (E'_1, \dots, E'_t)$ obtained by setting $E'_j = E_j$ for all $j \in [t] \setminus \{c\}$ and sampling $E'_c \sim \mathrm{Thin}_\beta(E_c)$.

The following definition captures the distribution of $(\mathcal{U},\mathcal{E}',\mathcal{U}')$ conditioned on $\mathcal{E}$ in Definition 5.20. Note that to describe our final PCP, we will not need the various advice features we incorporated (they are crucial however for the soundness analysis).

Definition 6.9. Let $\mathcal{E} = (E_1, \dots, E_t)$ be a sequence of tripled sets. Fix a parameter $\beta \in [0,1]$ and a nonnegative integer ℓ. We let $\mathcal{J}_{\mathsf{in}}(\mathcal{E}, \beta, \ell)$ denote the output distribution of the following process:

1. Sample a uniformly random index $c \in [t]$. Sample $(E'_c, \texttt{empty}, \texttt{empty}, U_c) \sim \mathcal{J}_{\mathsf{mid}}(E_c, 0, 0, \beta, \ell)$. Sample a random vector $x \in \mathbb{F}_2^{E'_c}$ and let $U'_c = U_c + \mathrm{span}(x)$.
2. For each $j \in [t] \setminus \{c\}$, sample $(E_j^{\mathrm{ghost}}, \texttt{empty}, \texttt{empty}, U_j) \sim \mathcal{J}_{\mathsf{mid}}(E_j, 0, 0, \beta, \ell)$. Let $E'_j = E_j$ and $U'_j = U_j$.
3. Let $\mathcal{E}' = (E'_1, \ldots, E'_t)$, $\mathcal{U} = (U_1, \ldots, U_t)$ and $\mathcal{U}' = (U'_1, \ldots, U'_t)$.
4. Output the tuple $(c, \mathcal{U}, \mathcal{E}', \mathcal{U}')$.

We will make use of the following simple observation:

Proposition 6.10. In the setting and notation of Definition 6.9, the random subspaces $U_1, \ldots, U_t$ are mutually independent. Furthermore, they are jointly independent of the random index c.

Proof. Regardless of whether $j = c$ or $j \neq c$, the subspace U_j is spanned by ℓ independent vectors supported on a random β-thinning of E_j. □

6.2 Preparation: Low-Rank Grassmann

As discussed in Section 2.2.6, in the inner PCP we want to use linear subspaces of $\mathbb{F}_2^{E_c} \otimes \mathbb{F}_2^{E_c}$ that consist only of low-rank vectors. As stated in Definition 2.6, these linear subspaces will form a subset $\mathsf{LGr}(\mathbb{F}_2^{E_c}, \ell)$ of $\mathsf{Gr}(\mathbb{F}_2^{E_c} \otimes \mathbb{F}_2^{E_c}, \ell)$, and the subgraph it induces from the Grassmann graph is called the *low-rank Grassmann graph*. In this subsection, we give formal definitions of the low-rank Grassmann and prove a number of elementary facts about it. We will primarily be interested in the basic "statistical" properties of $\mathsf{LGr}(\mathbb{F}_2^{E_c}, \ell)$, such as its zoom-out probabilities being close to those of the full Grassmann graph:

$$\mathop{\mathbb{P}}_{L \in \mathsf{LGr}(V,\ell)} [L \subseteq W] \approx \mathop{\mathbb{P}}_{L \in \mathsf{Gr}(V \otimes V, \ell)} [L \subseteq W], \tag{6.1}$$

for any vector space V and subspace $W \subseteq V \otimes V$ with small codimension. More sophisticated analysis of the low-rank Grassmann graph, such as its spectral expansion (see [Gol23]), will not be used in this paper.

6.2.1 Construction from Hadamard Code

The definition of the low-rank Grassmann will use linear combinations of vectors given by the Hadamard matrix, defined as follows.

Definition 6.11. Let ℓ be a positive integer. For each $i \in \{0, 1, \ldots, 2^\ell - 1\}$ and $b \in \{0, 1, \ldots, \ell - 1\}$, define $\mathrm{Bit}_\ell(i, b) \in \mathbb{F}_2$ to be the mod-2 remainder of the integer $\lfloor i \cdot 2^{-b} \rfloor$. Equivalently, the sequence

$$\big(\mathrm{Bit}_\ell(i, \ell - 1), \mathrm{Bit}_\ell(i, \ell - 2), \ldots, \mathrm{Bit}_\ell(i, 0)\big)$$

is the length-ℓ binary representation of i, with the most significant bit listed first.

For each integer $i \in \{0, 1, \ldots, 2^\ell - 1\}$ and vector $u \in \mathbb{F}_2^\ell$, define

$$\mathrm{Had}_\ell(u, i) = \sum_{b=1}^{\ell} u_b \, \mathrm{Bit}_\ell(i, \ell - b) \in \mathbb{F}_2,$$

to be the "inner product" of u with the binary representation of i (viewed as a vector in $\mathbb{F}_2^\ell$).

We next define the span of a list of vectors according to the Hadamard matrix.

Definition 6.12. Let ℓ, s be positive integers, and let $q = 2s(2^\ell - 1)$. Suppose E' is a finite set, and $A \in (\mathbb{F}_2^{E'})^q$ is a sequence of vectors

$$A = \left(a^{(1)}, \ldots, a^{(q)}\right).$$

For each $u \in \mathbb{F}_2^\ell$, define a vector

$$\mathrm{Lvec}(A, \ell, u) = \sum_{i=0}^{2^\ell - 2} \left(\mathrm{Had}_\ell(u, 2^\ell - i - 1) \sum_{j=1}^{s} a^{(2si+2j-1)} \otimes a^{(2si+2j)} \right) \in \mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}. \tag{6.2}$$

Since $u \mapsto \mathrm{Had}_\ell(u, 2^\ell - i - 1)$ is a linear map for each i, the map $u \mapsto \mathrm{Lvec}(A, \ell, u)$ from $\mathbb{F}_2^\ell$ to $\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}$ is also linear. For any linear subspace $U \subseteq \mathbb{F}_2^\ell$, we denote the image of the latter linear map restricted to U by

$$\mathrm{Lspan}(A, \ell, U) = \{\mathrm{Lvec}(A, \ell, u) \,|\, u \in U\} \subseteq \mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}.$$

Let us make a few remarks about Equation (6.2). To get some intuition it is useful to consider a vector defined along the lines of Equation (6.2), except that we replace the sum of rank 1 vectors with arbitrary vectors $x^{(i)} \in \mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}$, getting

$$\sum_{i=0}^{2^\ell - 2} \mathrm{Had}_\ell(u, 2^\ell - i - 1) x^{(i)}. \tag{6.3}$$

Ranging over all $u \in \mathbb{F}_2^\ell$, the resulting collection of vectors is easily seen to be a linear subspace of $\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}$ of dimension at most ℓ. Moreover, if the vectors $x^{(i)}$ are taken uniformly at random, then, conditioned on this subspace having dimension ℓ, it is uniformly distributed among all ℓ-dimensional subspaces of $\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}$.

It follows that the space $\mathrm{Lspan}(A, \ell, U)$ is also a subspace of $\mathbb{F}_2^{E'} \otimes \mathbb{F}_2^{E'}$ with at most ℓ dimensions. However, it is not distributed uniformly when the vectors in the list A are taken uniformly, and this fact is crucial for us. Indeed, it is easy to see that each vector in $\mathrm{Lspan}(A, \ell, U)$ has rank at most $2^{\ell-1}s$; this is the source of the term "low-rank" in the name. In the following remark, we argue that if the list A is generic, then the rank of each nonzero vector is exactly $2^{\ell-1}s$ (which will be convenient for us for technical reasons).

Remark 6.13. In the setting and notation of Definition 6.12, suppose that the odd-indexed vectors $a^{(1)}, a^{(3)}, \ldots, a^{(q-1)}$ are linearly independent, and that the even-indexed vectors $a^{(2)}, a^{(4)}, \ldots, a^{(q)}$ are also linearly independent. Then for any $u \in \mathbb{F}_2^\ell \setminus \{0\}$, since $\mathrm{Had}_\ell(u, i) = 1$ for exactly $2^{\ell-1}$ of the i's in $\{1, 2, \ldots, 2^\ell - 1\}$, it follows from Equation (6.2) that

$$\mathrm{rank}\left(\mathrm{Lvec}(A, \ell, u)\right) = 2^{\ell-1}s.$$

Furthermore, it is easy to see that $\dim(\mathrm{Lspan}(A, \ell, \mathbb{F}_2^\ell)) = \ell$.

6.2.2 Mixed-Type Advice Sequences

Our inner PCP will include a tensor product of the low-rank Grassmann graph (on the critical block) with the standard Grassmann graph (on non-critical blocks). In this section, we present a few notations that will be useful in presenting and working with the resulting product space.

Definition 6.14. Given a sequence of finite sets $\mathcal{E}' = (E'_1, \ldots, E'_t)$, a subset of indices $J \subseteq [t]$ and positive integers ℓ, s, we let $\mathscr{A}_J(\mathcal{E}', \ell, s)$ be the collection of advice sequences $(A_1, \ldots, A_t)$ such that:

(1) For each $j \in [t] \setminus J$, the component A_j is a sequence of ℓ vectors $A_j = (a^{(j,1)}, \ldots, a^{(j,\ell)})$, where

$$a^{(j,i)} \in \mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j} \qquad \text{for all } i \in [\ell].$$

(2) For each $c \in J$, the component A_c is a sequence of $q = 2s(2^\ell - 1)$ vectors $A_c = (a^{(c,1)}, \ldots, a^{(c,q)})$, where

$$a^{(c,i)} \in \mathbb{F}_2^{E'_c} \qquad \text{for all } i \in [q].$$

If both sequences $\left(a^{(c,1)}, a^{(c,3)}, \ldots, a^{(c,q-1)}\right)$ and $\left(a^{(c,2)}, a^{(c,4)}, \ldots, a^{(c,q)}\right)$ are linearly independent, we say that A_c is *saturated*.

We let $\mathscr{A}^*_J(\mathcal{E}', \ell, s)$ denote the sub-collection of advice sequences $(A_1, \ldots, A_t) \in \mathscr{A}_J(\mathcal{E}', \ell, s)$ such that A_c is saturated for each $c \in J$.

Remark 6.15. A few remarks are in order:

(1) The set of coordinates in J in Definition 6.14 should be thought of as the potentially critical coordinates, hence the collection there is specified by vectors from the base space $\mathbb{F}_2^{E'_j}$. The coordinates outside J are not critical, and the vectors there are completely arbitrary vectors from $\mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j}$.

(2) An advice sequence in $\mathscr{A}_J(\mathcal{E}', \ell, s)$ should be thought of as the analog of a spanning set of a subspace. In other words, there may be linear dependencies between vectors in a member from $\mathscr{A}_J(\mathcal{E}', \ell, s)$. In our setting of parameters this will be rare (analogously to the fact that a random set of $o(n)$ independently chosen vectors in $\mathbb{F}_2^n$ are linearly independent with high probability), and $\mathscr{A}^*_J(\mathcal{E}', \ell, s) \subseteq \mathscr{A}_J(\mathcal{E}', \ell, s)$ is the subcollection that corresponds to tuples with no linear dependencies.

The following definition associates vectors from $\mathcal{V}(\mathcal{E}')$ with each advice sequence. In words, on each potentially critical coordinate we take the low-rank Hadamard span as per Definition 6.12, and on non-critical coordinates we take the span for arbitrary vectors as per Equation (6.3).

Definition 6.16. Fix a sequence of finite sets $\mathcal{E}' = (E'_1, \ldots, E'_t)$, a proper subset of indices $J \subsetneq [t]$ and positive integers ℓ, s. Given an advice sequence $\mathcal{A} = (A_1, \ldots, A_t) \in \mathscr{A}_J(\mathcal{E}', \ell, s)$, write $A_j = \left(a^{(j,1)}, \ldots, a^{(j,\ell)}\right)$ for each $j \in [t] \setminus J$. For each $u \in \mathbb{F}_2^\ell$, define[12]

$$\mathrm{Mvec}_J(\mathcal{A}, u) = \sum_{c \in J} \mathrm{Lvec}(A_c, \ell, u) + \sum_{j \in [t] \setminus J} \left(\sum_{b=1}^{\ell} u_b \cdot a^{(j,b)} \right) \in \mathcal{V}(\mathcal{E}'). \tag{6.4}$$

For any linear subspace $U \subseteq \mathbb{F}_2^\ell$, we denote

$$\mathrm{Mspan}_J(\mathcal{A}, U) = \{\mathrm{Mvec}_J(A, u) \,|\, u \in U\} \subseteq \mathcal{V}(\mathcal{E}'). \tag{6.5}$$

[12] We suppress the parameters ℓ, s in the notation $\mathrm{Mvec}_J(\mathcal{A}, u)$; indeed, the values of ℓ and s can be inferred from $\mathcal{A}$, as long as $J \neq \emptyset$ and $J \neq [t]$

Notation 6.17. In the remainder of the paper, we will often use the notation $\mathrm{Mspan}_J(\mathcal{A}, U)$ (defined in Equation (6.5)) with $U = \mathbb{F}_2^\ell$. When J is clear from context, we abbreviate

$$\sigma(\mathcal{A}) := \mathrm{Mspan}_J\big(\mathcal{A}, \mathbb{F}_2^\ell\big).$$

We also write

$$\rho(\mathcal{A}) := \big(\mathrm{Mvec}_J(\mathcal{A}, e_1), \ldots, \mathrm{Mvec}_J(\mathcal{A}, e_\ell)\big) \in (\mathcal{V}(\mathcal{E}'))^\ell,$$

where $e_1, \ldots, e_\ell$ denote the standard basis vectors of $\mathbb{F}_2^\ell$. Thus, $\sigma(\mathcal{A})$ is the linear span of the vectors in the sequence $\rho(\mathcal{A})$, and we write $\sigma(\mathcal{A}) = \mathrm{span}(\rho(\mathcal{A}))$.

Given an advice sequence $\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)$, the subspace $\sigma(\mathcal{A})$ is "mixed-type" in the sense that its projection on a critical block $c \in J$ consists of low-rank vectors in $\mathbb{F}_2^{E'_c} \otimes \mathbb{F}_2^{E'_c}$, while its projection on a non-critical block $j \in [t] \setminus J$ consists of generic vectors in $\mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j}$.

Next, we will want to facilitate notions analogous to subspace containment for mixed-type subspaces. Towards this end, it is useful to note that if we considered Equation (6.3) and only looked at vectors u with $u_\ell = 0$, then we would get an $(\ell-1)$-dimensional space which is a subspace of the larger ℓ-dimensional space resulting from taking all u. Moreover, this subspace would not depend on what is $x^{(2^\ell-2)}$ (as its coefficient will always be 0). Generalizing this intuition, we get the following analogous notion of sub-advice.

Definition 6.18. Fix a sequence of finite sets $\mathcal{E}' = (E'_1, \ldots, E'_t)$, a proper subset of indices $J \subsetneq [t]$, positive integers ℓ, s and a nonnegative integer $q_0 \leqslant \ell$. Given an advice sequence $\mathcal{A} = (A_1, \ldots, A_t) \in \mathscr{A}_J(\mathcal{E}', \ell, s)$, we define a *sub-advice* $\mathcal{A}_{[q_0]} = (A'_1, \ldots, A'_t) \in \mathscr{A}_J(\mathcal{E}', q_0, 2^{\ell-q_0}s)$ as follows:

(1) For each $j \in [t] \setminus J$, write $A_j = \big(a^{(j,1)}, \ldots, a^{(j,\ell)}\big)$ and let $A'_j = \big(a^{(j,1)}, \ldots, a^{(j,q_0)}\big)$.

(2) For each $c \in J$, write $A_c = \big(a^{(c,1)}, \ldots, a^{(c,q)}\big)$, where $q = 2s(2^\ell - 1)$. Denote $q_1 = 2s(2^\ell - 2^{\ell-q_0})$ and let $A'_c = \big(a^{(c,1)}, \ldots, a^{(c,q_1)}\big)$.

The following proposition formalizes the intuition that taking a sub-advice corresponds to spanning a subspace of the original space.

Proposition 6.19. In the setting and notation of Definition 6.18, we have[13]

$$\mathrm{Mvec}_J(\mathcal{A}, u) = \mathrm{Mvec}_J(\mathcal{A}_{[q_0]}, u) \qquad \text{for all } u \in \mathbb{F}_2^{q_0},$$

Consequently, by Equation (6.5) we have

$$\mathrm{Mspan}_J(\mathcal{A}, \mathbb{F}_2^{q_0}) = \mathrm{Mspan}_J(\mathcal{A}_{[q_0]}, \mathbb{F}_2^{q_0}).$$

Proof. Denote $s_1 = 2^{\ell-q_0}s$. As in Definition 6.18, let $q = 2s(2^\ell - 1)$ and $q_1 = 2s(2^\ell - 2^{\ell-q_0})$.

Let i be an integer ranging in $\{0, 1, \ldots, 2^\ell - 2\}$. For each $u \in \mathbb{F}_2^{q_0}$ considered as an ℓ-bit string, since its last $(\ell - q_0)$ bits are all 0's, the value of

$$\mathrm{Had}_\ell(u, 2^\ell - i - 1) = \sum_{b=1}^{q_0} u_b(1 - \mathrm{Bit}_\ell(i, \ell - b))$$

does not depend on the last $(\ell - q_0)$ bits of the binary representation of i. Furthermore, if the first q_0 bits of i are all 1's, then $\mathrm{Had}_\ell(u, 2^\ell - i - 1) = 0$. Therefore, if we define $\pi(i) = 2^{\ell-q_0}\lfloor i/2^{\ell-q_0} \rfloor$ for

[13] Here, we identify $\mathbb{F}_2^{q_0}$ with the subspace $\mathbb{F}_2^{q_0} \times \{0\}^{\ell-q_0} \subseteq \mathbb{F}_2^\ell$; that is, a q_0-bit string is extended to an ℓ-bit string by appending $\ell - q_0$ zeros.

each $i \in \{0, 1, \ldots, 2^\ell - 2^{\ell-q_0} - 1\}$, for each $c \in J$ and $u \in \mathbb{F}_2^{q_0}$ we can deduce from Equation (6.2) that

$$\begin{aligned}
\mathrm{Lvec}(A_c, \ell, u) &= \sum_{i=0}^{2^\ell - 2^{\ell-q_0}-1} \left(\mathrm{Had}_\ell(u, 2^\ell - \pi(i) - 1) \sum_{j=1}^{s} a^{(c,2si+2j-1)} \otimes a^{(c,2si+2j)} \right) \\
&= \sum_{i=0}^{2^{q_0}-2} \left(\mathrm{Had}_\ell(u, 2^\ell - 2^{\ell-q_0} i - 1) \sum_{j=1}^{s_1} a^{(c,2s_1 i+2j-1)} \otimes a^{(c,2s_1 i+2j)} \right) \\
&= \sum_{i=0}^{2^{q_0}-2} \left(\mathrm{Had}_{q_0}(u, 2^{q_0} - i - 1) \sum_{j=1}^{s_1} a^{(c,2s_1 i+2j-1)} \otimes a^{(c,2s_1 i+2j)} \right) = \mathrm{Lvec}(A'_c, q_0, u).
\end{aligned}$$

It then follows from Equation (6.4) that

$$\mathrm{Mvec}_J(\mathcal{A}, u) = \sum_{c \in J} \mathrm{Lvec}(A'_c, q_0, u) + \sum_{j \in [t] \setminus J} \left(\sum_{b=1}^{q_0} u_b \cdot a^{(j,b)} \right) = \mathrm{Mvec}_J(\mathcal{A}_{[q_0]}, u),$$

for any $u \in \mathbb{F}_2^{q_0}$. □

Notation 6.20. Throughout the remainder of the paper, the set $J \subseteq [t]$ of critical indices will always satisfy $|J| \in \{1, 2\}$. When $J = \{c_1, c_2\}$ with $c_1 \neq c_2$, we suppress the braces and write c_1, c_2 in place of $\{c_1, c_2\}$ whenever J appears as a subscript. For example, we write

$$\mathscr{A}_{c_1,c_2}(\mathcal{E}', \ell, s) \qquad \text{instead of} \qquad \mathscr{A}_{\{c_1,c_2\}}(\mathcal{E}', \ell, s).$$

Similarly, when $J = \{c\}$, we write, for instance,

$$\mathrm{Mspan}_c(\mathcal{A}, U) \qquad \text{instead of} \qquad \mathrm{Mspan}_{\{c\}}(\mathcal{A}, U).$$

6.2.3 Zoom-out Probabilities

We next prove a few elementary properties of the "zoom-out probabilities" associated with mixed-type advice sequences. Specifically, given a linear subspace $W \subseteq \mathcal{V}(\mathcal{E}')$ of small codimension, we aim to show that the probability that $\sigma(\mathcal{A}) \subseteq W$, where $\mathcal{A}$ is sampled at random from $\mathscr{A}_J(\mathcal{E}', \ell, s)$, is close to the probability that a random subspace $L \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}'), \ell)$ is contained in W.

For technical reasons, we also need a "zoomed-in" version of this comparison. Specifically, we consider a random advice sequence $\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)$ conditioned on $\sigma(\mathcal{A}_{[q_0]}) = Q$, where $Q \subseteq \mathcal{V}(\mathcal{E}')$ is a fixed q_0-dimensional subspace. This should be analogous to sampling a random subspace $L \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}'), \ell)$ conditioned on $Q \subseteq L$. Our main goal is to argue that

$$\mathop{\mathbb{P}}_{\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)} \left[\sigma(\mathcal{A}) \subseteq W \,\middle|\, \sigma(\mathcal{A}_{[q_0]}) = Q \right] \approx \mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}'), \ell)} \left[L \subseteq W \,\middle|\, Q \subseteq L \right].$$

The first step is to show that conditioned on a fixed sub-advice $\mathcal{A}_{[q_0]}$, the vectors in $\sigma(\mathcal{A}) \setminus \sigma(\mathcal{A}_{[q_0]})$ still behave randomly with respect to any test function $f : \mathcal{V}(\mathcal{E}') \to \mathbb{F}_2$.

Proposition 6.21. Fix a sequence of non-empty finite sets $\mathcal{E}' = (E'_1, \ldots, E'_t)$, a subset of indices $J \subsetneq [t]$, positive integers ℓ, s and a nonnegative integer $q_0 < \ell$. Let $f : \mathcal{V}(\mathcal{E}') \to \mathbb{F}_2$ be an arbitrary linear functional that is not identically zero. For any fixed advice sequence $\mathcal{A}' \in \mathscr{A}_J(\mathcal{E}', q_0, 2^{\ell-q_0} s)$ and any vector $u \in \mathbb{F}_2^\ell \setminus \mathbb{F}_2^{q_0}$, we have

$$\frac{1}{2} - 2^{-s-1} \leqslant \mathop{\mathbb{P}}_{\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)} \left[f\big(\mathrm{Mvec}_J(\mathcal{A}, u)\big) = 1 \,\middle|\, \mathcal{A}_{[q_0]} = \mathcal{A}' \right] \leqslant \frac{1}{2} + 2^{-s-1}.$$

Proof. As usual, we denote $\mathcal{A} = (A_1, \dots, A_t)$ and $A_j = \big(a^{(j,1)}, a^{(j,2)}, \dots\big)$ for each $j \in [t]$. Since f is not identically zero, there exists an index $j \in [t]$ such that the restriction of f to $\mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j}$ is nonzero. Depending on whether this index lies in J, we divide into two cases:

Case 1: the restriction of f to $\mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j}$ is nonzero for some $j \in [t] \setminus J$. In this case, if we let $b \in \{q_0 + 1, \dots, \ell\}$ be an index such that $u_b = 1$, even after conditioning on all vectors in $\mathcal{A}$ except for $a^{(j,b)}$, the value of $f\big(\mathrm{Mvec}_J(\mathcal{A}, u)\big)$ (see Equation (6.4)) is still an unbiased Bernoulli random variable purely due to the randomness of the vector $a^{(j,b)} \in \mathbb{F}_2^{E'_j} \otimes \mathbb{F}_2^{E'_j}$. Since $a^{(j,b)}$ is not included in the sub-advice $\mathcal{A}_{[q_0]}$, we conclude that

$$\mathop{\mathbb{P}}_{\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)} \big[f\big(\mathrm{Mvec}_J(\mathcal{A}, u)\big) = 1 \,\big|\, \mathcal{A}_{[q_0]} = \mathcal{A}' \big] = \frac{1}{2}.$$

Case 2: the restriction of f to $\mathbb{F}_2^{E'_c} \otimes \mathbb{F}_2^{E'_c}$ is nonzero for some $c \in J$. As in case 1, we let $b \in \{q_0 + 1, \dots, \ell\}$ be an index such that $u_b \neq 0$. Now let $i = 2^\ell - 2^{\ell - b} - 1$, whose ℓ-bit binary representation has a 0 in the b-th position and 1's in all other positions. Even after conditioning on all vectors in $\mathcal{A}$ except for the $2s$ vectors (none of which is included in the sub-advice $\mathcal{A}_{[q_0]}$)

$$a^{(c, 2si+1)}, a^{(c, 2si+2)}, \dots, a^{(c, 2si+2s)},$$

the value of $f\big(\mathrm{Mvec}_J(\mathcal{A}, u)\big)$ (see Equations (6.2) and (6.4)) is still the Bernoulli random variable

$$X = \sum_{j=1}^{s} f\big(a^{(c, 2si+2j-1)} \otimes a^{(c, 2si+2j)}\big)$$

or its negation. By independence among the vectors in $\mathcal{A}$, it is easy to see that $\mathbb{P}[X = 1]$ lies in the interval $[\frac{1}{2} - 2^{-s-1}, \frac{1}{2} + 2^{-s-1}]$, and hence we have

$$\frac{1}{2} - 2^{-s-1} \leqslant \mathop{\mathbb{P}}_{\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)} \big[f\big(\mathrm{Mvec}_J(\mathcal{A}, u)\big) = 1 \,\big|\, \mathcal{A}_{[q_0]} = \mathcal{A}' \big] \leqslant \frac{1}{2} + 2^{-s-1}. \qquad \square$$

Recall that if $W \in \mathsf{Gr}(\mathbb{F}_2^n, n - w)$, $L' \in \mathsf{Gr}(W, \ell - 1)$, and we choose an ℓ-dimensional subspace $L \subseteq \mathbb{F}_2^n$ uniformly at random containing L', then $L \subseteq W$ with probability $2^{-w} + o(1)$. The following proposition establishes an analogous fact, with the notion of sub-advice replacing the containment $L' \subseteq L$.

Proposition 6.22. Fix a sequence of non-empty finite sets $\mathcal{E}' = (E'_1, \dots, E'_t)$, a subset of indices $J \subsetneq [t]$, and positive integers ℓ, s. Let $W \subseteq \mathcal{V}(\mathcal{E}')$ be a linear subspace with codimension w. For any fixed advice sequence $\mathcal{A}' \in \mathscr{A}_J(\mathcal{E}', \ell - 1, 2s)$ such that $\sigma(\mathcal{A}') \subseteq W$, we have

$$2^{-w} - 2^{-s} \leqslant \mathop{\mathbb{P}}_{\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)} \big[\sigma(\mathcal{A}) \subseteq W \,\big|\, \mathcal{A}_{[\ell-1]} = \mathcal{A}' \big] \leqslant 2^{-w} + 2^{-s}.$$

Proof. We fix an arbitrary vector $u \in \mathbb{F}_2^\ell \setminus \mathbb{F}_2^{\ell-1}$. Since $\sigma(\mathcal{A}') \subseteq W$ and

$$\sigma(\mathcal{A}) = \sigma(\mathcal{A}_{[\ell-1]}) + \mathrm{span}(\mathrm{Mvec}_J(\mathcal{A}, u)),$$

the desired inequality is equivalent to

$$2^{-w} - 2^{-s} \leqslant \mathop{\mathbb{P}}_{\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)} \big[\mathrm{Mvec}_J(\mathcal{A}, u) \in W \,\big|\, \mathcal{A}_{[\ell-1]} = \mathcal{A}' \big] \leqslant 2^{-w} + 2^{-s}. \tag{6.6}$$

Let $\mathcal{F}$ be the collection of linear functionals $f : \mathcal{V}(\mathcal{E}') \to \mathbb{F}_2$ such that $W \subseteq \ker(f)$, and denote

$$p = \mathop{\mathbb{E}}_{f\in\mathcal{F}} \left[\mathop{\mathbb{P}}_{\mathcal{A}\in\mathscr{A}_J(\mathcal{E}',\ell,s)} \left[f\big(\mathrm{Mvec}_J(\mathcal{A},u)\big) = 1 \,\middle|\, \mathcal{A}_{[\ell-1]} = \mathcal{A}' \right] \right]. \tag{6.7}$$

Note that $|\mathcal{F}| = 2^w$, and for each $f \in \mathcal{F}$ that is not identically zero we may apply Proposition 6.21 to estimate the probability expression on the right-hand side of Equation (6.7), which yields

$$\frac{2^w-1}{2^w}\left(\frac{1}{2} - 2^{-s-1}\right) \leqslant p \leqslant \frac{2^w-1}{2^w}\left(\frac{1}{2} + 2^{-s-1}\right). \tag{6.8}$$

On the other hand, by exchanging the expectation and the probability on the right-hand side of Equation (6.7) (i.e. double counting), we have

$$p = \frac{1}{2}\left(1 - \mathop{\mathbb{P}}_{\mathcal{A}\in\mathscr{A}_J(\mathcal{E}',\ell,s)} \left[\mathrm{Mvec}_J(\mathcal{A},u) \in W \,\middle|\, \mathcal{A}_{[\ell-1]} = \mathcal{A}' \right]\right). \tag{6.9}$$

Comparing Equations (6.8) and (6.9) immediately yields the desired Equation (6.6). □

Given $\mathcal{A}' \in \mathscr{A}_J(\mathcal{E}', \ell-1, 2s)$ such that $\sigma(\mathcal{A}') \subseteq W$, Proposition 6.22 estimates the probability of that random extension $\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)$ of $\mathcal{A}'$ still "stays in" W, in the sense that $\sigma(\mathcal{A}) \subseteq W$. Roughly speaking, this means that "random extension" by 1 dimension from $\sigma(\mathcal{A}_{[\ell-1]})$ to $\sigma(\mathcal{A})$ behaves nicely with respect to zoom-outs. This allows us to iterate the process and show that a random extension by $\ell - q_0$ dimensions from $\sigma(\mathcal{A}_{[q_0]})$ to $\sigma(\mathcal{A})$ also behaves as expected with respect to zoom-outs.

Corollary 6.23. *Fix a sequence of non-empty finite sets $\mathcal{E}' = (E'_1, \dots, E'_t)$, a subset of indices $J \subsetneq [t]$, and positive integers ℓ, s. Let $W \subseteq \mathcal{V}(\mathcal{E}')$ be a linear subspace with codimension w, and suppose that $s \geqslant \ell + w$. Then for any nonnegative integer $q_0 < \ell$ and any fixed advice sequence $\mathcal{A}' \in \mathscr{A}_J(\mathcal{E}', q_0, 2^{\ell-q_0}s)$ such that $\sigma(\mathcal{A}') \subseteq W$, we have*

$$2^{-w(\ell-q_0)-1} \leqslant \mathop{\mathbb{P}}_{\mathcal{A}\in\mathscr{A}_J(\mathcal{E}',\ell,s)} \left[\sigma(\mathcal{A}) \subseteq W \,\middle|\, \mathcal{A}_{[q_0]} = \mathcal{A}' \right] \leqslant 2^{-w(\ell-q_0)+1}.$$

Proof. We have

$$\mathop{\mathbb{P}}_{\mathcal{A}\in\mathscr{A}_J(\mathcal{E}',\ell,s)} \left[\sigma(\mathcal{A}) \subseteq W \,\middle|\, \mathcal{A}_{[q_0]} \right] = \mathbb{E}\left[\prod_{i=q_0}^{\ell-1} \mathop{\mathbb{P}}_{\mathcal{A}_{[i+1]}} \left[\sigma(\mathcal{A}_{[i+1]}) \subseteq W \,\middle|\, \mathcal{A}_{[i]} \right] \right],$$

where on the right-hand side, $\mathcal{A}_{[i+1]}$ is taken uniformly at random from $\mathscr{A}_J(\mathcal{E}', i+1, 2^{\ell-i-1}s)$ conditioned on $\mathcal{A}_{[i]}$. Since we can apply Proposition 6.22 to every factor on the right-hand side, the conclusion easily follows. Note that the condition $s \geqslant \ell + w$ guarantees

$$(2^{-w} - 2^{-s})^{\ell-q_0} \geqslant 2^{-w(\ell-q_0)}\left(1 - (\ell-q_0)2^{-s+w}\right) \geqslant 2^{-w(\ell-q_0)}(1 - 2^{\ell-1-s+w}) \geqslant 2^{-w(\ell-q_0)-1},$$

and similarly for the upper bound. □

6.2.4 Mixed-Type Grassmann Graph

So far, we have mainly worked with uniformly random mixed-type advice sequences $\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)$, the vectors in which may have linear dependencies (though unlikely). It will be useful later in the analysis of our inner PCP to consider those $\mathcal{A}$ with no linear dependency among their vectors, spanning actual ℓ-dimensional subspaces $\sigma(\mathcal{A}) \subseteq \mathcal{V}(\mathcal{E}')$.

This naturally leads us to the sub-collection $\mathcal{A}^*_J(\mathcal{E}', \ell, s)$ defined in Definition 6.14. For sequences $\mathcal{A} = (A_1, \ldots, A_t) \in \mathcal{A}^*_J(\mathcal{E}', \ell, s)$, the lists A_c in the critical blocks $c \in J$ are always saturated, and by Remark 6.13, it follows that $\dim \sigma(\mathcal{A}) = \ell$ (as long as J is nonempty). Therefore, these subspaces $\sigma(\mathcal{A})$ lie in the domain $\mathsf{Gr}(\mathcal{V}(\mathcal{E}'), \ell)$, and they induce a subgraph of the Grassmann graph $\mathsf{Gr}(\mathcal{V}(\mathcal{E}'), \ell)$. We call this subgraph a "mixed-type Grassmann graph," as defined below.

Definition 6.24. Fix a sequence of finite sets $\mathcal{E}' = (E'_1, \ldots, E'_t)$, a nonempty subset of indices $J \subsetneq [t]$ and positive integers ℓ, s. Define $\mathsf{MGr}_J(\mathcal{E}', \ell, s)$ to be the following collection of linear subspaces:

$$\mathsf{MGr}_J(\mathcal{E}', \ell, s) = \left\{ \sigma(\mathcal{A}) \,\middle|\, \mathcal{A} \in \mathscr{A}^*_J(\mathcal{E}', \ell, s) \right\} \subseteq \mathsf{Gr}(\mathcal{V}(\mathcal{E}'), \ell).$$

An important fact (whose proof we omit) is that for different subspaces $L \in \mathsf{MGr}_J(\mathcal{E}', \ell, s)$, the number of $\mathcal{A} \in \mathscr{A}^*_J(\mathcal{E}', \ell, s)$ such that $\sigma(\mathcal{A}) = L$ is the same.[14] Therefore, if we sample a uniformly random $\mathcal{A} \in \mathscr{A}^*_J(\mathcal{E}', \ell, s)$, the span $\sigma(\mathcal{A})$ is uniformly random over $\mathsf{MGr}_J(\mathcal{E}', \ell, s)$. The next proposition asserts that even if one chose $\mathcal{A}$ from $\mathscr{A}_J(\mathcal{E}', \ell, s)$, the resulting distribution of $\sigma(\mathcal{A})$ is still close to uniform over $\mathsf{MGr}_J(\mathcal{E}', \ell, s)$, since $|\mathcal{A}^*_J(\mathcal{E}', \ell, s)| = (1 - o(1)) \cdot |\mathcal{A}_J(\mathcal{E}', \ell, s)|$.

Proposition 6.25. Fix a sequence of non-empty finite sets $\mathcal{E}' = (E'_1, \ldots, E'_t)$, a nonempty subset of indices $J \subsetneq [t]$, positive integers ℓ, s and a nonnegative integer $q_0 < \ell$. Assume that $|E'_c| \geqslant 2^{\ell+2}s$ for each $c \in J$. For any fixed advice sequence $\mathcal{A}' \in \mathscr{A}^*_J(\mathcal{E}', q_0, 2^{\ell-q_0}s)$, if we sample $\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)$ conditioned on $\sigma(\mathcal{A}_{[q_0]}) = \sigma(\mathcal{A}')$, the resulting distribution of $\sigma(\mathcal{A})$ has total variation distance at most

$$\sum_{c \in J} 2^{-|E'_c|/2}$$

to the uniform distribution over the set $\left\{ L \in \mathsf{MGr}_J(\mathcal{E}', \ell, s) \,\middle|\, \sigma(\mathcal{A}') \subseteq L \right\}$.

Proof. It is not hard to see that if we sample $\mathcal{A} \in \mathscr{A}^*_J(\mathcal{E}', \ell, s)$ conditioned on $\sigma(\mathcal{A}_{[q_0]}) = \sigma(\mathcal{A}')$, the resulting distribution of $\sigma(\mathcal{A})$ is exactly the uniform distribution over $\left\{ L \in \mathsf{MGr}_J(\mathcal{E}', \ell, s) \,\middle|\, \sigma(\mathcal{A}') \subseteq L \right\}$. On the other hand, writing $q = 2s(2^\ell - 1)$ and $q_1 = 2s(2^\ell - 2^{\ell - q_0})$, for any $\mathcal{A}' \in \mathscr{A}^*_J(\mathcal{E}', q_0, 2^{\ell-q_0}s)$ we have

$$\begin{aligned}
&\mathop{\mathbb{P}}_{\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)} \left[\mathcal{A} \in \mathscr{A}^*_J(\mathcal{E}', \ell, s) \,\middle|\, \mathcal{A}_{[q_0]} = \mathcal{A}' \right] \\
&= \mathbb{P}\left[\left(a^{(c,1)}, \ldots, a^{(c,q)}\right) \text{ is saturated for each } c \in J \,\middle|\, \text{fixed saturated sequences } \left(a^{(c,1)}, \ldots, a^{(c,q_1)}\right) \right] \\
&\qquad \text{(where } a^{(c,i)} \text{ (for } c \in J \text{ and } i \in [q]) \text{ are sampled independently and uniformly from } \mathbb{F}_2^{E'_c}) \\
&\geqslant \prod_{c \in J} \left(1 - 2^{(2^\ell - 1)s - |E'_c|}\right)^2 \geqslant 1 - \sum_{c \in J} 2^{-|E'_c|/2}.
\end{aligned}$$

□

The next proposition asserts that for a valid zoom-in and zoom-out pair Q, W (i.e., $Q \subseteq W$), among the subspaces in $\mathsf{MGr}_J(\mathcal{E}', \ell, s)$ containing Q, the fraction of them contained in W behaves similarly to the standard Grassmann domain.

[14] To prove that, one needs to use the fact that for any $\mathbb{F}_2$-vector space V and any $x \in V \otimes V$ of rank s, the number of ways to write $x = a^{(1)} \otimes a^{(2)} + \cdots + a^{(2s-1)} \otimes a^{(2s)}$, where $a^{(i)} \in V$ for each $i \in [2s]$, depends only on s.

Proposition 6.26. Fix a sequence of non-empty finite sets $\mathcal{E}' = (E'_1, \dots, E'_t)$, a subset of indices $J \subsetneq [t]$ such that $1 \leqslant |J| \leqslant 2$, positive integers ℓ, s and nonnegative integers q_0, w. Assume

$$s \geqslant \ell + w, \qquad q_0 < \ell, \qquad \text{and} \qquad |E'_c| \geqslant 2^{\ell+2} s \text{ for each } c \in J.$$

Then for any fixed subspaces

$$Q \in \mathsf{MGr}_J(\mathcal{E}', q_0, 2^{\ell-q_0} s) \qquad \text{and} \qquad W \in \mathsf{Gr}\big(\mathcal{V}(\mathcal{E}'), \dim \mathcal{V}(\mathcal{E}') - w\big)$$

such that $Q \subseteq W$, we have

$$2^{-w(\ell-q_0)-2} \leqslant \mathop{\mathbb{P}}_{L \in \mathsf{MGr}_J(\mathcal{E}', \ell, s)} [L \subseteq W \mid Q \subseteq L] \leqslant 2^{-w(\ell-q_0)+2}.$$

Proof. Pick $\mathcal{A}'$ uniformly at random from the set

$$\left\{ \mathcal{A}' \in \mathscr{A}_J(\mathcal{E}', q_0, 2^{\ell-q_0} s) \,\middle|\, \sigma(\mathcal{A}') = Q \right\}.$$

It follows from Proposition 6.25 that

$$\left| \mathop{\mathbb{P}}_{L \in \mathsf{MGr}_J(\mathcal{E}', \ell, s)} [L \subseteq W \mid Q \subseteq L] - \mathop{\mathbb{E}}_{\mathcal{A}'} \left[\mathop{\mathbb{P}}_{\mathcal{A} \in \mathscr{A}_J(\mathcal{E}', \ell, s)} \left[\sigma(\mathcal{A}) \subseteq W \,\middle|\, \mathcal{A}_{[q_0]} = \mathcal{A}' \right] \right] \right| \leqslant \sum_{c \in J} 2^{-|E'_c|/2} \leqslant 2^{-w\ell-2}.$$

Combining the above with Corollary 6.23 yields the desired conclusion. □

6.2.5 Extension of Agreement

The next somewhat technical result is needed in Section 6.5.4, and to explain it we first give the analog in the simpler Grassmann setting. Suppose we have a table F that assigns to each subspace in $\mathsf{Gr}(\mathbb{F}_2^n, \ell+1)$ a linear function $F[L]$ (or `nil`), a subspace $W \subseteq \mathbb{F}_2^n$ of constant codimension, and a linear function $f \colon W \to \mathbb{F}_2$ that largely agrees with F in the sense that

$$\mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell+1)} \left[f|_{L \cap W} = F[L]|_{L \cap W} \right] \geqslant \delta.$$

We wish to extend f into a linear function $g : \mathbb{F}_2^n \to \mathbb{F}_2$ that agrees with F not only on the intersection with W. Then a random extension would suffice: for any L such that $f|_{L\cap W} = F[L]|_{L\cap W}$ we will have that $g|_L = F[L]$ with constant probability. The following lemma establishes an analogous statement in the context of mixed-type subspaces.

Lemma 6.27. *Fix a sequence of non-empty finite sets $\mathcal{E}' = (E'_1, \dots, E'_t)$, a critical index $c \in [t]$, positive integers ℓ, s, nonnegative integers q_0, w and a parameter $\delta \in (0, 1)$. Assume*

$$s \geqslant 2\ell \qquad \text{and} \qquad \ell > 2q_0 + 2w.$$

Let $W \subseteq \mathcal{V}(\mathcal{E}')$ be a linear subspace with codimension w, and $Q \subseteq W$ a linear subspace such that $\sigma(\mathcal{A}) = Q$ for some $\mathcal{A} \in \mathscr{A}_c(\mathcal{E}', q_0, 2^{\ell+1-q_0} s)$. Suppose $F[\cdot]$ is a table where $F[L] \in L^* \cup \{\texttt{nil}\}$ for each linear subspace $L \subseteq \mathcal{V}(\mathcal{E}')$, and there exists a linear function $f : W \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{\mathcal{A} \in \mathscr{A}_c(\mathcal{E}', \ell+1, s)} \left[F[\sigma(\mathcal{A})]|_{\sigma(\mathcal{A}_{[\ell]})} = f|_{\sigma(\mathcal{A}_{[\ell]})} \,\middle|\, \sigma(\mathcal{A}_{[q_0]}) = Q \text{ and } \sigma(\mathcal{A}_{[\ell]}) \subseteq W \right] \geqslant \delta. \tag{6.10}$$

Then there exists a subspace $W' \subseteq \mathcal{V}(\mathcal{E}')$ containing W and a linear function $g : W' \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{\mathcal{A} \in \mathscr{A}_c(\mathcal{E}', \ell+1, s)} \left[F[\sigma(\mathcal{A})] = g|_{\sigma(\mathcal{A})} \,\middle|\, \sigma(\mathcal{A}_{[q_0]}) = Q \text{ and } \sigma(\mathcal{A}) \subseteq W' \right] \geqslant \frac{\delta}{5}. \tag{6.11}$$

Proof. We let η be the maximum possible value of the left-hand side of Equation (6.11) when W' ranges in the set of linear subspaces containing W and g ranges in $(W')^*$. The goal is to show that $5\eta \geqslant \delta$. Throughout the proof, we let $L, L' \subseteq \mathcal{V}(\mathcal{E}')$ be random subspaces sampled by the following process:

1. Sample $\mathcal{A} \in \mathscr{A}_c(\mathcal{E}', \ell+1, s)$ conditioned on $\sigma(\mathcal{A}_{[q_0]}) = Q$.
2. Let $L = \sigma(\mathcal{A}_{[\ell]})$ and $L' = \sigma(\mathcal{A})$.

The most important fact about the joint distribution of (L, L') that we are going to use is the following: conditioned on L', for any linear subspace such that $Q \subseteq L_0 \subseteq L'$ and $\dim L_0 = \dim L' - 1$, the probability that $L = L_0$ is exactly $(2^{\ell+1-q_0}-1)^{-1}$.

Let $p = \mathbb{P}[L' \subseteq W]$ and denote

$$\gamma_0 = \mathbb{P}\left[F[L'] = f|_{L'} \,\middle|\, L' \subseteq W\right],$$
$$\gamma_1 = \mathbb{P}\left[F[L']\big|_Q = f|_Q \text{ but } F[L'] \neq f|_{L'} \,\middle|\, L' \subseteq W\right].$$

By the definitions of γ_0 and η, we immediately have $\eta \geqslant \gamma_0$. Furthermore, by picking a random $g : W \to \mathbb{F}_2$ conditioned on $g|_Q = f|_Q$, we have

$$\mathop{\mathbb{E}}_g\left[\mathbb{P}\left[F[L'] = g|_{L'} \text{ and } L' \subseteq W\right]\right] \geqslant 2^{-(\ell+1-q_0)} \cdot (\gamma_0 p + \gamma_1 p),$$

and hence

$$\eta \geqslant \mathop{\mathbb{E}}_g\left[\mathbb{P}\left[F[L'] = g|_{L'} \,\middle|\, L' \subseteq W\right]\right] \geqslant 2^{-(\ell+1-q_0)} \cdot (\gamma_0 + \gamma_1). \tag{6.12}$$

We denote by $\mathcal{W}$ the collection of linear subspaces $W' \subseteq \mathcal{V}(\mathcal{E}')$ such that $\dim W' = 1 + \dim W$ and $W \subseteq W'$. For each $W' \in \mathcal{W}$, we let

$$p_{W'} = \mathbb{P}\left[L' \subseteq W' \text{ but } L' \not\subseteq W\right], \text{ and}$$
$$\gamma_{W'} = \begin{cases} \mathbb{P}\left[F[L']|_{L'\cap W} = f|_{L'\cap W} \,\middle|\, L' \subseteq W' \text{ but } L' \not\subseteq W\right], & \text{if } p_{W'} > 0 \\ 0, & \text{if } p_{W'} = 0. \end{cases}$$

For each $W' \in \mathcal{W}$, there are exactly two linear extensions of f to W'. If we choose $g : W' \to \mathbb{F}_2$ to be a uniformly random one of the two, then it is easy to see that

$$\mathop{\mathbb{E}}_g\left[\mathbb{P}\left[F[L'] = g|_{L'} \text{ and } L' \subseteq W'\right]\right] = \gamma_0 p + \frac{1}{2}\gamma_{W'} p_{W'},$$

and hence

$$\eta \geqslant \mathop{\mathbb{E}}_g\left[\mathbb{P}\left[F[L'] = g|_{L'} \,\middle|\, L' \subseteq W'\right]\right] = \frac{\gamma_0 p + \gamma_{W'} p_{W'}/2}{p + p_{W'}}. \tag{6.13}$$

It is not hard to see that

$$\mathbb{P}[L \subseteq W] = p + \frac{1}{2^{\ell+1-q_0}-1}\sum_{W\in\mathcal{W}} p_{W'}, \tag{6.14}$$

and furthermore,

$$\mathbb{P}\left[F[L']\big|_L = f|_L \text{ and } L \subseteq W\right]$$

$$
\begin{aligned}
&= \gamma_0 p + \frac{1}{2^{\ell+1-q_0}-1}\left(\gamma_1 p + \sum_{W'\in\mathcal{W}} \gamma_{W'} p_{W'}\right) \\
&\leqslant \gamma_0 p + 2\cdot 2^{-(\ell+1-q_0)}\cdot(\gamma_0+\gamma_1)p + \frac{2}{2^{\ell+1-q_0}-1}\sum_{W'\in\mathcal{W}}\left(\gamma_0 p + \frac{1}{2}\gamma_{W'} p_{W'}\right) \\
&\leqslant 3\eta\cdot p + 2\eta\cdot\frac{1}{2^{\ell+1-q_0}-1}\sum_{W'\in\mathcal{W}}(p+p_{W'}) \qquad \text{(using } \gamma_0\leqslant\eta \text{ and Equations (6.12) and (6.13))} \\
&\leqslant 5\eta\cdot\mathbb{P}\left[L\subseteq W\right]. \qquad \text{(using Equation (6.14) and } |\mathcal{W}| = 2^w-1 \leqslant 2^{\ell+1-q_0}-1)
\end{aligned}
$$

This gives the desired inequality

$$
5\eta \geqslant \mathbb{P}\left[F[L']\big|_L = f|_L \,\middle|\, L\subseteq W\right] \geqslant \delta,
$$

where we used Equation (6.10) in the last transition. □

6.3 The 4-to-1 Game Construction

In this subsection, we use the framework prepared in Sections 6.1 and 6.2 to construct our final sub-4-to-1 game. Before defining the sub-4-to-1 game, we first rewrite the description of Game2b (defined in Section 5.4) using the notations introduced in Sections 6.1 and 6.2.

Definition 6.28. Let H be a 3-uniform hypergraph. Given positive integers k and t, we let $\mathscr{E}(H,k,t)$ denote the collection of all sequences $\mathcal{E}=(E_1,\ldots,E_t)$ such that $E_j\in\mathscr{E}(H,k)$ for all $j\in[t]$.

We can now rewrite the definition of $\mathsf{Game2b}[H,k,\ell_2,t,q_0,q_1,\beta]$ as follows.

Definition 6.29. Given positive integers k,ℓ_2,t,s_1, a nonnegative integer q_0, a parameter $\beta\in[0,1)$ and a 3-uniform hypergraph H, let $\mathsf{Game2c}[H,k,\ell_2,t,s_1,q_0,\beta]$ be a 2-prover 1-round game where the verifier executes the following:

1. Sample a uniformly random sequence $\mathcal{E}=(E_1,\ldots,E_t)\in\mathscr{E}(H,k,t)$.
2. Sample $(c,\mathcal{U},\mathcal{E}',\mathcal{U}')\sim\mathcal{J}_{\mathsf{in}}(\mathcal{E},\beta,\ell_2)$.
3. Sample a uniformly random sequence $\mathcal{A}=(A_1,\ldots,A_t)\in\mathscr{A}_c(\mathcal{E}',q_0,s_1)$.[15]
4. Send $(\mathcal{E},\mathcal{A},\mathcal{U})$ to Alice and $(c,\mathcal{E}',\mathcal{A},\mathcal{U}')$ to Bob.
5. Alice must return a linear map $f:\mathcal{V}(\mathcal{E})\to\mathbb{F}_2$ such that $f|_{\mathcal{N}(\mathcal{E},\mathcal{U})}$ is SNR-satisfying; Bob must return a linear map $g:\mathcal{V}(\mathcal{E}')\to\mathbb{F}_2$. Accept if $f|_{\mathcal{V}(\mathcal{E}')}=g$.

It is easy to observe that Definition 6.29 defines the same game as substituting $q_1=2s_1(2^{q_0}-1)$ in Definition 5.20 does.

Proposition 6.30. For any positive integers k,ℓ_2,t,s_1, nonnegative integer q_0, parameter $\beta\in[0,1)$ and 3-uniform hypergraph H, we have

$$
\mathrm{val}\Big(\mathsf{Game2b}[H,k,\ell_2,t,q_0,2s_1(2^{q_0}-1),\beta]\Big) = \mathrm{val}\Big(\mathsf{Game2c}[H,k,\ell_2,t,s_1,q_0,\beta]\Big).
$$

[15] If $q_0=0$, we take $\mathcal{A}=\texttt{empty}$ to be an empty sequence.

Proof. It is easy to see the equivalence between Game2b and Game2c by comparing Definitions 5.20 and 6.29. □

We are now ready to define our sub-4-to-1 game based on Game2c. The main idea here is that instead of providing the functions f, g fully as in the game in Game2c, the provers are expected to provide an encoding of them based on the mixed-type Grassmann graph. More specifically, each player receives a linear subspace of their ambient space, and is requested to respond by a linear function on it. In order to make the game sub-4-to-1, the subspace Alice receives is contained in the subspace Bob receives and is of 1 less dimension, and additionally we incorporate inside it the local tensor tests as well as the side-condition equations.

Definition 6.31. Given positive integers k, ℓ_1, ℓ_2, t, s, a parameter $\beta \in [0, 1)$ and a 3-uniform hypergraph H, let $\mathsf{Game3}[H, k, \ell_1, \ell_2, t, s, \beta]$ be a 2-prover 1-round game where the verifier executes the following:

1. Sample a uniformly random sequence $\mathcal{E} = (E_1, \dots, E_t) \in \mathscr{E}(H, k, t)$.
2. Sample $(c, \mathcal{U}, \mathcal{E}', \mathcal{U}') \sim \mathcal{J}_{\mathsf{in}}(\mathcal{E}, \beta, \ell_2)$.
3. Sample a uniformly random sequence $\mathcal{A} = (A_1, \dots, A_t) \in \mathscr{A}_c(\mathcal{E}', 2\ell_1, s)$. Let
$$L = \sigma(\mathcal{A}_{[\ell_1]}) \subseteq \mathcal{V}(\mathcal{E}') \qquad \text{and} \qquad L' = \sigma(\mathcal{A}_{[\ell_1+1]}) \subseteq \mathcal{V}(\mathcal{E}').$$
4. Send $(\mathcal{E}, \mathcal{U}, L + \mathcal{N}(\mathcal{E}, \mathcal{U}))$ to Alice and $(c, \mathcal{E}', \mathcal{U}', L' + \bigoplus \mathcal{U}')$ to Bob.
5. Alice must return a linear map $f : (L + \mathcal{N}(\mathcal{E}, \mathcal{U})) \to \mathbb{F}_2$ such that $f|_{\mathcal{N}(\mathcal{E}, \mathcal{U})}$ is SNR-satisfying; Bob must return a linear map $g : (L' + \bigoplus \mathcal{U}') \to \mathbb{F}_2$ such that $g|_{\mathcal{U}'_i \otimes \mathcal{U}'_i}$ is symmetric and rank at most 1 for each i. Accept if $g|_{(L + \bigoplus \mathcal{U})} = f|_{(L + \bigoplus \mathcal{U})}$.

The game defined in Definition 6.31 is clearly an "at most 4-to-1" game, in the sense that for any response of Alice there are at most 4 responses for Bob which would make the verifier accept. Indeed, this is because $L \subseteq L'$ is of one less dimension, and $\mathcal{U}, \mathcal{U}'$ are identical on all coordinates, except for the critical coordinate $i \in [t]$ where $\mathcal{U}_i \subseteq \mathcal{U}'_i$ has one less dimension. Thus, Bob has at most 2 more bits of freedom than Alice. It is not difficult to turn this "at most 4-to-1" game to an actual 4-to-1 game as per Definition 1.1; see Appendix B.

The completeness of Game3 is straightforward and is given in the following lemma.

Lemma 6.32 (Game3 completeness)**.** *For any positive integers k, ℓ_1, ℓ_2, t, s, a parameter $\beta \in [0, 1)$ and a 2-colorable 3-uniform hypergraph H, we have*
$$\mathrm{val}\Big(\mathsf{Game3}[H, k, \ell_1, \ell_2, t, s, \beta]\Big) = 1.$$

Proof. By Lemma 5.21 and Proposition 6.30, there is a set of strategies for Alice and Bob that achieve value 1 in $\mathsf{Game2c}[H, k, \ell_2, t, s_1, 0, \beta]$ (here we may take $s_1 = 1$, but it is irrelevant anyway). Note that since the second-to-last parameter in $\mathsf{Game2c}[H, k, \ell_2, t, s_1, 0, \beta]$ is set to be zero, there is no advice in this game. Suppose the strategies are represented by a table $F[\cdot]$ for Alice and a table $G[\cdot]$ for Bob, where $F[\mathcal{E}, \mathcal{U}] : \mathcal{V}(\mathcal{E}) \to \mathbb{F}_2$ is a linear map for each Alice-question $(\mathcal{E}, \mathcal{U})$, and $G[c, \mathcal{E}', \mathcal{U}'] : \mathcal{V}(\mathcal{E}') \to \mathbb{F}_2$ is a linear map for each Bob-question $(c, \mathcal{E}', \mathcal{U}')$.

We can then construct a strategy for Alice in the game $\mathsf{Game3}[H, k, \ell_1, \ell_2, t, s, \beta]$ as follows:

1. Receive the question $(\mathcal{E}, \mathcal{U}, L + \mathcal{N}(\mathcal{E}, \mathcal{U}))$.

2. Let f be the restriction of $F[\mathcal{E},\mathcal{U}]$ to $L+\mathcal{N}(\mathcal{E},\mathcal{U})$, and return f. Note that by Definition 6.29, the restriction of $F[\mathcal{E},\mathcal{U}]$ to $\mathcal{N}(\mathcal{E},\mathcal{U})$ must be SNR-satisfying.

Similarly, the strategy for Bob is:

1. Receive the question $(c,\mathcal{E}',\mathcal{U}',L'+\bigoplus\mathcal{U}')$.
2. Let g be the restriction of $G[c,\mathcal{E}',\mathcal{U}']$ to $L'+\bigoplus\mathcal{U}'$.

Since we always have $F[\mathcal{E},\mathcal{U}]|_{\mathcal{V}(\mathcal{E}')}=G[c,\mathcal{E}',\mathcal{U}']$, it is clear that we also have $f|_{(L+\bigoplus\mathcal{U})}=g|_{(L+\bigoplus\mathcal{U})}$. Moreover, we choose F,G to be the strategies induced from the proper 2-coloring of the original graph through the series of reductions. The strategies are symmetric and have rank at most 1 restricted to each block. Therefore, under this set of strategies, the verifier of $\mathsf{Game3}[H,k,\ell_1,\ell_2,t,s,\beta]$ accepts with probability 1. □

The soundness of Game3 is stated in the following lemma.

Lemma 6.33 (Game3 soundness)**.** *Assume Condition 3.12. Suppose that in every 4^r-coloring of H, at least ε fraction of the edges are monochromatic. Then we have*

$$\mathrm{val}\Big(\mathsf{Game3}[H,k,\ell_1,\ell_2,t,s,\beta]\Big)\leqslant\delta.$$

The proof of Lemma 6.33 is significantly more involved, and spans the rest of this section. Towards this end, we first establish preliminary results in Section 6.5, and then use them in Section 6.6 to complete the proof of Lemma 6.33.

6.4 Overview of Soundness Analysis

In this section we discuss the overview of the analysis of the construction from Definition 6.31. Let F, G be strategies of Alice and Bob respectively that achieve value at least δ. Our goal is to show how to derive from F and G strategies for Game2c from Definition 6.29 achieving a significant value. At a high-level, the proof proceeds via the following steps:

1. **A decoding strategy for Alice:** recall that for each $(\mathcal{E},\mathcal{U},L+\mathcal{N}(\mathcal{E},\mathcal{U}))$, Alice's strategy provides a linear function $F[\mathcal{E},\mathcal{U},L+\mathcal{N}(\mathcal{E},\mathcal{U})]\colon L+\mathcal{N}(\mathcal{E},\mathcal{U})\to\mathbb{F}_2$ satisfying the side conditions on $\mathcal{N}(\mathcal{E},\mathcal{U})$. Thus, for each fixing of $\mathcal{E},\mathcal{U}$, we can uniquely define $\widetilde{F}_{\mathcal{E},\mathcal{U}}$ that maps a linear subspace L such that $L\cap\mathcal{N}(\mathcal{E},\mathcal{U})=\{0\}$, to $F[\mathcal{E},\mathcal{U},L+\mathcal{N}(\mathcal{E},\mathcal{U})]|_L$. Using the consistency between F and G and the Cauchy-Schwarz inequality in a similar way to Lemma 5.15, it can be shown that on average, $\widetilde{F}_{\mathcal{E},\mathcal{U}}$ passes the Grassmann agreement test with noticeable probability. At this point, we would like to apply Theorem 3.5 to be able to decode a global linear function for Alice (inside some zoom-in/zoom-out pair), but the distribution over L induced by the construction is not exactly uniform over $\mathcal{V}(\mathcal{E})$ (due to the critical coordinate). However, it turns out that this distribution is close to being uniform over subspaces in $\mathcal{V}(\mathcal{E})$; we refer to this as *covering property for the mixed-type subspaces*.

   Ultimately, we conclude that there is a set of constant dimension zoom ins $\mathcal{Q}$ in $\mathcal{V}(\mathcal{E})$ of noticeable fractional size, such that for each $Q\in\mathcal{Q}$ there is a zoom-out $W\subseteq\mathcal{V}(\mathcal{E})$ of constant codimension, such that $\widetilde{F}_{\mathcal{E},\mathcal{U}}$ agrees with some global linear function $f_{\mathcal{E},\mathcal{U},Q,W}\colon W\to\mathbb{F}_2$ on a noticeable fraction of $Q\subseteq L\subseteq W$. Translating this information back to $F[\mathcal{E},\mathcal{U},\cdot]$ gives a global linear function that also satisfies the side conditions and the tensor tests.

   Alice in Game2c can utilize the above observation to construct a strategy. More precisely, she chooses Q using her Advice, and chooses W for which a global linear function as above

exists. Strictly speaking, she has to choose W as maximal subspace with respect to inclusion such that a global linear function as above exists; this is done so as to ensure that for a fixed Q, we have that the list of possible answers Alice gives is small.

2. **A decoding strategy for Bob:** by appropriate pruning, it is easy to ensure that a decoding strategy for Alice automatically gives a decoding strategy for Bob. Indeed, if we only keep entries $(\mathcal{E},\mathcal{U}, L+\mathcal{N}(\mathcal{E},\mathcal{U}))$ of Alice such that conditioned on them, Alice and Bob win with probability at least $\delta/2$, then any decoding of Alice automatically has quite good agreement with the strategy of Bob but in a little bit of a strange way. Indeed, because Alice's space has 1 less dimension than Bob's, we conclude that Bob's table G and Alice's global function f satisfy the premise of Lemma 6.27, and using it we are able to get a decoding for Bob.

3. **Consistency of the decodings:** let us assume now that we have fixed a zoom-in Q, so that Alice has a list of possible zoom-out global function pairs $(f_1, W_1), \ldots, (f_m, W_m)$, and Bob has the list $(g_1, W'_1), \ldots, (g_N, W'_N)$. A natural idea to prove the consistency between decodings is to show that the list of legitimate decodings of Alice and Bob are of bounded size. Such a result for Alice's table was proved in [DKK+25], but it is much more complicated for Bob since the structure of mixed-type Grassmann is quite different from the ordinary Grassmann. While we believe this is true, we do not address this problem directly, and instead bypass this difficulty using the following idea.

   Using the same argument as in step 2 (but in the reverse direction), we know that for a large number of Alice-packets $(\mathcal{E},\mathcal{U})$, (g_i, W'_i) also has good consistency with Alice's table, but also in the similar mixed-type Grassmann domain instead of our desired ordinary Grassmann domain as described in Alice's strategy. This is formalized by Definition 6.59 and Claim 6.61. The heart of the argument is to show that, because there are many such Bob-packets, there must be some global consistency between them. More precisely, we show that there is some i such that g_i (or its linear extension) has consistency with Alice's table in the ordinary Grassmann domain rather than mixed-type Grassmann, and we elaborate on this step in Section 6.8.

   Using this observation, while it does not say that Bob's list is bounded, it does say that we can conclude that for every Alice-packet $(\mathcal{E},\mathcal{U})$ and advice Q, if its table has noticeable consistency with respect to mixed-type Grassmann for a lot of Bob-packets $(\mathcal{E}',\mathcal{U}')$ and the corresponding decodings (f, W), then, many of them would also have consistency with $F[\mathcal{E},\mathcal{U},\cdot]$ in the ordinary Grassmann domain, so the responses of Alice and Bob are consistent with noticeable probability.

6.5 Preparatory Lemmas for Soundness Analysis

In this subsection, we fix a 3-uniform hypergraph H, positive integers k, ℓ_1, ℓ_2, t, s, a nonnegative integer q_0 and a parameter $\beta \in (0,1)$. We consider the game $\mathsf{Game3}[H, k, \ell_1, \ell_2, t, s, \beta]$ as defined in Definition 6.31.

6.5.1 Notations

The following notational convention will be used throughout Section 6.5. The first two notational definitions below are parallels of Notations 5.11 and 5.12 in Section 5.3.

Notation 6.34. We use $\mathcal{D}_{\mathsf{in}}$ to denote the joint distribution of $(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',\mathcal{A})$ sampled by the verifier in $\mathsf{Game3}[H, k, \ell_1, \ell_2, t, s, \beta]$. We also need a variant of $\mathcal{D}_{\mathsf{in}}$ where the five components are

expanded into seven components: denote $\mathcal{D}^{+}_{\mathsf{in}}$ to be the joint distribution of $(\mathcal{E},\mathcal{U},L,Q,\mathcal{E}',\mathcal{U}',L')$, where $(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',\mathcal{A})$ is sampled from $\mathcal{D}_{\mathsf{in}}$ and

$$Q := \sigma(\mathcal{A}_{[q_0]}), \qquad L := \sigma(\mathcal{A}_{[\ell_1]}), \qquad L' = \sigma(\mathcal{A}_{[\ell_1+1]}).$$

We use the symbol $*$ to denote marginalization over the corresponding component. For instance, sampling from the marginal distribution of $(\mathcal{U},\mathcal{E}')$ under $\mathcal{D}_{\mathsf{in}}$ is denoted by $(*,\mathcal{U},\mathcal{E}',*,*) \sim \mathcal{D}_{\mathsf{in}}$.

Notation 6.35. We will often condition the distribution $\mathcal{D}_{\mathsf{in}}$ on some subset of the five components $(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',\mathcal{A})$ and consider the induced joint distribution of the remaining components. For instance, if $\mathcal{E}$ and $\mathcal{U}'$ are fixed, then the conditional joint distribution of $(\mathcal{U},\mathcal{E}')$ is denoted by

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}_{\mathsf{in}}} \left[\cdot \,\middle|\, \mathcal{E},\mathcal{U}'\right].$$

Similar conditioning notation will be used for the seven-component joint distribution $\mathcal{D}^{+}_{\mathsf{in}}$.

The notions of *Alice-table* and *Bob-table* to be defined in Notation 6.37 are slightly more nuanced than their counterparts in Notation 5.13. In particular, the entries of an Alice-table do not exactly correspond to the possible questions Alice may receive.

Notation 6.36. For any possible sample $(\mathcal{E},\mathcal{U},*,*,*) \sim \mathcal{D}_{\mathsf{in}}$, the pair $(\mathcal{E},\mathcal{U})$ is called an *Alice-packet*. Similarly, a pair $(\mathcal{E}',\mathcal{U}')$ is called a *Bob-packet* if $(*,*,\mathcal{E}',\mathcal{U}',*)$ has nonzero probability under $\mathcal{D}_{\mathsf{in}}$. For any possible sample $(\mathcal{E},\mathcal{U},L,*,*,*,*) \sim \mathcal{D}^{+}_{\mathsf{in}}$, the tuple $(\mathcal{E},\mathcal{U},L)$ is called an *unfolded-Alice-question*. Similarly, a tuple $(\mathcal{E}',\mathcal{U}',L')$ is called an *unfolded-Bob-question* if $(*,*,*,*,\mathcal{E}',\mathcal{U}',L')$ has nonzero probability under $\mathcal{D}^{+}_{\mathsf{in}}$.

Notation 6.37. A table $F[\cdot]$ is called an Alice-table if for each Alice-packet $(\mathcal{E},\mathcal{U})$ and each linear subspace $L \subseteq \mathcal{V}(\mathcal{E})$, the entry $F[\mathcal{E},\mathcal{U},L]$ is either `nil` or a linear functional $F[\mathcal{E},\mathcal{U},L] : L \to \mathbb{F}_2$. Similarly, a table $G[\cdot]$ is called a Bob-table if for each Bob-packet $(\mathcal{E}',\mathcal{U}')$ and each linear subspace $L' \subseteq \mathcal{V}(\mathcal{E}')$, the entry $G[\mathcal{E}',\mathcal{U}',L']$ is either `nil` or a linear functional $G[\mathcal{E}',\mathcal{U}',L'] : L' \to \mathbb{F}_2$.

Notation 6.38. Fix an Alice-table $F[\cdot]$ and a Bob-table $G[\cdot]$. For any tuple $(\mathcal{E},\mathcal{U},L,*,\mathcal{E}',\mathcal{U}',L')$ with nonzero probability under $\mathcal{D}^{+}_{\mathsf{in}}$, we define

$$\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},L,\mathcal{E}',\mathcal{U}',L') = \begin{cases} 1, & \text{if } \; G[\mathcal{E}',\mathcal{U}',L']\big|_L = F[\mathcal{E},\mathcal{U},L], \\ 0, & \text{otherwise.} \end{cases}$$

We will often sample $(\mathcal{E},\mathcal{U},L,*,\mathcal{E}',\mathcal{U}',L')$ from $\mathcal{D}^{+}_{\mathsf{in}}$ conditioned on some subset of its components. For instance, if $\mathcal{E}$ and $\mathcal{U}'$ are fixed, then the conditional expectation of $\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},L,\mathcal{E}',\mathcal{U}',L')$ is denoted by

$$\mathrm{Vrf}_{F,G}(\mathcal{E},*,*,*,\mathcal{U}',*) = \mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},L,*,\mathcal{E}',\mathcal{U}',L')\sim\mathcal{D}^{+}_{\mathsf{in}}} \left[G[\mathcal{E}',\mathcal{U}',L']\big|_L = F[\mathcal{E},\mathcal{U},L] \,\middle|\, \mathcal{E},\mathcal{U}'\right].$$

6.5.2 Covering Property of Mixed-Type Subspaces

Note that in Definition 6.31, the subspace L that is being sent to Alice is not uniformly chosen from her space $\mathcal{V}(\mathcal{E})$, and instead it is chosen from $\mathcal{V}(\mathcal{E}')$ with a random critical coordinate c. For the test we perform to make sense from Alice's perspective (as well as for the advice feature we arranged to work) it is crucial that the distribution of L is statistically close to being uniform over subspaces of her ambient space. The following definition and lemma make this assertion more precise, and in fact they show that the two distributions are close even conditioning on a small number of fixed vectors being contained in L.

Definition 6.39. For a fixed Alice-packet $(\mathcal{E},\mathcal{U})$ and a sequence of vectors

$$B = (a^{(1)},\dots,a^{(q_0)}) \in (\mathcal{V}(\mathcal{E}))^{q_0},$$

let $\mathrm{TVD}_{\mathcal{E},\mathcal{U}}(B)$ denote the total variation distance between

(1) the distribution of $\rho(\mathcal{A})$ when $(\mathcal{E},\mathcal{U},*,*,\mathcal{A}) \sim \mathcal{D}_{\mathsf{in}}$ is sampled conditioned on the fixed Alice-packet $(\mathcal{E},\mathcal{U})$ and the event $\{\rho(\mathcal{A}_{[q_0]}) = B\}$ (if this event happens with zero probability, let $\mathrm{TVD}_{\mathcal{E},\mathcal{U}}(B) = 1$); and

(2) the uniform distribution over the set

$$\left\{ \left(a^{(1)},\dots,a^{(q_0)},b^{(q_0+1)},\dots,b^{(2\ell_1)}\right) \,\middle|\, b^{(q_0+1)},\dots,b^{(2\ell_1)} \in \mathcal{V}(\mathcal{E}) \right\} \subseteq (\mathcal{V}(\mathcal{E}))^{2\ell_1}.$$

Lemma 6.40. *Assume Condition 3.12. For any fixed Alice-packet $(\mathcal{E},\mathcal{U})$, we have*

$$\mathop{\mathbb{E}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}} \left[\mathrm{TVD}_{\mathcal{E},\mathcal{U}}\left(\rho(\mathcal{A}_{[q_0]})\right) \,\middle|\, \mathcal{E},\mathcal{U} \right] \leqslant t^{-1/3}.$$

Proof. Towards the goal of applying Proposition F.1, we set up some notation. Write

$$\mathcal{E} = (E_1,\dots,E_t), \qquad \mathcal{U} = (U_1,\dots,U_t) \qquad \text{and} \qquad q_1 = 2(2^{2\ell_1}-1)s.$$

For each $j \in [t]$, define sets $\mathscr{A}_j, \mathscr{B}_j$ and a map $\pi_j : \mathscr{A}_j \to \mathscr{B}_j$ by

$$\mathscr{A}_j = \left\{ \left(a^{(1)},\dots,a^{(2\ell_1)}\right) \,\middle|\, a^{(i)} \in \mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j} \text{ for each } i \in [2\ell_1] \right\} = \left(\mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j}\right)^{2\ell_1},$$
$$\mathscr{B}_j = \left(\mathbb{F}_2^{E_j} \otimes \mathbb{F}_2^{E_j}\right)^{q_0}, \qquad \text{and} \qquad \pi_j : \left(a^{(1)},\dots,a^{(2\ell_1)}\right) \mapsto \left(a^{(1)},\dots,a^{(q_0)}\right).$$

For each $j \in [t]$, let μ_j be the output distribution of the following process:

1. Sample $E'_j \sim \mathcal{J}_{\mathsf{rev}}(E_j, U_j, \beta, \ell_2)$.
2. Pick $q = 2(2^{2\ell_1}-1)s$ vectors $a^{(j,1)},\dots,a^{(j,q)}$ from $\mathbb{F}_2^{E'_j}$ independently and uniformly at random. Let $A = \left(a^{(j,1)},\dots,a^{(j,q)}\right)$.
3. Output the sequence

$$\left(\mathrm{Lvec}(A, 2\ell_1, e_1),\dots,\mathrm{Lvec}(A, 2\ell_1, e_{2\ell_1})\right) \in \mathscr{A}_j,$$

where $e_1,\dots,e_{2\ell_1} \in \mathbb{F}_2^{2\ell_1}$ are the standard basis vectors.

We can then define a distribution μ over $\prod_{j=1}^t \mathscr{A}_j$ as in the statement of Proposition F.1; it is easy to see (using Proposition 6.10) that μ is identical to the distribution of $\rho(\mathcal{A})$ when $(\mathcal{E},\mathcal{U},*,*,\mathcal{A}) \sim \mathcal{D}_{\mathsf{in}}$ is sampled conditioned on $(\mathcal{E},\mathcal{U})$.

Now we can apply Proposition F.1 to $\mathscr{A}_j, \mathscr{B}_j$, π_j and μ_j as defined above. The conclusion of Proposition F.1 says that

$$\mathop{\mathbb{E}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}} \left[\mathrm{TVD}_{\mathcal{E},\mathcal{U}}\left(\rho(\mathcal{A}_{[q_0]})\right) \,\middle|\, \mathcal{E},\mathcal{U} \right] \leqslant \sqrt{\frac{2^{18k^2\ell_1}}{4t}} \leqslant t^{-1/3},$$

where we used Condition 3.12 in the last transition. □

While Definition 6.39 deals with the vector list $\rho(\mathcal{A})$, we eventually need to study Alice's space L, which is spanned by the first ℓ_1 vectors in $\rho(\mathcal{A})$. The next definition is a variation of Definition 6.39 that directly addresses the distribution of L (the reason we started with Definition 6.39 was purely to facilitate the application of Proposition F.1 in Lemma 6.40).

Definition 6.41. For a fixed Alice-packet $(\mathcal{E},\mathcal{U})$, a linear subspace $Q \subseteq \mathcal{V}(\mathcal{E})$ and a linear subspace $W \subseteq \mathcal{V}(\mathcal{E})$ containing Q, let $\mathrm{TVD}_{\mathcal{E},\mathcal{U},W}(Q)$ be the total variation distance between

(1) the distribution of $\sigma(\mathcal{A}_{[\ell_1]})$ when $(\mathcal{E},\mathcal{U},*,*,\mathcal{A}) \sim \mathcal{D}_{\mathsf{in}}$ is sampled conditioned on the fixed Alice-packet $(\mathcal{E},\mathcal{U})$ and the event $\{\sigma(\mathcal{A}_{[q_0]}) = Q\} \cap \{\sigma(\mathcal{A}_{[\ell_1]}) \subseteq W\}$ (if this event happens with zero probability, let $\mathrm{TVD}_{\mathcal{E},\mathcal{U},W}(Q) = 1$); and

(2) the uniform distribution over the set $\{L \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1) \,|\, Q \subseteq L \subseteq W\}$.

If $W = \mathcal{V}(\mathcal{E})$, we omit the subscript W and write $\mathrm{TVD}_{\mathcal{E},\mathcal{U}}(Q) = \mathrm{TVD}_{\mathcal{E},\mathcal{U},\mathcal{V}(\mathcal{E})}(Q)$.

The following three lemmas about Definition 6.41 are easy to prove.

Lemma 6.42. *Assume Condition 3.12. Fix an Alice-packet $(\mathcal{E},\mathcal{U})$, a linear subspace $Q \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}),q_0)$ and a linear subspace $W \subseteq \mathcal{V}(\mathcal{E})$ containing Q. Let $w = \operatorname{codim} W$, and suppose $\dim W \geqslant \ell_1$. We always have*

$$\mathrm{TVD}_{\mathcal{E},\mathcal{U},W}(Q) \leqslant 2^{\ell_1(w+1)} \cdot \mathrm{TVD}_{\mathcal{E},\mathcal{U}}(Q).$$

Proof. This is simply due to

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)} \left[L \subseteq W \,\middle|\, Q \subseteq L\right] \geqslant 2^{-\ell_1(w+1)}. \qquad \square$$

Lemma 6.43. *Fix an Alice-packet $(\mathcal{E},\mathcal{U})$ and a linear subspace $Q \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}),q_0)$. We have*

$$\mathrm{TVD}_{\mathcal{E},\mathcal{U}}(Q) \leqslant \mathop{\mathbb{E}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}} \left[\mathrm{TVD}_{\mathcal{E},\mathcal{U}}\big(\rho(\mathcal{A}_{[q_0]})\big) \,\middle|\, \mathcal{E},\mathcal{U} \text{ and } \{\sigma(\mathcal{A}_{[q_0]}) = Q\}\right] + 2^{\ell_1-9k^2t}.$$

Proof. Use the following fact to pass from the uniform distribution over $(\mathcal{V}(\mathcal{E}))^{\ell_1}$ to the uniform distribution over $\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)$: for any sequence of vectors $B = (a^{(1)},\dots,a^{(q_0)}) \in (\mathcal{V}(\mathcal{E}))^{q_0}$ such that $\operatorname{span}(B) = Q$, if $b^{(q_0+1)},\dots,b^{(\ell_1)} \in \mathcal{V}(\mathcal{E})$ are sampled independently and uniformly at random, the probability that $a^{(1)},\dots,a^{(q_0)},b^{(q_0+1)},\dots,b^{(\ell_1)}$ are not linearly independent is at most $2^{\ell_1-9k^2t}$. $\square$

Lemma 6.44. *Assume Condition 3.12. For any fixed Alice-packet $(\mathcal{E},\mathcal{U})$, we have*

$$\mathop{\mathbb{E}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}} \left[\mathrm{TVD}_{\mathcal{E},\mathcal{U}}\big(\sigma(\mathcal{A}_{[q_0]})\big) \,\middle|\, \mathcal{E},\mathcal{U}\right] \leqslant t^{-1/4}.$$

Proof. Using Lemma 6.43 followed by Lemma 6.40, we have

$$\begin{aligned}\mathop{\mathbb{E}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}} \left[\mathrm{TVD}_{\mathcal{E},\mathcal{U}}\big(\sigma(\mathcal{A}_{[q_0]})\big) \,\middle|\, \mathcal{E},\mathcal{U}\right] &\leqslant \mathop{\mathbb{E}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}} \left[\mathrm{TVD}_{\mathcal{E},\mathcal{U}}\big(\rho(\mathcal{A}_{[q_0]})\big) \,\middle|\, \mathcal{E},\mathcal{U}\right] + 2^{\ell_1-9k^2t} \\ &\leqslant t^{-1/3} + 2^{\ell_1-9k^2t} \leqslant t^{-1/4},\end{aligned}$$

where we used Condition 3.12 in the last transition. $\square$

To summarize the discussion, we define the notion of a smooth zoom-in, which is a subspace Q such that conditioned on $(\mathcal{E},\mathcal{U},Q)$, the distribution of L in $(\mathcal{E},\mathcal{U},L,Q,*,*,*) \sim \mathcal{D}^{+}_{\mathsf{in}}$ is close to the uniform distribution over the set $\{L \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1) \,|\, Q \subseteq L\}$.

Definition 6.45. For a fixed Alice-packet $(\mathcal{E},\mathcal{U})$, a linear subspace $Q \subseteq \mathcal{V}(\mathcal{E})$ is said to be *smooth* with respect to $(\mathcal{E},\mathcal{U})$ if $\dim Q = q_0$ and $\mathrm{TVD}_{\mathcal{E},\mathcal{U}}(Q) \leqslant t^{-1/8}$.

The following corollary shows that Q is smooth with probability close to 1.

Corollary 6.46. *Assume Condition 3.12. For any fixed Alice-packet $(\mathcal{E},\mathcal{U})$, we have*

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,Q,*,*,*)\sim\mathcal{D}_{\mathsf{in}}^{+}}\left[Q \text{ is smooth with respect to } (\mathcal{E},\mathcal{U}) \,\middle|\, \mathcal{E},\mathcal{U}\right] \geqslant 1 - t^{-1/8}.$$

Proof. Immediate from Lemma 6.44 and Markov's inequality. □

Finally, we record a covering property without advice, meaning that the distribution of L in $(\mathcal{E},\mathcal{U},L,*,*,*,*) \sim \mathcal{D}_{\mathsf{in}}^{+}$ (without conditioning on Q) is close to uniform over $\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)$.

Proposition 6.47. Assume Condition 3.12. Fix a positive integer $\ell \leqslant 2\ell_1$. For any fixed Alice-packet $(\mathcal{E},\mathcal{U})$, if we sample $(\mathcal{E},\mathcal{U},*,*,\mathcal{A}) \sim \mathcal{D}_{\mathsf{in}}$ conditioned on $(\mathcal{E},\mathcal{U})$, the resulting distribution of $\sigma(\mathcal{A}_{[\ell]})$ has total variation distance at most $t^{-1/4}$ to the uniform distribution over $\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell)$.

Proof. The proof is entirely analogous to the (combined) proofs of Lemmas 6.40 and 6.43, and is hence omitted. The main difference is that when one invokes Proposition F.1, this time one can take $\mathscr{B}_j$ to be a singleton set for each $j \in [t]$. □

6.5.3 Alice's Existence of Decoding

The following lemma, analogous to Lemma 5.15, shows that Alice's table passes the Grassmann agreement test with noticeable probability.

Lemma 6.48. *Assume Condition 3.12. Let $F[\cdot]$ be an Alice-table and $G[\cdot]$ be a Bob-table. For any Alice-packet $(\mathcal{E},\mathcal{U})$, we have*

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathcal{V}(\mathcal{E}),\ell_1)}\left[F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2} = F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}\right] \geqslant \mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},*,*,*,*)^2 - 2^{-\ell_1}.$$

Proof. We organize the proof into the following two steps.

Step 1: fixing $(\mathcal{E}',\mathcal{U}',L')$. Fix a Bob-packet $(\mathcal{E}',\mathcal{U}')$ and a subspace $L' \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}'),\ell_1+1)$ such that

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}}\left[\sigma(\mathcal{A}_{[\ell_1+1]}) = L' \,\middle|\, \mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}'\right] > 0.$$

By Definition 6.31, it is clear that if we sample $(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',\mathcal{A}) \sim \mathcal{D}_{\mathsf{in}}$ conditioned on $\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}'$ and the event $\{\sigma(\mathcal{A}_{[\ell_1+1]}) = L'\}$, the resulting distribution of $\sigma(\mathcal{A}_{[\ell_1]})$ is uniform over the set $\mathsf{Gr}(L',\ell_1)$. Therefore, we have

$$\begin{aligned}
&\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},*,\mathcal{E}',\mathcal{U}',L')^2 \\
&= \mathop{\mathbb{E}}_{(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}}\left[\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},*,\mathcal{E}',\mathcal{U}',L') \,\middle|\, \mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}' \text{ and } \{\sigma(\mathcal{A}_{[\ell_1+1]}) = L'\}\right]^2 \\
&= \mathop{\mathbb{E}}_{L_1,L_2\in\mathsf{Gr}(L',\ell_1)}\left[\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},L_1,\mathcal{E}',\mathcal{U}',L')\,\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},L_2,\mathcal{E}',\mathcal{U}',L')\right]. \qquad (6.15)
\end{aligned}$$

Expanding the right-hand side of Equation (6.15), we get

$$\begin{aligned}
&\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},*,\mathcal{E}',\mathcal{U}',L')^2 \\
&= \mathop{\mathbb{P}}_{L_1,L_2\in\mathsf{Gr}(L',\ell_1)}\left[F[\mathcal{E},\mathcal{U},L_1] = G[\mathcal{E}',\mathcal{U}',L']|_{L_1} \text{ and } F[\mathcal{E},\mathcal{U},L_2] = G[\mathcal{E}',\mathcal{U}',L']|_{L_2}\right] \\
&\leqslant \mathop{\mathbb{P}}_{L_1,L_2\in\mathsf{Gr}(L',\ell_1)}\left[F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2} = F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}\right]. \qquad (6.16)
\end{aligned}$$

Step 2: $(\mathcal{E}',\mathcal{U}',L')$ unfixed. By Proposition 6.47, we know that conditioned on $(\mathcal{E},\mathcal{U})$, the distribution of $\sigma(\mathcal{A}_{[\ell_1+1]})$ in the sample $(\mathcal{E},\mathcal{U},*,*,\mathcal{A}) \sim \mathcal{D}_{\mathsf{in}}$ is $t^{-1/4}$-close to uniform over $\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1+1)$. Therefore, we have

$$\begin{aligned}
&\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},*,*,*,*)^2 \\
&= \mathop{\mathbb{E}}_{(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}} \left[\mathrm{Vrf}_{F,G}\left(\mathcal{E},\mathcal{U},*,\mathcal{E}',\mathcal{U}',\sigma(\mathcal{A}_{[\ell_1+1]})\right) \,\middle|\, \mathcal{E},\mathcal{U}\right]^2 && \text{(by definition)} \\
&\leqslant \mathop{\mathbb{E}}_{(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}} \left[\mathrm{Vrf}_{F,G}\left(\mathcal{E},\mathcal{U},*,\mathcal{E}',\mathcal{U}',\sigma(\mathcal{A}_{[\ell_1+1]})\right)^2 \,\middle|\, \mathcal{E},\mathcal{U}\right] && \text{(by Cauchy-Schwarz)} \\
&\leqslant \mathop{\mathbb{E}}_{L'\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1+1)} \left[\mathop{\mathbb{E}}_{(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}} \left[\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},*,\mathcal{E}',\mathcal{U}',L')^2 \,\middle|\, \mathcal{E},\mathcal{U} \text{ and } \{\sigma(\mathcal{A}_{[\ell_1+1]}) = L'\}\right]\right] + t^{-1/4}.
\end{aligned}$$

We can now apply Equation (6.16) to the above and get

$$\begin{aligned}
&\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},*,*,*,*)^2 \\
&\leqslant \mathop{\mathbb{E}}_{L'\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1+1)} \left[\mathop{\mathbb{P}}_{L_1,L_2\in\mathsf{Gr}(L',\ell_1)} \left[F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2} = F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}\right]\right] + t^{-1/4} \\
&\leqslant \mathop{\mathbb{E}}_{(L_1,L_2)\sim\mathcal{G}(\mathcal{V}(\mathcal{E}),\ell_1)} \left[F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2} = F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}\right] + \frac{1}{2^{\ell_1+1}-1} + t^{-1/4} \\
&\leqslant \mathop{\mathbb{E}}_{(L_1,L_2)\sim\mathcal{G}(\mathcal{V}(\mathcal{E}),\ell_1)} \left[F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2} = F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}\right] + 2^{-\ell_1}, \qquad \text{(using Condition 3.12)}
\end{aligned}$$

as desired. □

We now define the type of Alice-packets on which Alice from Game2c can come up with a response.

Definition 6.49. Let $F[\cdot]$ be an Alice table, and let $(\mathcal{E},\mathcal{U})$ be an Alice-packet. For any linear subspace $Q \subseteq \mathcal{V}(\mathcal{E})$ with $\dim Q \leqslant q_0$, we say that the tuple $(\mathcal{E},\mathcal{U},Q)$ is *answerable* with respect to $F[\cdot]$ if $\dim Q = q_0$ and the following statement holds: there exists a linear subspace $W \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}), 9k^2t - q_0)$ and a linear map $f : W \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)} \left[F[\mathcal{E},\mathcal{U},L] = f|_L \,\middle|\, Q \subseteq L \subseteq W\right] \geqslant \exp(-q_0^3).$$

Combining Lemma 6.48 and the Grassmann decoding theorem, Theorem 3.5, shows that there are many Alice-packets for which Alice from Game2c can come up with a response.

Lemma 6.50. *Assume Condition 3.12. Let $F[\cdot]$ be an Alice-table and let $G[\cdot]$ be a Bob-table. For any Alice-packet $(\mathcal{E},\mathcal{U})$ such that $\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},*,*,*,*) \geqslant 2^{-q_0/3}$, we have*

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,Q,*,*,*)\sim\mathcal{D}^+_{\mathsf{in}}} \left[(\mathcal{E},\mathcal{U},Q) \text{ is answerable with respect to } F[\cdot] \,\middle|\, \mathcal{E},\mathcal{U}\right] \geqslant 4^{-\ell_1^2}.$$

Proof. Since $\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},*,*,*,*) \geqslant 2^{-q_0/3}$, Lemma 6.48 implies

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathcal{V}(\mathcal{E}),\ell_1)} \left[F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2} = F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}\right] \geqslant 2^{-2q_0/3} - 2^{-\ell_1} \geqslant 2^{-q_0+5},$$

where we used Condition 3.12 in the last transition. Applying Theorem 3.5 to the table $F[\mathcal{E},\mathcal{U},L]$, where L ranges in $\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)$, yields a collection of subspaces $\mathcal{Q} \subseteq \mathsf{Gr}(\mathcal{V}(\mathcal{E}),q_0)$ such that:

(1) $|\mathcal{Q}| \geqslant 2^{-q_0-\ell_1^2}\,|\mathsf{Gr}(\mathcal{V}(\mathcal{E}), q_0)|$;

(2) for each $Q \in \mathcal{Q}$, there exists a linear subspace $W_Q \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}), 9k^2t - q_0)$ and a linear map $f_Q : W_Q \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\left[F[\mathcal{E},\mathcal{U},L] = f_Q|_L \;\middle|\; Q \subseteq L \subseteq W_Q\right] \geqslant \exp(-Cq_0^2) \geqslant \exp(-q_0^3),$$

where C is an absolute constant and the last transition is due to Condition 3.12. By Definition 6.49, this means that for each $Q \in \mathcal{Q}$, the tuple $(\mathcal{E},\mathcal{U},Q)$ is answerable with respect to $F[\cdot]$.

Therefore, it now suffices to show that

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}}\left[\sigma(\mathcal{A}_{[q_0]}) \in \mathcal{Q} \;\middle|\; \mathcal{E},\mathcal{U}\right] \geqslant 4^{-\ell_1^2}.$$

But it follows immediately from Proposition 6.47 that

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}}\left[\sigma(\mathcal{A}_{[q_0]}) \in \mathcal{Q} \;\middle|\; \mathcal{E},\mathcal{U}\right] \geqslant \mathop{\mathbb{P}}_{Q\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),q_0)}[Q \in \mathcal{Q}] - t^{-1/4} \geqslant 2^{-q_0-\ell_1^2} - t^{-1/4} \geqslant 4^{-\ell_1^2},$$

where we used Condition 3.12 in the last transition. □

In conclusion, we have shown that if the F, G are Alice and Bob tables that agree with noticeable probability, then with noticeable probability Alice can use her zoom-in Q to come up with a global linear function f agreeing. We next show a similar assertion to Bob.

6.5.4 Bob's Existence of Decoding

We intend to show that Bob is able to come up with a response using the fact that Alice can, and that Alice's space "covers" Bob's space. Towards this end, we make the following definitions. The first definition gives a precise formulation of the packets of Bob that "inherit" good agreement from a global linear function of Alice.

Definition 6.51. Let $G[\cdot]$ be a Bob table, and let $(\mathcal{E}',\mathcal{U}')$ be a Bob-packet with critical index $c \in [t]$. For any linear subspace $Q \subseteq \mathcal{V}(\mathcal{E}')$ with $\dim Q \leqslant q_0$, we say that the tuple $(\mathcal{E}',\mathcal{U}',Q)$ is *pre-answerable* with respect to $G[\cdot]$ if the following statements hold:

1. The tuple $(*,*,*,Q,\mathcal{E}',\mathcal{U}',*)$ has nonzero probability under $\mathcal{D}^+_{\mathsf{in}}$. In other words, $Q = \sigma(\mathcal{A})$ for some $\mathcal{A} \in \mathscr{A}_c(\mathcal{E}', q_0, 2^{2\ell_1-q_0}s)$.
2. There exists a linear subspace $W \subseteq \mathcal{V}(\mathcal{E}')$ with $\operatorname{codim} W \leqslant q_0$ and a linear map $f : W \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{\mathcal{A}\in\mathscr{A}_c(\mathcal{E}',\ell_1+1,s')}\left[G[\mathcal{E}',\mathcal{U}',\sigma(\mathcal{A})] = f|_{\sigma(\mathcal{A})} \;\middle|\; \sigma(\mathcal{A}_{[q_0]}) = Q \text{ and } \sigma(\mathcal{A}) \subseteq W\right] \geqslant \exp(-q_0^4),$$

where $s' = 2^{\ell_1-1}s$.

The next definition formalizes the type of pruning we need for the strategies of Alice and Bob to be able to convert a global function of Alice to Bob and vice versa.

Definition 6.52. Let $\eta \in (0,1)$ be a threshold parameter. An Alice-table $F[\cdot]$ and a Bob-table $G[\cdot]$ are said to be *η-transferable* (with each other) if the following statements hold:

(1) For any unfolded-Alice-question $(\mathcal{E},\mathcal{U},L)$ such that $F[\mathcal{E},\mathcal{U},L] \neq \texttt{nil}$, we have

$$\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},L,*,*,*) \geqslant \eta.$$

(2) For any unfolded-Bob-question $(\mathcal{E}',\mathcal{U}',L')$ such that $G[\mathcal{E}',\mathcal{U}',L'] \neq \texttt{nil}$, we have

$$\mathrm{Vrf}_{F,G}(*,*,*,\mathcal{E}',\mathcal{U}',L') \geqslant \eta.$$

The following lemma shows that if F, G are transferable, Q is a typical zoom-in, and Alice has a global linear function consistent with her table, then on average over Bob's challenge, he has a global linear function satisfying as per the notion of pre-answerable.

Lemma 6.53. *Assume Condition 3.12. Let $F[\cdot]$ be an Alice-table, and let $G[\cdot]$ be a Bob-table that is 2^{-q_0}-transferable with $F[\cdot]$. Fix an Alice-packet $(\mathcal{E},\mathcal{U})$ and a linear subspace $Q \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}), q_0)$ such that*

(1) Q is smooth with respect to $(\mathcal{E},\mathcal{U})$ (as per Definition 6.45), and

(2) $(\mathcal{E},\mathcal{U},Q)$ is answerable with respect to $F[\cdot]$ (as per Definition 6.49).

Then

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}^+_{\mathsf{in}}}\Big[(\mathcal{E}',\mathcal{U}',Q) \text{ is pre-answerable with respect to } G[\cdot] \,\Big|\, \mathcal{E},\mathcal{U},Q\Big] \geqslant 4^{-\ell_1^2}. \tag{6.17}$$

Proof. We organize the proof into the following three steps.

Step 1: transference. Since $(\mathcal{E},\mathcal{U},Q)$ is answerable with respect to $F[\cdot]$, by Definition 6.49 there exists a linear subspace $W \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}), 9k^2t - q_0)$ and a linear map $f : W \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\Big[F[\mathcal{E},\mathcal{U},L] = f|_L \,\Big|\, Q \subseteq L \subseteq W\Big] \geqslant \exp(-q_0^3).$$

We then have

$$\begin{aligned}
&\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},L,Q,*,*,*)\sim\mathcal{D}^+_{\mathsf{in}}}\Big[F[\mathcal{E},\mathcal{U},L] = f|_L \,\Big|\, \mathcal{E},\mathcal{U},Q \text{ and } \{L \subseteq W\}\Big] \\
&\geqslant \mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\Big[F[\mathcal{E},\mathcal{U},L] = f|_L \,\Big|\, Q \subseteq L \subseteq W\Big] - \mathrm{TVD}_{\mathcal{E},\mathcal{U},W}(Q) && \text{(by Definition 6.41)} \\
&\geqslant \exp(-q_0^3) - 2^{\ell_1(q_0+1)}\,\mathrm{TVD}_{\mathcal{E},\mathcal{U}}(Q) && \text{(by Lemma 6.42)} \\
&\geqslant \exp(-q_0^3) - 2^{\ell_1(q_0+1)}t^{-1/8} \geqslant \exp(-2q_0^3). && \text{(by Definition 6.45 and Condition 3.12)}
\end{aligned}$$

By the definition of $\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},L,\mathcal{E}',\mathcal{U}',L')$ in Notation 6.38, we have

$$\begin{aligned}
&\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},L,Q,\mathcal{E}',\mathcal{U}',L')\sim\mathcal{D}^+_{\mathsf{in}}}\Big[G[\mathcal{E}',\mathcal{U}',L']\big|_L = f|_L \,\Big|\, \mathcal{E},\mathcal{U},Q \text{ and } \{L \subseteq W\}\Big] \\
&\geqslant \mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},L,Q,\mathcal{E}',\mathcal{U}',L')\sim\mathcal{D}^+_{\mathsf{in}}}\Big[F[\mathcal{E},\mathcal{U},L] = f|_L \text{ and } \mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},L,\mathcal{E}',\mathcal{U}',L') = 1 \,\Big|\, \mathcal{E},\mathcal{U},Q \text{ and } \{L \subseteq W\}\Big] \\
&\geqslant 2^{-q_0} \cdot \mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},L,Q,*,*,*)\sim\mathcal{D}^+_{\mathsf{in}}}\Big[F[\mathcal{E},\mathcal{U},L] = f|_L \,\Big|\, \mathcal{E},\mathcal{U},Q \text{ and } \{L \subseteq W\}\Big] \\
&\qquad\qquad \text{(since } F[\cdot] \text{ and } G[\cdot] \text{ are } 2^{-q_0} \text{ transferable)} \\
&\geqslant 2^{-q_0}\exp(-2q_0^3) \geqslant \exp(-3q_0^3).
\end{aligned}$$

Step 2: locating good Bob-packets. For any Bob-packet $(\mathcal{E}',\mathcal{U}')$ such that $(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*)$ has nonzero probability under $\mathcal{D}^+_{\mathsf{in}}$, we say it is "good" if

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},L,Q,\mathcal{E}',\mathcal{U}',L')\sim\mathcal{D}^+_{\mathsf{in}}}\Big[G[\mathcal{E}',\mathcal{U}',L']\big|_L = f|_L \,\Big|\, \mathcal{E},\mathcal{U},Q,\mathcal{E}',\mathcal{U}' \text{ and } \{L\subseteq W\}\Big] \geqslant \exp(-3q_0^3)/2. \tag{6.18}$$

By the conclusion from Step 1 of the proof, it follows that

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},L,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}^+_{\mathsf{in}}}\Big[(\mathcal{E}',\mathcal{U}') \text{ is good} \,\Big|\, \mathcal{E},\mathcal{U},Q \text{ and } \{L\subseteq W\}\Big] \geqslant \exp(-3q_0^3)/2. \tag{6.19}$$

Since Q is smooth with respect to $(\mathcal{E},\mathcal{U})$, by Definitions 6.41 and 6.45 we have

$$\begin{aligned}\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},L,Q,*,*,*)\sim\mathcal{D}^+_{\mathsf{in}}}\Big[L\subseteq W \,\Big|\, \mathcal{E},\mathcal{U},Q\Big] &\geqslant \mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\big[L\subseteq W \,\big|\, Q\subseteq L\big] - \mathrm{TVD}_{\mathcal{E},\mathcal{U}}(Q)\\ &\geqslant 2^{-\ell_1(q_0+1)} - t^{-1/8} \geqslant 2^{-\ell_1^2},\end{aligned}$$

where we used Condition 3.12 in the last transition. Consequently, Equation (6.19) implies

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}^+_{\mathsf{in}}}\Big[(\mathcal{E}',\mathcal{U}') \text{ is good} \,\Big|\, \mathcal{E},\mathcal{U},Q\Big] \geqslant 2^{-\ell_1^2}\cdot\exp(-3q_0^3)/2 \geqslant 4^{-\ell_1^2}, \tag{6.20}$$

where we used Condition 3.12 in the last transition.

Step 3: extension. By Definition 6.31, the condition Equation (6.18) is equivalent to

$$\mathop{\mathbb{P}}_{\mathcal{A}\in\mathscr{A}_c(\mathcal{E}',\ell_1+1,s')}\Big[G[\mathcal{E}',\mathcal{U}',\sigma(\mathcal{A})]\big|_{\sigma(\mathcal{A}_{[\ell_1]})} = f|_{\sigma(\mathcal{A}_{[\ell_1]})} \,\Big|\, \sigma(\mathcal{A}_{[q_0]}) = Q \text{ and } \sigma(\mathcal{A}_{[\ell_1]})\subseteq W\Big] \geqslant \exp(-3q_0^3)/2,$$

where c is the critical index of $\mathcal{E}'$ and $s' = 2^{\ell_1-1}s$. By Lemma 6.27, this implies

$$\mathop{\mathbb{P}}_{\mathcal{A}\in\mathscr{A}_c(\mathcal{E}',\ell_1+1,s')}\Big[G[\mathcal{E}',\mathcal{U}',\sigma(\mathcal{A})] = g|_{\sigma(\mathcal{A})} \,\Big|\, \sigma(\mathcal{A}_{[q_0]}) = Q \text{ and } \sigma(\mathcal{A})\subseteq W'\Big] \geqslant \exp(-3q_0^3)/10 \geqslant \exp(-q_0^4),$$

for some linear subspace $W'\subseteq\mathcal{V}(\mathcal{E}')$ containing $W\cap\mathcal{V}(\mathcal{E}')$ and some linear function $g: W'\to\mathbb{F}_2$. Since the codimension of W' in $\mathcal{V}(\mathcal{E}')$ is at most

$$\dim\mathcal{V}(\mathcal{E}') - \dim(W\cap\mathcal{V}(\mathcal{E}')) \leqslant \dim\mathcal{V}(\mathcal{E}) - \dim W = q_0,$$

this means that $(\mathcal{E}',\mathcal{U}',Q)$ is pre-answerable with respect to $G[\cdot]$ whenever $(\mathcal{E}',\mathcal{U}')$ is good. Therefore, Equation (6.20) implies the desired Equation (6.17). □

The following definition is Bob's analog for Alice's notion of answerable packet.

Definition 6.54. Let $G[\cdot]$ be a Bob table, and let $(\mathcal{E}',\mathcal{U}')$ be a Bob-packet with critical index $c\in[t]$. For any linear subspace $Q\subseteq\mathcal{V}(\mathcal{E}')$ with $\dim Q\leqslant q_0$, we say that the tuple $(\mathcal{E}',\mathcal{U}',Q)$ is *answerable* with respect to $G[\cdot]$ if $Q\in\mathsf{MGr}_c(\mathcal{E}',q_0,2^{2\ell_1-q_0}s)$ and the following statement holds: there exists a linear subspace $W\subseteq\mathcal{V}(\mathcal{E}')$ with $\operatorname{codim} W\leqslant q_0$ and a linear map $f: W\to\mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L'\in\mathsf{MGr}(\mathcal{E}',\ell_1+1,s')}\Big[G[\mathcal{E}',\mathcal{U}',L'] = f|_{L'} \,\Big|\, Q\subseteq L'\subseteq W\Big] \geqslant \exp(-q_0^5),$$

where $s' = 2^{\ell_1-1}s$. More specifically, we also say that $(\mathcal{E}',\mathcal{U}',Q)$ is *answerable with respect to* $G[\cdot]$ *using* (f,W).

The following proposition shows that a pre-answerable packet is typically also an answerable packet.

Proposition 6.55. Assume Condition 3.12. Let $F[\cdot]$ be an Alice-table, and let $G[\cdot]$ be a Bob-table that is 2^{-q_0}-transferable with $F[\cdot]$. Suppose $(\mathcal{E}',\mathcal{U}')$ is a Bob-packet with critical index c and $Q \in \mathsf{MGr}_c(\mathcal{E}', q_0, 2^{2\ell_1-q_0}s)$ is a subspace such that

(1) $|E'_c| \geqslant 3k/2$, and

(2) $(\mathcal{E}',\mathcal{U}',Q)$ is pre-answerable with respect to $G[\cdot]$.

Then $(\mathcal{E}',\mathcal{U}',Q)$ is answerable with respect to $G[\cdot]$.

Proof. By Proposition 6.25, the total variation distance between the following two distributions of L' is at most, say, k^{-2}:

(1) sample $L' \in \mathsf{MGr}_c(\mathcal{E}', \ell_1+1, 2^{\ell_1-1}s)$ conditioned on $Q \subseteq L'$; and

(2) sample $\mathcal{A} \in \mathscr{A}_c(\mathcal{E}', \ell_1+1, 2^{\ell_1-1}s)$ conditioned on $\sigma(\mathcal{A}_{[q_0]}) = Q$, and let $L' = \sigma(\mathcal{A})$.

We know from Proposition 6.26 that even if we further condition both distributions on $\{L' \subseteq W\}$, for any linear subspace $W \subseteq \mathcal{V}(\mathcal{E}')$ of codimension at most q_0, the resulting distributions are still k^{-1}-close in total variation distance. By definition, this allows one to convert pre-answerability to answerability. □

The following lemma, summarizing this current section, shows that Bob's packet together with his zoom-in are answerable with noticeable probability.

Lemma 6.56. *Assume Condition 3.12. Let $F[\cdot]$ be an Alice-table, and let $G[\cdot]$ be a Bob-table that is 2^{-q_0}-transferable with $F[\cdot]$. Then for any Alice-packet $(\mathcal{E},\mathcal{U})$ such that $\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},*,*,*,*) \geqslant 2^{-q_0/3}$, we have*

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}^+_{\mathsf{in}}}\left[(\mathcal{E}',\mathcal{U}',Q) \text{ is answerable with respect to } G[\cdot] \,\middle|\, \mathcal{E},\mathcal{U}\right] \geqslant \exp(-6\ell_1^2).$$

Proof. First of all, combining Corollary 6.46 with Lemmas 6.50 and 6.53 yields

$$\begin{aligned}\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}^+_{\mathsf{in}}}\left[(\mathcal{E}',\mathcal{U}',Q) \text{ is pre-answerable with respect to } G[\cdot] \,\middle|\, \mathcal{E},\mathcal{U}\right] &\geqslant (4^{-\ell_1^2} - t^{-1/8})\cdot 4^{-\ell_1^2} \\ &\geqslant \exp(-5\ell_1^2), \qquad (6.21)\end{aligned}$$

where we used Condition 3.12 in the last transition.

For each Bob-packet $(\mathcal{E}',\mathcal{U}')$ with critical index $c \in [t]$, we let $\|\mathcal{E}'\|_{\mathsf{min}} := |E'_c|$. Using Condition 3.12, it is not hard to see (cf. Propositions 4.7 and 5.5)

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{E}',*,*)\sim\mathcal{D}^+_{\mathsf{in}}}\left[\|\mathcal{E}'\|_{\mathsf{min}} \geqslant 3k/2\right] \geqslant 1-k^{-1}, \qquad (6.22)$$

and (cf. Proposition 6.25)

$$\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',*,*)\sim\mathcal{D}^+_{\mathsf{in}}}\left[Q \in \mathsf{MGr}(\mathcal{E}', q_0, 2^{2\ell_1-q_0}s) \,\middle|\, \|\mathcal{E}'\|_{\mathsf{min}} \geqslant 3k/2\right] \geqslant 1-k^{-1}. \qquad (6.23)$$

For any Bob-packet $(\mathcal{E}',\mathcal{U}')$ with $\|\mathcal{E}'\|_{\mathsf{min}} \geqslant 3k/2$ and any $Q \in \mathsf{MGr}(\mathcal{E}', q_0, 2^{2\ell_1-q_0}s)$, the pre-answerability of $(\mathcal{E}',\mathcal{U}',Q)$ can be converted to answerability, due to Proposition 6.55. Therefore, combining Equations (6.21) to (6.23) yields the desired conclusion. □

6.6 Soundness Analysis

Now, we have all ingredients to analyze the soundness of Game3.

Proof of Lemma 6.33. Assume that

$$\mathrm{val}\Big(\mathsf{Game3}[H,k,\ell_1,\ell_2,t,s,\beta]\Big) > \delta,$$

the goal is to show that

$$\mathrm{val}\Big(\mathsf{Game2c}[H,k,\ell_2,t,s_1,q_0,\beta]\Big) > 2^{-q_2/4},$$

which leads to a contradiction to Lemma 5.22.

Recall that the answers returned by Alice and Bob in Game3 are "folded". We define the following folded tables $\widetilde{F},\widetilde{G}$ for Alice and Bob. Suppose that the verifier samples $(\mathcal{E},\mathcal{U},L,*,\mathcal{E}',\mathcal{U}',L') \sim \mathcal{D}^{+}_{\mathsf{in}}$. Then the question sent to Alice is $(\mathcal{E},\mathcal{U},L+\mathcal{N}(\mathcal{E},\mathcal{U}))$ and she returns a linear function f on $L+\mathcal{N}(\mathcal{E},\mathcal{U})$. Then, we define $\widetilde{F}[\mathcal{E},\mathcal{U},L+\mathcal{N}(\mathcal{E},\mathcal{U})] := f$. Similarly, $\widetilde{G}$ is defined by $\widetilde{G}[\mathcal{E}',\mathcal{U}',L'+\bigoplus\mathcal{U}'] := g$ if g is the linear function returned by Bob upon receiving the question $(\mathcal{E}',\mathcal{U}',L'+\bigoplus\mathcal{U}')$.

Getting unfolded tables. To make use of the decoding lemmas (Lemmas 6.50 and 6.53), we need to define "unfolded" tables $F^*[\cdot],G^*[\cdot]$

$$F^*[\mathcal{E},\mathcal{U},L] := \widetilde{F}[\mathcal{E},\mathcal{U},L+\mathcal{N}(\mathcal{E},\mathcal{U})]|_L,$$

and

$$G^*[\mathcal{E}',\mathcal{U}',L'] := \widetilde{G}\left[\mathcal{E}',\mathcal{U}',L'+\bigoplus\mathcal{U}'\right]|_{L'},$$

for every possible sample $(\mathcal{E},\mathcal{U},L,*,\mathcal{E}',\mathcal{U}',L') \sim \mathcal{D}^{+}_{\mathsf{in}}$. For other unfolded questions $(\mathcal{E},\mathcal{U},L),(\mathcal{E}',\mathcal{U}',L')$, we simply define $F^*[\mathcal{E},\mathcal{U},L] := \mathtt{nil}$ and $G^*[\mathcal{E}',\mathcal{U}',L'] := \mathtt{nil}$.

Getting transferable tables via pruning. To use Lemma 6.53, we also want our table to be transferable. We use the following process to get transferable tables $F[\cdot],G[\cdot]$ from F^*,G^*.

Definition 6.57 (Pruned tables)**.** We use the following process to define our final tables $F[\cdot],G[\cdot]$ used for decoding.

1. Initialize with $F := F^*$, $G := G^*$.
2. If there exists an unfolded-Alice-question $(\mathcal{E},\mathcal{U},L)$ such that

   $$\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},L,*,*,*) < 2^{-q_0},$$

   let $F[\mathcal{E},\mathcal{U},L] := \mathtt{nil}$. Similarly, if there exists an unfolded-Bob-question $(\mathcal{E}',\mathcal{U}',L')$ such that

   $$\mathrm{Vrf}_{F,G}(*,*,*,\mathcal{E}',\mathcal{U}',L') < 2^{-q_0},$$

   let $G[\mathcal{E}',\mathcal{U}',L'] := \mathtt{nil}$. Repeat this process until for every unfolded-Alice-question $(\mathcal{E},\mathcal{U},L)$ such that $F[\mathcal{E},\mathcal{U},L] \neq \mathtt{nil}$, we have $\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},L,*,*,*) \geqslant 2^{-q_0}$ and for every unfolded-Bob-question $(\mathcal{E}',\mathcal{U}',L')$ such that $G[\mathcal{E}',\mathcal{U}',L'] \neq \mathtt{nil}$, we have $\mathrm{Vrf}_{F,G}(*,*,*,\mathcal{E}',\mathcal{U}',L') \geqslant 2^{-q_0}$.

By definitions, $F[\cdot], G[\cdot]$ obtained are 2^{-q_0}-transferable.

Now, we want to argue that when we sample question $(\mathcal{E},\mathcal{U},L,*,\mathcal{E}',\mathcal{U}',L') \sim \mathcal{D}^{+}_{\mathsf{in}}$, the probability that $\mathrm{Vrf}_{F,G}(\mathcal{E},\mathcal{U},L,\mathcal{E}',\mathcal{U}',L') = 1$ holds is non-negligible. Due to the condition that

$$\mathrm{val}\Big(\mathsf{Game3}[H,k,\ell_1,\ell_2,t,s,\beta]\Big) > \delta,$$

and our definition F^*, G^*, we know that

$$\mathrm{Vrf}_{F^*,G^*}(*,*,*,*,*,*) > \delta.$$

It is easy to see from the definitions of $F[\cdot], G[\cdot]$, the probability of agreement would decrease by at most 2^{-q_0+1} since when each unfolded question of Alice or Bob is pruned, its conditional contribution to the agreement is at most 2^{-q_0}. As a result, we get

$$\mathrm{Vrf}_{F,G}(*,*,*,*,*,*) \geqslant \mathrm{Vrf}_{F^*,G^*}(*,*,*,*,*,*) - 2^{-q_0+1} \geqslant \delta/2, \tag{6.24}$$

under Condition 3.12.

Strategies. Now, we are ready to explain our strategy for

$$\mathsf{Game2c}[H,k,\ell_2,t,s_1,q_0,\beta]$$

and defer the analysis of its success probability to later sections.

Definition 6.58 (Induced strategies for $\mathsf{Game2c}$)**.** Assume that the verifier sends $(\mathcal{E},\mathcal{A},\mathcal{U})$ to Alice, sends $(\mathcal{E}',\mathcal{A},\mathcal{U}')$ to Bob and we use Q to denote $\sigma\left(\mathcal{A}_{[q_0]}\right)$ here. The strategies are,

- Let $\vec{\tau} := (\tau_0,\ldots,\tau_{2q_0})$ with $\tau_0,\ldots,\tau_{2q_0}$ determined in Condition 3.12. Note that by Condition 3.12, $\vec{\tau}$ satisfies the constraints of Lemma 3.9. Alice first picks a subspace $Q' \subseteq Q$ of codimension at most 1 *uniformly at random*. If $\mathsf{LIST}^{\vec{\tau}}_{Q'}(F[\mathcal{E},\mathcal{U},\cdot]) \neq \emptyset$, Alice picks a *uniformly random* element $(f,W) \in \mathsf{LIST}^{\vec{\tau}}_{Q'}(F[\mathcal{E},\mathcal{U},\cdot])$ and *randomly* extends f to a linear function on $\mathcal{V}(\mathcal{E})$ as its answer; if $\mathsf{LIST}^{\vec{\tau}}_{Q'}(F[\mathcal{E},\mathcal{U},\cdot])$ is empty, Alice returns an arbitrary answer.
- Bob first picks a *uniformly random* integer $q^* \in \{0,\ldots,q_0\}$, and then, picks an *arbitrary* pair (f,W) where W is a subspace of $\mathcal{V}(\mathcal{E}')$ of codimension q^* and $f \in W^*$ among those satisfying

$$\mathop{\mathbb{P}}_{L'\in\mathsf{MGr}(\mathcal{E}',\ell_1+1,2^{\ell_1-1}\cdot s)}\Big[G[\mathcal{E}',\mathcal{U}',L'] = f|_{L'} \,\Big|\, Q \subseteq L' \subseteq W\Big] \geqslant \exp(-q_0^5),$$

  if there exists such a pair; then, Bob also randomly extends the function to a linear function on $\mathcal{V}(\mathcal{E}')$ and returns it. If there does not exist such a pair (f,W), Bob returns an arbitrary answer.

It suffices to prove this strategy succeeds with probability larger than $2^{-q_2/4}$.

Applying decoding lemma for Bob. In this part, we prove that

$$\mathbb{P}_{(*,*,*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}^{+}_{\mathsf{in}}}\left[(\mathcal{E}',\mathcal{U}',Q) \text{ is answerable with respect to } G[\cdot]\right] \geqslant \exp(-7\ell_1^2). \tag{6.25}$$

Recall that answerable is defined in Definition 6.54.

By an averaging argument and Equation (6.24), we know that

$$\mathbb{P}_{(\mathcal{E},\mathcal{U},*,*,*,*,*)\sim\mathcal{D}_{\mathsf{in}}^{+}}\left[\mathrm{Vrf}(\mathcal{E},\mathcal{U},*,*,*,*)\geqslant 2^{-q_0/3}\right]\geqslant \delta/2-2^{-q_0/3}\geqslant \delta/4,$$

under Condition 3.12. After fixing an Alice-packet $(\mathcal{E},\mathcal{U})$ with $\mathrm{Vrf}(\mathcal{E},\mathcal{U},*,*,*,*)\geqslant 2^{-q_0/3}$, Lemma 6.56 gives

$$\mathbb{P}_{(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}_{\mathsf{in}}^{+}}\left[(\mathcal{E}',\mathcal{U}',Q)\text{ is answerable with respect to }G[\cdot]\,\middle|\,\mathcal{E},\mathcal{U}\right]\geqslant \exp(-6\ell_1^2).$$

Combining the facts above, we conclude that

$$\mathbb{P}_{(*,*,*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}_{\mathsf{in}}^{+}}\left[(\mathcal{E}',\mathcal{U}',Q)\text{ is answerable with respect to }G[\cdot]\right]\geqslant \frac{\delta}{4}\cdot\exp(-6\ell_1^2)\geqslant\exp(-7\ell_1^2), \tag{6.26}$$

under Condition 3.12.

We define a partial list of pairs $\mathscr{G}[\cdot]$ as follows, for every possible sample $(\mathcal{E}',\mathcal{U}',Q)$ which is answerable with respect to $G[\cdot]$ and let (f,W) be an arbitrary pair that satisfies (1) $\mathrm{codim}_{\mathcal{V}(\mathcal{E}')}(W)\leqslant q_0$, and $f\in W^*$, (2)

$$\mathbb{P}_{L'\in\mathsf{MGr}(\mathcal{E}',\ell_1+1,2^{\ell_1-1}\cdot s)}\left[G[\mathcal{E}',\mathcal{U}',L']=f|_{L'}\,\middle|\,Q\subseteq L'\subseteq W\right]\geqslant\exp(-q_0^5),$$

define $\mathscr{G}[\mathcal{E}',\mathcal{U}',Q]:=(f,W)$. For other possible samples $(\mathcal{E}',\mathcal{U}',Q)$, define $\mathscr{G}[\mathcal{E}',\mathcal{U}',Q]:=\mathtt{nil}$. Note that $\mathscr{G}$ actually stands for any possible table that Bob might answer for answerable $(\mathcal{E}',\mathcal{U}',Q)$ in his strategy before randomly extending it to a linear function on $\mathcal{V}(\mathcal{E}')$. Conditioned on any possible list $\mathscr{G}$ of Bob, we want to show that the strategy still succeeds with probability larger than $2^{-q_2/4}$.

6.6.1 Good tuples and excellent tuples

We first introduce several important definitions, starting with the notion of good tuples.

Definition 6.59 (Good tuples)**.** We say a sample $(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}_{\mathsf{in}}^{+}$ is *good* if

1. $(\mathcal{E}',\mathcal{U}',Q)$ is answerable, and let $(f,W)=\mathscr{G}[\mathcal{E}',\mathcal{U}',Q]$;
2. there exists a subspace $Q'\subseteq Q$ with $e:=\mathrm{codim}_Q(Q')\in\{0,1\}$ such that the probability

$$\mathbb{P}_{L\in\mathsf{MGr}(\mathcal{E}',\ell_1,2^{\ell_1}\cdot s)}\left[F[\mathcal{E},\mathcal{U},L]=f|_L\,\middle|\,Q'\subseteq L\subseteq W\wedge Q+L\in\mathsf{MGr}(\mathcal{E}',\ell_1+e,2^{\ell_1-e}\cdot s)\right]$$

   is at least $\exp(-q_0^6)$.

The first item in definition asserts that the pair (f,W) is a legitimate answer for Bob, in the sense that he is able to choose it in the strategies defined in Definition 6.58. The second item asserts that the same function f also agrees with Alice's table $F[\mathcal{E},\mathcal{U},\cdot]$, but the guarantee is only with respect to subspaces L from Bob's mixed-type Grassmann domain (which is much smaller domain compared to the domain of Alice, which is a standard Grassmann domain).[16]

[16] Moreover, the agreement is defined with the same zoom-out W, a possibly smaller zoom-in $Q'\subseteq Q$, and an additional constraint of $Q+L\in\mathsf{MGr}(\mathcal{E}',\ell_1+e,2^{\ell_1-e}\cdot s)$, but these differences are more minor.

It is not very difficult to prove that a randomly chosen tuple is good with noticeable probability (see Claim 6.61), but goodness is insufficient for our soundness analysis to go through. Indeed, it is a priori possible that pairs (f, W) that Bob is able to choose have noticeable agreement with Alice's table on Bob's subdomain but that they cannot be extended to (f', W') that agree with Alice's table on her own larger domain. In other words, it still may be the case that (f, W) is not consistent with any assignment that Alice may decode. To address this gap, we introduce the notion of excellent tuples.

Definition 6.60 (Excellent tuples)**.** We say a sample $(\mathcal{E}, \mathcal{U}, *, Q, \mathcal{E}', \mathcal{U}', *) \sim \mathcal{D}^{+}_{\mathsf{in}}$ is *excellent* if

1. $(\mathcal{E}', \mathcal{U}', Q)$ is answerable, and let $(f, W) = \mathscr{G}[\mathcal{E}', \mathcal{U}', Q]$;
2. there exists a subspace $Q' \subseteq Q$ with $\mathrm{codim}_Q(Q') \leqslant 1$ and a pair $(f', W') \in \mathsf{LIST}^{\vec{\tau}}_{Q'}(F[\mathcal{E}, \mathcal{U}, \cdot])$ such that $W \subseteq W'$ and $f'|_W = f$.

The first item in the definition is the same as in Definition 6.59. The second item however is very different, and morally asserts that (f, W) can be extended to (f', W'), which is a viable response of Alice from her list, possibly needing to shrink the zoom-in set Q to some $Q' \subseteq Q$, and/or enlarge the zoom out set to some $W' \supseteq W$. Thus, the function (f', W') is a legit answer for Alice in her strategy when she makes the correct guesses of Q' and the codimension of W', and this is the reason we have defined Alice's strategy in this way.

Intuitively, it is clear that excellent tuples are very beneficial for our soundness analysis, and indeed we show in Claim 6.63 that if the challenges of the players form an excellent tuple, then they will make the verifier accept with noticeable probability. This is due to the fact the list of responses of the players is bounded (after fixing choices of Q and W), and they have at least one pair of choices that make the verifier accept.

Thus, it suffices to prove that a randomly chosen tuple is excellent with noticeable probability. This turns out to be true; the precise statement is given in Claim 6.62. This is the most challenging part of the soundness analysis, and we defer the proof to Sections 6.7 and 6.8.

6.6.2 A Lower Bound on the Fraction of Good Tuples

The following claim shows that the probability a randomly chosen tuple is good is noticeable.

Claim 6.61. $\mathbb{P}_{(\mathcal{E}, \mathcal{U}, *, Q, \mathcal{E}', \mathcal{U}', *) \sim \mathcal{D}^{+}_{\mathsf{in}}} [(\mathcal{E}, \mathcal{U}, Q, \mathcal{E}', \mathcal{U}') \text{ is good}] \geqslant \exp(-8\ell_1^2)$.

Proof. For every possible sample

$$(*, *, *, Q, \mathcal{E}', \mathcal{U}', *) \sim \mathcal{D}^{+}_{\mathsf{in}}$$

where $(\mathcal{E}', \mathcal{U}', Q)$ is answerable, we know that

1. $\mathbb{P}_{L' \in \mathsf{MGr}(\mathcal{E}', \ell_1 + 1, 2^{\ell_1 - 1} \cdot s)} \left[G[\mathcal{E}', \mathcal{U}', L'] = f|_{L'} \,\middle|\, Q \subseteq L' \subseteq W \right] \geqslant \exp(-q_0^5)$, where we let $(f, W) = \mathscr{G}[\mathcal{E}', \mathcal{U}', Q]$;
2. for every L' such that $G[\mathcal{E}', \mathcal{U}', L'] = f|_{L'} \neq \mathtt{nil}$, it holds

   $$\mathrm{Vrf}_{F,G}(*, *, *, \mathcal{E}', \mathcal{U}', L') \geqslant 2^{-q_0},$$

   since tables $F[\cdot], G[\cdot]$ are 2^{-q_0}-transferable.

Combining the two facts above, for the following distributions defined in two steps (1) sample $L' \in \mathsf{MGr}(\mathcal{E}', \ell_1+1, 2^{\ell_1-1}\cdot s) \mid Q \subseteq L' \subseteq W$; (2) sample $(\mathcal{E},\mathcal{U},L,*,\mathcal{E}',\mathcal{U}',L') \sim \mathcal{D}^+_{\mathsf{in}} \mid (\mathcal{E}',\mathcal{U}',L')$, we have

$$\mathop{\mathbb{E}}_{\substack{L'\in\mathsf{MGr}(\mathcal{E}',\ell_1+1,2^{\ell_1-1}\cdot s)\mid Q\subseteq L'\subseteq W\\ (\mathcal{E},\mathcal{U},L,*,\mathcal{E}',\mathcal{U}',L')\sim\mathcal{D}^+_{\mathsf{in}}\mid(\mathcal{E}',\mathcal{U}',L')}} \left[F[\mathcal{E},\mathcal{U},L] = G[\mathcal{E}',\mathcal{U}',L']|_L = f|_L\right] \geqslant \exp(-q_0^5 - q_0).$$

Observe that the distribution of $(\mathcal{E},\mathcal{U},Q,L)$ in

$$(\mathcal{E},\mathcal{U},L,Q,\mathcal{E}',\mathcal{U}',L') \sim \mathcal{D}^+_{\mathsf{in}} \mid (\mathcal{E}',\mathcal{U}'),$$

has the following property: $(\mathcal{E},\mathcal{U})$ and (Q,L,L') are independent. As a result, we can rewrite the two step sampling process as follows (1) sample

$$(\mathcal{E},\mathcal{U}) \sim (\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*) \sim \mathcal{D}^+_{\mathsf{in}} \mid (\mathcal{E}',\mathcal{U}',Q)$$

(2) sample

$$L' \in \mathsf{MGr}(\mathcal{E}', \ell_1+1, 2^{\ell_1-1}\cdot s) \mid Q \subseteq L' \subseteq W$$

(3) sample $L \subseteq L'$. Then, by Markov's inequality, we get

$$\mathop{\mathbb{E}}_{\substack{L'\in\mathsf{MGr}(\mathcal{E}',\ell_1+1,2^{\ell_1-1}\cdot s)\mid Q\subseteq L'\subseteq W\\ L\subseteq L'}} [F[\mathcal{E},\mathcal{U},L] = f|_L] \geqslant \exp(-q_0^5 - q_0)/2, \tag{6.27}$$

for at least $\exp(-q_0^5 - q_0)/2$ fraction of Alice-packets $(\mathcal{E},\mathcal{U})$ sampled from $(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*) \sim \mathcal{D}^+_{\mathsf{in}} \mid (\mathcal{E}',\mathcal{U}',Q)$.

Now, we prove that for every fixed Alice-packet $(\mathcal{E},\mathcal{U})$ for which Equation (6.27) holds,

$$(\mathcal{E},\mathcal{U},Q,\mathcal{E}',\mathcal{U}')$$

is good. This would conclude that

$$\begin{aligned}
&\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}^+_{\mathsf{in}}} \left[(\mathcal{E},\mathcal{U},Q,\mathcal{E}',\mathcal{U}') \text{ is good}\right] \\
&\geqslant \mathop{\mathbb{P}}_{(*,*,*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}^+_{\mathsf{in}}} \left[(\mathcal{E}',\mathcal{U}',Q) \text{ is answerable}\right] \cdot \exp(-q_0^5 - q_0)/2 \\
&\geqslant \exp(-7\ell_1^2)\cdot\exp(-q_0^5 - q_0)/2 \geqslant \exp(-8\ell_1^2). \qquad \text{(using Equation (6.26))}
\end{aligned}$$

After taking a closer look at the distribution in LHS of Equation (6.27), we know that the sampled L satisfies $L \in \mathsf{MGr}(\mathcal{E}', \ell_1, 2^{\ell_1}\cdot s)$ by the definition of MGr. Furthermore, we claim that this distribution over $\mathsf{MGr}(\mathcal{E}', \ell_1, 2^{\ell_1}\cdot s)$ is $2^{-\Omega(\ell_1)}$-close to the following one under Condition 3.12:

- with probability 2^{-q_0} let $Q' := Q$ and uniformly sample $L \in \mathsf{MGr}(\mathcal{E}', \ell_1, 2^{\ell_1}\cdot s)$ conditioned on $Q' \subseteq L \subseteq W$;
- otherwise, uniformly sample $Q' \subseteq Q$ with $\mathrm{codim}_Q(Q') = 1$ and uniformly sample $L \in \mathsf{MGr}(\mathcal{E}', \ell_1, 2^{\ell_1}\cdot s)$ conditioned on $Q' \subseteq L \subseteq W$ and $Q + L \in \mathsf{MGr}(\mathcal{E}', \ell_1+1, 2^{\ell_1-1}\cdot s)$.

This can be verified using Proposition 6.21.

Combining the fact above and an averaging argument, we conclude that there exists $Q' \subseteq Q$ with $e := \mathrm{codim}_Q(Q') \leqslant 1$ such that

$$\begin{aligned}
&\mathop{\mathbb{P}}_{L\in\mathsf{MGr}(\mathcal{E}',\ell_1,2^{\ell_1}\cdot s)} \left[F[\mathcal{E},\mathcal{U},L] = f|_L \,\middle|\, Q' \subseteq L \subseteq W \wedge Q + L \in \mathsf{MGr}(\mathcal{E}', \ell_1+e, 2^{\ell_1-e}\cdot s)\right] \\
&\geqslant \exp(-q_0^5 - q_0)/2 - 2^{-\Omega(\ell_1)} \geqslant \exp(-q_0^6),
\end{aligned}$$

under our Condition 3.12. □

In the last step of our soundness analysis, we prove that samples are excellent with non-negligible probability. Formally, we prove the following claim.

Claim 6.62. $\mathbb{P}_{(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}^+_{\mathsf{in}}}[(\mathcal{E},\mathcal{U},Q,\mathcal{E}',\mathcal{U}') \text{ is excellent}] \geqslant \exp(-30\ell_1^2)$.

Proof. Deferred to Section 6.7. □

6.6.3 Completing the Soundness Analysis

Before proving the claim, we first show how to complete the soundness analysis using Claim 6.62. Assume that in $\mathsf{Game2c}[H,k,\ell_2,t,s_1,q_0,\beta]$, the verifier samples a tuple $(\mathcal{E},\mathcal{U},\mathcal{A},\mathcal{E}',\mathcal{U}')$ and it sends $(\mathcal{E},\mathcal{U},\mathcal{A})$ to Alice and sends $(\mathcal{E}',\mathcal{U}',\mathcal{A})$ to Bob. Let $Q := \sigma(\mathcal{A}_{[q_0]})$. We prove the following claim on the success probability of our strategy on an excellent tuple.

Claim 6.63. *If $(\mathcal{E},\mathcal{U},Q,\mathcal{E}',\mathcal{U}')$ is excellent, the strategy wins with probability at least* $\exp(-3Cq_0^2\ell_1)$.

Proof. Let $(f_B,W_B) = \mathscr{G}[\mathcal{E}',\mathcal{U}',Q]$. Note that by our assumption and strategies, upon getting the tuple $(\mathcal{E},\mathcal{U},Q,\mathcal{E}',\mathcal{U}')$, Bob first guesses the codimension of W_B, and if he makes a correct guess (which happens with probability $1/(q_0+1)$), he randomly extends it to a linear function $\widetilde{f}_B$ on $\mathcal{V}(\mathcal{E}')$. Alice first randomly samples a subspace $Q' \subseteq Q$ of codimension at most 1 and then randomly picks a pair $(f_A,W_A) \in \mathsf{LIST}^{\vec{\tau}}_{Q'}(F[\mathcal{E},\mathcal{U},\cdot])$ (which is non-empty since $(\mathcal{E},\mathcal{U},Q,\mathcal{E}',\mathcal{U}')$ is excellent), also randomly extends it to a linear function $\widetilde{f}_A$ on $\mathcal{V}(\mathcal{E})$. The strategy wins when all the following events happen,

1. $W_B \subseteq W_A$, $f_A|_{W_B} = f_B$;
2. $\widetilde{f}_A$ is SNR-satisfying;
3. $\widetilde{f}_A|_{\mathcal{V}(\mathcal{E}')} = \widetilde{f}_B$.

Due to Lemma 3.9 and the fact that $(\mathcal{E},\mathcal{U},Q,\mathcal{E}',\mathcal{U}')$ is excellent, with probability at least $2^{-q_0}\cdot\exp(-Cq_0^2\ell_1) \geqslant \exp(-2Cq_0^2\ell_1)$, we have $W_B \subseteq W_A$ and $f_A|_{W_B} = f_B$.

Next, we prove that for every $(f_A,W_A) \in \mathsf{LIST}^{\vec{\tau}}_{Q'}(F[\mathcal{E},\mathcal{U},\cdot])$, there exists a good linear function $f' \in \mathcal{V}(\mathcal{E})^*$ such that $f'|_{W_A} = f_A$ and $f'|_{\mathcal{N}(\mathcal{E},\mathcal{U})}$ is SNR-satisfying, which implies that conditioned on the first event, the second event happens with probability at least 2^{-2q_0} since $\mathrm{codim}_{\mathcal{V}(\mathcal{E})}(W_A) \leqslant 2q_0$. Let $w_A = \mathrm{codim}_{\mathcal{V}(\mathcal{E})}(W_A) \leqslant 2q_0$. By the definition of $\mathsf{LIST}^{\vec{\tau}}_{Q'}(F[\mathcal{E},\mathcal{U},\cdot])$, we have

$$\mathbb{P}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\left[F[\mathcal{E},\mathcal{U},L] = f_A|_L \mid Q' \subseteq L \subseteq W_A\right] \geqslant \tau_{w_A}.$$

As a result, we get the same lower bound for the folded table $\widetilde{F}$,

$$\mathbb{P}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\left[\widetilde{F}[\mathcal{E},\mathcal{U},L+\mathcal{N}(\mathcal{E},\mathcal{U})]|_L = f_A|_L \mid Q' \subseteq L \subseteq W_A\right] \geqslant \tau_{w_A}.$$

By an averaging argument, there exists a subspace $\widetilde{L} = L+\mathcal{N}(\mathcal{E},\mathcal{U})$, where $L \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1), Q' \subseteq L \subseteq W_A$ such that

$$\mathbb{P}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\left[\widetilde{F}[\mathcal{E},\mathcal{U},\widetilde{L}]|_L = f_A|_L \mid Q' \subseteq L \subseteq \widetilde{L}\cap W_A \wedge L+\mathcal{N}(\mathcal{E},\mathcal{U}) = \widetilde{L}\right] \geqslant \tau_{w_A}/2. \tag{6.28}$$

Note that $\widetilde{F}[\mathcal{E},\mathcal{U},\widetilde{L}] \in \widetilde{L}^*$ is a linear function and its restriction $\widetilde{F}[\mathcal{E},\mathcal{U},\widetilde{L}]|_{\mathcal{N}(\mathcal{E},\mathcal{U})}$ is SNR-satisfying. If f_A does not admit such good extension, we then have $f_A|_{W_A\cap\widetilde{L}} \neq \widetilde{F}[\mathcal{E},\mathcal{U},\widetilde{L}]|_{W_A\cap\widetilde{L}}$, which means

$$\begin{aligned}&\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\left[\widetilde{F}[\mathcal{E},\mathcal{U},\widetilde{L}]|_L = f_A|_L \mid Q'\subseteq L\subseteq\widetilde{L}\cap W_A \wedge L+\mathcal{N}(\mathcal{E},\mathcal{U})=\widetilde{L}\right]\\ \leqslant\ &\frac{\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\left[\widetilde{F}[\mathcal{E},\mathcal{U},\widetilde{L}]|_L = f_A|_L \mid Q'\subseteq L\subseteq\widetilde{L}\cap W_A\right]}{\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\left[L+\mathcal{N}(\mathcal{E},\mathcal{U})=\widetilde{L} \mid Q'\subseteq L\subseteq\widetilde{L}\cap W_A\right]} \leqslant 2^{-\Omega(\ell_1)} < \tau_{w_A}/2\end{aligned}$$

under Condition 3.12, leading to a contradiction. Here, we use the fact that

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\left[L+\mathcal{N}(\mathcal{E},\mathcal{U})=\widetilde{L} \mid Q'\subseteq L\subseteq\widetilde{L}\cap W_A\right] \geqslant \Omega(1),$$

which is justified as follows. Letting $T := \mathcal{N}(\mathcal{E},\mathcal{U}) + Q'$, since $L+\mathcal{N}(\mathcal{E},\mathcal{U}) = \widetilde{L}$ we see that $d := \operatorname{codim}_{\widetilde{L}}(T) \leqslant \operatorname{codim}_L(Q') = \ell_1 - \dim(Q')$. One may sample L by sampling vectors $u_1,\dots,u_{\ell_1-\dim(Q')}$ from $\widetilde{L}\cap W_A$ sequentially, where u_i is sampled uniformly conditioned on $u_i \notin Q'+\operatorname{span}\{u_1,\dots,u_{i-1}\}$, and then taking the span of those vectors and Q' as L. It is easy to see that $L+\mathcal{N}(\mathcal{E},\mathcal{U}) = \widetilde{L}$ if for every $i\in[d]$, $u_i \notin T+\operatorname{span}\{u_1,\dots,u_{i-1}\}$ and this happens with probability at least

$$\prod_{i=1}^{d}(1-2^{-i}) \geqslant \Omega(1).$$

Finally, it is easy to see that conditioned on the first two events, the last event happens with probability at least 2^{-q_0} since $\operatorname{codim}_{\mathcal{V}(\mathcal{E}')}(W_B) \leqslant q_0$. Combining facts above, we conclude that the strategy wins with probability at least $\exp(-2Cq_0^2\ell_1)2^{-3q_0}/(q_0+1) \geqslant \exp(-3Cq_0^2\ell_1)$ under Condition 3.12. □

Combining Claim 6.62 and Claim 6.63, we conclude that the strategy wins with probability at least

$$\exp(-30\ell_1^2)\cdot\exp(-3Cq_0^2\ell_1) > 2^{-q_2/4},$$

under Condition 3.12, giving a contradiction to Lemma 5.22. □

6.7 Proof of Claim 6.62

In this section, we prove Claim 6.62. As discussed in Section 6.4, the proof builds on the intuition that given a subspace $Q\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),q_0)$ and an Alice-packet $(\mathcal{E},\mathcal{U})$, if the table $F[\mathcal{E},\mathcal{U},\cdot]$ has noticeable consistency on the mixed-Grassmann domain of a lot of Bob-packets $(\mathcal{E}',\mathcal{U}')$ and corresponding decodings (f,W) (captured by the definition of good tuples), then there exists a Bob-packet $(\mathcal{E}',\mathcal{U}')$ and decoding (f,W) has the ordinary Grassmann-type consistency with table $F[\mathcal{E},\mathcal{U},\cdot]$ (captured by the definition of excellent tuple). This intuition is formalized by the following lemma.

Lemma 6.64. *Assume Condition 3.12. Let $F[\cdot]$, $G[\cdot]$ be the pruned tables of Alice and Bob (see Definition 6.57). Fix an Alice-packet $(\mathcal{E},\mathcal{U})$ and a linear subspace $Q\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),q_0)$. Suppose $m\geqslant\exp(1000\cdot k^5)$ is a positive integer and $c_1,\dots,c_m\in[t]$ are pairwise distinct indices. For each $i\in[m]$, assume $(\mathcal{E}_i',\mathcal{U}_i')$ is a c_i-critical Bob-packet such that the following hold:*

(1) For every $i\in[m]$, $(\mathcal{E},\mathcal{U},Q,\mathcal{E}_i',\mathcal{U}_i')$ is good; let (f_i,W_i) denote $\mathscr{G}[\mathcal{E}_i',\mathcal{U}_i',Q]$.

(2) Writing $\mathcal{E}'_i = (E_{i,1}, \ldots, E_{i,t})$ for each $i \in [m]$, we have $|E_{1,c_1}| = |E_{2,c_2}| = \cdots = |E_{m,c_m}|$.

Then there exist linear subspaces $W \subseteq \mathcal{V}(\mathcal{E})$ with $\operatorname{codim}_{\mathcal{V}(\mathcal{E})}(W) \leqslant 2q_0$ and $Q' \subseteq Q, \operatorname{codim}_Q(Q') \leqslant 1$ and a linear map $f : W \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)} \left[F[\mathcal{E},\mathcal{U},L] = f|_L \,\middle|\, Q' \subseteq L \subseteq W \right] \geqslant \exp(-100 \cdot q_0^7).$$

Furthermore, for some $i \in [m]$ we have $W_i \subseteq W$ and $f|_{W_i} = f_i$.

Note that one of the conditions in this lemma requires Bob-packets to be "regular", namely, their critical coordinates have to be different and the size of the critical coordinate has to be the same. Motivated by this, we introduce a useful definition.

Definition 6.65. Given an integer $k^* \in \{k, \ldots, 3k\}$ and a tuple $(\mathcal{E},\mathcal{U},Q)$, we define $\mathcal{C}_{\mathcal{E},\mathcal{U},Q,k^*} \subseteq [t]$ to be the set of coordinates such that for every $j \in \mathcal{C}_{\mathcal{E},\mathcal{U},Q,k^*}$, there exists a j-critical Bob-packet $(\mathcal{E}',\mathcal{U}')$ such that (1) $\|\mathcal{E}'\|_{\min} = k^*$, (2) the tuple $(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',Q)$ is good but not excellent. Define $\mathcal{C}_{\mathcal{E},\mathcal{U},Q} := \bigcup_{k^* \in \{k,\ldots,3k\}} \mathcal{C}_{\mathcal{E},\mathcal{U},Q,k^*}$.

In words, this definition captures, for a given tuple $(\mathcal{E},\mathcal{U},Q)$, the set of possible coordinates $j \in [t]$ that support a j-critical Bob-packet $(\mathcal{E}',\mathcal{U}')$ such that $(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',Q)$ is good but not excellent. It is not hard to see that Lemma 6.64 helps to bound the size of $\mathcal{C}_{\mathcal{E},\mathcal{U},Q,k^*}$. We present formal arguments below.

The proof of Claim 6.62 then consists of two parts. The first part is to prove that for every fixed tuple $(\mathcal{E},\mathcal{U},Q)$ and thinning size $k^* \in \{k, \ldots 3k\}$, we have

$$|\mathcal{C}_{\mathcal{E},\mathcal{U},Q,k^*}| \leqslant \exp(1000k^5), \tag{6.29}$$

using Lemma 6.64. Furthermore, we get

$$|\mathcal{C}_{\mathcal{E},\mathcal{U},Q}| \leqslant 3k \cdot \exp(1000k^5).$$

The second part is, building upon Equation (6.29), to prove that for every Alice-packet $(\mathcal{E},\mathcal{U})$, we have

$$\mathbb{P}_{(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*) \sim \mathcal{D}^+_{\mathsf{in}}|(\mathcal{E},\mathcal{U})} \left[(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',Q) \text{ is good but not excellent} \right] \leqslant \frac{3k \cdot 2^{9q_0k^2} \cdot \exp(1000k^5)}{t}. \tag{6.30}$$

Combining Claim 6.61 and a union bound, Claim 6.62 follows as a direct corollary.

Bound the size of $\mathcal{C}_{\mathcal{E},\mathcal{U},Q,k^*}$. We prove Equation (6.29) here. Fix an arbitrary tuple $(\mathcal{E},\mathcal{U},Q)$ and thinning size $k^* \in \{k, \ldots 3k\}$. If the size of $\mathcal{C}_{\mathcal{E},\mathcal{U},Q,k^*}$ is larger than $\exp(1000k^5)$, we are able to find Bob-packets $(\mathcal{E}'_1,\mathcal{U}'_1), \ldots, (\mathcal{E}'_m,\mathcal{U}'_m)$, where $m > \exp(1000k^5)$, satisfying the condition of Lemma 6.64 and any of those packets has its critical coordinate lying in the set $\mathcal{C}_{\mathcal{E},\mathcal{U},Q,k^*}$; and none of those tuples $(\mathcal{E},\mathcal{U},\mathcal{E}'_i,\mathcal{U}'_i,Q)$ is excellent. Lemma 6.64 then shows that one of those packets gives an excellent tuple, leading to a contradiction.

Bound the probability of good but not excellent tuples. It suffices to prove Equation (6.30). When we sample $(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*) \sim \mathcal{D}_{\mathsf{in}}^{+} \mid (\mathcal{E},\mathcal{U})$, let $c \in [t]$ be the critical coordinate that we use as an auxiliary random variable in the sampling procedure. By definitions, we have

$$
\begin{aligned}
&\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,Q,\mathcal{E}',\mathcal{U}',*)\sim\mathcal{D}_{\mathsf{in}}^{+}|(\mathcal{E},\mathcal{U})} \left[(\mathcal{E},\mathcal{U},\mathcal{E}',\mathcal{U}',Q) \text{ is good but not excellent}\right] \\
&\leqslant \mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}|(\mathcal{E},\mathcal{U})} \left[c \in \mathcal{C}_{\mathcal{E},\mathcal{U},\sigma(\mathcal{A}_{[q_0]})}\right] && \text{(seeing } Q \text{ as } \sigma(\mathcal{A}_{[q_0]})\text{)} \\
&= \mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}|(\mathcal{E},\mathcal{U})} \left[\sum_{j=1}^{t} (c=j) \wedge (j \in \mathcal{C}_{\mathcal{E},\mathcal{U},\sigma(\mathcal{A}_{[q_0]})})\right] \\
&= \frac{1}{t}\sum_{j=1}^{t} \mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}|(\mathcal{E},\mathcal{U})} \left[j \in \mathcal{C}_{\mathcal{E},\mathcal{U},\sigma(\mathcal{A}_{[q_0]})} \mid c=j\right] && \text{(the marginal of } c \text{ is uniform)} \\
&= \frac{1}{t}\sum_{j=1}^{t} \sum_{\rho'\in\mathcal{V}(\mathcal{E})^{q_0}} \mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}|(\mathcal{E},\mathcal{U})} \left[j \in \mathcal{C}_{\mathcal{E},\mathcal{U},\sigma(\mathcal{A}_{[q_0]})}, \rho(\mathcal{A}_{[q_0]}) = \rho' \mid c=j\right] \\
&\leqslant \frac{3k\cdot\exp(1000k^5)}{t} \sum_{\rho'\in\mathcal{V}(\mathcal{E})_0^q} \max_{j\in[t]} \mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}|(\mathcal{E},\mathcal{U})} \left[\rho(\mathcal{A}_{[q_0]}) = \rho' \mid c=j\right]. \\
& && \text{(using Equation (6.29))}
\end{aligned}
$$

It suffices to prove the following upper bound

$$
\sum_{\rho'\in\mathcal{V}(\mathcal{E})^{q_0}} \max_{j\in[t]} \mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}|(\mathcal{E},\mathcal{U})} \left[\rho(\mathcal{A}_{[q_0]}) = \rho' \mid c=j\right] \leqslant 2^{9q_0k^2}.
$$

Let $N := 2^{9q_0k^2}$. It is easy to see that the number of valid ρ' is at most N^t. Also, for every fixed $j \in [t]$ and $\rho' \in \mathcal{V}(\mathcal{E})^{q_0}$, we have

$$
\mathop{\mathbb{P}}_{(\mathcal{E},\mathcal{U},*,*,\mathcal{A})\sim\mathcal{D}_{\mathsf{in}}|(\mathcal{E},\mathcal{U})} \left[\rho(\mathcal{A}_{[q_0]}) = \rho' \mid c=j\right] \leqslant N^{-t+1}
$$

since $\rho(\mathcal{A}_{[q_0]})$ projected on $\mathbb{F}_2^{E'_{j'}} \otimes \mathbb{F}_2^{E'_{j'}}$ are uniform and independent for non-critical coordinates $j' \neq j$. Combining the two facts, we have the desired estimate.

6.8 Proof of Lemma 6.64

In this section, we prove Lemma 6.64, thereby completing the proof of Lemma 6.33.

Proof Overview. We begin with giving an overview of the argument, which is inspired by an argument in [DKK+25, Min22], and for simplicity of presentation we ignore the issue of zoom-ins. Suppose there are many Bob-packets $(\mathcal{E}'_i,\mathcal{U}'_i)$ and function zoom-out pairs (f_i,W_i) that are good and their critical coordinates are all distinct. We view the $W_i \subseteq \mathcal{V}(\mathcal{E}'_i)$ as subspaces of the larger ambient space of Alice $\mathcal{V}(\mathcal{E})$, and note that their co-dimension becomes very large (roughly growing from q_0 with respect to $\mathcal{V}(\mathcal{E}'_i)$ to $q_0+\Theta(k^2)$ with respect to $\mathcal{V}(\mathcal{E})$). However, since their critical coordinates are distinct, we have that $\mathcal{V}(\mathcal{E}'_i)+\mathcal{V}(\mathcal{E}'_{i'}) = \mathcal{V}(\mathcal{E})$ for all $i \neq i'$, and thus $\operatorname{codim}_{\mathcal{V}(\mathcal{E})}(W_i+W_{i'}) \leqslant 2q_0$ for $i \neq i'$ is constant.

1. **Finding a sunflower:** we use Lemma 3.11 to find a collection of W_i that form a sunflower. By that, we mean a subspace $W \subseteq \mathcal{V}(\mathcal{E})$ that contains many of the W_i, and furthermore these contained subspaces satisfy that $\operatorname{codim}_W(W_i) = w$ and $\operatorname{codim}(W_i \cap W_{i'}) = 2w$ for some fixed w. Informally, the subspaces W_i should be thought of as looking "random" with respect to the ambient space W. Our next goal is to show that Alice can find a linear function $f \in W^*$ that has a significant agreement with her table.

2. **Linearity testing:** consider the linearity tester that picks two subspaces $L_1, L_2 \subseteq W$ of dimension ℓ_1 that intersect in dimension $\ell_1/10$ uniformly at random, and then checks that $F[\mathcal{E},\mathcal{U},L_1]$ and $F[\mathcal{E},\mathcal{U},L_2]$ agree on $L_1 \cap L_2$. It was shown in [DKK+25] that if F passes this linearity test with noticeable probability, then there exists a linear function $f \in W^*$ with noticeable agreement with the table $F[\mathcal{E},\mathcal{U},\cdot]$. Thus, our goal is to show that F indeed passes the linearity tester with noticeable probability.

3. **Local passing:** we note that since $(\mathcal{E},\mathcal{U},\mathcal{E}'_i,\mathcal{U}'_i)$ is good, Alice's table has noticeable agreement with f_i on subspaces from W_i. This implies that if we were to sample L_1, L_2 as above conditioned on being contained in W_i, then with noticeable probability $F[\mathcal{E},\mathcal{U},L_1]$, $F[\mathcal{E},\mathcal{U},L_2]$ would both agree with f_i, and thus agree between themselves.

4. **From local passing to global passing:** since the number of Bob-packets is very large, with large probability over the choice of L_1, L_2, there will be many i such that $L_1, L_2 \subseteq W_i$. Intuitively, the item above asserts that each such i gives a noticeable chance for $F[\mathcal{E},\mathcal{U},L_1]$, $F[\mathcal{E},\mathcal{U},L_2]$ to agree, so it should follow that F indeed passes the ℓ_1 vs $\ell_1/10$ linearity test with noticeable probability over W. We show that this is essentially correct, but for technical reasons we proceed somewhat differently, using the inclusion-exclusion principle on the event that L_1, L_2 are contained in W_i and agree with f_i.[17]

5. **Agreement with some f_i:** so far, we have shown that if there are many good Bob-packets, then Alice can decode some global linear function that has a large agreement with her table. To finish the proof we have to argue that she can decode a global linear function f that agrees with Bob's global function f_i for some i. Towards this end, we use a list-decoding argument to decode all f's for Alice that exhibit non-trivial agreement with her table, so that all of the passing probability of the linearity test above can be attributed to the list. We then show that with noticeable probability, Alice's assignment agrees with both some function in her list, as well as with f_i of Bob for some i, and in particular f and f_i agree. A quick calculation, plus the fact that linear functions either fully agree or only agree on half of the space, shows that f and f_i must fully agree for some i.

We now proceed to the formal proof.

Proof of Lemma 6.64. Since for every $i \in [m]$, $(\mathcal{E},\mathcal{U},Q,\mathcal{E}'_i,\mathcal{U}'_i)$ is good, we know that there exists a subspace $Q_i \subseteq Q$ with $e_i := \operatorname{codim}_Q(Q_i) \leqslant 1$ such that

$$\mathop{\mathbb{P}}_{L \in \mathsf{MGr}(\mathcal{E}'_i,\ell_1,2^{\ell_1}\cdot s)} \left[F[\mathcal{E},\mathcal{U},L] = f_i|_L \,\middle|\, Q_i \subseteq L \subseteq W_i \wedge Q + L \in \mathsf{MGr}(\mathcal{E}'_i, \ell_1 + e_i, 2^{\ell_1 - e_i} \cdot s) \right] \geqslant \exp(-q_0^6) \tag{6.31}$$

[17] These technical reasons do not seem very fundamental. They amount to the fact that many calculations over the standard Grassmann graph, such as the probability that a random subspace L is contained in a fixed subspace W of constant codimension, become a bit trickier over the mixed-type Grassmann. Indeed, we use results such as Proposition 6.22 and Corollary 6.23 in such situations.

By an averaging argument, there exists a subspace $Q' \subseteq Q$ of dimension q^*, where we let $e := q_0 - q^* \in \{0,1\}$, and an integer $w \in \{0,\dots,q_0\}$ such that

$$\left|\{i : Q_i = Q' \wedge \operatorname{codim}_{\mathcal{V}(\mathcal{E}'_i)}(W_i) = w\}\right| \geqslant 2^{-2q_0^2} \cdot m.$$

Let $m^* = \left\lceil \exp(-q_0^7)/(100 \cdot \lambda \cdot 2^{-(D-q^*)9k^2}) \right\rceil$, where $\lambda > \exp(-20\ell_1 k^2)$ is a real number to be determined later. Applying Lemma 3.11 to the collection of subspaces W_i[18] that satisfy $Q_i = Q'$ and $\operatorname{codim}_{\mathcal{V}(\mathcal{E}'_i)}(W_i) = w$, we know that there exist distinct indices $i_1,\dots,i_{m*} \in [m]$ and a subspace $W \subseteq \mathcal{V}(\mathcal{E})$ such that

1. for every $j \in [m^*]$, $Q_{i_j} = Q'$, $\operatorname{codim}_{\mathcal{V}(\mathcal{E}'_{i_j})}(W_{i_j}) = w$;
2. $W_1 + \cdots + W_{i_{m^*}} = W$;
3. there exists an integer $0 \leqslant w' \leqslant 9k^2$ such that (1) for every $j \in [m^*]$, $\operatorname{codim}_W(W_{i_j}) = w'$ and (2) for every distinct $j, j' \in [m^*]$, $\operatorname{codim}_W(W_{i_j} \cap W_{i_{j'}}) = 2w'$.

Note that $\operatorname{codim}_{\mathcal{V}(\mathcal{E})}(W) \leqslant 2w \leqslant 2q_0$. Indeed, as $m^* \geqslant 2$, the second item above implies

$$\operatorname{codim}_{\mathcal{V}(\mathcal{E})}(W) \leqslant \operatorname{codim}_{\mathcal{V}(\mathcal{E})}(W_{i_1} + W_{i_2}) \leqslant \operatorname{codim}_{\mathcal{V}(\mathcal{E}'_{i_1})}(W_{i_1}) + \operatorname{codim}_{\mathcal{V}(\mathcal{E}'_{i_2})}(W_{i_2}) \leqslant 2w,$$

where in the second inequality we used the fact that $c_{i_1} \neq c_{i_2}$ and thus $\mathcal{V}(\mathcal{E}'_{i_1}) + \mathcal{V}(\mathcal{E}'_{i_2}) = \mathcal{V}(\mathcal{E})$.

Now, we define $m' = \left\lceil \exp(-q_0^7)/(100 \cdot \lambda \cdot 2^{-(D-q^*)w'}) \right\rceil$, and only focus on the first m' indices in $[m^*]$ in the following proofs. Actually, the λ we pick later assures that $m' \gg 1$. We introduce the definition m^* only to avoid the circular definition, since the definition m' depends on the codimension w' which is a parameter of the resulting sunflower we find.

In the remaining proofs, we show that there exists a linear function $f \in W^*$ and $j \in [m']$, such that

$$\mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)} \left[F[\mathcal{E},\mathcal{U},L] = f|_L \,\middle|\, Q' \subseteq L \subseteq W \right] \geqslant \exp(-100q_0^7). \tag{6.32}$$

and $f_{i_j} = f|_{W_{i_j}}$. The plan is to first show that there exists an $f \in W^*$ that satisfies Equation (6.32) and then prove the full statement.

Existence of decoding. To prove that there exists a linear function $f \in W^*$ such that

$$\mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)} \left[F[\mathcal{E},\mathcal{U},L] = f|_L \,\middle|\, Q' \subseteq L \subseteq W \right] \geqslant \exp(-100q_0^7),$$

we consider the *Grassmann linearity test* proposed by [KMS25, DKK+25]. To be more specific, we consider the following test over W. Sample two subspaces $Q' \subseteq L_1, L_2 \subseteq W$ of dimension ℓ_1 conditioned on $\dim(L_1 \cap L_2) = d$, where we define $d = \lfloor \ell_1/10 \rfloor$ and $D = 2\ell_1 - d$, and check if $F[\mathcal{E},\mathcal{U},L_1]|_{L_1 \cap L_2} = F[\mathcal{E},\mathcal{U},L_2]|_{L_1 \cap L_2}$. We define $\mathsf{agree}(\mathcal{E},\mathcal{U},W,Q')$ to denote the probability that the test passes. We use the following variant of a decoding lemma proved by [KMS25, DKK+25].

Corollary 6.66. *There is an absolute constant $C \geqslant 40$ such that the following holds. Fix positive integers n, ℓ, b and a nonnegative integer q such that $n \geqslant 2\ell$ and $C(q+1) \leqslant b \leqslant \ell/4$. Let $Q \in$*

[18] Here, we view W_i as subspaces of the larger vector space $\mathcal{V}(\mathcal{E})$ when applying the sunflower lemma.

$\mathsf{Gr}(\mathbb{F}_2^n, q)$ *be a subspace. Suppose* $F[\cdot]$ *is a table where* $F[L] \in L^* \cup \{\mathtt{nil}\}$ *for each* $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$*, such that*

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell,b)}\left[F[L_1]|_{L_1\cap L_2} = F[L_2]|_{L_1\cap L_2} \,\middle|\, Q \subseteq L_1, L_2\right] \geqslant \varepsilon, \tag{6.33}$$

for some parameter $\varepsilon \in [2^{-b/8}, 1]$*. Then there exists a linear map* $f : \mathbb{F}_2^n \to \mathbb{F}_2$ *such that*

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[F[L] = f|_L \,\middle|\, Q \subseteq L\right] \geqslant C^{-1} \cdot 2^{-3q}\varepsilon^3.$$

We include its proof in Appendix E. To avoid a dependency issue in the future, we consider the following process of sampling a matrix $\mathcal{M} \in (\mathcal{V}(\mathcal{E}))^D$, where each row $x \in [D], \mathcal{M}_x$ is a vector in $\mathcal{V}(\mathcal{E})$:

- for each $x \in [D] \setminus [q^*]$, independently and uniformly sample $\mathcal{M}_x$ from W;
- let $\{y_1, \ldots, y_{q^*}\}$ be an arbitrary fixed basis of Q', let $\mathcal{M}_x := y_x$ for every $x \in [q^*]$.

Given $\mathcal{M}$, we define

$$L_1 := \mathrm{span}\{\mathcal{M}_x : x \in [\ell_1]\}, \qquad L_2 := \mathrm{span}\{\mathcal{M}_x : x \in [d] \cup ([D] \setminus [\ell_1])\}.$$

Use $\mathcal{P}$ to denote the joint distribution of L_1, L_2. It is easy to see that $\mathcal{P}$ is $\exp(-\Omega(k))$-close to the distribution in the definition of $\mathsf{agree}(\mathcal{E}, \mathcal{U}, W, Q')$. Thus, it suffices to prove that

$$\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2} = F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}\right] \geqslant \exp(-5q_0^7). \tag{6.34}$$

We also define auxiliary distributions $\mathcal{P}_j$ for every $j \in [m']$. The distribution $\mathcal{P}_j$ is defined by the same sampling process as $\mathcal{P}$, except that, for every $x \in [D] \setminus [q^*]$, $\mathcal{M}_x$ is sampled uniformly from $\mathcal{V}(\mathcal{E}'_{i_j})$ rather than from W.

First-moment calculations. For every $j \in [m']$, let $\mathscr{X}_j$ denote the event that $L_1, L_2 \subseteq W_{i_j}$; it is easy to see that

$$\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{X}_j\right] = 2^{-(D-q^*)w'}. \tag{6.35}$$

We let $\mathscr{Y}_j$ denote the event that $Q + L_1 + L_2 \in \mathsf{MGr}(\mathcal{E}'_{i_j}, D+e, 2^{d-e}\cdot s)$. We want to calculate the probability that

$$\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{Y}_j | \mathscr{X}_j\right].$$

To this end, we define $\lambda := \mathbb{P}_{(L_1,L_2)\sim\mathcal{P}_j}\left[\mathscr{Y}_j\right]$. It is not hard to see that the λ defined here satisfies $\lambda > \exp(-20\ell_1 k^2)$, since the total number of subspaces of $\mathbb{F}_2^{E_{i_j,c_{i_j}}} \otimes \mathbb{F}_2^{E_{i_j,c_{i_j}}}$, with at most D dimensions, is at most $2^{20\ell_1 k^2}$. Instead of explicitly computing the value

$$\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{Y}_j | \mathscr{X}_j\right],$$

we prove that

$$1/4 \leqslant \mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{Y}_j | \mathscr{X}_j\right]/\lambda \leqslant 4. \tag{6.36}$$

Note that

$$\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{Y}_j|\mathscr{X}_j\right]=\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}_j}\left[\mathscr{Y}_j|\mathscr{X}_j\right]=\frac{\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}_j}\left[\mathscr{Y}_j\wedge\mathscr{X}_j\right]}{\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}_j}\left[\mathscr{X}_j\right]}.$$

It is easy to see that the denominator equals

$$\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}_j}\left[\mathscr{X}_j\right]=2^{-(D-q^*)\cdot w}.$$

Also, we have

$$\begin{aligned}&\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}_j}\left[\mathscr{Y}_j\wedge\mathscr{X}_j\right]\\&=\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}_j}\left[\mathscr{Y}_j\right]\cdot\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}_j}\left[\mathscr{X}_j\mid\mathscr{Y}_j\right]\\&=\lambda\cdot\mathop{\mathbb{P}}_{L\in\mathsf{MGr}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)}\left[L\subseteq W_{i_j}\mid Q\subseteq L\right].\end{aligned}$$

By Proposition 6.26, we get

$$1/4\leqslant\frac{\mathop{\mathbb{P}}_{L\in\mathsf{MGr}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)}\left[L\subseteq W_{i_j}\mid Q\subseteq L\right]}{2^{-w(D-q^*)}}\leqslant 4,$$

which completes the proof of Equation (6.36) under Condition 3.12.

Define $\mathscr{Z}_j$ to be the event that $F[\mathcal{E},\mathcal{U},L_1]|_{L_1}=f_{i_j}|_{L_1}\wedge F[\mathcal{E},\mathcal{U},L_2]|_{L_2}=f_{i_j}|_{L_2}$. We want to calculate the conditional probability

$$\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}_j}\left[\mathscr{Z}_j\mid\mathscr{X}_j\wedge\mathscr{Y}_j\right].$$

To this end, we introduce the auxiliary spaces $L,T\subseteq\mathcal{V}(\mathcal{E}'_{i_j})$, where T represents L_1+L_2 while L represents $Q+T$. We then define the following sampling procedure:

- uniformly sample a subspace $L\in\mathsf{MGr}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)$ conditioned on $Q\subseteq L\subseteq W_{i_j}$;
- sample a D-dimensional subspace $T\subseteq L$ with $\operatorname{codim}_L(T)=e$ conditioned on $T\cap Q=Q'$;
- conditional on L,T, sample two subspaces $L_1,L_2\in\mathsf{MGr}(\mathcal{E}'_{i_j},\ell_1,2^{\ell_1}\cdot s)$ independently and uniformly among those subspaces of T containing Q'.

By definitions, the original conditional distribution $(L_1,L_2)\sim\mathcal{P}_j\mid\mathscr{X}_j\wedge\mathscr{Y}_j$ is equivalent to the above sampling procedure if we add the condition that $L_1+L_2=T$. Thus, the statistical distance between the two distributions is upper bounded by $2^{-\Omega(\ell_1)}$ (since for any D-dimensional subspaces T, the probability uniformly sampled two subspaces $Q'\subseteq L_1,L_2\subseteq T$ span T $1-2^{-\Omega(\ell_1)}$).

Using this fact, we conclude that

$$\begin{aligned}&\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}_j}\left[\mathscr{Z}_j\mid\mathscr{X}_j\wedge\mathscr{Y}_j\right]\\&\geqslant\mathop{\mathbb{P}}_{\substack{L\in\mathsf{MGr}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)\mid Q\subseteq L\subseteq W_{i_j}\\T\subseteq L,T\cap Q=Q'\\Q'\subseteq L_1,L_2\subseteq T}}\left[F[\mathcal{E},\mathcal{U},L_1]=f_{i_j}|_{L_1}\wedge F[\mathcal{E},\mathcal{U},L_2]=f_{i_j}|_{L_2}\right]-2^{-\Omega(\ell_1)}\\&\geqslant\mathop{\mathbb{E}}_{\substack{L\in\mathsf{MGr}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)\mid Q\subseteq L\subseteq W_{i_j}\\T\subseteq L,T\cap Q=Q'}}\left[\mathop{\mathbb{P}}_{Q'\subseteq L_1\subseteq T}\left[F[\mathcal{E},\mathcal{U},L_1]=f_{i_j}|_{L_1}\right]^2\right]-2^{-\Omega(\ell_1)}\end{aligned}$$

$$\geqslant \mathop{\mathbb{P}}_{\substack{L\in\mathsf{MGr}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)\mid Q\subseteq L\subseteq W_{i_j}\\ T\subseteq L, T\cap Q=Q'\\ Q'\subseteq L_1\subseteq T}} \left[F[\mathcal{E},\mathcal{U},L_1]=f_{i_j}|_{L_1}\right]^2 - 2^{-\Omega(\ell_1)}, \tag{6.37}$$

where we use Cauchy-Schwarz in the last transition. Our next goal is to show that the term

$$\mathop{\mathbb{P}}_{\substack{L\in\mathsf{MGr}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)\mid Q\subseteq L\subseteq W_{i_j}\\ T\subseteq L, T\cap Q=Q'\\ Q'\subseteq L_1\subseteq T}} \left[F[\mathcal{E},\mathcal{U},L_1]=f_{i_j}|_{L_1}\right]$$

is lower bounded from the consistency between table $F[\mathcal{E},\mathcal{U},\cdot]$ and f_{i_j} guaranteed by Equation (6.31).

To do that, it suffices to prove that the following two distributions are $2^{-\Omega(\ell_1)}$-close in statistical distance (and then use Equation (6.31) to get the desired lower bound). The first distribution is defined by uniformly sampling a subspace $L_1\in\mathsf{MGr}(\mathcal{E}'_{i_j},\ell_1,2^{\ell_1}\cdot s)$ conditioned on $Q'\subseteq L_1\subseteq W_{i_j}$ and $Q+L_1\in\mathsf{MGr}(\mathcal{E}'_{i_j},\ell_1+e,2^{\ell_1-e}\cdot s)$, and outputting L_1. The second distribution is the one that appears in the RHS of Equation (6.37): uniformly sample a subspace $L\in\mathsf{MGr}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)$ conditioned on $Q\subseteq L\subseteq W_{i_j}$, then uniformly sample a D-dimensional subspace $T\subseteq L$ conditioned on $T\cap Q=Q'$ and finally uniformly sample an ℓ_1-dimensional subspace $Q'\subseteq L_1\subseteq T$. Note that the second distribution can be simplified by directly sampling $L_1\subseteq L$ conditioned on $Q'\subseteq L_1$ and $Q\cap L_1=Q'$ without sampling an intermediate T. We will actually prove the following stronger pointwise estimate: for every $L_1\in\mathsf{MGr}(\mathcal{E}'_{i_j},\ell_1,2^{\ell_1}\cdot s)$ that satisfies $Q+L_1\in\mathsf{MGr}(\mathcal{E}'_{i_j},\ell_1+e,2^{\ell_1-e}\cdot s)$ and $Q'\subseteq L_1\subseteq W_{i_j}$, we have

$$\left(1-2^{-\Omega(\ell_1)}\right)2^{-(D-\ell_1)w}\leqslant \mathop{\mathbb{P}}_{L\in\mathsf{MGr}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)}\left[L\subseteq W_{i_j}\mid Q+L_1\subseteq L\right]\leqslant\left(1+2^{-\Omega(\ell_1)}\right)2^{-(D-\ell_1)w}. \tag{6.38}$$

To see this, let $\mathcal{A}'$ be an advice sequence that satisfies $\sigma(\mathcal{A}')=Q+L_1$. Then, using Proposition 6.22 $(D-\ell_1)$ times, we get

$$\left(2^{-w}-2^{-2^{d-e}s}\right)^{D-\ell_1}\leqslant \mathop{\mathbb{P}}_{\mathcal{A}\in\mathscr{A}_{c_{i_j}}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)}\left[\sigma(\mathcal{A})\subseteq W_{i,j}\mid \mathcal{A}_{[\ell_1+e]}=\mathcal{A}'\right]\leqslant\left(2^{-w}+2^{-2^{d-e}s}\right)^{D-\ell_1},$$

where we use $c_{i_j}\in[t]$ to denote the critical coordinate of the Bob-packet $(\mathcal{E}'_{i_j},\mathcal{U}'_{i_j})$. Together with the statistical distance bound of Proposition 6.25, it gives Equation (6.38), which implies that the two distributions above are $2^{-\Omega(\ell_1)}$-close. Together with Equation (6.37), we get

$$\begin{aligned}
&\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}_j}\left[\mathscr{Z}_j\mid\mathscr{X}_j\wedge\mathscr{Y}_j\right]\\
&\geqslant \mathop{\mathbb{P}}_{\substack{L\in\mathsf{MGr}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)\mid Q\subseteq L\subseteq W_{i_j}\\ T\subseteq L, T\cap Q=Q'\\ Q'\subseteq L_1\subseteq T}} \left[F[\mathcal{E},\mathcal{U},L_1]=f_{i_j}|_{L_1}\right]^2 - 2^{-\Omega(\ell_1)}\\
&\geqslant \left(\mathop{\mathbb{P}}_{\substack{L_1\in\mathsf{MGr}(\mathcal{E}'_{i_j},\ell_1,2^{\ell_1}\cdot s)\\ \Big| \\ Q'\subseteq L_1\subseteq W_{i_j}\wedge Q+L_1\in\mathsf{MGr}(\mathcal{E}'_{i_j},\ell_1+e,2^{\ell_1-e}\cdot s)}} \left[F[\mathcal{E},\mathcal{U},L_1]=f_{i_j}|_{L_1}\right]-2^{-\Omega(\ell_1)}\right)^2-2^{-\Omega(\ell_1)}\\
&\geqslant \exp(-q_0^7),
\end{aligned} \tag{6.39}$$

under Condition 3.12, where we used Equation (6.31) in the last inequality.

Second-moment calculations. Towards Equation (6.34), we show that for every two distinct indices $j, j' \in [m']$, we have

$$2^{-2(D-q^*)w'-2}\lambda^2 \leqslant \mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{X}_j \wedge \mathscr{X}_{j'} \wedge \mathscr{Y}_j \wedge \mathscr{Y}_{j'}\right] \leqslant 2^{-2(D-q^*)w'+2}\lambda^2 \tag{6.40}$$

Again, it is easy to see that $\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{X}_j \wedge \mathscr{X}_{j'}\right] = 2^{-2(D-q^*)w'}$. It suffices to calculate the probability

$$\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{Y}_j \wedge \mathscr{Y}_{j'} \mid \mathscr{X}_j \wedge \mathscr{X}_{j'}\right]$$

The goal is to prove that

$$\lambda^2/4 \leqslant \mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{Y}_j \wedge \mathscr{Y}_{j'} \mid \mathscr{X}_j \wedge \mathscr{X}_{j'}\right] \leqslant 4\lambda^2. \tag{6.41}$$

We use a similar argument as the proof of Equation (6.36). We first define auxiliary sets $\widetilde{\mathcal{E}} = (\widetilde{E}_1, \ldots, \widetilde{E}_t)$ as follows

- for every $c \in [t] \setminus \{c_{i_j}, c_{i_{j'}}\}$, let $\widetilde{E}_c = E_{j,c}$;
- let $\widetilde{E}_{c_{i_j}} = E'_{i_j,c_{i_j}}$ and $\widetilde{E}_{c_{i_{j'}}} = E'_{i_{j'},c_{i_{j'}}}$.

We define the auxiliary distribution $\widetilde{\mathcal{P}}$ sampled from the same process as $\mathcal{P}$ except that for every $x \in [D] \setminus [q^*]$, $\mathcal{M}_x$ is uniformly sampled from $\mathcal{V}(\widetilde{\mathcal{E}})$. By an argument similar to the first-moment calculation, we have

$$\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{Y}_j \wedge \mathscr{Y}_{j'} \mid \mathscr{X}_j \wedge \mathscr{X}_{j'}\right] = \frac{\mathbb{P}_{(L_1,L_2)\sim\widetilde{\mathcal{P}}}\left[\mathscr{Y}_j \wedge \mathscr{Y}_{j'} \wedge \mathscr{X}_j \wedge \mathscr{X}_{j'}\right]}{\mathbb{P}_{(L_1,L_2)\sim\widetilde{\mathcal{P}}}\left[\mathscr{X}_j \wedge \mathscr{X}_{j'}\right]}.$$

First, it is easy to see that $W_{i_j} \cap W_{i_{j'}} \subseteq \mathcal{V}(\widetilde{\mathcal{E}})$, and let $\widetilde{w} = \operatorname{codim}_{\mathcal{V}(\widetilde{\mathcal{E}})}(W_{i_j} \cap W_{i_{j'}}) \leqslant 2w$. Then, we have

$$\mathbb{P}_{(L_1,L_2)\sim\widetilde{\mathcal{P}}}\left[\mathscr{X}_j \wedge \mathscr{X}_{j'}\right] = 2^{-(D-q^*)\widetilde{w}}.$$

Also, we have

$$\begin{aligned}
&\mathbb{P}_{(L_1,L_2)\sim\widetilde{\mathcal{P}}}\left[\mathscr{Y}_j \wedge \mathscr{Y}_{j'} \wedge \mathscr{X}_j \wedge \mathscr{X}_{j'}\right] \\
&= \mathop{\mathbb{P}}_{(L_1,L_2)\sim\widetilde{\mathcal{P}}}\left[\mathscr{Y}_j \wedge \mathscr{Y}_{j'}\right] \cdot \mathop{\mathbb{P}}_{(L_1,L_2)\sim\widetilde{\mathcal{P}}}\left[\mathscr{X}_j \wedge \mathscr{X}_{j'} \mid \mathscr{Y}_j \wedge \mathscr{Y}_{j'}\right] \\
&= \lambda^2 \cdot \mathop{\mathbb{P}}_{(L_1,L_2)\sim\widetilde{\mathcal{P}}}\left[\mathscr{X}_j \wedge \mathscr{X}_{j'} \mid \mathscr{Y}_j \wedge \mathscr{Y}_{j'}\right].
\end{aligned}$$

Here, the second equality comes from the fact that the projections of L_1, L_2 on coordinates $c_{i_j}, c_{i_{j'}}$ are independent under the distribution $\widetilde{\mathcal{P}}$. Applying Proposition 6.26 and using a similar argument in the first-moment calculations, we get

$$2^{-(D-q^*)\widetilde{w}-2} \leqslant \mathop{\mathbb{P}}_{(L_1,L_2)\sim\widetilde{\mathcal{P}}}\left[\mathscr{X}_j \wedge \mathscr{X}_{j'} \mid \mathscr{Y}_j \wedge \mathscr{Y}_{j'}\right] \leqslant 2^{-(D-q^*)\widetilde{w}+2},$$

under Condition 3.12. This also concludes the proof of Equation (6.40).

Inclusion-exclusion. We now use the first moments and second moment calculations to establish Equation (6.34). Indeed, by the inclusion-exclusion principle we have

$$\begin{aligned}
&\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2}=F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}\right] \qquad (6.42)\\
&\geqslant \sum_{j\in[m']}\left(\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\right]-\sum_{j'\in[m']\setminus\{j\}}\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{Y}_j\wedge\mathscr{Y}_{j'}\wedge\mathscr{X}_j\wedge\mathscr{X}_{j'}\right]\right)\\
&\geqslant m'\cdot 2^{-(D-q^*)w'}\cdot\lambda\cdot\exp(-q_0^7)/4-(2m'\cdot 2^{-(D-q^*)w'}\lambda)^2\\
&\geqslant \exp(-5q_0^7). \qquad \text{(plugging in the choice of } m')
\end{aligned}$$

Applying Corollary 6.66 and using the fact that

$$\mathsf{agree}(\mathcal{E},\mathcal{U},W,Q')\geqslant\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2}=F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}\right]-\exp(-\Omega(k))\geqslant\exp(-6q_0^7),$$

we conclude that there exists $f\in W^*$ such that

$$\mathbb{P}_{L\in\mathsf{Gr}(\mathcal{V}(\mathcal{E}),\ell_1)}\left[F[\mathcal{E},\mathcal{U},L]=f|_L\,\middle|\,Q'\subseteq L\subseteq W\right]\geqslant\exp(-100\cdot q_0^7).$$

Existence of a consistent decoding. In the first part of the proof, we have shown the existence of a linear function $f\in W^*$ satisfying Equation (6.32). In the second part, we further show that there exists $f\in W^*$ and index $j\in[m']$ such that f satisfies Equation (6.32) and $f|_{W_{i_j}}=f_{i_j}$ at the same time (possibly different from the one we have shown above). Towards this end, we shall use the following variant of a list-decoding result from [DKK+25].

Claim 6.67 (Corollary of Lemma 3.15, [DKK+25])**.** *Assume Condition 3.12. Let $\{f^{\mathsf{i}}\in W^*\}_{\mathsf{i}\in[p]}$ be the list of all linear functions that satisfy Equation (6.32). We then have (1) $p\leqslant\exp(105\cdot q_0^7)$; and (2)*

$$\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\begin{smallmatrix}F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2}=F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}\\ \wedge F[\mathcal{E},\mathcal{U},L_1]\notin\{f^{\mathsf{i}}|_{L_1}\}_{\mathsf{i}\in[p]}\wedge F[\mathcal{E},\mathcal{U},L_2]\notin\{f^{\mathsf{i}}|_{L_2}\}_{\mathsf{i}\in[p]}\end{smallmatrix}\right]$$

is at most $\exp(-20\cdot q_0^7)$.

For every $j\in[m']$, we define an event $\mathscr{W}_j$ by

$$\mathscr{W}_j:=\wedge_{j'\in[m'],j'\neq j}\neg\left(\mathscr{X}_{j'}\wedge\mathscr{Y}_{j'}\right).$$

Then, we have

$$\begin{aligned}
&\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\begin{smallmatrix}\left(\vee_{j\in[m']}\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\wedge\mathscr{W}_j\right)\\ \wedge\left(F[\mathcal{E},\mathcal{U},L_1]\in\{(f^{\mathsf{i}})|_{L_1}\}_{\mathsf{i}\in[p]}\vee F[\mathcal{E},\mathcal{U},L_2]\in\{(f^{\mathsf{i}})|_{L_2}\}_{\mathsf{i}\in[p]}\right)\end{smallmatrix}\right]\\
&\geqslant\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\left(\vee_{j\in[m']}\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\wedge\mathscr{W}_j\right)\right]\\
&\quad-\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\begin{smallmatrix}\left(\vee_{j\in[m']}\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\wedge\mathscr{W}_j\right)\\ \wedge\left(F[\mathcal{E},\mathcal{U},L_1]\notin\{(f^{\mathsf{i}})|_{L_1}\}_{\mathsf{i}\in[p]}\wedge F[\mathcal{E},\mathcal{U},L_2]\notin\{(f^{\mathsf{i}})|_{L_2}\}_{\mathsf{i}\in[p]}\right)\end{smallmatrix}\right]
\end{aligned}$$

As shown in Equation (6.42) we have that

$$\begin{aligned}
&\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\left(\vee_{j\in[m']}\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\wedge\mathscr{W}_j\right)\right]\\
&\geqslant\sum_{j\in[m']}\left(\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\right]-\sum_{j'\in[m']\setminus\{j\}}\mathbb{P}_{(L_1,L_2)\sim\mathcal{P}}\left[\mathscr{Y}_j\wedge\mathscr{Y}_{j'}\wedge\mathscr{X}_j\wedge\mathscr{X}_{j'}\right]\right)
\end{aligned}$$

$$\geqslant \exp(-5q_0^7).$$

Also, notice that $\left(\vee_{j\in[m']}\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\wedge\mathscr{W}_j\right)$ implies $F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2}=F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}$ by definitions and thus

$$\begin{aligned}
&\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{P}}\left[\begin{smallmatrix}\left(\vee_{j\in[m']}\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\wedge\mathscr{W}_j\right)\\ \wedge\left(F[\mathcal{E},\mathcal{U},L_1]\notin\{(f^{\mathsf{i}})|_{L_1}\}_{\mathsf{i}\in[p]}\wedge F[\mathcal{E},\mathcal{U},L_2]\notin\{(f^{\mathsf{i}})|_{L_2}\}_{\mathsf{i}\in[p]}\right)\end{smallmatrix}\right]\\
&\leqslant \mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{P}}\left[\begin{smallmatrix}F[\mathcal{E},\mathcal{U},L_1]|_{L_1\cap L_2}=F[\mathcal{E},\mathcal{U},L_2]|_{L_1\cap L_2}\\ \wedge F[\mathcal{E},\mathcal{U},L_1]\notin\{(f^{\mathsf{i}})|_{L_1}\}_{\mathsf{i}\in[p]}\wedge F[\mathcal{E},\mathcal{U},L_2]\notin\{(f^{\mathsf{i}})|_{L_2}\}_{\mathsf{i}\in[p]}\end{smallmatrix}\right]\\
&\leqslant \exp(-20q_0^7). \qquad \text{(by Claim 6.67)}
\end{aligned}$$

Combining the two estimates, we conclude that

$$\begin{aligned}
&\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{P}}\left[\begin{smallmatrix}\left(\vee_{j\in[m']}\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\wedge\mathscr{W}_j\right)\\ \wedge\left(F[\mathcal{E},\mathcal{U},L_1]\in\{(f^{\mathsf{i}})|_{L_1}\}_{\mathsf{i}\in[p]}\vee F[\mathcal{E},\mathcal{U},L_2]\in\{(f^{\mathsf{i}})|_{L_2}\}_{\mathsf{i}\in[p]}\right)\end{smallmatrix}\right]\\
&\geqslant \exp(-5q_0^7)-\exp(-20q_0^7)\geqslant\exp(-6q_0^7),
\end{aligned}$$

under Condition 3.12.

By the symmetry of L_1, L_2 and the union bound, we get

$$\begin{aligned}
&\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{P}}\left[\begin{smallmatrix}\left(\vee_{j\in[m']}\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\wedge\mathscr{W}_j\right)\\ \wedge\left(F[\mathcal{E},\mathcal{U},L_1]\in\{(f^{\mathsf{i}})|_{L_1}\}_{\mathsf{i}\in[p]}\vee F[\mathcal{E},\mathcal{U},L_2]\in\{(f^{\mathsf{i}})|_{L_2}\}_{\mathsf{i}\in[p]}\right)\end{smallmatrix}\right]\\
&\leqslant 2\sum_{j\in[m']}\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{P}}\left[(\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\wedge\mathscr{W}_j)\wedge\left(F[\mathcal{E},\mathcal{U},L_1]\in\{(f^{\mathsf{i}})|_{L_1}\}_{\mathsf{i}\in[p]}\right)\right]\\
&\leqslant 2\sum_{j\in[m']}\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{P}}\left[(\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\wedge\mathscr{W}_j)\wedge\left(f_{i_j}|_{L_1}\in\{(f^{\mathsf{i}})|_{L_1}\}_{\mathsf{i}\in[p]}\right)\right].
\end{aligned}$$

Here, the last inequality comes from the fact that $(\mathscr{X}_j\wedge\mathscr{Y}_j\wedge\mathscr{Z}_j\wedge\mathscr{W}_j)$ implies that $(f_{i_j})|_{L_1}=F[\mathcal{E},\mathcal{U},L_1]$. By an averaging argument, there exists $j\in[m']$, such that

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{P}}\left[(\mathscr{Z}_j\wedge\mathscr{W}_j)\wedge\left(f_{i_j}|_{L_1}\in\{(f^{\mathsf{i}})|_{L_1}\}_{\mathsf{i}\in[p]}\right)\mid\mathscr{X}_j\wedge\mathscr{Y}_j\right]\geqslant\frac{\exp(-6q_0^7)}{2m'\cdot\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{P}}[\mathscr{X}_j\wedge\mathscr{Y}_j]}.$$

By our choice of m', the denominator satisfies $2m'\cdot\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{P}}[\mathscr{X}_j\wedge\mathscr{Y}_j]\leqslant 1$, and we get

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{P}}\left[f_{i_j}|_{L_1}\in\{(f^{\mathsf{i}})|_{L_1}\}_{\mathsf{i}\in[p]}\mid\mathscr{X}_j\wedge\mathscr{Y}_j\right]\geqslant\exp(-6q_0^7).$$

By an averaging argument again, we conclude that there exists $\mathsf{i}\in[p]$ such that

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{P}}\left[f_{i_j}|_{L_1}=(f^{\mathsf{i}})|_{L_1}\mid\mathscr{X}_j\wedge\mathscr{Y}_j\right]\geqslant\exp(-6q_0^7)/p\geqslant\exp(-200q_0^7). \tag{6.43}$$

It suffices to prove $f_{i_j}=f^{\mathsf{i}}|_{W_{i_j}}$ using this probability lower bound. Assume that $f_{i_j}\neq f^{\mathsf{i}}|_{W_{i_j}}$ and define $h=f_{i_j}-f^{\mathsf{i}}|_{W_{i_j}}\in W_{i_j}^*$. Let $W'_{i_j}:=\ker(h)\subseteq W_{i_j}$. It is easy to see that $\operatorname{codim}_{\mathcal{V}(\mathcal{E}'_{i_j})}(W'_{i_j})=w+1$. Recall that the distribution of L_1 when we sample $(L_1,L_2)\sim\mathcal{P}\mid\mathscr{X}_j\wedge\mathscr{Y}_j$ is defined by the three-step procedure

- uniformly sample a subspace $L\in\mathsf{MGr}(\mathcal{E}'_{i_j},D+e,2^{d-e}\cdot s)$ conditioned on $Q\subseteq L\subseteq W_{i_j}$;

- sample a D-dimensional subspace $T \subseteq L$ with $\mathrm{codim}_L(T) = e$ conditioned on $T \cap Q = Q'$;
- conditioned on L, T, sample a subspace $L_1 \in \mathsf{MGr}(\mathcal{E}'_{i_j}, \ell_1, 2^{\ell_1} \cdot s)$ independently and uniformly among those subspaces of T containing Q'.

Let ν_j be the joint distribution of (L, T, L_1) defined above. Next, we prove that

$$\mathop{\mathbb{P}}_{(L,T,L_1)\sim\nu_j}\left[L_1 \subseteq W'_{i_j}\right] \leqslant 2^{-(D-q^*)+4} + 2^{-(\ell_1-q^*)},$$

which gives a contradiction to Equation (6.43).

We first prove the statement when $Q \subseteq W'_{i_j}$. By Proposition 6.26, we have

$$\mathop{\mathbb{P}}_{(L,T,L_1)\sim\nu_j}\left[L \subseteq W'_{i_j}\right] \leqslant 2^{-(D-q^*)+4}.$$

If $L \nsubseteq W'_{i_j}$, we also have $T \nsubseteq W'_{i_j}$ and $\dim(W'_{i_j} \cap T) = D - 1$, and thus the probability that $L_1 \subseteq W'_{i_j}$ is at most $2^{-(\ell_1-q^*)}$.

When $Q \nsubseteq W'_{i_j}$, we have $L \nsubseteq W'_{i_j}$ and then the probability that $T \subseteq L$ is contained in W'_{i_j} is at most $2^{-(D-q^*)}$ since the only one valid hyperplane is $W'_{i_j} \cap L$. Again, if $T \nsubseteq W'_{i_j}$, the probability that $L_1 \subseteq W'_{i_j}$ is at most $2^{-(\ell_1-q^*)}$.

Combining the two cases above, we complete the proof. □

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A Basic Linear Algebra

In this appendix, we collect some basic linear algebraic facts about bilinear maps. Recall from Section 3.1 that given an $\mathbb{F}_2$-vector space V and a bilinear map $f : V \otimes V \to \mathbb{F}_2$, we use T_f to denote the associated linear map from V to V^*. The rank of f is defined to be the rank of T_f. The first proposition below analyzes how the rank of f changes after restricted to a subspace $U \otimes U \subseteq V \otimes V$.

Proposition A.1. Let V be a finite dimensional $\mathbb{F}_2$-vector space and let $U \subseteq V$ be a linear subspace. For any bilinear form $f : V \otimes V \to \mathbb{F}_2$, we have

$$\operatorname{rank}(f) - 2 \operatorname{codim} U \leqslant \operatorname{rank}(f|_{U\otimes U}) \leqslant \operatorname{rank}(f).$$

Proof. Let $\iota : U \hookrightarrow V$ be the embedding map, and let $g = f|_{U\otimes U}$. Then we have the commutative diagram

$$U \xrightarrow{\iota} V \xrightarrow{T_f} V^* \xrightarrow{\iota^\dagger} U^*.$$
$$\qquad\qquad\qquad\qquad\quad \overset{T_g}{\curvearrowright}$$

Both composing with ι on the right and composing with $\iota^\dagger$ on the left reduce the rank by at most $\operatorname{codim} U$, so we have

$$\operatorname{rank}(T_f) - 2 \operatorname{codim} U \leqslant \operatorname{rank}(T_g) \leqslant \operatorname{rank}(T_f),$$

as desired. □

The next two corollaries of Proposition A.1 are both used in Lemma 4.26.

Corollary A.2. *Let V be a finite dimensional $\mathbb{F}_2$-vector space and let $U \subseteq V$ be a linear subspace. Suppose $f : V \otimes V \to \mathbb{F}_2$ is a bilinear form such that $\operatorname{rank}(f|_{U\otimes U}) = \operatorname{rank}(f)$. Then for any $v \in V$, there exists $u \in U$ such that*

$$f(v \otimes x) = f(u \otimes x) \qquad \text{for all } x \in V.$$

In other words, letting $g = f|_{U\otimes U}$, we have $h|_U \in \operatorname{image}(T_g)$ for each $h \in \operatorname{image}(T_f)$.

Proof. Let $\iota : U \hookrightarrow V$ be the embedding map. Since $\operatorname{rank}(f|_{U\otimes U}) = \operatorname{rank}(f)$, it follows from the proof of Proposition A.1 that

$$\operatorname{image}(T_f \circ \iota) = \operatorname{image}(T_f),$$

which is exactly the desired conclusion. □

Corollary A.3. *For any positive integers n, r and any symmetric bilinear form $f : \mathbb{F}_2^n \otimes \mathbb{F}_2^n \to \mathbb{F}_2$ with rank r, there exists an r-element subset $S \subseteq [n]$ such that the restriction of f to $\mathbb{F}_2^S \otimes \mathbb{F}_2^S$ has rank r.*

Proof. Since $\dim \ker(T_f) = n - r$, there exists an r-element subset $S \subseteq [n]$ such that

$$\mathbb{F}_2^S + \ker(T_f) = \mathbb{F}_2^n. \tag{A.1}$$

Let $\iota : \mathbb{F}_2^S \to \mathbb{F}_2^n$ be the embedding map. To show that the restriction of f to $\mathbb{F}_2^S \otimes \mathbb{F}_2^S$ has rank r, it suffices to show that

$$\operatorname{rank}(\iota^\dagger \circ T_f \circ \iota) = \operatorname{rank}(T_f).$$

The symmetry of f implies

$$(T_f \circ \iota)^\dagger = \iota^\dagger \circ T_f,$$

so it suffices to show that $\operatorname{rank}(T_f \circ \iota) = \operatorname{rank}(T_f)$. But this is immediate due to Equation (A.1). □

The next lemma is a standard decomposition of bilinear forms into symmetric rank-1 forms. It is used also in Lemma 4.26.

Lemma A.4. *Let V be a finite dimensional $\mathbb{F}_2$-vector space, and let r be a positive integer. For any symmetric bilinear form $f : V \otimes V \to \mathbb{F}_2$ with rank r, there exist a nonnegative integer $k \leqslant 3r/2$ and k vectors $v_1, \ldots, v_k \in V$ such that*

$$f(x \otimes y) = \sum_{i=1}^{k} f(x \otimes v_i) f(y \otimes v_i) \qquad \text{for all } x, y \in V.$$

In other words, any symmetric bilinear form of rank r has a tensor decomposition (see Definition 4.13) into at most $3r/2$ linear functionals.

Proof. We proceed by induction on r. The conclusion clearly holds for $r = 0$. In the rest of the proof, assume $r \geqslant 1$ and the conclusion is true for all smaller values of r. Consider the following two cases.

Case 1: there exists a vector $v \in V$ such that $f(v \otimes v) = 1$. Define a subspace

$$U = \{u \in V \mid f(u \otimes v) = 0\} \subseteq V.$$

It is clear that $\ker(T_f) \subseteq U$, and since $v \notin U$ we also have $\dim U \leqslant \dim V - 1$. Therefore,

$$\operatorname{rank}(f|_{U\otimes U}) \leqslant \dim U - \dim\ker(T_f) \leqslant \dim V - \dim\ker(T_f) - 1 = \operatorname{rank}(f) - 1.$$

Define a linear map $\pi : V \to V$ by letting $\pi(x) = x - f(x \otimes v)v$ for all $x \in V$. It is clear that

$$f(\pi(x) \otimes v) = f(x \otimes v) - f(x \otimes v)^2 = 0 \qquad \text{for all } x \in V,$$

and hence the image of π is contained in U. Furthermore, for each $x, y \in V$ we have

$$\begin{aligned} f\big(\pi(x) \otimes \pi(y)\big) &= f(x \otimes y) - 2f(x \otimes v)f(y \otimes v) + f(x \otimes v)f(y \otimes v)f(v \otimes v) \\ &= f(x \otimes y) - f(x \otimes v)f(y \otimes v). \end{aligned} \tag{A.2}$$

Applying the induction hypothesis to the bilinear form $f\big(\pi(\cdot) \otimes \pi(\cdot)\big)$ (which has rank at most $r - 1$ because $\operatorname{image}(\pi) \subseteq U$ and $\operatorname{rank}(f|_{U\otimes U}) \leqslant r - 1$), we get k vectors $v_1, \ldots, v_k \in V$ with $k \leqslant 3(r-1)/2$ such that

$$f\big(\pi(x) \otimes \pi(y)\big) = \sum_{i=1}^{k} f(x \otimes v_i)f(y \otimes v_i) \qquad \text{for all } x, y \in V. \tag{A.3}$$

Combining Equations (A.2) and (A.3), we get

$$f(x \otimes y) = f(x \otimes v)f(y \otimes v) + \sum_{i=1}^{k} f(x \otimes v_i)f(y \otimes v_i) \qquad \text{for all } x, y \in V,$$

as desired (noting that $k + 1 \leqslant 3r/2$).

Case 2: $f(v \otimes v) = 0$ holds for all $v \in V$. Since $\operatorname{rank}(f) \geqslant 1$, there exist $v, w \in V$ such that $f(v \otimes w) = 1$. It follows from the case assumption that v and w are linearly independent. Define a subspace

$$U = \{u \in V \mid f(u \otimes v) = f(u \otimes w) = 0\} \subseteq V.$$

The rest of the argument is entirely analogous to Case 1. Since $U \cap \operatorname{span}\{v, w\} = \{0\}$ we have $\dim U \leqslant \dim V - 2$. Due to $\ker(T_f) \subseteq U$, we then have $\operatorname{rank}(f|_{U\otimes U}) \leqslant r - 2$. Define a linear map $\pi : V \to V$ by letting

$$\pi(x) = x - f(x \otimes v)w - f(x \otimes w)v \qquad \text{for all } x \in V.$$

Similarly to Case 1, it is easy to verify that

$$f(\pi(x) \otimes v) = f(\pi(x) \otimes w) = 0 \qquad \text{for all } x \in V,$$

and hence the image of π is contained in U. Furthermore, for each $x, y \in V$ one gets

$$\begin{aligned} &f\big(\pi(x) \otimes \pi(y)\big) \\ &= f(x \otimes y) - f(x \otimes v)f(y \otimes w) - f(x \otimes w)f(y \otimes v) \\ &= f(x \otimes y) - f(x \otimes v)f(y \otimes v) - f(x \otimes w)f(y \otimes w) - f(x \otimes (v+w))f(y \otimes (v+w)). \end{aligned}$$

As in Case 1, we then apply the induction hypothesis to decompose $f\big(\pi(x) \otimes \pi(y)\big)$ into the sum of at most $3(r-2)/2$ terms of the form $f(x \otimes v_i)f(y \otimes v_i)$. Thus $f(x \otimes y)$ can be written as the sum of at most $3(r-2)/2 + 3 = 3r/2$ such terms, as desired. □

The next lemma is the key to the success of the "local tensor tests" in our middle PCP, and it is used in Lemma 5.16. In words, it asserts that if a bilinear form f has rank at least $2+2\mathrm{codim}W$, then (1) restricting it to $W\otimes W$ drops the rank by at most $2\mathrm{codim}W$, and (2) with high probability, restricting it to a random subspace $U\subseteq W$ of sufficiently large dimension does not drop the rank to at most 1.

Lemma A.5. *Let V be a finite dimensional $\mathbb{F}_2$-vector space, and let $Q, W\subseteq V$ be linear subspaces such that $Q\subseteq W$. Denote $q=\dim Q$. Let ℓ be a positive integer such that $q+2\leqslant \ell\leqslant \dim W$. Then for any bilinear form $f: V\otimes V\to\mathbb{F}_2$ such that $\mathrm{rank}(f)\geqslant 2+2\operatorname{codim} W$, we have*

$$\mathop{\mathbb{P}}_{U\in\mathsf{Gr}(W,\ell)}\left[\mathrm{rank}(f|_{U\otimes U})\leqslant 1 \,\middle|\, Q\subseteq U\right]\leqslant 2^{4-(\ell-q)/2}.$$

Proof. Let $g=f|_{W\otimes W}$. By Proposition A.1, the assumption that $\mathrm{rank}(f)\geqslant 2+2\operatorname{codim} W$ implies $\mathrm{rank}(g)\geqslant 2$. It suffices to work with the restricted map g in the rest of the proof.

Let $\ell'=\lfloor(\ell-q)/2\rfloor\geqslant 1$. To pick a uniformly random $U\in\mathsf{Gr}(W,\ell)$ that contains Q, we first pick uniformly random subspaces $U_1,U_2\in\mathsf{Gr}(W,\ell')$ independently, and then pick a uniformly random $U\in\mathsf{Gr}(W,\ell)$ conditioned on $Q,U_1,U_2\subseteq U$. Since we always have

$$\mathrm{rank}(f|_{U\otimes U})=\mathrm{rank}(g|_{U\otimes U})\geqslant \mathrm{rank}(g|_{U_1\otimes U_2}),$$

it suffices to show that

$$\mathop{\mathbb{P}}_{U_1,U_2\in\mathsf{Gr}(W,\ell')}\left[\mathrm{rank}(g|_{U_1\otimes U_2})\leqslant 1\right]\leqslant 2^{3-\ell'}.$$

As in the proof of Proposition A.1, we let $\iota_{U_1}: U_1\hookrightarrow W$ and $\iota_{U_2}: U_2\hookrightarrow W$ be the two canonically defined embedding maps. Then we have

$$\mathrm{rank}(g|_{U_1\otimes U_2})=\mathrm{rank}\left(\iota_{U_2}^\dagger\circ T_g\circ\iota_{U_1}\right).$$

Note that

$$\mathrm{rank}(T_g\circ\iota_{U_1})=\dim(U_1+\ker(T_g))-\dim\ker(T_g).$$

Therefore, denoting $w=\dim W$ and $r=\mathrm{rank}(g)=w-\dim\ker(T_g)$, we can calculate

$$\begin{aligned}
&\mathop{\mathbb{P}}_{U_1\in\mathsf{Gr}(W,\ell')}\left[\mathrm{rank}(T_g\circ\iota_{U_1})\leqslant 1\right]\\
&=\mathop{\mathbb{P}}_{U_1\in\mathsf{Gr}(W,\ell')}\left[\dim(U_1+\ker(T_g))-\dim\ker(T_g)\leqslant 1\right]\\
&\leqslant\sum_{K\in\mathsf{Gr}(W,w-r+1),\ K\supseteq\ker(T_g)}\ \mathop{\mathbb{P}}_{U_1\in\mathsf{Gr}(W,\ell')}\left[U_1\subseteq K\right]\\
&\leqslant(2^r-1)\cdot\begin{bmatrix}w-r+1\\ \ell'\end{bmatrix}_2\begin{bmatrix}w\\ \ell'\end{bmatrix}_2^{-1}\\
&=(2^r-1)\cdot\prod_{i=0}^{\ell'-1}\frac{2^{w-r-i+1}-1}{2^{w-i}-1}\\
&\leqslant 2^r\cdot(2^{-r+1})^{\ell'}\\
&\leqslant 2^{-\ell'+2}. && \text{(using } r\geqslant 2 \text{ and } \ell'\geqslant 1)
\end{aligned}$$

Similarly, since

$$\mathrm{rank}\left(\iota_{U_2}^\dagger\circ T_g\circ\iota_{U_1}\right)=\dim\left(U_2+\ker\left(\iota_{U_1}^\dagger\circ T_g^\dagger\right)\right)-\dim\ker\left(\iota_{U_1}^\dagger\circ T_g^\dagger\right),$$

an analogous calculation shows

$$\mathop{\mathbb{P}}_{U_2 \in \mathsf{Gr}(W,\ell')}\left[\operatorname{rank}\left(\iota_{U_2}^\dagger \circ T_g \circ \iota_{U_1}\right) \leqslant 1\right] \leqslant 2^{-\ell'+2} \qquad \text{whenever} \qquad \dim\ker\left(\iota_{U_1}^\dagger \circ T_g^\dagger\right) \leqslant \dim W - 2.$$

But $\dim\ker\left(\iota_{U_1}^\dagger \circ T_g^\dagger\right) \leqslant \dim W - 2$ happens exactly when $\operatorname{rank}(T_g \circ \iota_{U_1}) \geqslant 2$, the probability of which has already been shown to be at least $1 - 2^{-\ell'+2}$. Therefore, we conclude that

$$\mathop{\mathbb{P}}_{U_1,U_2 \in \mathsf{Gr}(W,\ell')}\left[\operatorname{rank}\left(\iota_{U_2}^\dagger \circ T_g \circ \iota_{U_1}\right) \leqslant 1\right] \leqslant 2 \cdot 2^{-\ell'+2} = 2^{-\ell'+3},$$

as desired. □

B From Sub-4-to-1 to Exact 4-to-1 Games

In this appendix, we show that the hardness of approximating sub-4-to-1 games implies the exact 4-to-1 games theorem.

Proof of Theorem 1.6 assuming Theorem 3.3. Fix an arbitrary parameter $\varepsilon \in (0,1)$ and a positive integer r. Write $k = \lceil 4/\varepsilon \rceil$ and $r' = kr$. We will prove Theorem 1.6 by constructing a polynomial-time reduction from $\mathsf{GapS\text{-}4\text{-}to\text{-}1}_r(1, \varepsilon/2)$ to $\mathsf{Gap\text{-}4\text{-}to\text{-}1}_{r'}(1,\varepsilon)$.

Given a sub-4-to-1 game instance Ψ with underlying graph $(L \sqcup R, E)$, alphabets Σ_L, Σ_R and constraints $\{\Phi_e \subseteq \Sigma_L \times \Sigma_R\}_{e\in E}$, we construct a 4-to-1 game instance Ψ' as follows:

1. Let Ψ' have the same constraint graph $(L \sqcup R, E)$ as Ψ.
2. The left-alphabet of Ψ' is $\Sigma'_L = \Sigma_L \times (\mathbb{Z}/k\mathbb{Z})$, and the right-alphabet of Ψ' is $\Sigma'_R = \Sigma_R \times (\mathbb{Z}/k\mathbb{Z})$.
3. For each edge $e = (u,v) \in E$, according to Definition 3.1, there exists a partition $\Sigma_L = \Sigma_{L,1} \sqcup \cdots \sqcup \Sigma_{L,t}$ into sets of size 4 and a partition $\Sigma_R = \Sigma_{R,1} \sqcup \cdots \sqcup \Sigma_{R,t}$ into singleton sets such that

$$\Phi_e = \bigcup_{i=1}^{t} \Gamma_{L,i} \times \Gamma_{R,i}, \qquad \text{for some subsets } \Gamma_{L,i} \subseteq \Sigma_{L,i} \text{ and } \Gamma_{R,i} \subseteq \Sigma_{R,i}. \tag{B.1}$$

   For each $j, \ell \in \mathbb{Z}/k\mathbb{Z}$ and $i \in [t]$, we let

$$\Sigma^{(\ell)}_{L,i,j} = \left(\Gamma_{L,i} \times \{j\}\right) \cup \left((\Sigma_{L,i} \setminus \Gamma_{L,i}) \times \{j+\ell\}\right) \subseteq \Sigma_L \times (\mathbb{Z}/k\mathbb{Z}) = \Sigma'_L, \tag{B.2}$$

$$\Sigma^{(\ell)}_{R,i,j} = \left(\Gamma_{R,i} \times \{j\}\right) \cup \left((\Sigma_{R,i} \setminus \Gamma_{R,i}) \times \{j-\ell\}\right) \subseteq \Sigma_R \times (\mathbb{Z}/k\mathbb{Z}) = \Sigma'_R. \tag{B.3}$$

   We then define a constraint $\Phi'_{e,\ell} \subseteq \Sigma'_L \times \Sigma'_R$

$$\Phi'_{e,\ell} = \bigcup_{i=1}^{t} \bigcup_{j\in\mathbb{Z}/k\mathbb{Z}} \Sigma^{(\ell)}_{L,i,j} \times \Sigma^{(\ell)}_{R,i,j}. \tag{B.4}$$

   Since for each $i \in [t]$, the sets $\Sigma^{(\ell)}_{L,i,1}, \ldots, \Sigma^{(\ell)}_{L,i,k}$ have size $|\Sigma_{L,i}| = 4$ and form a partition of $\Sigma_{L,i} \times (\mathbb{Z}/k\mathbb{Z})$, and similarly the sets $\Sigma^{(\ell)}_{R,i,1}, \ldots, \Sigma^{(\ell)}_{R,i,k}$ have size $|\Sigma_{R,i}| = 1$ and form a partition of $\Sigma_{R,i} \times (\mathbb{Z}/k\mathbb{Z})$, it is clear that $\Phi'_{e,\ell}$ is a 4-to-1 constraint.
4. For each edge $e \in E$, we put in Ψ' the k constraints $\Phi'_{e,1}, \ldots, \Phi'_{e,k}$. The total number of constraints in Ψ' is therefore equal to $k|E|$.

It is easy to see that Ψ' can be constructed from Ψ in polynomial time. We then show the completeness and soundness of the reduction.

Completeness: if $\mathrm{val}(\Psi) = 1$, there exist assignment maps $A_L : L \to \Sigma_L$ and $A_R : R \to \Sigma_R$ that satisfy all constraints Φ_e, for $e \in E$. Then it is clear that the map $A'_L : L \to \Sigma'_L$ defined by $A'_L(u) = (A_L(u), 1)$ and the map $A'_R : R \to \Sigma'_R$ defined by $A'_R(v) = (A_R(v), 1)$ jointly satisfy all constraints $\Phi'_{e,\ell}$, for $e \in E$ and $\ell \in \mathbb{Z}/k\mathbb{Z}$. Consequently, we have $\mathrm{val}(\Psi') = 1$.

Soundness: suppose we have assignment maps $A'_L : L \to \Sigma'_L$ and $A'_R : R \to \Sigma'_R$ given by

$$A'_L(u) = (A_L(u), K_L(u)) \qquad \text{and} \qquad A'_R(v) = (A_R(v), K_R(v)).$$

For any edge $e = (u, v) \in E$, we claim that if $(A_L(u), A_R(v)) \notin \Phi_e$, then at most 2 of the k constraints $\Phi'_{e,1}, \ldots, \Phi'_{e,k}$ are satisfied by (A'_L, A'_R). This means

$$\mathrm{val}_{\Psi'}(A'_L, A'_R) \leqslant \mathrm{val}_{\Psi}(A_L, A_R) + 2/k \leqslant \mathrm{val}_{\Psi}(A_L, A_R) + \varepsilon/2,$$

and by the arbitrariness of (A'_L, A'_R) it follows that

$$\mathrm{val}(\Psi') \leqslant \mathrm{val}(\Psi) + \varepsilon/2,$$

as desired.

We now fix an edge $e = (u, v) \in E$ such that $(A_L(u), A_R(v)) \notin \Phi_e$, and show that at most 2 of the k constraints $\Phi'_{e,1}, \ldots, \Phi'_{e,k}$ are satisfied by (A'_L, A'_R). Let the subsets

$$\Gamma_{L,i}, \Sigma_{L,i} \subseteq \Sigma_L, \qquad \Gamma_{R,i}, \Sigma_{R,i} \subseteq \Sigma_R, \qquad \Sigma^{(\ell)}_{L,i,j} \subseteq \Sigma'_L, \qquad \text{and} \qquad \Sigma^{(\ell)}_{R,i,j} \subseteq \Sigma'_R$$

be as defined in Equations (B.1) to (B.3). We divide into the following five cases.

Case 1: $(A_L(u), A_R(v)) \notin \Sigma_{L,i} \times \Sigma_{R,i}$ for any $i \in [t]$. In this case, it is easy to see from Equations (B.2) to (B.4) that whichever values $K_L(u)$ and $K_R(v)$ take, we always have

$$(A'_L(u), A'_R(v)) \notin \bigcup_{\ell=1}^{k} \Phi'_{e,\ell}.$$

Therefore, none of the constraints $\Phi'_{e,1}, \ldots, \Phi'_{e,k}$ are satisfied by (A'_L, A'_R).

Case 2: $(A_L(u), A_R(v)) \in (\Sigma_{L,i} \setminus \Gamma_{L,i}) \times \Gamma_{R,i}$ for some $i \in [t]$. In this case, $(A'_L(u), A'_R(v)) \in \Phi'_{e,\ell}$ if and only if $K_L(u) = K_R(v) + \ell$, and hence exactly one of the constraints $\Phi'_{e,1}, \ldots, \Phi'_{e,\ell}$ is satisfied by (A'_L, A'_R).

Case 3: $(A_L(u), A_R(v)) \in \Gamma_{L,i} \times (\Sigma_{R,i} \setminus \Gamma_{R,i})$ for some $i \in [t]$. This case is symmetric to the previous case, and exactly one of the constraints $\Phi'_{e,1}, \ldots, \Phi'_{e,\ell}$ is satisfied by (A'_L, A'_R).

Case 4: $(A_L(u), A_R(v)) \in (\Sigma_{L,i} \setminus \Gamma_{L,i}) \times (\Sigma_{R,i} \setminus \Gamma_{R,i})$ for some $i \in [t]$. In this case, we have $(A'_L(u), A'_R(v)) \in \Phi'_{e,\ell}$ if and only if $K_L(u) - \ell = K_R(v) + \ell$, and hence at most two of the constraints $\Phi'_{e,1}, \ldots, \Phi'_{e,\ell}$ are satisfied by (A'_L, A'_R).

Case 5: $(A_L(u), A_R(v)) \in \Gamma_{L,i} \times \Gamma_{R,i}$ for some $i \in [t]$. This case cannot happen because if it does then (A_L, A_R) satisfies Φ_e, violating the assumption. □

C Approximate Coloring of Hypergraphs

In this appendix, we present a proof of the "starting point" theorem (restated below) used in our reduction to 4-to-1 games. The proof of this theorem is essentially due to [DGKR05, DRS05], and we claim no originality here. The inclusion of the proof here is purely for the sake of expositional completeness.

Theorem 3.4 ([DRS05]). *For every fixed positive integer $c \geqslant 2$, there exists a constant $\varepsilon \geqslant \exp(-\exp(O(c^3)))$ such that given a 3-uniform hypergraph H, it is* **NP***-hard to distinguish between the following cases:*

(1) Yes case: H is 2-colorable.

(2) No case: in every c-coloring of H, at least ε fraction of the edges are monochromatic.

To prove Theorem 3.4, we need to use the PCP theorem of [ALM+98, AS98] and the parallel repetition theorem of [Raz98]. More specifically, we apply the parallel repetition theorem to the equation-vs.-variable game associated to 3SAT5 instances.[19] This step is standard in the hardness of approximation literature, and we state the result (Theorem C.2) below without proof.

Definition C.1. Let d_0 and d_1 be positive integers. A (d_0, d_1)*-biregular projection label cover instance* $\Psi = (V_0, V_1, E, \Sigma_0, \Sigma_1, (\pi_e)_{e \in E})$ consists of the following data:

(1) two vertex sets V_0 and V_1;

(2) a set E of edges between V_0 and V_1 such that in the bipartite graph $(V_0 \sqcup V_1, E)$, every vertex in V_0 has degree d_0 and every vertex in V_1 has degree d_1;

(3) two finite alphabets Σ_0, Σ_1; and

(4) a collection of constraints $(\pi_e)_{e \in E}$, where each π_e is a map from Σ_0 to Σ_1.

The value of Ψ, denoted by $\mathrm{val}(\Psi)$, is the maximum value of

$$\frac{1}{|E|} \sum_{(v_0, v_1) \in E} \mathbb{1}\left[\pi_{(v_0,v_1)}(A_0(v_0)) = A_1(v_1)\right]$$

achievable by assignment maps $A_0 : V_0 \to \Sigma_0$ and $A_1 : V_1 \to \Sigma_1$.

Theorem C.2 ([ALM+98, AS98, Raz98]). *There exists an absolute constant $\gamma \in (0, 1)$ such that for every fixed positive integer u, given a $(3^u, 5^u)$-biregular projection label cover instance $\Psi = (V_0, V_1, E, \Sigma_0, \Sigma_1, (\pi_e)_{e \in E})$ such that $|\Sigma_0| = 7^u$ and $|\Sigma_1| = 2^u$, it is* **NP***-hard to distinguish between the case $\mathrm{val}(\Psi) = 1$ and the case $\mathrm{val}(\Psi) \leqslant 2^{-\gamma u}$.*

Assuming Theorem C.2, we now present a proof of Theorem 3.4.

Proof of Theorem 3.4. We proceed by a reduction from Theorem C.2. Let $\gamma \in (0, 1)$ be as in Theorem C.2, and let $u = \lceil 40\gamma^{-2}c \rceil$. We fix a $(3^u, 5^u)$-biregular projection label cover instance $\Psi = (V_0, V_1, E, \Sigma_0, \Sigma_1, (\pi_e)_{e \in E})$ with $|\Sigma_0| = 7^u$ and $|\Sigma_1| = 2^u$; we will show how to map it to a 3-uniform hypergraph H while linking the value of Ψ to the colorability of H.

The proof is divided into 8 steps. We first construct the hypergraph H in Steps 1 and 2. The completeness of the reduction is proved in Step 3, while Steps 4 to 8 will be devoted to proving the soundness of the reduction. Except for Step 6, the proof is almost identical to the one in [DRS05].[20] Throughout the proof, we let $\ell = 2c^2$.

[19] By 3SAT5, we mean 3SAT instances where each variable is contained in exactly 5 clauses; see [Fei98, Proposition 2.1.2].

[20] The arguments in Steps 1, 5 and 8 are due to [DGKR05].

Step 1: multi-layered label cover. Following the ideas of [DGKR05, DRS05], the first step is to perform a "multi-layered" parallel repetition on Ψ. For each $i \in [\ell]$, let $\mathcal{V}_i$ and R_i be the cartesian products

$$\mathcal{V}_i = V_1^{i-1} \times V_0^{\ell-i} \qquad \text{and} \qquad R_i = \Sigma_1^{i-1} \times \Sigma_0^{\ell-i}.$$

The $(\ell-1)$-tuples $\mathbf{v} = (v_1, \ldots, v_{\ell-1}) \in \mathcal{V}_i$ will be the "blown-up vertices" in the i-th layer, and R_i can be viewed as their alphabets. For each $i, j \in [\ell]$ such that $i < j$, let $\mathcal{E}_{i,j}$ be the collection of pairs

$$(\mathbf{v}, \mathbf{v}') = \big((v_1, \ldots, v_{\ell-1}), (v'_1, \ldots, v'_{\ell-1})\big) \in \mathcal{V}_i \times \mathcal{V}_j$$

such that:

(1) $v_s = v'_s$ for all $s \in \{1, 2, \ldots, i-1\} \cup \{j, j+1, \ldots, \ell-1\}$; and

(2) $(v_s, v'_s) \in E$ for all $s \in \{i, i+1, \ldots, j-1\}$.

Given such a pair $(\mathbf{v}, \mathbf{v}') \in \mathcal{E}_{i,j}$, we define a projection map $\pi_{(\mathbf{v},\mathbf{v}')} : R_i \to R_j$ by

$$\pi_{(\mathbf{v},\mathbf{v}')}(\sigma_1, \ldots, \sigma_{\ell-1}) = \Big(\sigma_1, \ldots, \sigma_{i-1}, \pi_{(v_i, v'_i)}(\sigma_i), \ldots, \pi_{(v_{j-1}, v'_{j-1})}(\sigma_{j-1}), \sigma_j, \ldots, \sigma_{\ell-1}\Big),$$

for all $\sigma_1, \ldots, \sigma_{i-1} \in \Sigma_1$ and $\sigma_i, \ldots, \sigma_{\ell-1} \in \Sigma_0$.

For each $i, j \in [\ell]$ such that $i < j$, we have

$$|\mathcal{E}_{i,j}| = |\mathcal{V}_j| \cdot (5^u)^{j-i} = (5^u)^{j-i} \cdot |V_1|^{j-1} |V_0|^{\ell-j} = 5^{u(j-i)} \cdot (5/3)^{u(\ell-j)} \cdot |V_1|^{\ell-1}, \tag{C.1}$$

where we used the $(3^u, 5^u)$-biregularity of Ψ and its consequence $|V_0|/|V_1| = 5^u/3^u$.

Step 2: construction of H. Following [DRS05], we construct the hypergraph H by composing the multi-layered label cover from Step 1 with an inner gadget based on the Kneser graph.

For each $i \in [\ell]$, we let $r_i = \lceil (|R_i| - c)/2 \rceil$. The vertex set of H is the disjoint union of cartesian products $\bigsqcup_{i=1}^{\ell} \left(\mathcal{V}_i \times \binom{R_i}{r_i}\right)$, and the hyperedges of H are the collection of unordered triples $\big\{(\mathbf{v}, S_1), (\mathbf{v}, S_2), (\mathbf{v}', S_3)\big\}$ such that:

(1) $\mathbf{v} \in \mathcal{V}_i$ and $\mathbf{v}' \in \mathcal{V}_j$ for some $i, j \in [\ell]$ such that $i < j$;

(2) $(\mathbf{v}, \mathbf{v}') \in \mathcal{E}_{i,j}$;

(3) $S_1, S_2 \in \binom{R_i}{r_i}$ and $S_3 \in \binom{R_j}{r_j}$; and

(4) $S_1 \cap S_2 = \emptyset$ and $\pi_{(\mathbf{v},\mathbf{v}')}(R_i \setminus (S_1 \cup S_2)) \subseteq S_3$.

Using Equation (C.1), we can upper-bound the number of edges in H by

$$\sum_{i,j \in [\ell],\, i<j} |\mathcal{E}_{i,j}| \cdot 2^{2|R_i|} \cdot 2^{|R_j|} \leqslant 8^{7^{u(\ell-1)}} \ell^2 \cdot \max_{i,j \in [\ell],\, i<j} |\mathcal{E}_{i,j}| \leqslant 5^{u(\ell-1)} 8^{7^{u(\ell-1)}} \ell^2 \cdot |V_1|^{\ell-1}. \tag{C.2}$$

It is clear that H can be constructed from Ψ in polynomial time. It suffices to prove the following statements:

(1) Completeness: if $\mathrm{val}(\Psi) = 1$, then H is 2-colorable.

(2) Soundness: if under some c-coloring of H, less than $\exp(-\exp(\Omega(c^3)))$ fraction of edges of H are monochromatic, then $\mathrm{val}(\Psi) > 2^{-\gamma u}$.

The first statement will be proved in Step 3, and the second will be proved in Steps 4 to 8.

Step 3: completeness. In this step, we assume $\mathrm{val}(\Psi) = 1$ and show that H is 2-colorable.

Suppose $A_0 : V_0 \to \Sigma_0$ and $A_1 : V_1 \to \Sigma_1$ are assignments that together satisfy every constraint of Ψ. Then for any $i \in [\ell]$, we can define a lifted assignment map $\widetilde{A}_i : \mathcal{V}_i \to R_i$ by letting

$$\widetilde{A}_i(\mathbf{v}) = \big(A_1(v_1), \dots, A_1(v_{i-1}), A_0(v_i), \dots, A_0(v_{\ell-1})\big), \qquad \text{for all } \mathbf{v} = (v_1, \dots, v_{\ell-1}) \in \mathcal{V}_i.$$

For any vertex $(\mathbf{v}, S) \in \mathcal{V}_i \times \binom{R_i}{r_i}$, we color it red if $\widetilde{A}_i(\mathbf{v}) \in S$, and otherwise color it blue.

Now, for any hyperedge $\big\{(\mathbf{v}, S_1), (\mathbf{v}, S_2), (\mathbf{v}', S_3)\big\}$ included in H (see Step 2), we have:

(1) The vertices $(\mathbf{v}, S_1)$ and $(\mathbf{v}, S_2)$ cannot be both red, because $S_1 \cap S_2 = \emptyset$.

(2) Let $i, j \in [\ell]$ be the indices such that $\mathbf{v} \in \mathcal{V}_i$ and $\mathbf{v}' \in \mathcal{V}_j$. If $(\mathbf{v}, S_1)$ and $(\mathbf{v}, S_2)$ are both colored blue, then $\widetilde{A}_i(\mathbf{v}) \in R_i \setminus (S_1 \cup S_2)$ and hence

$$\pi_{(\mathbf{v},\mathbf{v}')}\big(\widetilde{A}_i(\mathbf{v})\big) \in S_3, \qquad \text{due to} \qquad \pi_{(\mathbf{v},\mathbf{v}')}(R_i \setminus (S_1 \cup S_2)) \subseteq S_3.$$

By the definitions of $\pi_{(\mathbf{v},\mathbf{v}')}$ and $\widetilde{A}_i, \widetilde{A}_j$, we have $\pi_{(\mathbf{v},\mathbf{v}')}\big(\widetilde{A}_i(\mathbf{v})\big) = \widetilde{A}_j(\mathbf{v}')$. Therefore, if $(\mathbf{v}, S_1)$ and $(\mathbf{v}, S_2)$ are both colored blue, then $(\mathbf{v}', S_3)$ must be colored red.

In conclusion, the hyperedge $\big\{(\mathbf{v}, S_1), (\mathbf{v}, S_2), (\mathbf{v}', S_3)\big\}$ cannot be monochromatic under the proposed coloring. This means that H is 2-colorable as long as $\mathrm{val}(\Psi) = 1$.

Step 4: finding the leading color. In the rest of the proof, we fix a c-coloring of H, denoted by the map

$$f : \bigsqcup_{i=1}^{\ell} \left(\mathcal{V}_i \times \binom{R_i}{r_i}\right) \to [c].$$

We suppose that at most ε fraction of edges of H are monochromatic under f, for some $\varepsilon \in (0, 1)$ to be specified later.

For each $i \in [\ell]$ and each $\mathbf{v} \in \mathcal{V}_i$, we apply [DRS05, Lemma 2.2] to the vertices in the set $\left\{(\mathbf{v}, S) \,\middle|\, S \in \binom{R_i}{r_i}\right\}$ to find a color $c_{\mathbf{v}} \in [c]$ such that:

(1) The fraction of sets $S \in \binom{R_i}{r_i}$ such that $f(\mathbf{v}, S) = c_{\mathbf{v}}$ is at least $|R_i|^{-c-1}$.

(2) There exist disjoint sets $S_1, S_2 \in \binom{R_i}{r_i}$ such that $f(\mathbf{v}, S_1) = f(\mathbf{v}, S_2) = c_{\mathbf{v}}$. The set $R_i \setminus (S_1 \cup S_2)$ will be denoted by $B(\mathbf{v})$ and used later in the proof.

For each $i \in [\ell]$, let $c^{(i)} \in [c]$ be a color that maximizes the number of tuples $\mathbf{v} \in \mathcal{V}_i$ with $c_{\mathbf{v}} = c^{(i)}$. Since $\ell = 2c^2$, by the pigeonhole principle there exists a color $c^* \in [c]$ such that $c^{(i)} = c^*$ for at least $2c$ distinct indices $i \in [\ell]$. Pick exactly $2c$ indices among them, and let I be the set of these $2c$ indices.

Step 5: locating two critical layers. Observe that the bipartite graph $(\mathcal{V}_i \sqcup \mathcal{V}_{i+1}, \mathcal{E}_{i,i+1})$ is bi-regular for all $i \in [\ell - 1]$, due to the bi-regularity of Ψ. Consider the following random walk:

1. Start at a uniformly random element $\mathbf{v}^{(1)} \in \mathcal{V}_1$.
2. For each $i \in [\ell - 1]$, let $\mathbf{v}^{(i+1)}$ be a uniformly random element of $\mathcal{V}_{i+1}$ conditioned on $(\mathbf{v}^{(i)}, \mathbf{v}^{(i+1)}) \in \mathcal{E}_{i,i+1}$.

Due to the bi-regularity of the bipartite graphs $(\mathcal{V}_i \sqcup \mathcal{V}_{i+1}, \mathcal{E}_{i,i+1})$, this random walk produces a random sequence $(\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(\ell)}) \in \mathcal{V}_1 \times \cdots \times \mathcal{V}_\ell$ that is marginally uniform over each $\mathcal{V}_i$. Furthermore, by the definition of the edge sets $\mathcal{E}_{i,j}$, it is easy to see that for any $i, j \in [\ell]$ such that $i < j$, the distribution of the pair $(\mathbf{v}^{(i)}, \mathbf{v}^{(j)})$ is uniform over $\mathcal{E}_{i,j}$.

For each $i \in [\ell]$, let Γ_i be the event that $c_{\mathbf{v}^{(i)}} = c^{(i)}$. By the marginal uniformity and the choice of the $c^{(i)}$'s we have $\mathbb{P}[\Gamma_i] \geqslant c^{-1}$. By the inclusion-exclusion principle, we have

$$\begin{aligned} 1 &\geqslant \mathbb{P}\left[\bigcup\nolimits_{i \in I} \Gamma_i\right] \geqslant \sum_{i \in I} \mathbb{P}[\Gamma_i] - \sum_{i,j \in I,\, i<j} \mathbb{P}[\Gamma_i \cap \Gamma_j] \\ &\geqslant |I| \cdot c^{-1} - \binom{|I|}{2} \cdot \max_{i,j \in I,\, i<j} \mathbb{P}[\Gamma_i \cap \Gamma_j]\,. \end{aligned}$$

Therefore, there exist $i, j \in I$ with $i < j$ such that

$$\mathbb{P}[\Gamma_i \cap \Gamma_j] \geqslant \binom{|I|}{2}^{-1} \cdot (c^{-1}|I| - 1) = \binom{2c}{2}^{-1} \geqslant \frac{1}{2c^2}.$$

By the uniformity of $(\mathbf{v}^{(i)}, \mathbf{v}^{(j)})$ over $\mathcal{E}_{i,j}$, this means that

$$\mathbb{P}_{(\mathbf{v},\mathbf{v}') \in \mathcal{E}_{i,j}}\left[c_{\mathbf{v}} = c^{(i)} \text{ and } c_{\mathbf{v}'} = c^{(j)}\right] \geqslant \frac{1}{2c^2}.$$

Since $i, j \in I$, by the definition of I, the above is equivalent to

$$\mathbb{P}_{(\mathbf{v},\mathbf{v}') \in \mathcal{E}_{i,j}}[c_{\mathbf{v}} = c_{\mathbf{v}'} = c^*] \geqslant \frac{1}{2c^2}. \tag{C.3}$$

In the remainder of the proof, we fix the indices i and j identified above (satisfying Equation (C.3)).

Step 6: discarding monochromatic edges. We let $\mathcal{E}^*$ be the collection of all pairs $(\mathbf{v}, \mathbf{v}') \in \mathcal{E}_{i,j}$ such that:

(1) $c_{\mathbf{v}} = c_{\mathbf{v}'} = c^*$; and

(2) no hyperedge of H of the form $\{(\mathbf{v}, S_1), (\mathbf{v}, S_2), (\mathbf{v}', S_3)\}$ is colored monochromatically by f.

By Equation (C.2), the total number of monochromatic edges of H under the coloring f is at most

$$\varepsilon \cdot 5^{u(\ell-1)} 8^{7^{u(\ell-1)}} \ell^2 \cdot |V_1|^{\ell-1} \leqslant \varepsilon \cdot 5^{u(\ell-1)} 8^{7^{u(\ell-1)}} \ell^2 \cdot |\mathcal{E}_{i,j}| \leqslant \frac{1}{4c^2} |\mathcal{E}_{i,j}|\,,$$

where we chose a sufficiently small $\varepsilon = \exp(-\exp(O(c^3)))$ that justifies the last transition. It then follows from Equation (C.3) and the definition of $\mathcal{E}^*$ that

$$|\mathcal{E}^*| \geqslant \frac{1}{2c^2} |\mathcal{E}_{i,j}| - \frac{1}{4c^2} |\mathcal{E}_{i,j}| = \frac{1}{4c^2} |\mathcal{E}_{i,j}|\,. \tag{C.4}$$

Step 7: strategy for the multi-layered label cover. The goal of this step is to decode from f a "good" assignment map for the two critical layers (identified in Step 5) of the ℓ-layered label cover instance (constructed in Step 1).

For each $\mathbf{v}' \in \mathcal{V}_j$, let $N^*(\mathbf{v}')$ be the collection of tuples $\mathbf{v} \in \mathcal{V}_i$ such that $(\mathbf{v}, \mathbf{v}') \in \mathcal{E}^*$.

In this paragraph, we define a deterministic map $\widetilde{A}_j : \mathcal{V}_j \to R_j$ using the coloring f. Fix a tuple $\mathbf{v}' \in \mathcal{V}_j$. If $c_{\mathbf{v}'} \neq c^*$, we assign $\widetilde{A}_j(\mathbf{v}')$ arbitrarily. Otherwise, for each $\mathbf{v} \in N^*(\mathbf{v}')$, we denote

$$B(\mathbf{v}, \mathbf{v}') := \pi_{(\mathbf{v},\mathbf{v}')}(B(\mathbf{v})).$$

Recall from Step 4 that, the set $R_i \setminus B(\mathbf{v}) \subseteq R_i$ can be decomposed as the disjoint union of two sets $S_1, S_2 \in \binom{R_i}{r_i}$ such that $f(\mathbf{v}, S_1) = f(\mathbf{v}, S_2) = c^*$. For each $S \in \binom{R_j}{r_j}$ such that $B(\mathbf{v}, \mathbf{v}') \subseteq S$, by the construction in Step 2, the hyperedge

$$\big\{(\mathbf{v}, S_1), (\mathbf{v}, S_2), (\mathbf{v}', S)\big\}$$

is included in H. By the definition of $\mathcal{E}^*$, this hyperedge cannot be monochromatic under the coloring f, and therefore $f(\mathbf{v}', S) \neq c^*$. However, recall also from Step 4 that the fraction of $S \in \binom{R_j}{r_j}$ such that $f(\mathbf{v}', S) = c^*$ is at least $|R_j|^{-c-1}$. Therefore, at least an $|R_j|^{-c-1}$ fraction of $S \in \binom{R_j}{r_j}$ contain none of the sets $B(\mathbf{v}, \mathbf{v}')$, where $\mathbf{v}$ ranges in $N^*(\mathbf{v}')$. We can then apply Lemma C.3 to find[21] an element of R_j that is contained in $B(\mathbf{v}, \mathbf{v}')$ for at least

$$\eta = \frac{6^{-c}}{c(c+1)\ln|R_j|}$$

fraction of the tuples $\mathbf{v} \in N^*(\mathbf{v}')$. Define $\widetilde{A}_j(\mathbf{v}')$ to be such an element of R_j.

We now define a random assignment map $\widetilde{\mathcal{A}}_i : \mathcal{V}_i \to R_i$ by picking each $\widetilde{\mathcal{A}}_i(\mathbf{v})$ independently and uniformly at random from $B(\mathbf{v})$. By the definition of $B(\mathbf{v})$ (in Step 4), we have $|B(\mathbf{v})| \leqslant |R_i| - 2r_i \leqslant c$. Therefore, we have

$$\begin{aligned}
&\mathbb{E}_{\widetilde{\mathcal{A}}_i}\left[\mathop{\mathbb{P}}_{(\mathbf{v},\mathbf{v}')\in\mathcal{E}^*}\left[\pi_{(\mathbf{v},\mathbf{v}')}\big(\widetilde{\mathcal{A}}_i(\mathbf{v})\big) = \widetilde{A}_j(\mathbf{v}')\right]\right] \\
&= \mathop{\mathbb{E}}_{(\mathbf{v},\mathbf{v}')\in\mathcal{E}^*}\left[\mathbb{P}_{\widetilde{\mathcal{A}}_i}\left[\pi_{(\mathbf{v},\mathbf{v}')}\big(\widetilde{\mathcal{A}}_i(\mathbf{v})\big) = \widetilde{A}_j(\mathbf{v}')\right]\right] \\
&\geqslant \eta \cdot \mathop{\mathbb{E}}_{(\mathbf{v},\mathbf{v}')\in\mathcal{E}^*}\left[\mathbb{P}_{\widetilde{\mathcal{A}}_i}\left[\pi_{(\mathbf{v},\mathbf{v}')}\big(\widetilde{\mathcal{A}}_i(\mathbf{v})\big) = \widetilde{A}_j(\mathbf{v}')\right] \,\middle|\, \widetilde{A}_j(\mathbf{v}') \in B(\mathbf{v}, \mathbf{v}')\right] \qquad \text{(by the choice of } \widetilde{A}_j(\mathbf{v}')) \\
&\geqslant \eta \cdot \mathop{\mathbb{E}}_{(\mathbf{v},\mathbf{v}')\in\mathcal{E}^*}\left[\frac{1}{|B(\mathbf{v})|}\right] \geqslant \frac{\eta}{c}. \qquad \text{(by the definition of } B(\mathbf{v}, \mathbf{v}'))
\end{aligned}$$

Therefore, there exists a deterministic instantiation $\widetilde{A}_i : \mathcal{V}_i \to R_i$ of $\widetilde{\mathcal{A}}_i$ such that

$$\mathop{\mathbb{P}}_{(\mathbf{v},\mathbf{v}')\in\mathcal{E}^*}\left[\pi_{(\mathbf{v},\mathbf{v}')}\big(\widetilde{A}_i(\mathbf{v})\big) = \widetilde{A}_j(\mathbf{v}')\right] \geqslant \frac{\eta}{c}.$$

By Equation (C.4), it follows that

$$\mathop{\mathbb{P}}_{(\mathbf{v},\mathbf{v}')\in\mathcal{E}_{i,j}}\left[\pi_{(\mathbf{v},\mathbf{v}')}\big(\widetilde{A}_i(\mathbf{v})\big) = \widetilde{A}_j(\mathbf{v}')\right] \geqslant \frac{\eta}{4c^3}. \tag{C.5}$$

In the rest of the proof, we fix $\widetilde{A}_i$ to be an instantiation of $\widetilde{\mathcal{A}}_i$ that satisfies Equation (C.5).

[21] Note that for any $\mathbf{v} \in N^*(\mathbf{v}')$, we have $|B(\mathbf{v}, \mathbf{v}')| = \big|\pi_{(\mathbf{v},\mathbf{v}')}(B(\mathbf{v}))\big| \leqslant |B(\mathbf{v})| \leqslant |R_i| - 2r_i \leqslant c$.

Step 8: strategy for the original label cover. We define a random pair of (correlated) embedding maps $\varphi_0 : V_0 \hookrightarrow \mathcal{V}_i$ and $\varphi_1 : V_1 \hookrightarrow \mathcal{V}_j$ as follows:

1. For each $s \in \{1, 2, \ldots, i-1\}$, let $v_s = v'_s$ be a uniformly random element of V_1.
2. For each $s \in \{i+1, i+2, \ldots, j-1\}$, pick a pair $(v_s, v'_s) \in E$ uniformly at random.
3. For each $s \in \{j, j+1, \ldots, \ell-1\}$, let $v_s = v'_s$ be a uniformly random element of V_0.
4. For each $w \in V_0$, let $\varphi_0(w) = (v_1, \ldots, v_{i-1}, w, v_{i+1}, \ldots, v_{\ell-1}) \in \mathcal{V}_i$.
5. For each $w' \in V_1$, let $\varphi_1(w') = (v'_1, \ldots, v'_{i-1}, w', v'_{i+1}, \ldots, v'_{\ell-1}) \in \mathcal{V}_j$.

If we first sample the random maps φ_0, φ_1 and then independently sample a pair $(w, w') \in E$, it is clear that $(\varphi_0(w), \varphi_1(w'))$ is uniformly distributed over $\mathcal{E}_{i,j}$. Therefore, it follows from Equation (C.5) that

$$\mathop{\mathbb{E}}_{(\varphi_0,\varphi_1)} \left[\mathop{\mathbb{P}}_{(w,w')\in E} \left[\pi_{(\varphi_0(w),\varphi_1(w'))} \left(\widetilde{A}_i(\varphi_0(w)) \right) = \widetilde{A}_j(\varphi_1(w')) \right] \right] \geqslant \frac{\eta}{4c^3}. \tag{C.6}$$

We now define a random pair of (correlated) assignment maps $\mathcal{A}_0 : V_0 \to \Sigma_0$ and $\mathcal{A}_1 : V_1 \to \Sigma_1$ as follows:

1. Sample the pair of embedding maps φ_0, φ_1 as described earlier.
2. For each $w \in V_0$, let $\mathcal{A}_0(w)$ be the i-th coordinate of $\widetilde{A}_i(\varphi_0(w))$.
3. For each $w' \in V_1$, let $\mathcal{A}_1(w')$ be the i-th coordinate of $\widetilde{A}_j(\varphi_1(w'))$.

By the definition of $\pi_{(\varphi_0(w),\varphi_1(w'))}$ in Step 1, it follows from Equation (C.6) that

$$\mathop{\mathbb{E}}_{(\mathcal{A}_0,\mathcal{A}_1)} \left[\mathop{\mathbb{P}}_{(w,w')\in E} \left[\pi_{(w,w')} \big(\mathcal{A}_0(w) \big) = \mathcal{A}_1(w') \right] \right] \geqslant \frac{\eta}{4c^3}.$$

Therefore, there exists a deterministic instantiation (A_0, A_1) of the random pair of maps $(\mathcal{A}_0, \mathcal{A}_1)$ that satisfies at least $\eta/(4c^3)$ fraction of the constraints in Ψ. Thus we have

$$\mathrm{val}(\Psi) \geqslant \frac{\eta}{4c^3} \geqslant \frac{6^{-c}}{4c^4(c+1)\ln|R_j|} \geqslant \frac{6^{-c}}{4c^4(c+1)(\ell-1)\ln(7^u)} \geqslant \frac{1}{2^{20c} \cdot u} > 2^{-\gamma u},$$

where we used $u \geqslant 40\gamma^{-2}c$ in the last transition. □

The following combinatorial lemma (adapted from [DRS05, Claim 4.5]) is used in Step 7 of the preceding proof.

Lemma C.3. *Fix a positive integer c. Let R be a nonempty finite set with $|R| \geqslant 6c$, and let r be an integer such that $|R|/3 \leqslant r \leqslant 2|R|/3$. Suppose $B_1, \ldots, B_m \subseteq R$ are a sequence of subsets such that $|B_i| \leqslant c$ for each $i \in [m]$. If at least an $|R|^{-c-1}$ fraction of r-element subsets $S \in \binom{R}{r}$ contain none of the sets $B_1, \ldots, B_m$, then there exists an element $a \in R$ contained in at least*

$$\frac{6^{-c}}{c(c+1)\ln|R|}$$

fraction of the sets $B_1, \ldots, B_m$.

Proof. Let

$$\eta := \max_{a \in R} \frac{\left|\{i \in [m] \mid a \in B_i\}\right|}{m}.$$

Without loss of generality, suppose $B_1, \ldots, B_q$ are a maximum pairwise disjoint subfamily among the sets $B_1, \ldots, B_m$, for some integer $q \in [m]$. By maximality, every B_i contains some element of

$$U := B_1 \cup \cdots \cup B_q.$$

Therefore

$$m \leqslant \sum_{a \in U} \left|\{i \in [m] \mid a \in B_i\}\right| \leqslant |U| \cdot \eta m \leqslant cq \cdot \eta m,$$

and hence $\eta \geqslant 1/(cq)$.

Next we upper-bound q using the assumptions in the lemma. Let S be chosen uniformly at random from $\binom{R}{r}$. For each $j \in [q]$, we have

$$\mathop{\mathbb{P}}_{S \in \binom{R}{r}}[B_j \subseteq S] = \binom{|R| - |B_j|}{r - |B_j|} \Big/ \binom{|R|}{r} = \prod_{t=0}^{|B_j|-1} \frac{r-t}{|R|-t} \geqslant \left(\frac{1}{6}\right)^{|B_j|} \geqslant 6^{-c},$$

where we used the assumptions $r \geqslant |R|/3$ and $|B_j| \leqslant c \leqslant |R|/6$ in the third transition.

Because the sets $B_1, \ldots, B_q$ are pairwise disjoint, the events $\{B_j \subseteq S\}$, for $j \in [q]$, are negatively correlated under the uniform distribution over $\binom{R}{r}$. Hence

$$\mathop{\mathbb{P}}_{S \in \binom{R}{r}}\left[B_j \nsubseteq S \text{ for all } j \in [q]\right] \leqslant \prod_{j=1}^{q} \mathop{\mathbb{P}}_{S \in \binom{R}{r}}[B_j \nsubseteq S] \leqslant (1 - 6^{-c})^q.$$

Therefore, by the assumption in the lemma, we have

$$|R|^{-c-1} \leqslant \mathop{\mathbb{P}}_{S \in \binom{R}{r}}\left[B_i \nsubseteq S \text{ for all } i \in [m]\right] \leqslant \mathop{\mathbb{P}}_{S \in \binom{R}{r}}\left[B_j \nsubseteq S \text{ for all } j \in [q]\right] \leqslant (1 - 6^{-c})^q.$$

Taking logarithms gives

$$q \leqslant \frac{(c+1)\ln|R|}{-\ln(1-6^{-c})} \leqslant 6^c(c+1)\ln|R|.$$

Combining this with $\eta \geqslant 1/(cq)$, we obtain the desired inequality

$$\eta \geqslant \frac{6^{-c}}{c(c+1)\ln|R|}.$$

□

D Grassmann Decoding Theorems

In this appendix, we explain how the Grassmann decoding theorems (Theorems 3.5 and 3.6) used in this paper can be derived from the slightly different versions in the existing literature.

D.1 The Expansion Theorem

To prove Grassmann decoding theorems, the standard approach is to first establish some small-set expansion properties of the Grassmann graph. It is observed in [KMS25, DKK+25] that small vertex sets may not expand very well in the Grassmann graph, and the correct approach is to consider sets that remain small even after zoom-ins and zoom-outs. This is formalized in the next definition.

Definition D.1. Let n, ℓ, q be nonnegative integers such that $n - q \geqslant \ell \geqslant q + 1$, and let $\varepsilon \in (0, 1)$. Suppose $\mathcal{L} \subseteq \mathsf{Gr}(\mathbb{F}_2^n, \ell)$ is a set of subspaces. We say that $\mathcal{L}$ is (q, ε)-*pseudorandom* if for any subspaces $Q \in \mathsf{Gr}(\mathbb{F}_2^n, q)$ and $W \in \mathsf{Gr}(\mathbb{F}_2^n, n - q)$ such that $Q \subseteq W$, we have

$$\mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)} \left[L \in \mathcal{L} \,\middle|\, Q \subseteq L \subseteq W \right] \leqslant \varepsilon.$$

To facilitate Fourier analysis, it is useful to "lift" the Grassmann graph to a Cayley graph over an Abelian group. The "short-code graph" is a Cayley graph that serves this purpose very well, with the matrix space $\mathbb{F}_2^{n \times \ell}$ as the additive group and the rank-1 matrices as the generating set. The counterpart of Definition D.1 in the short-code graph is given as follows.

Definition D.2. Let n, ℓ, q be nonnegative integers such that $n - q \geqslant \ell \geqslant q + 1$, and let $\varepsilon \in (0, 1)$. A function $f : \mathbb{F}_2^{n \times \ell} \to \{0, 1\}$ is said to be (q, ε)-*pseudorandom* if for any matrices

$$Y_1 \in \mathbb{F}_2^{\ell \times q}, \qquad Z_1 \in \mathbb{F}_2^{n \times q}, \qquad Y_2 \in \mathbb{F}_2^{q \times n} \qquad \text{and} \qquad Z_2 \in \mathbb{F}_2^{q \times \ell},$$

such that $\left\{ X \in \mathbb{F}_2^{n \times \ell} \,\middle|\, XY_1 = Z_1 \text{ and } Y_2 X = Z_2 \right\}$ is nonempty, we have

$$\mathop{\mathbb{E}}_{X \in \mathbb{F}_2^{n \times \ell}} \left[f(X) \,\middle|\, XY_1 = Z_1 \text{ and } Y_2 X = Z_2 \right] \leqslant \varepsilon.$$

It is not hard to relate Definitions D.1 and D.2 together; we use the following lemma from [MZ26].

Lemma D.3 ([MZ26, Lemma A.18])**.** *Let n, ℓ, q be nonnegative integers such that $n - q \geqslant \ell \geqslant q + 1$, and let $\varepsilon \in (0, 1)$. Suppose $\mathcal{L} \subseteq \mathsf{Gr}(\mathbb{F}_2^n, \ell)$ is (q, ε)-pseudorandom (in the sense of Definition D.1). Define a function $f : \mathbb{F}_2^{n \times \ell} \to \{0, 1\}$ by*

$$f(X) = \begin{cases} 1, & \text{if } \operatorname{rank}(X) = \ell \text{ and } \operatorname{col}(X) \in \mathcal{L}, \\ 0, & \text{otherwise.} \end{cases}$$

Then f is $(q, 4\varepsilon)$-pseudorandom (in the sense of Definition D.2).

The adjacency matrix of the short-code graph is represented by the following linear operator.

Definition D.4. Given a function $f : \mathbb{F}_2^{n \times \ell} \to \mathbb{R}$, we let $\mathcal{T} f : \mathbb{F}_2^{n \times \ell} \to \mathbb{R}$ be the function defined by

$$\mathcal{T} f(X) = \mathop{\mathbb{E}}_{Y_1 \in \mathbb{F}_2^{n \times 1},\, Y_2 \in \mathbb{F}_2^{1 \times \ell}} \left[f(X + Y_1 Y_2) \right]. \tag{D.1}$$

We are now ready to state the main expansion theorem from [KMS23, EKL23], which shows that pseudorandom sets have good expansion in the short-code graph.

Lemma D.5 ([EKL23, Theorem 69])**.** *There is an absolute constant $C > 0$ such that the following holds. Let n, ℓ, q be nonnegative integers such that $n - q \geqslant \ell \geqslant q + 1$. Then any $(q, 2^{-Cq^2})$-pseudorandom function $f : \mathbb{F}_2^{n \times \ell} \to \{0, 1\}$ satisfies*[22]

$$\langle f, \mathcal{T} f \rangle \leqslant 2^{-q+1} \mathbb{E}[f].$$

We then transfer the expansion theorem from the short-code graph to the Grassmann graph.

[22] Here the inner product $\langle f, \mathcal{T} f \rangle$ should be interpreted as $\mathbb{E}_{X \in \mathbb{F}_2^{n \times \ell}} [f(X) \mathcal{T} f(X)]$.

Lemma D.6. *There is an absolute constant $C > 0$ such that the following holds. Let n, ℓ, q be nonnegative integers such that $n - q - 3 \geqslant \ell \geqslant q + 3$. Then any $(q, 2^{-Cq^2})$-pseudorandom subset $\mathcal{L} \subseteq \mathsf{Gr}(\mathbb{F}_2^n, \ell)$ satisfies*

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)} \left[L_1, L_2 \in \mathcal{L}\right] \leqslant 2^{-q+2} \cdot \mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)} \left[L \in \mathcal{L}\right].$$

Proof. If $q = 0$ the conclusion obviously holds. We assume $q \geqslant 1$, and let C be the constant from Lemma D.5. Suppose $\mathcal{L} \subseteq \mathsf{Gr}(\mathbb{F}_2^n, \ell)$ is $(q, 2^{-(C+2)q^2})$-pseudorandom. Define a function $f : \mathbb{F}_2^{n\times\ell} \to \{0, 1\}$ from $\mathcal{L}$ as in Lemma D.3. By Lemma D.3 we know that f is $(q, 2^{-Cq^2})$-pseudorandom (in the sense of Definition D.2).

If $X \in \mathbb{F}_2^{n\times\ell}$, $Y_1 \in \mathbb{F}_2^{n\times 1}$ and $Y_2 \in \mathbb{F}_2^{1\times\ell}$ are independent and uniformly random, then conditioned on the event

$$\{\mathrm{rank}(X) = \ell\} \cap \{Y_1 \notin \mathrm{col}(X)\} \cap \{Y_2 \neq 0\}, \tag{D.2}$$

the pair $\big(\mathrm{col}(X), \mathrm{col}(X + Y_1Y_2)\big)$ is a uniformly random edge of the Grassmann graph $\mathsf{Gr}(\mathbb{F}_2^n, \ell)$. Note that the probability of the event Equation (D.2) is at least $1 - 2^{\ell-n} - 2^{\ell-n} - 2^{-\ell} \geqslant 1/2$. Therefore, we have

$$\mathop{\mathbb{P}}_{X\in\mathbb{F}_2^{n\times\ell}, Y_1\in\mathbb{F}_2^{n\times 1}, Y_2\in\mathbb{F}_2^{1\times\ell}} \left[f(X) = f(X + Y_1Y_2) = 1\right] \geqslant \frac{1}{2} \cdot \mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)} \left[L_1, L_2 \in \mathcal{L}\right] \tag{D.3}$$

Since the left-hand side of Equation (D.3) is exactly $\langle f, \mathcal{T}f\rangle$, using Lemma D.5 we can conclude that

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)} \left[L_1, L_2 \in \mathcal{L}\right] \leqslant 2\langle f, \mathcal{T}f\rangle \leqslant 2^{-q+2} \cdot \mathbb{E}\left[f\right] \leqslant 2^{-q+2} \cdot \mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)} \left[L \in \mathcal{L}\right]. \qquad \square$$

D.2 From Expansion to Decoding

We now use an argument, originally due to [BKS19], to deduce the decoding statement from the expansion statement.

Theorem D.7. *There is an absolute constant $C > 0$ such that the following holds. Let n, ℓ, q be nonnegative integers such that $n-q-3 \geqslant \ell \geqslant q+3$. Suppose $F[\cdot]$ is a table where $F[L] \in L^* \cup \{\mathtt{nil}\}$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, such that*

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)} \left[F[L_1]|_{L_1\cap L_2} = F[L_2]|_{L_1\cap L_2}\right] \geqslant 2^{-q+4}. \tag{D.4}$$

Then there exist a linear map $f : \mathbb{F}_2^n \to \mathbb{F}_2$ and subspaces $Q \in \mathsf{Gr}(\mathbb{F}_2^n, q)$, $W \in \mathsf{Gr}(\mathbb{F}_2^n, n - q)$ such that $Q \subseteq W$ and

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)} \left[F[L] = f|_L \;\middle|\; Q \subseteq L \subseteq W\right] \geqslant 2^{-Cq^2}.$$

Proof. For each linear functional $f : \mathbb{F}_2^n \to \mathbb{F}_2$, let $\mathcal{L}_f$ denote the set of subspaces $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$ such that $F[L] = f|_L$. We have

$$\mathop{\mathbb{E}}_{f\in(\mathbb{F}_2^n)^*} \left[\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)} \left[L \in \mathcal{L}_f\right]\right] = \mathop{\mathbb{E}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)} \left[\mathop{\mathbb{P}}_{f\in(\mathbb{F}_2^n)^*} \left[f|_L = F[L]\right]\right] \leqslant 2^{-\ell}. \tag{D.5}$$

Furthermore,

$$\mathop{\mathbb{E}}_{f\in(\mathbb{F}_2^n)^*} \left[\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)} \left[L_1, L_2 \in \mathcal{L}_f\right]\right] = \mathop{\mathbb{E}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)} \left[\mathop{\mathbb{P}}_{f\in(\mathbb{F}_2^n)^*} \left[f|_{L_1} = F[L_1] \text{ and } f|_{L_2} = F[L_2]\right]\right]. \tag{D.6}$$

Note that for any edge (L_1, L_2) of the Grassmann graph $\mathsf{Gr}(\mathbb{F}_2^n, \ell)$ such that

$$F[L_1]|_{L_1 \cap L_2} = F[L_2]|_{L_1 \cap L_2},$$

we have

$$\mathop{\mathbb{P}}_{f \in (\mathbb{F}_2^n)^*} \left[f|_{L_1} = F[L_1] \text{ and } f|_{L_2} = F[L_2] \right] = 2^{-\ell-1}.$$

Therefore, combining Equations (D.4) and (D.6) yields

$$\mathop{\mathbb{E}}_{f \in (\mathbb{F}_2^n)^*} \left[\mathop{\mathbb{P}}_{(L_1, L_2) \sim \mathcal{G}(\mathbb{F}_2^n, \ell)} \left[L_1, L_2 \in \mathcal{L}_f \right] \right] \geqslant 2^{-\ell-1} \cdot 2^{-q+4} = 2^{-q-\ell+3}. \tag{D.7}$$

Combining Equations (D.5) and (D.7), we know that there exists a linear functional $f \in (\mathbb{F}_2^n)^*$ such that

$$\mathop{\mathbb{P}}_{(L_1, L_2) \sim \mathcal{G}(\mathbb{F}_2^n, \ell)} \left[L_1, L_2 \in \mathcal{L}_f \right] > 2^{-q+2} \cdot \mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)} \left[L \in \mathcal{L}_f \right].$$

It then follows from Lemma D.6 that $\mathcal{L}_f$ is not $(q, 2^{-Cq^2})$-pseudorandom, which is exactly the desired conclusion. □

We now restate and prove the precise decoding result stated in Section 3.3.2. It is almost identical to Theorem D.7, except that we require the feasible zoom-ins Q to be abundant.

Theorem 3.5 ([KMS23, EKL23])**.** *There is an absolute constant $C > 0$ such that the following holds. Let n, ℓ, q be nonnegative integers such that $n - q - 3 \geqslant \ell \geqslant q + 3$. Suppose $F[\cdot]$ is a table where $F[L] \in L^* \cup \{\mathtt{nil}\}$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, such that*

$$\mathop{\mathbb{P}}_{(L_1, L_2) \sim \mathcal{G}(\mathbb{F}_2^n, \ell)} \left[F[L_1]|_{L_1 \cap L_2} = F[L_2]|_{L_1 \cap L_2} \right] \geqslant 2^{-q+5}. \tag{3.1}$$

Then for at least $2^{-q-\ell^2}$ fraction of subspaces $Q \in \mathsf{Gr}(\mathbb{F}_2^n, q)$, there exist a subspace $W \in \mathsf{Gr}(\mathbb{F}_2^n, n-q)$ containing Q and a linear map $f : W \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)} \left[F[L] = f|_L \,\middle|\, Q \subseteq L \subseteq W \right] \geqslant 2^{-Cq^2}.$$

Proof. We take C to be the same constant as in Theorem D.7. Let $\mathcal{Q}$ be the collection of subspaces $Q \in \mathsf{Gr}(\mathbb{F}_2^n, q)$ that satisfy the desired property, and we will prove that

$$|\mathcal{Q}| \geqslant 2^{-q-\ell^2} |\mathsf{Gr}(\mathbb{F}_2^n, q)|.$$

In the rest of the proof, we assume on the contrary that $|\mathcal{Q}| < 2^{-q-\ell^2} |\mathsf{Gr}(\mathbb{F}_2^n, q)|$.

We define a table $F'[\cdot]$, where $F'[L] \in L^* \cup \{\mathtt{nil}\}$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, as follows:

$$F'[L] = \begin{cases} \mathtt{nil}, & \text{if } Q \subseteq L \text{ for some } Q \in \mathcal{Q}, \\ F[L], & \text{otherwise.} \end{cases}$$

We thus have

$$\begin{aligned} \mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)} \left[F[L] \neq F'[L] \right] &\leqslant \sum_{Q \in \mathcal{Q}} \mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)} \left[Q \subseteq L \right] = |\mathcal{Q}| \cdot \begin{bmatrix} n-q \\ \ell-q \end{bmatrix}_2 \begin{bmatrix} n \\ \ell \end{bmatrix}_2^{-1} \\ &= |\mathcal{Q}| \cdot \begin{bmatrix} n \\ q \end{bmatrix}_2^{-1} \begin{bmatrix} \ell \\ q \end{bmatrix}_2 \leqslant 2^{-q-\ell^2} \cdot 2^{\ell^2} = 2^{-q}. \end{aligned}$$

Therefore, it follows from the assumption Equation (3.1) that

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)}\left[F'[L_1]\big|_{L_1\cap L_2} = F'[L_2]\big|_{L_1\cap L_2}\right] \geqslant 2^{-q+5} - 2\cdot 2^{-q} \geqslant 2^{-q+4}.$$

Applying Theorem D.7 to the table F' yields subspaces $Q \in \mathsf{Gr}(\mathbb{F}_2^n, q)$, $W \in \mathsf{Gr}(\mathbb{F}_2^n, n-q)$ such that $Q \subseteq W$ and

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[F'[L] = f|_L \;\middle|\; Q \subseteq L \subseteq W\right] \geqslant 2^{-Cq^2}. \tag{D.8}$$

By the definition of $\mathcal{Q}$, this implies that $Q \in \mathcal{Q}$. However, as $F'[L] = \texttt{nil}$ for any $Q \in \mathcal{Q}$, the left-hand side of Equation (D.8) must be 0, which is a contradiction. □

We now move on to the second decoding theorem we need. The following lemma says that if a function F on the vertices of the Grassmann graph is locally constant on its edges, then it is globally constant inside some zoom-in/zoom-out combination.

Lemma D.8. *There is an absolute constant $C > 0$ such that the following holds. Let n, ℓ, q be nonnegative integers such that $n - q - 3 \geqslant \ell \geqslant q + 3$. Suppose Σ is a finite set and $F[\cdot]$ is a table where $F[L] \in \Sigma \cup \{\texttt{nil}\}$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, such that*

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)}\left[F[L_1] = F[L_2] \neq \texttt{nil}\right] \geqslant 2^{-q+3}. \tag{D.9}$$

Then there exist an element $\sigma \in \Sigma$ and subspaces $Q \in \mathsf{Gr}(\mathbb{F}_2^n, q)$, $W \in \mathsf{Gr}(\mathbb{F}_2^n, n-q)$ such that $Q \subseteq W$ and

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[F[L] = \sigma \;\middle|\; Q \subseteq L \subseteq W\right] \geqslant 2^{-Cq^2}.$$

Proof. For each element $\sigma \in \Sigma$, let $\mathcal{L}_\sigma$ denote the set of subspaces $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$ such that $F[L] = \sigma$. We have

$$\mathop{\mathbb{E}}_{\sigma\in\Sigma}\left[\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[L \in \mathcal{L}_\sigma\right]\right] = \mathop{\mathbb{E}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[\mathop{\mathbb{P}}_{\sigma\in\Sigma}\left[\sigma = F[L]\right]\right] \leqslant \frac{1}{|\Sigma|}. \tag{D.10}$$

Furthermore, due to the assumption of the lemma, we have

$$\begin{aligned}\mathop{\mathbb{E}}_{\sigma\in\Sigma}\left[\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)}\left[L_1, L_2 \in \mathcal{L}_\sigma\right]\right] &= \mathop{\mathbb{E}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)}\left[\mathop{\mathbb{P}}_{\sigma\in\Sigma}\left[\sigma = F[L_1] = F[L_2]\right]\right] \\ &= \frac{1}{|\Sigma|}\cdot\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)}\left[F[L_1] = F[L_2] \neq \texttt{nil}\right] \geqslant \frac{2^{-q+3}}{|\Sigma|}.\end{aligned} \tag{D.11}$$

Therefore, combining Equations (D.10) and (D.11), we know that there exists an element $\sigma \in \Sigma$ such that

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)}\left[L_1, L_2 \in \mathcal{L}_\sigma\right] > 2^{-q+2}\cdot\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[L \in \mathcal{L}_\sigma\right].$$

It then follows from Lemma D.6 that $\mathcal{L}_\sigma$ is not $(q, 2^{-Cq^2})$-pseudorandom, which is exactly the desired conclusion. □

Our second decoding theorem (restated below) is a similar variant of Lemma D.8 with "abundant zoom-ins."

Theorem 3.6. *There is an absolute constant $C > 0$ such that the following holds. Let n, ℓ, q be nonnegative integers such that $n - q - 3 \geqslant \ell \geqslant q + 3$. Suppose Σ is a finite set and $F[\cdot]$ is a table where $F[L] \in \Sigma \cup \{\mathtt{nil}\}$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, such that*

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell)} \Big[F[L_1] = F[L_2] \neq \mathtt{nil}\Big] \geqslant 2^{-q+4}. \tag{3.2}$$

Then for at least $2^{-q-\ell^2}$ fraction of subspaces $Q \in \mathsf{Gr}(\mathbb{F}_2^n, q)$, there exist an element $\sigma \in \Sigma$ and a subspace $W \in \mathsf{Gr}(\mathbb{F}_2^n, n-q)$ containing Q such that

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)} \Big[F[L] = \sigma \,\Big|\, Q \subseteq L \subseteq W\Big] \geqslant 2^{-Cq^2}.$$

Proof. One can use Lemma D.8 to prove Theorem 3.6 in an entirely analogous way to how Theorem D.7 is used to prove Theorem 3.5. □

E Linearity Testing and List Decoding Bounds

The following lemma is from [KMS25, Theorem B.1] (see also [DKK+25, Lemma 3.1]).

Lemma E.1 ([KMS25, DKK+25])**.** *There is an absolute constant $C > 0$ such that the following holds. Let n, ℓ, b be positive integers such that $n \geqslant 2\ell$ and $40 \leqslant b \leqslant \ell/4$. Suppose $F[\cdot]$ is a table where $F[L] \in L^*$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, such that*

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell,b)} \Big[F[L_1]|_{L_1\cap L_2} = F[L_2]|_{L_1\cap L_2}\Big] \geqslant \varepsilon,$$

for some parameter $\varepsilon \in [2^{4-b/4}, 1]$. Then there exists a linear map $f : \mathbb{F}_2^n \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)} \Big[F[L] = f|_L\Big] \geqslant C^{-1}\varepsilon^3.$$

The following lemma is a slight variant where some entries are allowed to be $\mathtt{nil}$.

Lemma E.2. *There is an absolute constant $C \geqslant 40$ such that the following holds. Let n, ℓ, b be positive integers such that $n \geqslant 2\ell$ and $C \leqslant b \leqslant \ell/4$. Suppose $F[\cdot]$ is a table where $F[L] \in L^* \cup \{\mathtt{nil}\}$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, such that*

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell,b)} \Big[F[L_1]|_{L_1\cap L_2} = F[L_2]|_{L_1\cap L_2}\Big] \geqslant \varepsilon, \tag{E.1}$$

for some parameter $\varepsilon \in [2^{4-b/4}, 1]$. Then there exists a linear map $f : \mathbb{F}_2^n \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)} \Big[F[L] = f|_L\Big] \geqslant C^{-1}\varepsilon^3.$$

Proof. We define a random table $F'[\cdot]$ as follows: for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$,

(1) if $F[L] \neq \mathtt{nil}$, let $F'[L] = F[L]$;

(2) if $F[L] = \mathtt{nil}$, pick a uniformly random linear functional $h \in L^*$ and let $F'[L] = h$.

Therefore, for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$ the entry $F'[L]$ always takes value in the space L^*.

For any fixed linear map $f : \mathbb{F}_2^n \to \mathbb{F}_2$, the random variables

$$Z_f(L) = \mathbb{1}\left[F'[L] = f|_L \text{ and } F[L] = \mathtt{nil}\right] \qquad \text{for } L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$$

take values in $\{0,1\}$, have expected values at most $2^{-\ell}$, and are mutually independent. Therefore, by Chernoff bound we have

$$\begin{aligned}\mathop{\mathbb{P}}_{F'}\left[\frac{1}{|\mathsf{Gr}(\mathbb{F}_2^n,\ell)|}\sum_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)} Z_f(L) \geqslant 2^{1-\ell}\right] &\leqslant \exp\left(-\frac{1}{3}\cdot 2^{-\ell}\cdot|\mathsf{Gr}(\mathbb{F}_2^n,\ell)|\right)\\ &\leqslant \exp\left(-\frac{1}{3}\cdot 2^{\ell(n-\ell-1)}\right) < 2^{-n},\end{aligned}$$

where in the last transition we used the condition that $n \geqslant 2\ell \geqslant 320$. By a union bound over all linear maps $f : \mathbb{F}_2^n \to \mathbb{F}_2$, it follows that with nonzero probability over F' we have

$$\mathop{\mathbb{E}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[Z_f(L)\right] \leqslant 2^{1-\ell}.$$

In other words, there exists a realization of the random table $F'[\cdot]$, which we denote by $F_0[\cdot]$, such that

(1) for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, we have $F_0[L] \in L^*$;

(2) for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, we have $F_0[L] = F[L]$ whenever $F[L] \neq \mathtt{nil}$;

(3) $\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[F_0[L] = f|_L \text{ and } F[L] = \mathtt{nil}\right] \leqslant 2^{1-\ell}$ for any linear map $f : \mathbb{F}_2^n \to \mathbb{F}_2$.

Since $F_0[\cdot]$ only replaces the $\mathtt{nil}$ values in $F[\cdot]$, it is clear that Equation (E.1) also holds for F_0, i.e.

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell,b)}\left[F_0[L_1]|_{L_1\cap L_2} = F_0[L_2]|_{L_1\cap L_2}\right] \geqslant \varepsilon,$$

and hence we may apply Lemma E.1 to obtain a linear map $f : \mathbb{F}_2^n \to \mathbb{F}_2$ such that

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[F_0[L] = f|_L\right] \geqslant C_0^{-1}\varepsilon^3,$$

for some absolute constant $C_0 > 0$. Combining this with the second and third properties of $F_0[\cdot]$ listed above, it follows that

$$\mathop{\mathbb{P}}_{L\in\mathsf{Gr}(\mathbb{F}_2^n,\ell)}\left[F[L] = f|_L\right] \geqslant C_0^{-1}\varepsilon^3 - 2^{1-\ell} \geqslant (2C_0)^{-1}\varepsilon^3,$$

where the last transition holds as long as $\varepsilon^3 \cdot 2^\ell$ is sufficiently large. □

The next corollary is a version of Lemma E.2 that also takes zoom-ins into account.

Corollary 6.66. *There is an absolute constant $C \geqslant 40$ such that the following holds. Fix positive integers n, ℓ, b and a nonnegative integer q such that $n \geqslant 2\ell$ and $C(q+1) \leqslant b \leqslant \ell/4$. Let $Q \in \mathsf{Gr}(\mathbb{F}_2^n, q)$ be a subspace. Suppose $F[\cdot]$ is a table where $F[L] \in L^* \cup \{\mathtt{nil}\}$ for each $L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)$, such that*

$$\mathop{\mathbb{P}}_{(L_1,L_2)\sim\mathcal{G}(\mathbb{F}_2^n,\ell,b)}\left[F[L_1]|_{L_1\cap L_2} = F[L_2]|_{L_1\cap L_2} \,\middle|\, Q \subseteq L_1, L_2\right] \geqslant \varepsilon, \tag{6.33}$$

for some parameter $\varepsilon \in [2^{-b/8}, 1]$. *Then there exists a linear map* $f : \mathbb{F}_2^n \to \mathbb{F}_2$ *such that*

$$\mathop{\mathbb{P}}_{L \in \mathsf{Gr}(\mathbb{F}_2^n, \ell)} \left[F[L] = f|_L \,\middle|\, Q \subseteq L \right] \geqslant C^{-1} \cdot 2^{-3q} \varepsilon^3.$$

Proof. Since there are exactly 2^q linear functionals $Q \to \mathbb{F}_2$, there exists a linear functional $h \in Q^*$ such that

$$\mathop{\mathbb{P}}_{(L_1, L_2) \sim \mathcal{G}(\mathbb{F}_2^n, \ell, b)} \left[F[L_1]|_{L_1 \cap L_2} = F[L_2]|_{L_1 \cap L_2} \text{ and } F[L_1]|_Q = h \,\middle|\, Q \subseteq L_1, L_2 \right] \geqslant 2^{-q} \varepsilon, \tag{E.2}$$

Let $Q^{\mathsf{c}} \in \mathsf{Gr}(\mathbb{F}_2^n, n-q)$ be a linear subspace such that $Q \cap Q^{\mathsf{c}} = \{0\}$. Define a table $F'[\cdot]$ as follows: for each $L \in \mathsf{Gr}(Q^{\mathsf{c}}, \ell - q)$,

(1) if $F[L+Q] \neq \texttt{nil}$ and $F[L+Q]|_Q = h$, let $F'[L] = F[L+Q]|_L$;

(2) if $F[L+Q] = \texttt{nil}$ or $F[L+Q]|_Q \neq h$, let $F'[L] = \texttt{nil}$.

It then follows from Equation (E.2) that

$$\mathop{\mathbb{P}}_{(L_1, L_2) \sim \mathcal{G}(Q^{\mathsf{c}}, \ell - q, b - q)} \left[F'[L_1]\big|_{L_1 \cap L_2} = F'[L_2]\big|_{L_1 \cap L_2} \right] \geqslant 2^{-q} \varepsilon.$$

Applying Lemma E.2 to the table $F'[\cdot]$ and then translating the conclusion back to $F[\cdot]$ yields the desired result. □

F Covering Properties

Several covering-property-type statements were left unproved in Sections 4 to 6. In this section, we complete their proofs.

F.1 Covering in the Outer PCP

We first prove Proposition 4.7 (restated below), which analyzes the distribution of E' in a sample $(E', A_{\mathsf{in}}, A_{\mathsf{mid}}) \sim \mathcal{J}_{\mathsf{out}}(E, q_1, q_2, \beta)$ conditioned on $(A_{\mathsf{in}}, A_{\mathsf{mid}})$.

Proposition 4.7. Let E be a tripled set. Fix nonnegative integers q_1, q_2 and a parameter $\beta \in [0, 1)$. Fix arbitrary tuples

$$A_1 \in (\mathbb{F}_2^E)^{q_1} \qquad \text{and} \qquad A_2 \in (\mathbb{F}_2^E)^{q_2}.$$

Sample

$$(E', A_{\mathsf{in}}, A_{\mathsf{mid}}) \sim \mathcal{J}_{\mathsf{out}}(E, q_1, q_2, \beta)$$

conditioned on the event $\{A_{\mathsf{in}} = A_1\} \cap \{A_{\mathsf{mid}} = A_2\}$. Then the resulting marginal distribution of E' is independent over triples of E. Furthermore, each triple of E is entirely included in E' with conditional probability at least $1 - 4^{q_1+q_2}\beta$.

Proof. We write $q = q_1 + q_2$. Without loss of generality, we assume

$$E = [3k] \qquad \text{and} \qquad \mathrm{Triples}(E) = \big\{ \{3i-2, 3i-1, 3i\} \,\big|\, i \in [k] \big\}$$

for some positive integer k.

Let $P \subseteq E$ be the union of the supports of all vectors appearing in the fixed advice sequences A_1 and A_2. For each $i \in [k]$, let

$$P_i = \big\{ j \in \{1,2,3\} \,\big|\, 3i - 3 + j \in P \big\}.$$

Consider a sample

$$(E', A_{\mathsf{in}}, A_{\mathsf{mid}}) \sim \mathcal{J}_{\mathsf{out}}([3k], q_1, q_2, \beta)$$

conditioned on the event $\{A_{\mathsf{in}} = A_1\} \cap \{A_{\mathsf{mid}} = A_2\}$. Under this conditioning, the marginal distribution of E' is independent over the triples of E: for every $i \in [k]$,

(1) if $|P_i| \geqslant 2$, then $3i-2, 3i-1$ and $3i$ are all included in E' with probability 1;

(2) if P_i is a singleton set $\{j\}$, then $3i-2, 3i-1$ and $3i$ are all included in E' with probability $1 - \eta_1$, and with probability η_1 only $3i - 3 + j$ is included in E', where

$$\eta_1 = \frac{(\beta/3)(1-2^{-q})}{(\beta/3)(1-2^{-q}) + (1-\beta)(1-2^{-q})2^{-2q}} = \frac{2^{2q}\beta}{2^{2q}\beta + 3(1-\beta)} \leqslant 2^{2q}\beta; \tag{F.1}$$

(3) if $P_i = \emptyset$, then $3i-2, 3i-1$ and $3i$ are all included in E' with probability $1 - \eta_0$, and with probability η_0 only (a uniformly random) one of them is included in E', where

$$\eta_0 = \frac{\beta \cdot 2^{-q}}{\beta \cdot 2^{-q} + (1-\beta)2^{-3q}} = \frac{2^{2q}\beta}{2^{2q}\beta + (1-\beta)} \leqslant 2^{2q}\beta. \tag{F.2}$$

In each of the above three cases, the triple $\{3i-2, 3i-1, 3i\}$ is included entirely in E' with probability at least $1 - 2^{2q}\beta = 1 - 4^{q_1+q_2}\beta$. □

Building on the proof of Proposition 4.7, we can now prove the main covering property needed for the middle PCP (Proposition 4.8, restated below), which asserts that the distribution of A_{mid} is close to uniform over Alice's space even when conditioned on A_{in} (similar arguments appeared in [KS13, KMS25]).

Proposition 4.8. Let E be a tripled set. Fix nonnegative integers q_1, q_2 and a parameter $\beta \in [0, 4^{-q_1-q_2})$. Fix an arbitrary tuple $A_1 \in (\mathbb{F}_2^E)^{q_1}$. Sample

$$(E', A_{\mathsf{in}}, A_{\mathsf{mid}}) \sim \mathcal{J}_{\mathsf{out}}(E, q_1, q_2, \beta)$$

conditioned on the event $\{A_{\mathsf{in}} = A_1\}$. Then the resulting marginal distribution of A_{mid} has total variation distance at most $4^{q_1+q_2}\beta\sqrt{|E|/3}$ to the uniform distribution over $(\mathbb{F}_2^E)^{q_2}$.

Proof. As in the proof of Proposition 4.7, we assume

$$E = [3k] \qquad \text{and} \qquad \mathrm{Triples}(E) = \big\{ \{3i-2, 3i-1, 3i\} \,\big|\, i \in [k] \big\}$$

for some positive integer k. Let $P \subseteq E$ be the union of the supports of all vectors appearing in the fixed advice sequence A_1. For each $i \in [k]$, let $P_i = \big\{ j \in \{1,2,3\} \,\big|\, 3i - 3 + j \in P \big\}$. We organize the proof into the following two steps.

Step 1: description of the distribution. Similarly to Equations (F.1) and (F.2), we let

$$\eta_1 = \frac{2^{2q_1}\beta}{2^{2q_1}\beta + 3(1-\beta)} \leqslant 2^{2q_1}\beta \qquad \text{and} \qquad \eta_0 = \frac{2^{2q_1}\beta}{2^{2q_1}\beta + (1-\beta)} \leqslant 2^{2q_1}\beta.$$

Let $\mathcal{D}_0$ be the uniform distribution over

$$(\mathbb{F}_2^{q_2})^3 = \left\{ \left(x^{(1)}, x^{(2)}, x^{(3)}\right) \,\middle|\, x^{(1)}, x^{(2)}, x^{(3)} \text{ are vectors in } \mathbb{F}_2^{q_2} \right\}.$$

For each $j \in [3]$, let $\mathcal{D}_j$ be the uniform distribution over the subset

$$\left\{ \left(x^{(1)}, x^{(2)}, x^{(3)}\right) \in (\mathbb{F}_2^{q_2})^3 \,\middle|\, x^{(i)} = 0 \text{ for all } i \in [3] \setminus \{j\} \right\} \subseteq (\mathbb{F}_2^{q_2})^3.$$

For each $i \in [k]$, we define a distribution $\mathcal{D}^{(i)}$ over $(\mathbb{F}_2^{\ell'})^3$ as follows (according to the conditional distribution of $E' \cap \{3i-2, 3i-1, 3i\}$ described in the proof of Proposition 4.7):

(1) if $|P_i| \geqslant 2$, let $\mathcal{D}^{(i)} = \mathcal{D}_0$;

(2) if P_i is a singleton set $\{j\}$, let $\mathcal{D}^{(i)}$ be the mixture $(1-\eta_1)\mathcal{D}_0 + \eta_1\mathcal{D}_j$;

(3) if $P_i = \emptyset$, let $\mathcal{D}^{(i)}$ be the mixture $(1-\eta_0)\mathcal{D}_0 + \frac{1}{3}\eta_0\mathcal{D}_1 + \frac{1}{3}\eta_0\mathcal{D}_2 + \frac{1}{3}\eta_0\mathcal{D}_3$.

It is straightforward to verify that in each of the above three cases, for any tuple $X \in (\mathbb{F}_2^{q_2})^3$ we have

$$\left| \sqrt{\mathcal{D}^{(i)}(X)/\mathcal{D}_0(X)} - 1 \right| \leqslant \left| \mathcal{D}^{(i)}(X)/\mathcal{D}_0(X) - 1 \right| \leqslant 2^{2q_2} \max\{\eta_0, \eta_1\} \leqslant 2^{2q_2+2q_1}\beta. \tag{F.3}$$

Now consider a sample

$$(E', A_{\mathsf{in}}, A_{\mathsf{mid}}) \sim \mathcal{J}_{\mathsf{out}}([3k], q_1, q_2, \beta)$$

conditioned on the event $\{A_{\mathsf{in}} = A_1\}$. The conditional distribution of E' is as described in the proof of Proposition 4.7 and does not depend on q_2. It is then easy to see from the proof of Proposition 4.7 that the conditional distribution of A_{mid} is the output distribution of the following process:

1. For each $i \in [k]$, sample a tuple of three vectors $\left(x^{(3i-2)}, x^{(3i-1)}, x^{(3i)}\right) \sim \mathcal{D}^{(i)}$.
2. Form the $3k$-by-q_2 matrix $X \in \mathbb{F}_2^{3k \times q_2}$ whose i-th row is $x^{(i)}$ for each $i \in [3k]$. Let $\mathcal{M}_P$ denote the resulting distribution of X.
3. Output the q_2 columns of X.

Step 2: tensorization of Hellinger affinity. It now suffices to show that $\mathcal{M}_P$ is close to uniform over $\mathbb{F}_2^{3k \times q_2}$ (the space of $3k$-by-q_2 matrices).

For each $i \in [k]$, by Equation (F.3) we have that the squared Hellinger distance between $\mathcal{D}^{(i)}$ and $\mathcal{D}_0$ is

$$\frac{1}{2} \cdot \mathop{\mathbb{E}}_{X \sim \mathcal{D}_0} \left[\left(\sqrt{\mathcal{D}^{(i)}(X)/\mathcal{D}_0(X)} - 1 \right)^2 \right] \leqslant 2^{4q_1+4q_2-1}\beta^2,$$

and therefore

$$\sum_{X \in (\mathbb{F}_2^{q_2})^3} \sqrt{\mathcal{D}^{(i)}(X)\mathcal{D}_0(X)} = 1 - \frac{1}{2} \sum_{X \in (\mathbb{F}_2^{q_2})^3} \left(\sqrt{\mathcal{D}^{(i)}(X)} - \sqrt{\mathcal{D}_0(X)} \right)^2 \geqslant 1 - 2^{4q_1+4q_2-1}\beta^2.$$

Consider the distributions

$$\prod_{i=1}^{k} \mathcal{D}^{(i)} = \mathcal{M}_P \qquad \text{and} \qquad \prod_{i=1}^{k} \mathcal{D}_0 = \mathcal{D}_0^{\otimes k}.$$

over $\mathbb{F}_2^{3k\times q_2}$. By the tensorization we have

$$\begin{aligned}\sum_{X\in\mathbb{F}_2^{3k\times q_2}} \sqrt{\mathcal{M}_P(X)\mathcal{D}_0^{\otimes k}(X)} &= \prod_{i=1}^{k} \sum_{X\in(\mathbb{F}_2^{q_2})^3} \sqrt{\mathcal{D}^{(i)}(X)\mathcal{D}_0(X)} \\ &\geqslant (1-2^{4q_1+4q_2-1}\beta^2)^k \geqslant 1-2^{4q_1+4q_2-1}\beta^2 k.\end{aligned}$$

Consequently, the squared Hellinger distance between $\mathcal{M}_P$ and $\mathcal{D}_0^{\otimes k}$ is

$$\frac{1}{2}\sum_{X\in\mathbb{F}_2^{3k\times q_2}} \left(\sqrt{\mathcal{M}_P(X)} - \sqrt{\mathcal{D}_0^{\otimes k}(X)}\right)^2 \leqslant 2^{4q_1+4q_2-1}\beta^2 k,$$

and hence by Cauchy-Schwarz,[23] the total variation distance between $\mathcal{M}_P$ and $\mathcal{D}_0^{\otimes k}$ is

$$\frac{1}{2}\sum_{X\in\mathbb{F}_2^{3k\times q_2}} \left|\mathcal{M}_P(X) - \mathcal{D}_0^{\otimes k}(X)\right| \leqslant \sqrt{2}\cdot\sqrt{2^{4q_1+4q_2-1}\beta^2 k} = 2^{2q_1+2q_2}\beta\sqrt{k} \leqslant 2^{2q_1+2q_2}\beta\sqrt{k}. \qquad \square$$

F.2 Covering in the Middle PCP

In the middle PCP, we introduced a new joint distribution $\mathcal{J}_{\mathsf{mid}}(E, q_1, q_2, \beta, \ell)$ based on the "thinning with advice" distribution $\mathcal{J}_{\mathsf{out}}(E, q_1, q_2, \beta)$. Specifically, in addition to the tuple $(E', A_{\mathsf{in}}, A_{\mathsf{mid}})$ sampled in the latter joint distribution, a subspace $U \subseteq \mathbb{F}_2^{E'}$ is generated. Since the way we generated U in $\mathcal{J}_{\mathsf{mid}}(E, q_1, q_2, \beta, \ell)$ can be viewed as sampling "advice vectors" and taking their span, the following proposition follows rather easily from Proposition 4.8.

Proposition 5.4. Let E be a tripled set. Fix a parameter $\beta \in [0, 1)$ and nonnegative integers q_1, q_2, ℓ such that $|E| \geqslant \ell \geqslant q_2$. Fix arbitrary tuples

$$A_1 \in (\mathbb{F}_2^E)^{q_1} \qquad \text{and} \qquad A_2 \in (\mathbb{F}_2^E)^{q_2} \text{ such that } \dim \operatorname{span}(A_2) = q_2.$$

Sample

$$(E', A_{\mathsf{in}}, A_{\mathsf{mid}}, U) \sim \mathcal{J}_{\mathsf{mid}}(E, q_1, q_2, \beta, \ell)$$

conditioned on the event $\{A_{\mathsf{in}} = A_1\} \cap \{A_{\mathsf{mid}} = A_2\}$. Then the resulting marginal distribution of U has total variation distance at most

$$4^{q_1+\ell}\beta\sqrt{|E|/3} + 2^{\ell-|E|}$$

to the uniform distribution over the set $\left\{U \in \mathsf{Gr}(\mathbb{F}_2^E, \ell) \,\middle|\, \operatorname{span}(A_2) \subseteq U\right\}$.

Proof. We write $q = q_1 + q_2$ and $\ell' = \ell - q_2$.

According to Definition 5.1, when $(E', A_{\mathsf{in}}, A_{\mathsf{mid}}, U)$ is sampled from $\mathcal{J}_{\mathsf{mid}}(E, q_1, q_2, \beta, \ell)$, a vector sequence $A' \in (\mathbb{F}_2^{E'})^{\ell'}$ is generated, and then U is taken to be the sum $\operatorname{span}(A_{\mathsf{mid}}) + \operatorname{span}(A')$. Let

$$A_{\mathsf{in}} \uplus A_{\mathsf{mid}} \in (\mathbb{F}_2^{E'})^{q_1+q_2}$$

[23] For any two probability vectors $p, q \in \mathbb{R}^n$, we have $\|p-q\|_1 \leqslant \|\sqrt{p}-\sqrt{q}\|_2 \cdot \|\sqrt{p}+\sqrt{q}\|_2 \leqslant 2\|\sqrt{p}-\sqrt{q}\|_2$.

denote the concatenation of the two vector sequences A_{in} and A_{mid}. Note that before any conditioning, the distribution of $(E', A_{\mathsf{in}} \uplus A_{\mathsf{mid}}, A')$ is identical to $\mathcal{J}_{\mathsf{out}}(E, q, \ell', \beta)$. Therefore, it follows from Proposition 4.8 that after conditioning on $\{A_{\mathsf{in}} = A_1\} \cap \{A_{\mathsf{mid}} = A_2\}$,

$$\text{the distribution of } A' \text{ is } \left(4^{q+\ell'}\beta\sqrt{|E|/3}\right)\text{-close to uniform over } (\mathbb{F}_2^E)^{\ell'}. \tag{F.4}$$

Since $\dim \operatorname{span}(A_2) = q_2$, it is easy to see that

$$\mathop{\mathbb{P}}_{A \in (\mathbb{F}_2^E)^{\ell'}} \left[\dim\big(\operatorname{span}(A_2) + \operatorname{span}(A)\big) = \ell\right] \geqslant 1 - 2^{\ell - |E|}. \tag{F.5}$$

Furthermore, if $A \in (\mathbb{F}_2^E)^{\ell'}$ is a uniformly random vector sequence, then the distribution of $\big(\operatorname{span}(A_2) + \operatorname{span}(A)\big)$ conditioned on the event

$$\left\{\dim\big(\operatorname{span}(A_2) + \operatorname{span}(A)\big) = \ell\right\}$$

is uniform over the set $\left\{U \in \mathsf{Gr}(\mathbb{F}_2^E, \ell) \;\middle|\; \operatorname{span}(A_2) \subseteq U\right\}$. Therefore, it follows from Equations (F.4) and (F.5) that conditioned on $\{A_{\mathsf{in}} = A_1\} \cap \{A_{\mathsf{mid}} = A_2\}$, the distribution of $U = \big(\operatorname{span}(A_{\mathsf{mid}}) + \operatorname{span}(A')\big)$ has total variation distance at most

$$4^{q+\ell'}\beta\sqrt{|E|/3} + 2^{\ell - |E|} = 4^{q_1+\ell}\beta\sqrt{|E|/3} + 2^{\ell - |E|}$$

to the uniform distribution over $\left\{U \in \mathsf{Gr}(\mathbb{F}_2^E, \ell) \;\middle|\; \operatorname{span}(A_2) \subseteq U\right\}$. □

F.3 Covering in the Inner PCP

We finish by stating the following general probabilistic fact that establishes the covering property achieved by the lazy parallel repetition step. It is essentially the same as Claim 2.5, but stated more generally in a way that is suitable for us.

Proposition F.1. Fix positive integers k, t and finite sets $\mathscr{A}_1, \ldots, \mathscr{A}_t$ with $|\mathscr{A}_j| = k$ for each $j \in [t]$. Suppose that for each $j \in [t]$ there is a probability distribution μ_j over $\mathscr{A}_j$ and a surjective map $\pi_j : \mathscr{A}_j \to \mathscr{B}_j$ onto some finite set $\mathscr{B}_j$. Let $\pi : \prod_{j=1}^t \mathscr{A}_j \to \prod_{j=1}^t \mathscr{B}_j$ be the coordinate-wise map defined by

$$\pi(a_1, \ldots, a_t) = (\pi_1(a_1), \ldots, \pi_t(a_t)) \qquad \text{for all } (a_1, \ldots, a_t) \in \textstyle\prod_{j=1}^t \mathscr{A}_j.$$

Define a probability distribution μ on $\prod_{j=1}^t \mathscr{A}_j$ to be the output distribution of the following process:

1. Sample a uniformly random index $c \in [t]$.
2. Sample $a_c \sim \mu_c$. For each $j \in [t] \setminus \{c\}$, sample a_j uniformly at random from $\mathscr{A}_j$.
3. Output $(a_1, \ldots, a_t)$.

For each $b \in \prod_{j=1}^t \mathscr{B}_j$, let $\operatorname{dist}(b)$ be the total variation distance between

(1) the distribution of $a \sim \mu$ conditioned on the event $\{\pi(a) = b\}$ (if $\mathbb{P}_{a\sim\mu}[\pi(a) = b] = 0$, let $\operatorname{dist}(b) = 1$), and

(2) the uniform distribution over the subset $\pi^{-1}(\{b\}) \subseteq \prod_{j=1}^t \mathscr{A}_j$.

Then we have

$$\mathop{\mathbb{E}}_{a\sim\mu}\left[\mathrm{dist}(\pi(a))\right] \leqslant \sqrt{\frac{k}{4t}}.$$

Proof. For each $j \in [t]$, let $f_j : \mathscr{A}_j \to [0,+\infty)$ be the probability density function of μ_j; that is, let $f_j(a) = k \cdot \mu_j(\{a\})$ for all $a \in \mathscr{A}_j$. It is then easy to see that the probability density function of μ is given by

$$f(a_1,\ldots,a_t) := \frac{1}{t}\sum_{j=1}^{t} f_j(a_j), \qquad \text{for all } (a_1,\ldots,a_t) \in \textstyle\prod_{j=1}^{t} \mathscr{A}_j.$$

For each $j \in [t]$, let $g_j : \mathscr{B}_j \to [0,+\infty)$ be defined by

$$g_j(b_j) = \begin{cases} \mathbb{E}_{a_j\in\pi_j^{-1}(\{b_j\})}\left[f_j(a_j)\right], & \text{if } \pi_j^{-1}(\{b_j\}) \neq \emptyset, \\ 0, & \text{otherwise.} \end{cases}$$

For each $b \in \prod_{j=1}^{t} \mathscr{B}_j$, let

$$g(b) = \frac{1}{t}\sum_{j=1}^{t} g_j(b_j) = \mathop{\mathbb{E}}_{a\in\pi^{-1}(\{b\})}\left[f(a)\right].$$

For any $b \in \prod_{j=1}^{t} \mathscr{B}_j$ that lies in the support of g, the total variation distance between the distribution of $a \sim \mu$ conditioned on $\{\pi(a) = b\}$ and the uniform distribution over $\pi^{-1}(\{b\})$ is

$$\begin{aligned}
\frac{1}{2}\cdot\mathop{\mathbb{E}}_{a\in\pi^{-1}(\{b\})}\left[\left|\frac{f(a)}{g(b)}-1\right|\right] &\leqslant \frac{1}{2g(b)}\sqrt{\mathop{\mathrm{Var}}_{a\in\pi^{-1}(\{b\})}\left[f(a)\right]} && \text{(by Cauchy-Schwarz)} \\
&= \frac{1}{2t\cdot g(b)}\sqrt{\sum_{j=1}^{t}\mathop{\mathrm{Var}}_{a_j\in\pi_j^{-1}(b_j)}\left[f_j(a_j)\right]} \\
& && \text{(using the fact that } \pi^{-1}(\{b\}) \text{ is a product set)} \\
&\leqslant \frac{1}{2t\cdot g(b)}\sqrt{\sum_{j=1}^{t} k\cdot\mathop{\mathbb{E}}_{a_j\in\pi^{-1}(b_j)}\left[f_j(a_j)\right]} && (f_j \text{ is upper-bounded by } k) \\
&\leqslant \frac{1}{2}\sqrt{\frac{k}{t\cdot g(b)}}. && \text{(by the definition of } g(b))
\end{aligned}$$

Let ν be the distribution of $\pi(a)$ when a is sampled uniformly from $\prod_{j=1}^{t}\mathscr{A}_j$. We then have

$$\mathop{\mathbb{E}}_{a\sim\mu}\left[\mathrm{dist}(\pi(a))^2\right] = \mathop{\mathbb{E}}_{b\sim\nu}\left[g(b)\cdot\mathrm{dist}(b)^2\right] \leqslant \mathop{\mathbb{E}}_{b\sim\nu}\left[\frac{k}{4t}\right] = \frac{k}{4t},$$

and hence $\mathbb{E}_{a\sim\mu}\left[\mathrm{dist}(\pi(a))\right] \leqslant \sqrt{k/(4t)}$ by Cauchy-Schwarz. □

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